Material Characterization Report

NanoVNA UTN Toolkit

2026-10-23 18:55:57

Measurement Details:

- **Sample:** Muestra2-236c
- **Technique:** Open-ended coaxial probe (liquids)
- **Temperature:** 23.0 °C
- **Reference Liquids:** Distilled Water / Isopropyl Alcohol (IPA)
- **Probe:** 3 mm probe
- **Frequency Range:** 0.001–6.000 GHz, 1280 pts
- **Data sources:**
  - **Open:** imported
  - **Short:** imported (1 spike(s) replaced in the computation: 4885 MHz)
  - **Ref 1:** imported
  - **Ref 2:** imported
  - **DUT:** imported

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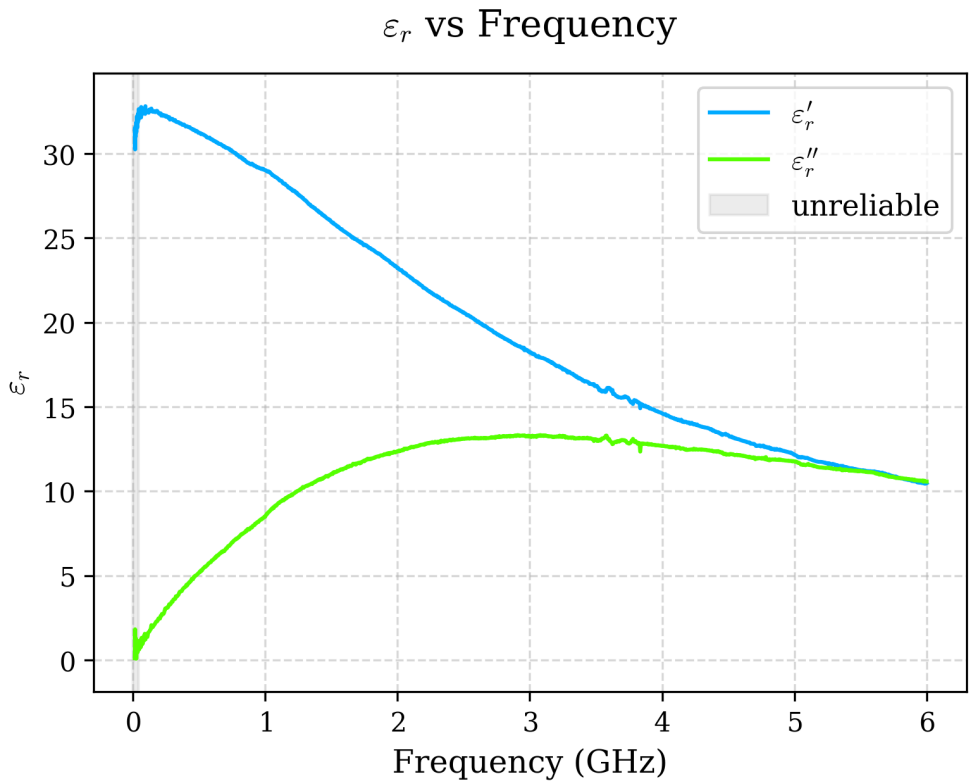
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1 Measurement Results

1.1 Complex Permittivity $\varepsilon_r(f)$



Points without a solution: 13 of 1280 (left blank in the plot and shown as — in the table).

Frequency (GHz)	ε_r'	ε_r''	$\tan\delta$
0.001	—	—	—
0.002	—	—	—
0.003	—	—	—
0.004	—	—	—
0.005	—	—	—
0.006	—	—	—
0.007	—	—	—
0.008	—	—	—
0.009	—	—	—
0.010	—	—	—
0.011	—	—	—
0.012	—	—	—
0.013	—	—	—
0.014	31.4386	0.1103	0.0035
0.015	31.5220	0.5047	0.0160
0.016	31.1325	1.0142	0.0326
0.017	30.2447	1.8208	0.0602

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Frequency (GHz)	ε_r'	ε_r''	$\tan\delta$
0.018	31.1010	1.2945	0.0416
0.019	31.4529	1.2665	0.0403
0.020	30.9690	0.4729	0.0153
0.021	31.2593	0.4103	0.0131
0.022	31.3820	1.3018	0.0415
0.023	31.6545	0.9843	0.0311
0.024	31.3996	0.7699	0.0245
0.025	31.8495	1.0373	0.0326
0.026	31.2646	0.0860	0.0027
0.027	31.6300	0.7503	0.0237
0.028	31.8010	0.5042	0.0159
0.029	31.6736	0.7091	0.0224
0.030	31.7636	0.6946	0.0219
0.031	32.2012	0.5502	0.0171
0.032	32.0168	0.4703	0.0147
0.033	31.6863	0.8952	0.0283
0.034	31.5378	1.0503	0.0333
0.035	32.1936	0.4463	0.0139
0.036	32.1581	0.7124	0.0222
0.037	31.8520	0.9359	0.0294
0.038	32.3389	0.9790	0.0303
0.039	31.9310	0.8718	0.0273
0.040	31.8998	0.7010	0.0220
0.041	32.3921	0.7842	0.0242
0.042	32.0550	0.7349	0.0229
0.043	32.1227	0.8280	0.0258
0.044	32.1910	1.0254	0.0319
0.045	32.6356	1.1177	0.0342
0.046	32.0492	0.9982	0.0311
0.047	32.0815	0.7301	0.0228
0.048	32.4381	0.6301	0.0194
0.049	32.5393	0.8891	0.0273
0.050	32.2006	0.7269	0.0226
0.051	32.2217	1.1458	0.0356
0.052	32.3773	0.7389	0.0228
0.053	32.5282	0.9653	0.0297
0.054	32.2906	0.7354	0.0228
0.055	32.3235	1.0590	0.0328
0.056	32.2644	0.7845	0.0243
0.057	32.3591	0.9569	0.0296
0.058	32.2805	0.8293	0.0257
0.059	32.2168	0.9869	0.0306
0.060	32.6009	0.7716	0.0237
0.061	32.3401	1.1251	0.0348
0.062	32.1369	1.1147	0.0347
0.063	32.7571	0.8819	0.0269

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Frequency (GHz)	ε_r'	ε_r''	$\tan\delta$
0.064	32.5016	0.9617	0.0296
0.065	32.2730	1.2089	0.0375
0.066	32.5822	1.3163	0.0404
0.067	32.5579	0.9025	0.0277
0.068	32.4434	1.1318	0.0349
0.069	32.4388	1.0719	0.0330
0.070	32.3624	1.1359	0.0351
0.071	32.4156	1.3035	0.0402
0.072	32.6526	1.0700	0.0328
0.073	32.5206	1.1093	0.0341
0.074	32.5961	1.2798	0.0393
0.075	32.6391	0.9749	0.0299
0.076	32.5665	1.1914	0.0366
0.077	32.5445	1.2575	0.0386
0.078	32.6203	1.3567	0.0416
0.079	32.5841	1.1202	0.0344
0.080	32.5427	1.2851	0.0395
0.081	32.6689	1.1301	0.0346
0.082	32.5661	1.1793	0.0362
0.083	32.6513	1.2645	0.0387
0.084	32.4737	1.3665	0.0421
0.085	32.4550	1.4328	0.0441
0.086	32.4133	1.4199	0.0438
0.087	32.4634	1.2331	0.0380
0.088	32.6593	1.3263	0.0406
0.089	32.5908	1.3276	0.0407
0.090	32.4315	1.5564	0.0480
0.091	32.5454	1.3765	0.0423
0.092	32.5950	1.2654	0.0388
0.093	32.3474	1.2134	0.0375
0.094	32.5742	1.4754	0.0453
0.095	32.8052	1.2319	0.0376
0.096	32.4346	1.4033	0.0433
0.097	32.5322	1.5099	0.0464
0.098	32.5068	1.3781	0.0424
0.099	32.5903	1.4930	0.0458
0.100	32.5740	1.4334	0.0440
0.105	32.5435	1.5441	0.0474
0.110	32.5380	1.3687	0.0421
0.115	32.5426	1.6108	0.0495
0.120	32.4204	1.6585	0.0512
0.125	32.6100	1.6959	0.0520
0.130	32.5215	1.7749	0.0546
0.135	32.6282	1.8261	0.0560
0.140	32.6650	2.0682	0.0633
0.145	32.4927	1.9326	0.0595

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Frequency (GHz)	ε_r'	ε_r''	$\tan\delta$
0.150	32.4772	1.9531	0.0601
0.155	32.5142	2.0085	0.0618
0.160	32.4675	2.0731	0.0639
0.165	32.5135	2.1222	0.0653
0.170	32.4532	2.1903	0.0675
0.175	32.4714	2.2696	0.0699
0.180	32.5316	2.2943	0.0705
0.185	32.5195	2.2990	0.0707
0.190	32.4336	2.4059	0.0742
0.195	32.3955	2.4733	0.0763
0.200	32.3502	2.4735	0.0765
0.205	32.3878	2.4591	0.0759
0.210	32.3821	2.6160	0.0808
0.215	32.2471	2.6181	0.0812
0.220	32.3144	2.7348	0.0846
0.225	32.2414	2.7253	0.0845
0.230	32.2364	2.7233	0.0845
0.235	32.2703	2.7832	0.0862
0.240	32.2620	2.9156	0.0904
0.245	32.2100	3.0552	0.0949
0.250	32.1789	3.0731	0.0955
0.255	32.2319	3.0704	0.0953
0.260	32.1696	3.1248	0.0971
0.265	32.1519	3.1171	0.0969
0.270	32.1017	3.2160	0.1002
0.275	32.1574	3.3064	0.1028
0.280	32.0113	3.2848	0.1026
0.285	32.0731	3.3113	0.1032
0.290	32.0576	3.3953	0.1059
0.295	32.0265	3.4663	0.1082
0.300	32.0537	3.4527	0.1077
0.305	31.9857	3.5626	0.1114
0.310	31.9505	3.6056	0.1128
0.315	31.9885	3.6047	0.1127
0.320	31.9798	3.6284	0.1135
0.325	31.8887	3.7774	0.1185
0.330	31.8723	3.7565	0.1179
0.335	31.8621	3.7833	0.1187
0.340	31.8522	3.8283	0.1202
0.345	31.8636	3.8720	0.1215
0.350	31.8271	3.9742	0.1249
0.355	31.7805	4.0538	0.1276
0.360	31.7973	4.0557	0.1275
0.365	31.7764	4.0724	0.1282
0.370	31.7409	4.0906	0.1289
0.375	31.7001	4.1781	0.1318

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Frequency (GHz)	ε_r'	ε_r''	$\tan\delta$
0.380	31.7167	4.2318	0.1334
0.385	31.7232	4.2660	0.1345
0.390	31.6934	4.3125	0.1361
0.395	31.6413	4.3270	0.1368
0.400	31.6008	4.3570	0.1379
0.405	31.6176	4.4622	0.1411
0.410	31.6176	4.4189	0.1398
0.415	31.5923	4.4929	0.1422
0.420	31.5658	4.5570	0.1444
0.425	31.5686	4.5726	0.1448
0.430	31.5619	4.6411	0.1470
0.435	31.5148	4.6495	0.1475
0.440	31.4872	4.7004	0.1493
0.445	31.4607	4.8081	0.1528
0.450	31.4585	4.8088	0.1529
0.455	31.4041	4.7926	0.1526
0.460	31.4220	4.9201	0.1566
0.465	31.3944	4.9246	0.1569
0.470	31.3769	4.9908	0.1591
0.475	31.3136	5.0042	0.1598
0.480	31.3487	5.0304	0.1605
0.485	31.3087	5.0604	0.1616
0.490	31.2999	5.0994	0.1629
0.495	31.2289	5.1451	0.1648
0.500	31.2311	5.1706	0.1656
0.505	31.2289	5.2337	0.1676
0.510	31.1974	5.2306	0.1677
0.515	31.1514	5.3111	0.1705
0.520	31.1747	5.2852	0.1695
0.525	31.1492	5.3497	0.1717
0.530	31.1349	5.4153	0.1739
0.535	31.1144	5.4485	0.1751
0.540	31.0976	5.4894	0.1765
0.545	31.0722	5.5330	0.1781
0.550	31.0317	5.5257	0.1781
0.555	31.0149	5.5748	0.1797
0.560	30.9542	5.6046	0.1811
0.565	30.9459	5.6481	0.1825
0.570	30.9297	5.7060	0.1845
0.575	30.9403	5.7420	0.1856
0.580	30.9461	5.7659	0.1863
0.585	30.8743	5.7833	0.1873
0.590	30.8659	5.8068	0.1881
0.595	30.8345	5.8861	0.1909
0.600	30.8366	5.9132	0.1918
0.605	30.7935	5.9310	0.1926

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Frequency (GHz)	ε_r'	ε_r''	$\tan\delta$
0.610	30.7583	5.9676	0.1940
0.615	30.7602	6.0202	0.1957
0.620	30.7331	6.0285	0.1962
0.625	30.6983	6.1050	0.1989
0.630	30.6959	6.1001	0.1987
0.635	30.6480	6.1884	0.2019
0.640	30.6510	6.1914	0.2020
0.645	30.5997	6.2008	0.2026
0.650	30.5825	6.2554	0.2045
0.655	30.5629	6.3035	0.2062
0.660	30.5661	6.3137	0.2066
0.665	30.5565	6.3563	0.2080
0.670	30.5108	6.4078	0.2100
0.675	30.4980	6.3879	0.2095
0.680	30.4633	6.4578	0.2120
0.685	30.4178	6.4709	0.2127
0.690	30.4390	6.4427	0.2117
0.695	30.3965	6.4982	0.2138
0.700	30.3498	6.5562	0.2160
0.705	30.3615	6.6171	0.2179
0.710	30.3382	6.6211	0.2182
0.715	30.3180	6.6792	0.2203
0.720	30.3143	6.7485	0.2226
0.725	30.2893	6.7657	0.2234
0.730	30.2783	6.7715	0.2236
0.735	30.1987	6.8301	0.2262
0.740	30.2178	6.8721	0.2274
0.745	30.1985	6.9212	0.2292
0.750	30.1195	6.9750	0.2316
0.755	30.1087	6.9844	0.2320
0.760	30.0857	7.0437	0.2341
0.765	30.0628	7.0867	0.2357
0.770	30.0320	7.1140	0.2369
0.775	30.0342	7.1281	0.2373
0.780	29.9782	7.1898	0.2398
0.785	29.9212	7.1816	0.2400
0.790	29.9375	7.2104	0.2408
0.795	29.9008	7.2573	0.2427
0.800	29.8461	7.2696	0.2436
0.805	29.7958	7.3251	0.2458
0.810	29.8012	7.3589	0.2469
0.815	29.8096	7.4174	0.2488
0.820	29.7523	7.4528	0.2505
0.825	29.7089	7.4488	0.2507
0.830	29.7266	7.4968	0.2522
0.835	29.6443	7.5098	0.2533

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Frequency (GHz)	ε_r'	ε_r''	$\tan\delta$
0.840	29.6226	7.5362	0.2544
0.845	29.6347	7.5652	0.2553
0.850	29.5677	7.5941	0.2568
0.855	29.5445	7.6671	0.2595
0.860	29.5001	7.6432	0.2591
0.865	29.4650	7.7386	0.2626
0.870	29.4579	7.7712	0.2638
0.875	29.4529	7.7410	0.2628
0.880	29.4102	7.7329	0.2629
0.885	29.3811	7.7824	0.2649
0.890	29.3881	7.8447	0.2669
0.895	29.3517	7.8415	0.2672
0.900	29.4071	7.8498	0.2669
0.905	29.3744	7.9373	0.2702
0.910	29.3436	7.9596	0.2713
0.915	29.2806	7.9898	0.2729
0.920	29.2746	7.9964	0.2732
0.925	29.2810	8.0846	0.2761
0.930	29.2531	8.0944	0.2767
0.935	29.1961	8.1152	0.2780
0.940	29.2043	8.1707	0.2798
0.945	29.1676	8.1800	0.2804
0.950	29.1790	8.1807	0.2804
0.955	29.1535	8.2557	0.2832
0.960	29.1202	8.2759	0.2842
0.965	29.1138	8.3210	0.2858
0.970	29.1181	8.3351	0.2863
0.975	29.0767	8.3373	0.2867
0.980	29.1063	8.3983	0.2885
0.985	29.0966	8.4161	0.2892
0.990	29.0433	8.4482	0.2909
0.995	29.0339	8.4975	0.2927
1.000	29.0250	8.4858	0.2924
1.005	28.9950	8.5690	0.2955
1.010	28.9889	8.6385	0.2980
1.015	28.9364	8.6496	0.2989
1.020	28.9326	8.7186	0.3013
1.025	28.9525	8.7583	0.3025
1.030	28.9002	8.7869	0.3040
1.035	28.8982	8.8318	0.3056
1.040	28.8537	8.8811	0.3078
1.045	28.8123	8.9226	0.3097
1.050	28.7756	8.9651	0.3116
1.055	28.7183	8.9554	0.3118
1.060	28.7118	9.0416	0.3149
1.065	28.6809	9.0081	0.3141

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Frequency (GHz)	ε_r'	ε_r''	$\tan\delta$
1.070	28.6627	9.0835	0.3169
1.075	28.6155	9.1418	0.3195
1.080	28.6071	9.1383	0.3194
1.085	28.5869	9.1509	0.3201
1.090	28.5621	9.2209	0.3228
1.095	28.5165	9.2569	0.3246
1.100	28.4943	9.3040	0.3265
1.105	28.4543	9.2999	0.3268
1.110	28.4016	9.3504	0.3292
1.115	28.3977	9.3838	0.3304
1.120	28.3918	9.3960	0.3309
1.125	28.3628	9.4207	0.3321
1.130	28.3010	9.4692	0.3346
1.135	28.2376	9.4784	0.3357
1.140	28.2244	9.4570	0.3351
1.145	28.1953	9.5339	0.3381
1.150	28.2149	9.5307	0.3378
1.155	28.1517	9.5976	0.3409
1.160	28.1163	9.5786	0.3407
1.165	28.0943	9.6430	0.3432
1.170	28.0596	9.6402	0.3436
1.175	27.9840	9.6766	0.3458
1.180	27.9772	9.7158	0.3473
1.185	27.9731	9.7118	0.3472
1.190	27.9173	9.7183	0.3481
1.195	27.9070	9.7665	0.3500
1.200	27.8915	9.8056	0.3516
1.205	27.8114	9.7703	0.3513
1.210	27.8283	9.8276	0.3532
1.215	27.7745	9.8591	0.3550
1.220	27.7997	9.8751	0.3552
1.225	27.7586	9.9090	0.3570
1.230	27.7358	9.9596	0.3591
1.235	27.6575	9.9558	0.3600
1.240	27.6809	9.9917	0.3610
1.245	27.6306	10.0191	0.3626
1.250	27.5586	10.0332	0.3641
1.255	27.5354	10.0505	0.3650
1.260	27.5332	10.0742	0.3659
1.265	27.4751	10.1188	0.3683
1.270	27.4579	10.1363	0.3692
1.275	27.4365	10.1625	0.3704
1.280	27.3842	10.1851	0.3719
1.285	27.3772	10.2547	0.3746
1.290	27.3189	10.2030	0.3735
1.295	27.2718	10.2478	0.3758

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Frequency (GHz)	ε_r'	ε_r''	$\tan\delta$
1.300	27.2518	10.2704	0.3769
1.305	27.2291	10.2975	0.3782
1.310	27.1833	10.3140	0.3794
1.315	27.1535	10.3319	0.3805
1.320	27.0944	10.3203	0.3809
1.325	27.0603	10.3541	0.3826
1.330	27.0032	10.3716	0.3841
1.335	26.9893	10.3898	0.3850
1.340	26.9771	10.4103	0.3859
1.345	26.9058	10.4233	0.3874
1.350	26.8845	10.4544	0.3889
1.355	26.8479	10.4257	0.3883
1.360	26.8510	10.4881	0.3906
1.365	26.8041	10.5019	0.3918
1.370	26.7783	10.5203	0.3929
1.375	26.7755	10.5592	0.3944
1.380	26.7144	10.5568	0.3952
1.385	26.6856	10.5912	0.3969
1.390	26.6456	10.6108	0.3982
1.395	26.6105	10.6554	0.4004
1.400	26.6160	10.6681	0.4008
1.405	26.5376	10.6709	0.4021
1.410	26.5401	10.7111	0.4036
1.415	26.4904	10.6796	0.4032
1.420	26.4784	10.7349	0.4054
1.425	26.4584	10.7109	0.4048
1.430	26.3790	10.7554	0.4077
1.435	26.3781	10.7409	0.4072
1.440	26.3375	10.7940	0.4098
1.445	26.3028	10.8020	0.4107
1.450	26.3021	10.8403	0.4121
1.455	26.2520	10.8615	0.4137
1.460	26.2122	10.9056	0.4160
1.465	26.1837	10.9043	0.4165
1.470	26.1476	10.9003	0.4169
1.475	26.1242	10.8644	0.4159
1.480	26.1140	10.9366	0.4188
1.485	26.0708	10.9449	0.4198
1.490	26.0655	10.9703	0.4209
1.495	26.0125	10.9862	0.4223
1.500	25.9546	11.0211	0.4246
1.505	25.9320	11.0510	0.4262
1.510	25.8890	11.0200	0.4257
1.515	25.9075	11.1152	0.4290
1.520	25.8311	11.0998	0.4297
1.525	25.8097	11.1420	0.4317

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Frequency (GHz)	ε_r'	ε_r''	$\tan\delta$
1.530	25.7980	11.1245	0.4312
1.535	25.7666	11.1374	0.4322
1.540	25.7222	11.1481	0.4334
1.545	25.6740	11.2092	0.4366
1.550	25.6905	11.2106	0.4364
1.555	25.6519	11.2002	0.4366
1.560	25.6124	11.2413	0.4389
1.565	25.5660	11.2652	0.4406
1.570	25.5400	11.2643	0.4410
1.575	25.5055	11.3052	0.4432
1.580	25.5154	11.2942	0.4426
1.585	25.4647	11.2843	0.4431
1.590	25.4228	11.3312	0.4457
1.595	25.4038	11.3394	0.4464
1.600	25.3760	11.3553	0.4475
1.605	25.3622	11.3803	0.4487
1.610	25.3278	11.3634	0.4487
1.615	25.2969	11.4105	0.4511
1.620	25.2770	11.4555	0.4532
1.625	25.2448	11.4792	0.4547
1.630	25.2053	11.4418	0.4539
1.635	25.1947	11.4873	0.4559
1.640	25.1754	11.5132	0.4573
1.645	25.1260	11.5109	0.4581
1.650	25.0801	11.4912	0.4582
1.655	25.0772	11.5178	0.4593
1.660	25.0033	11.5722	0.4628
1.665	25.0323	11.6292	0.4646
1.670	24.9865	11.6207	0.4651
1.675	24.9448	11.6079	0.4653
1.680	24.9569	11.6364	0.4663
1.685	24.9182	11.5754	0.4645
1.690	24.8925	11.6449	0.4678
1.695	24.8847	11.6397	0.4677
1.700	24.8411	11.6559	0.4692
1.705	24.8092	11.6818	0.4709
1.710	24.8005	11.6816	0.4710
1.715	24.8009	11.6472	0.4696
1.720	24.7204	11.7024	0.4734
1.725	24.7082	11.6821	0.4728
1.730	24.7342	11.7358	0.4745
1.735	24.6681	11.7540	0.4765
1.740	24.6358	11.7760	0.4780
1.745	24.5681	11.8022	0.4804
1.750	24.6123	11.7699	0.4782
1.755	24.5969	11.8348	0.4812

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Frequency (GHz)	ε_r'	ε_r''	$\tan\delta$
1.760	24.5657	11.8370	0.4818
1.765	24.5666	11.8702	0.4832
1.770	24.5072	11.8752	0.4846
1.775	24.5000	11.9294	0.4869
1.780	24.4715	11.8870	0.4858
1.785	24.4084	11.9055	0.4878
1.790	24.4136	11.8981	0.4874
1.795	24.3959	11.9318	0.4891
1.800	24.3667	11.9094	0.4888
1.805	24.3201	11.9422	0.4910
1.810	24.3179	11.9440	0.4912
1.815	24.2943	11.9561	0.4921
1.820	24.2716	11.9681	0.4931
1.825	24.2460	11.9928	0.4946
1.830	24.2001	11.9902	0.4955
1.835	24.1715	12.0307	0.4977
1.840	24.1461	12.0380	0.4986
1.845	24.1254	12.0528	0.4996
1.850	24.0951	12.0590	0.5005
1.855	24.0919	12.0647	0.5008
1.860	24.0546	12.0503	0.5010
1.865	24.0552	12.0907	0.5026
1.870	24.0299	12.1078	0.5039
1.875	23.9801	12.1232	0.5056
1.880	23.9562	12.1250	0.5061
1.885	23.9066	12.1203	0.5070
1.890	23.8971	12.1320	0.5077
1.895	23.8517	12.1487	0.5093
1.900	23.8120	12.1951	0.5121
1.905	23.8085	12.1783	0.5115
1.910	23.7760	12.1791	0.5122
1.915	23.7324	12.2156	0.5147
1.920	23.7194	12.2031	0.5145
1.925	23.6921	12.2460	0.5169
1.930	23.6741	12.2040	0.5155
1.935	23.6528	12.2504	0.5179
1.940	23.6069	12.1945	0.5166
1.945	23.5411	12.2707	0.5212
1.950	23.5187	12.2582	0.5212
1.955	23.4900	12.2810	0.5228
1.960	23.4870	12.2683	0.5223
1.965	23.4300	12.2788	0.5241
1.970	23.4171	12.2862	0.5247
1.975	23.3789	12.3436	0.5280
1.980	23.3335	12.3035	0.5273
1.985	23.3529	12.2998	0.5267

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Frequency (GHz)	ε_r'	ε_r''	$\tan\delta$
1.990	23.2997	12.3276	0.5291
1.995	23.2628	12.3467	0.5307
2.000	23.2064	12.3673	0.5329
2.005	23.2277	12.3593	0.5321
2.010	23.1655	12.4081	0.5356
2.015	23.1382	12.3664	0.5345
2.020	23.1049	12.3830	0.5359
2.025	23.0945	12.4149	0.5376
2.030	23.1001	12.4095	0.5372
2.035	23.0374	12.4473	0.5403
2.040	23.0183	12.4472	0.5408
2.045	22.9397	12.4301	0.5419
2.050	22.9478	12.5205	0.5456
2.055	22.9472	12.5014	0.5448
2.060	22.8759	12.5183	0.5472
2.065	22.8910	12.5317	0.5475
2.070	22.8276	12.5009	0.5476
2.075	22.7845	12.5047	0.5488
2.080	22.7627	12.5312	0.5505
2.085	22.7419	12.5518	0.5519
2.090	22.7176	12.5651	0.5531
2.095	22.6889	12.5665	0.5539
2.100	22.6315	12.6172	0.5575
2.105	22.6203	12.6021	0.5571
2.110	22.6126	12.6278	0.5584
2.115	22.5573	12.6157	0.5593
2.120	22.5184	12.6013	0.5596
2.125	22.4826	12.6319	0.5619
2.130	22.4667	12.6340	0.5623
2.135	22.4502	12.6564	0.5638
2.140	22.4311	12.6657	0.5646
2.145	22.3829	12.6574	0.5655
2.150	22.3899	12.6729	0.5660
2.155	22.3363	12.6832	0.5678
2.160	22.2887	12.6906	0.5694
2.165	22.2910	12.7218	0.5707
2.170	22.2341	12.7223	0.5722
2.175	22.2235	12.7527	0.5738
2.180	22.1903	12.7526	0.5747
2.185	22.1719	12.7622	0.5756
2.190	22.1296	12.7679	0.5770
2.195	22.0536	12.7588	0.5785
2.200	22.0441	12.7721	0.5794
2.205	22.0399	12.7592	0.5789
2.210	22.0089	12.8208	0.5825
2.215	21.9868	12.7996	0.5822

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Frequency (GHz)	ε_r'	ε_r''	$\tan\delta$
2.220	21.9633	12.8129	0.5834
2.225	21.9158	12.8491	0.5863
2.230	21.8800	12.8482	0.5872
2.235	21.8632	12.8571	0.5881
2.240	21.8537	12.8379	0.5874
2.245	21.7951	12.8749	0.5907
2.250	21.7896	12.8650	0.5904
2.255	21.7800	12.8696	0.5909
2.260	21.7192	12.8975	0.5938
2.265	21.7009	12.9104	0.5949
2.270	21.6970	12.8729	0.5933
2.275	21.6312	12.8866	0.5957
2.280	21.6245	12.8836	0.5958
2.285	21.6007	12.9319	0.5987
2.290	21.5706	12.9157	0.5988
2.295	21.5913	12.9057	0.5977
2.300	21.5543	12.8791	0.5975
2.305	21.4861	12.9355	0.6020
2.310	21.5044	12.8752	0.5987
2.315	21.4953	12.9322	0.6016
2.320	21.4356	12.9506	0.6042
2.325	21.4008	12.9006	0.6028
2.330	21.3837	12.9982	0.6079
2.335	21.3536	12.9704	0.6074
2.340	21.3775	12.9464	0.6056
2.345	21.2927	12.9740	0.6093
2.350	21.2702	12.9960	0.6110
2.355	21.2536	12.9403	0.6089
2.360	21.2359	13.0020	0.6123
2.365	21.2093	13.0105	0.6134
2.370	21.1885	12.9817	0.6127
2.375	21.1905	12.9624	0.6117
2.380	21.1549	13.0168	0.6153
2.385	21.1068	13.0109	0.6164
2.390	21.1121	13.0286	0.6171
2.395	21.0411	13.0522	0.6203
2.400	21.0105	13.0376	0.6205
2.405	21.0072	13.0053	0.6191
2.410	21.0019	13.0404	0.6209
2.415	20.9515	13.0404	0.6224
2.420	20.9304	13.0726	0.6246
2.425	20.9064	13.0344	0.6235
2.430	20.8936	13.0336	0.6238
2.435	20.8949	13.0444	0.6243
2.440	20.8522	13.0473	0.6257
2.445	20.8295	13.0654	0.6273

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Frequency (GHz)	ε_r'	ε_r''	$\tan\delta$
2.450	20.8221	13.0701	0.6277
2.455	20.7845	13.0589	0.6283
2.460	20.7668	13.0723	0.6295
2.465	20.7495	13.1030	0.6315
2.470	20.7102	13.0685	0.6310
2.475	20.6986	13.0751	0.6317
2.480	20.6647	13.0869	0.6333
2.485	20.6545	13.1112	0.6348
2.490	20.6235	13.0973	0.6351
2.495	20.5837	13.1080	0.6368
2.500	20.5968	13.1138	0.6367
2.505	20.5590	13.1008	0.6372
2.510	20.5361	13.1146	0.6386
2.515	20.5071	13.1142	0.6395
2.520	20.4825	13.1363	0.6413
2.525	20.4284	13.1173	0.6421
2.530	20.4262	13.1274	0.6427
2.535	20.4340	13.1485	0.6435
2.540	20.3985	13.1214	0.6433
2.545	20.3775	13.1142	0.6436
2.550	20.3356	13.1286	0.6456
2.555	20.3372	13.1340	0.6458
2.560	20.2941	13.1486	0.6479
2.565	20.2684	13.1400	0.6483
2.570	20.2540	13.1396	0.6487
2.575	20.2328	13.1528	0.6501
2.580	20.1858	13.1318	0.6505
2.585	20.1519	13.1745	0.6538
2.590	20.1459	13.1270	0.6516
2.595	20.1118	13.1550	0.6541
2.600	20.0736	13.1371	0.6544
2.605	20.0266	13.1255	0.6554
2.610	20.0567	13.1499	0.6556
2.615	20.0288	13.1202	0.6551
2.620	19.9726	13.1344	0.6576
2.625	19.9387	13.1303	0.6585
2.630	19.9365	13.1574	0.6600
2.635	19.9314	13.1389	0.6592
2.640	19.8617	13.1442	0.6618
2.645	19.8626	13.1452	0.6618
2.650	19.8135	13.1559	0.6640
2.655	19.8189	13.1354	0.6628
2.660	19.7673	13.1524	0.6654
2.665	19.7895	13.1563	0.6648
2.670	19.7526	13.1282	0.6646
2.675	19.7153	13.1446	0.6667

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Frequency (GHz)	ε_r'	ε_r''	$\tan\delta$
2.680	19.6960	13.1448	0.6674
2.685	19.6558	13.1423	0.6686
2.690	19.6224	13.1485	0.6701
2.695	19.6071	13.1598	0.6712
2.700	19.5821	13.1696	0.6725
2.705	19.5697	13.1655	0.6728
2.710	19.5297	13.2045	0.6761
2.715	19.5320	13.1755	0.6746
2.720	19.4888	13.1887	0.6767
2.725	19.4404	13.1746	0.6777
2.730	19.4008	13.1800	0.6794
2.735	19.3871	13.2121	0.6815
2.740	19.3949	13.1603	0.6785
2.745	19.3482	13.2258	0.6836
2.750	19.3206	13.2226	0.6844
2.755	19.2813	13.2191	0.6856
2.760	19.2878	13.2367	0.6863
2.765	19.2511	13.2048	0.6859
2.770	19.2435	13.2673	0.6894
2.775	19.1930	13.2101	0.6883
2.780	19.1910	13.2211	0.6889
2.785	19.1577	13.2367	0.6909
2.790	19.1571	13.2267	0.6904
2.795	19.1182	13.2637	0.6938
2.800	19.0999	13.2422	0.6933
2.805	19.0841	13.2626	0.6950
2.810	19.0309	13.2369	0.6955
2.815	19.0210	13.2588	0.6971
2.820	19.0194	13.2558	0.6970
2.825	18.9842	13.2340	0.6971
2.830	18.9708	13.2660	0.6993
2.835	18.9626	13.2637	0.6995
2.840	18.9109	13.2736	0.7019
2.845	18.8885	13.2818	0.7032
2.850	18.8953	13.2337	0.7004
2.855	18.8666	13.2848	0.7041
2.860	18.8274	13.2557	0.7041
2.865	18.7886	13.2729	0.7064
2.870	18.7740	13.2651	0.7066
2.875	18.7532	13.2508	0.7066
2.880	18.7316	13.2817	0.7091
2.885	18.7266	13.2839	0.7094
2.890	18.6674	13.3085	0.7129
2.895	18.6667	13.2677	0.7108
2.900	18.6567	13.2750	0.7115
2.905	18.6224	13.3114	0.7148

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Frequency (GHz)	ε_r'	ε_r''	$\tan\delta$
2.910	18.6056	13.3264	0.7163
2.915	18.5868	13.3015	0.7156
2.920	18.5812	13.3068	0.7161
2.925	18.5881	13.2741	0.7141
2.930	18.5469	13.2839	0.7162
2.935	18.5159	13.2940	0.7180
2.940	18.5232	13.2970	0.7179
2.945	18.4763	13.3075	0.7202
2.950	18.4413	13.2920	0.7208
2.955	18.5048	13.2937	0.7184
2.960	18.3499	13.3122	0.7255
2.965	18.4263	13.2739	0.7204
2.970	18.3641	13.2478	0.7214
2.975	18.3534	13.2520	0.7220
2.980	18.3636	13.2811	0.7232
2.985	18.3153	13.2590	0.7239
2.990	18.2692	13.2927	0.7276
2.995	18.2772	13.2424	0.7245
3.000	18.2108	13.2632	0.7283
3.005	18.2328	13.2649	0.7275
3.010	18.2112	13.2109	0.7254
3.015	18.1813	13.2242	0.7274
3.020	18.1259	13.2914	0.7333
3.025	18.1274	13.2402	0.7304
3.030	18.1385	13.2248	0.7291
3.035	18.1212	13.2747	0.7325
3.040	18.0617	13.2588	0.7341
3.045	18.0324	13.2428	0.7344
3.050	17.9849	13.2143	0.7347
3.055	17.9985	13.2960	0.7387
3.060	17.9706	13.2887	0.7395
3.065	18.0122	13.2695	0.7367
3.070	17.9299	13.3066	0.7421
3.075	17.9507	13.3089	0.7414
3.080	17.9273	13.3278	0.7434
3.085	17.9043	13.2722	0.7413
3.090	17.8984	13.3010	0.7431
3.095	17.8916	13.3067	0.7437
3.100	17.8919	13.3092	0.7439
3.105	17.8682	13.2704	0.7427
3.110	17.8511	13.3032	0.7452
3.115	17.8413	13.2906	0.7449
3.120	17.8170	13.3066	0.7469
3.125	17.7926	13.2790	0.7463
3.130	17.7763	13.2769	0.7469
3.135	17.7412	13.2588	0.7473

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Frequency (GHz)	ε_r'	ε_r''	$\tan\delta$
3.140	17.7512	13.2645	0.7472
3.145	17.6958	13.2453	0.7485
3.150	17.6958	13.2513	0.7488
3.155	17.6613	13.2513	0.7503
3.160	17.6679	13.2430	0.7495
3.165	17.5991	13.2484	0.7528
3.170	17.6340	13.2144	0.7494
3.175	17.5817	13.2065	0.7511
3.180	17.5676	13.2109	0.7520
3.185	17.5105	13.2366	0.7559
3.190	17.5048	13.2222	0.7553
3.195	17.4601	13.2303	0.7577
3.200	17.4159	13.2222	0.7592
3.205	17.4244	13.2479	0.7603
3.210	17.4021	13.2435	0.7610
3.215	17.4089	13.2234	0.7596
3.220	17.4200	13.2639	0.7614
3.225	17.3500	13.2927	0.7662
3.230	17.3601	13.2222	0.7616
3.235	17.3300	13.2560	0.7649
3.240	17.2896	13.2511	0.7664
3.245	17.3026	13.2119	0.7636
3.250	17.2074	13.2863	0.7721
3.255	17.2039	13.2343	0.7693
3.260	17.2312	13.2350	0.7681
3.265	17.2122	13.2683	0.7709
3.270	17.1707	13.1720	0.7671
3.275	17.1420	13.1922	0.7696
3.280	17.1451	13.2078	0.7704
3.285	17.0355	13.2025	0.7750
3.290	17.0952	13.1582	0.7697
3.295	17.0419	13.1623	0.7724
3.300	17.0726	13.1696	0.7714
3.305	17.0216	13.1912	0.7750
3.310	16.9969	13.1745	0.7751
3.315	16.9698	13.1490	0.7748
3.320	16.9625	13.1878	0.7775
3.325	16.9624	13.1713	0.7765
3.330	16.9116	13.1801	0.7794
3.335	16.8622	13.1240	0.7783
3.340	16.8698	13.1521	0.7796
3.345	16.8281	13.1174	0.7795
3.350	16.7815	13.1287	0.7823
3.355	16.8094	13.1282	0.7810
3.360	16.7346	13.1470	0.7856
3.365	16.7177	13.1349	0.7857

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Frequency (GHz)	ε_r'	ε_r''	$\tan\delta$
3.370	16.6807	13.1266	0.7869
3.375	16.7141	13.1658	0.7877
3.380	16.6840	13.1496	0.7882
3.385	16.6433	13.1435	0.7897
3.390	16.6319	13.1377	0.7899
3.395	16.6212	13.1384	0.7905
3.400	16.5517	13.1593	0.7950
3.405	16.5524	13.1299	0.7932
3.410	16.5466	13.1644	0.7956
3.415	16.4905	13.1471	0.7972
3.420	16.5076	13.1350	0.7957
3.425	16.5154	13.1379	0.7955
3.430	16.5000	13.1409	0.7964
3.435	16.4789	13.1748	0.7995
3.440	16.4717	13.1235	0.7967
3.445	16.4385	13.1326	0.7989
3.450	16.3638	13.1151	0.8015
3.455	16.3797	13.1479	0.8027
3.460	16.3582	13.1368	0.8031
3.465	16.3467	13.1088	0.8019
3.470	16.3365	13.1069	0.8023
3.475	16.3428	13.0837	0.8006
3.480	16.3165	13.0360	0.7989
3.485	16.3428	12.9738	0.7939
3.490	16.2342	13.0838	0.8059
3.495	16.2223	13.0847	0.8066
3.500	16.2207	13.0124	0.8022
3.505	16.2478	13.0624	0.8039
3.510	16.2144	12.9764	0.8003
3.515	16.1441	13.0257	0.8068
3.520	16.1162	12.9828	0.8056
3.525	15.9845	13.0378	0.8157
3.530	16.0197	13.0548	0.8149
3.535	15.9966	13.0023	0.8128
3.540	15.9217	13.0549	0.8199
3.545	15.9559	13.1241	0.8225
3.550	15.9019	13.1218	0.8252
3.555	15.9126	13.1151	0.8242
3.560	15.9252	13.1654	0.8267
3.565	15.9676	13.1658	0.8245
3.570	15.8810	13.1957	0.8309
3.575	15.8879	13.2918	0.8366
3.580	15.9701	13.2983	0.8327
3.585	16.1148	13.2217	0.8205
3.590	16.1033	13.1113	0.8142
3.595	16.1269	13.0619	0.8099

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Frequency (GHz)	ε_r'	ε_r''	$\tan\delta$
3.600	16.0763	13.0950	0.8146
3.605	16.1048	12.9786	0.8059
3.610	16.0591	12.9573	0.8069
3.615	15.9687	12.9207	0.8091
3.620	15.9128	12.8841	0.8097
3.625	16.0004	12.7872	0.7992
3.630	15.7533	12.8080	0.8130
3.635	15.7575	12.8771	0.8172
3.640	15.7696	12.9056	0.8184
3.645	15.7063	12.9193	0.8226
3.650	15.6589	12.9256	0.8254
3.655	15.6384	12.9039	0.8251
3.660	15.6119	12.9658	0.8305
3.665	15.6126	12.9377	0.8287
3.670	15.5607	12.9617	0.8330
3.675	15.5617	12.9399	0.8315
3.680	15.5610	12.9693	0.8334
3.685	15.5495	13.0140	0.8369
3.690	15.5343	13.0058	0.8372
3.695	15.5360	13.0360	0.8391
3.700	15.5572	13.0326	0.8377
3.705	15.5762	13.0323	0.8367
3.710	15.5465	13.0203	0.8375
3.715	15.5833	13.0232	0.8357
3.720	15.6257	12.9825	0.8308
3.725	15.5819	12.9889	0.8336
3.730	15.5855	12.9477	0.8308
3.735	15.5554	12.8751	0.8277
3.740	15.5535	12.8543	0.8265
3.745	15.5142	12.8424	0.8278
3.750	15.4272	12.8209	0.8311
3.755	15.3755	12.8391	0.8350
3.760	15.3169	12.8653	0.8399
3.765	15.2584	12.9106	0.8461
3.770	15.2975	13.0031	0.8500
3.775	15.2860	12.9840	0.8494
3.780	15.1632	13.1002	0.8639
3.785	15.3313	13.0165	0.8490
3.790	15.4001	12.9281	0.8395
3.795	15.3656	12.9144	0.8405
3.800	15.3357	12.9741	0.8460
3.805	15.3596	12.8943	0.8395
3.810	15.2990	12.8854	0.8422
3.815	15.2620	12.8126	0.8395
3.820	15.2835	12.8757	0.8425
3.825	15.2402	12.8074	0.8404

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Frequency (GHz)	ε_r'	ε_r''	$\tan\delta$
3.830	15.2203	12.8443	0.8439
3.835	14.9149	12.3483	0.8279
3.840	15.1728	12.8246	0.8452
3.845	15.1446	12.8312	0.8472
3.850	15.1409	12.8225	0.8469
3.855	15.0988	12.8335	0.8500
3.860	15.0829	12.7958	0.8484
3.865	15.0255	12.8053	0.8522
3.870	15.0761	12.8092	0.8496
3.875	15.0219	12.8049	0.8524
3.880	15.0408	12.7942	0.8506
3.885	14.9805	12.7983	0.8543
3.890	14.9857	12.8233	0.8557
3.895	14.9985	12.7688	0.8513
3.900	14.9565	12.7686	0.8537
3.905	14.8996	12.7972	0.8589
3.910	14.9248	12.7753	0.8560
3.915	14.9064	12.7677	0.8565
3.920	14.8721	12.7913	0.8601
3.925	14.8577	12.7540	0.8584
3.930	14.8504	12.7476	0.8584
3.935	14.8528	12.7515	0.8585
3.940	14.8129	12.7509	0.8608
3.945	14.8050	12.7350	0.8602
3.950	14.7701	12.7204	0.8612
3.955	14.7494	12.7344	0.8634
3.960	14.7425	12.7193	0.8628
3.965	14.7407	12.7187	0.8628
3.970	14.7052	12.7287	0.8656
3.975	14.6911	12.7155	0.8655
3.980	14.6814	12.7027	0.8652
3.985	14.6714	12.7116	0.8664
3.990	14.6589	12.7056	0.8667
3.995	14.6337	12.6844	0.8668
4.000	14.6117	12.6862	0.8682
4.005	14.5909	12.6812	0.8691
4.010	14.5626	12.6787	0.8706
4.015	14.5563	12.6951	0.8721
4.020	14.5483	12.6697	0.8709
4.025	14.5712	12.6606	0.8689
4.030	14.5040	12.6561	0.8726
4.035	14.4797	12.6672	0.8748
4.040	14.4929	12.6608	0.8736
4.045	14.4882	12.6356	0.8721
4.050	14.4533	12.6279	0.8737
4.055	14.4324	12.6090	0.8737

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Frequency (GHz)	ε_r'	ε_r''	$\tan\delta$
4.060	14.4031	12.6246	0.8765
4.065	14.3799	12.6146	0.8772
4.070	14.3803	12.6334	0.8785
4.075	14.3608	12.6092	0.8780
4.080	14.3401	12.6407	0.8815
4.085	14.3307	12.6106	0.8800
4.090	14.3303	12.6048	0.8796
4.095	14.2824	12.6116	0.8830
4.100	14.2921	12.6306	0.8837
4.105	14.2623	12.5406	0.8793
4.110	14.2691	12.6042	0.8833
4.115	14.2390	12.5806	0.8835
4.120	14.2059	12.5802	0.8856
4.125	14.2387	12.5810	0.8836
4.130	14.2282	12.5966	0.8853
4.135	14.1757	12.5501	0.8853
4.140	14.1955	12.5100	0.8813
4.145	14.2367	12.5298	0.8801
4.150	14.1322	12.5265	0.8864
4.155	14.1146	12.5385	0.8883
4.160	14.1464	12.5373	0.8863
4.165	14.0987	12.5015	0.8867
4.170	14.0928	12.4955	0.8867
4.175	14.1162	12.4626	0.8829
4.180	14.1156	12.4962	0.8853
4.185	14.0951	12.4465	0.8830
4.190	14.0595	12.4486	0.8854
4.195	14.0553	12.4518	0.8859
4.200	14.0772	12.4602	0.8851
4.205	14.0637	12.4669	0.8865
4.210	14.0000	12.4616	0.8901
4.215	14.0333	12.4721	0.8888
4.220	14.0109	12.4762	0.8905
4.225	14.0528	12.4810	0.8882
4.230	14.0045	12.4677	0.8903
4.235	13.9721	12.4633	0.8920
4.240	13.9600	12.5213	0.8969
4.245	13.9814	12.4858	0.8930
4.250	13.9583	12.4657	0.8931
4.255	13.9592	12.4992	0.8954
4.260	13.9118	12.4760	0.8968
4.265	13.9546	12.4748	0.8940
4.270	13.9022	12.4655	0.8967
4.275	13.8847	12.4641	0.8977
4.280	13.8769	12.4755	0.8990
4.285	13.8611	12.4757	0.9001

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Frequency (GHz)	ε_r'	ε_r''	$\tan\delta$
4.290	13.8423	12.4676	0.9007
4.295	13.8313	12.4593	0.9008
4.300	13.8267	12.4785	0.9025
4.305	13.8053	12.4714	0.9034
4.310	13.8029	12.4630	0.9029
4.315	13.7926	12.4618	0.9035
4.320	13.7905	12.4524	0.9030
4.325	13.7712	12.4562	0.9045
4.330	13.7711	12.4497	0.9040
4.335	13.7627	12.4517	0.9047
4.340	13.7623	12.4308	0.9032
4.345	13.7385	12.4253	0.9044
4.350	13.7359	12.4223	0.9044
4.355	13.7151	12.4280	0.9062
4.360	13.6975	12.4070	0.9058
4.365	13.6643	12.4034	0.9077
4.370	13.6855	12.3993	0.9060
4.375	13.6371	12.4097	0.9100
4.380	13.6383	12.3996	0.9092
4.385	13.5825	12.3653	0.9104
4.390	13.6006	12.3520	0.9082
4.395	13.5669	12.3592	0.9110
4.400	13.5474	12.3727	0.9133
4.405	13.5096	12.3633	0.9151
4.410	13.5208	12.3671	0.9147
4.415	13.5058	12.3372	0.9135
4.420	13.4839	12.3719	0.9175
4.425	13.4944	12.3782	0.9173
4.430	13.4674	12.3417	0.9164
4.435	13.4246	12.3135	0.9172
4.440	13.4334	12.3345	0.9182
4.445	13.3770	12.3074	0.9200
4.450	13.3918	12.2755	0.9166
4.455	13.3338	12.3053	0.9229
4.460	13.3673	12.2443	0.9160
4.465	13.3670	12.2033	0.9129
4.470	13.3810	12.2262	0.9137
4.475	13.3888	12.1959	0.9109
4.480	13.4054	12.2187	0.9115
4.485	13.3364	12.1643	0.9121
4.490	13.3502	12.1914	0.9132
4.495	13.3463	12.1808	0.9127
4.500	13.3201	12.1831	0.9146
4.505	13.2886	12.1702	0.9158
4.510	13.2341	12.1670	0.9194
4.515	13.2329	12.2100	0.9227

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Frequency (GHz)	ε_r'	ε_r''	$\tan\delta$
4.520	13.2246	12.2038	0.9228
4.525	13.2449	12.1993	0.9211
4.530	13.1856	12.1807	0.9238
4.535	13.1812	12.1945	0.9251
4.540	13.1849	12.1690	0.9230
4.545	13.1450	12.1609	0.9251
4.550	13.1504	12.1354	0.9228
4.555	13.1193	12.1475	0.9259
4.560	13.1230	12.1496	0.9258
4.565	13.1014	12.1623	0.9283
4.570	13.0913	12.1333	0.9268
4.575	13.0830	12.1365	0.9277
4.580	13.0784	12.1470	0.9288
4.585	13.0608	12.1382	0.9294
4.590	13.0457	12.1185	0.9289
4.595	13.0266	12.1290	0.9311
4.600	13.0192	12.1143	0.9305
4.605	12.9985	12.1019	0.9310
4.610	12.9897	12.1108	0.9323
4.615	12.9749	12.0955	0.9322
4.620	12.9721	12.0792	0.9312
4.625	12.9563	12.0853	0.9328
4.630	12.9518	12.0743	0.9322
4.635	12.9349	12.0512	0.9317
4.640	12.9097	12.0505	0.9334
4.645	12.9129	12.0410	0.9325
4.650	12.8932	12.0424	0.9340
4.655	12.8856	12.0061	0.9317
4.660	12.8919	12.0453	0.9343
4.665	12.8690	12.0253	0.9344
4.670	12.8140	12.0177	0.9379
4.675	12.8197	11.9786	0.9344
4.680	12.8130	11.9867	0.9355
4.685	12.8127	11.9764	0.9347
4.690	12.8008	12.0109	0.9383
4.695	12.8054	11.9541	0.9335
4.700	12.7735	11.9962	0.9391
4.705	12.7361	11.8967	0.9341
4.710	12.6799	11.9657	0.9437
4.715	12.7082	11.9635	0.9414
4.720	12.6912	11.9354	0.9404
4.725	12.6894	11.9732	0.9436
4.730	12.7002	11.9522	0.9411
4.735	12.7181	11.9729	0.9414
4.740	12.6989	11.9553	0.9414
4.745	12.6551	11.9516	0.9444

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Frequency (GHz)	ε_r'	ε_r''	$\tan\delta$
4.750	12.6918	11.9337	0.9403
4.755	12.6820	11.9769	0.9444
4.760	12.6603	11.9346	0.9427
4.765	12.7029	11.9583	0.9414
4.770	12.6853	11.9786	0.9443
4.775	12.6129	11.9016	0.9436
4.780	12.6281	11.9049	0.9427
4.785	12.6200	12.0224	0.9526
4.790	12.6268	11.8723	0.9402
4.795	12.5816	11.8822	0.9444
4.800	12.5835	11.8816	0.9442
4.805	12.6018	11.8398	0.9395
4.810	12.5562	11.8977	0.9476
4.815	12.5818	11.8315	0.9404
4.820	12.5072	11.8707	0.9491
4.825	12.5427	11.8766	0.9469
4.830	12.4884	11.8719	0.9506
4.835	12.4890	11.8866	0.9518
4.840	12.4988	11.8807	0.9505
4.845	12.5079	11.8695	0.9490
4.850	12.4566	11.8740	0.9532
4.855	12.4978	11.8958	0.9518
4.860	12.4713	11.8679	0.9516
4.865	12.5064	11.8589	0.9482
4.870	12.4615	11.8585	0.9516
4.875	12.4619	11.8703	0.9525
4.880	12.4562	11.8822	0.9539
4.885	12.4425	11.8472	0.9522
4.890	12.4753	11.8789	0.9522
4.895	12.4257	11.8468	0.9534
4.900	12.4341	11.8657	0.9543
4.905	12.4281	11.8459	0.9532
4.910	12.4014	11.8368	0.9545
4.915	12.3932	11.8419	0.9555
4.920	12.3896	11.8326	0.9550
4.925	12.3830	11.8284	0.9552
4.930	12.3584	11.8172	0.9562
4.935	12.3563	11.8184	0.9565
4.940	12.3514	11.8118	0.9563
4.945	12.3269	11.8100	0.9581
4.950	12.3257	11.8072	0.9579
4.955	12.3163	11.8127	0.9591
4.960	12.2985	11.7929	0.9589
4.965	12.2924	11.7964	0.9596
4.970	12.2760	11.7960	0.9609
4.975	12.2893	11.7936	0.9597

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Frequency (GHz)	ε_r'	ε_r''	$\tan\delta$
4.980	12.2427	11.7931	0.9633
4.985	12.2370	11.7646	0.9614
4.990	12.2218	11.7839	0.9642
4.995	12.1752	11.7791	0.9675
5.000	12.1747	11.7591	0.9659
5.005	12.1451	11.7616	0.9684
5.010	12.1254	11.7396	0.9682
5.015	12.1092	11.7541	0.9707
5.020	12.0905	11.7165	0.9691
5.025	12.0385	11.6941	0.9714
5.030	12.0118	11.6737	0.9719
5.035	12.0287	11.6864	0.9715
5.040	12.0047	11.6536	0.9708
5.045	12.0179	11.6196	0.9669
5.050	11.9924	11.5781	0.9655
5.055	12.0281	11.5943	0.9639
5.060	12.0014	11.5762	0.9646
5.065	12.0147	11.5529	0.9616
5.070	12.0001	11.5894	0.9658
5.075	11.9989	11.5714	0.9644
5.080	11.9741	11.5512	0.9647
5.085	11.9709	11.5416	0.9641
5.090	11.9754	11.5674	0.9659
5.095	11.9796	11.5830	0.9669
5.100	11.9692	11.5792	0.9674
5.105	11.9431	11.6224	0.9732
5.110	11.9266	11.5627	0.9695
5.115	11.9033	11.5521	0.9705
5.120	11.8979	11.5512	0.9709
5.125	11.8835	11.5548	0.9723
5.130	11.8753	11.5354	0.9714
5.135	11.8458	11.5266	0.9731
5.140	11.8390	11.5336	0.9742
5.145	11.8185	11.4865	0.9719
5.150	11.8136	11.5127	0.9745
5.155	11.8009	11.4842	0.9732
5.160	11.7896	11.4729	0.9731
5.165	11.7713	11.4667	0.9741
5.170	11.7673	11.4462	0.9727
5.175	11.7502	11.4318	0.9729
5.180	11.7496	11.4309	0.9729
5.185	11.7384	11.4195	0.9728
5.190	11.7402	11.4136	0.9722
5.195	11.7398	11.4093	0.9718
5.200	11.7371	11.4080	0.9720
5.205	11.7351	11.4094	0.9722

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Frequency (GHz)	ε_r'	ε_r''	$\tan\delta$
5.210	11.7260	11.3989	0.9721
5.215	11.7220	11.4047	0.9729
5.220	11.7099	11.3903	0.9727
5.225	11.6945	11.3894	0.9739
5.230	11.7009	11.3825	0.9728
5.235	11.6744	11.3933	0.9759
5.240	11.6630	11.3772	0.9755
5.245	11.6693	11.3822	0.9754
5.250	11.6449	11.3935	0.9784
5.255	11.6178	11.3911	0.9805
5.260	11.6570	11.3724	0.9756
5.265	11.6514	11.3947	0.9780
5.270	11.6148	11.3711	0.9790
5.275	11.6037	11.3471	0.9779
5.280	11.6065	11.3691	0.9795
5.285	11.6141	11.3530	0.9775
5.290	11.5855	11.3342	0.9783
5.295	11.6104	11.3165	0.9747
5.300	11.5786	11.3283	0.9784
5.305	11.5840	11.3356	0.9786
5.310	11.5274	11.3143	0.9815
5.315	11.5572	11.3241	0.9798
5.320	11.5353	11.2970	0.9793
5.325	11.5408	11.3057	0.9796
5.330	11.5225	11.3039	0.9810
5.335	11.4818	11.3028	0.9844
5.340	11.4776	11.3198	0.9863
5.345	11.4584	11.3373	0.9894
5.350	11.5063	11.2886	0.9811
5.355	11.4538	11.2943	0.9861
5.360	11.4497	11.2371	0.9814
5.365	11.5028	11.2890	0.9814
5.370	11.4321	11.2406	0.9832
5.375	11.4443	11.2676	0.9846
5.380	11.4490	11.2812	0.9853
5.385	11.4071	11.2424	0.9856
5.390	11.3999	11.2845	0.9899
5.395	11.4307	11.2737	0.9863
5.400	11.3946	11.2596	0.9882
5.405	11.3918	11.2447	0.9871
5.410	11.3819	11.2497	0.9884
5.415	11.3764	11.2170	0.9860
5.420	11.3549	11.2454	0.9904
5.425	11.3473	11.2532	0.9917
5.430	11.3637	11.2214	0.9875
5.435	11.3515	11.2145	0.9879

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Frequency (GHz)	ε_r'	ε_r''	$\tan\delta$
5.440	11.3102	11.2313	0.9930
5.445	11.3459	11.2219	0.9891
5.450	11.2989	11.2066	0.9918
5.455	11.3167	11.2241	0.9918
5.460	11.3030	11.1987	0.9908
5.465	11.3128	11.2299	0.9927
5.470	11.2559	11.1999	0.9950
5.475	11.2823	11.1884	0.9917
5.480	11.2559	11.2052	0.9955
5.485	11.2670	11.1813	0.9924
5.490	11.2242	11.2065	0.9984
5.495	11.2604	11.1872	0.9935
5.500	11.2390	11.1949	0.9961
5.505	11.2407	11.1945	0.9959
5.510	11.2373	11.1876	0.9956
5.515	11.2424	11.1777	0.9942
5.520	11.2179	11.1692	0.9957
5.525	11.2295	11.1707	0.9948
5.530	11.2157	11.1640	0.9954
5.535	11.2248	11.1564	0.9939
5.540	11.2024	11.1626	0.9964
5.545	11.1837	11.1396	0.9961
5.550	11.1894	11.1420	0.9958
5.555	11.1936	11.1317	0.9945
5.560	11.1817	11.1318	0.9955
5.565	11.1858	11.1309	0.9951
5.570	11.1830	11.1297	0.9952
5.575	11.1861	11.1240	0.9945
5.580	11.1777	11.1365	0.9963
5.585	11.1636	11.1254	0.9966
5.590	11.1761	11.1150	0.9945
5.595	11.1651	11.1091	0.9950
5.600	11.1742	11.1113	0.9944
5.605	11.1787	11.0903	0.9921
5.610	11.1582	11.0732	0.9924
5.615	11.1518	11.0459	0.9905
5.620	11.1451	11.0477	0.9913
5.625	11.1368	11.0583	0.9930
5.630	11.1279	11.0327	0.9914
5.635	11.1291	11.0187	0.9901
5.640	11.1220	10.9923	0.9883
5.645	11.0934	10.9859	0.9903
5.650	11.0982	11.0101	0.9921
5.655	11.0761	10.9871	0.9920
5.660	11.0615	10.9899	0.9935
5.665	11.1039	10.9855	0.9893

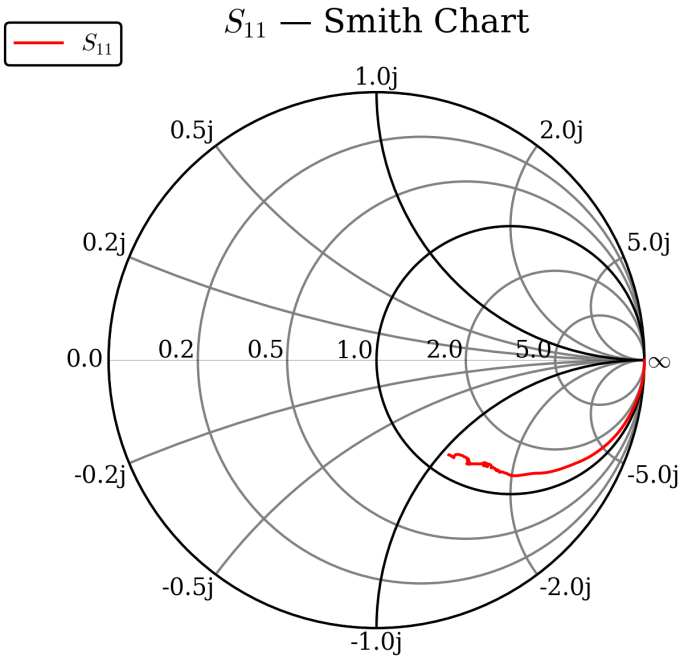
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Frequency (GHz)	ε_r'	ε_r''	$\tan\delta$
5.670	11.0201	10.9806	0.9964
5.675	11.0431	10.9470	0.9913
5.680	11.0198	10.9489	0.9936
5.685	11.0533	10.9315	0.9890
5.690	11.0241	10.9253	0.9910
5.695	11.0103	10.9522	0.9947
5.700	10.9951	10.8951	0.9909
5.705	10.9526	10.8920	0.9945
5.710	10.9936	10.8849	0.9901
5.715	10.9639	10.8554	0.9901
5.720	10.9605	10.8674	0.9915
5.725	10.9229	10.8661	0.9948
5.730	10.9235	10.8516	0.9934
5.735	10.9196	10.8441	0.9931
5.740	10.9042	10.8554	0.9955
5.745	10.8997	10.8663	0.9969
5.750	10.8723	10.8311	0.9962
5.755	10.8805	10.8316	0.9955
5.760	10.8482	10.8058	0.9961
5.765	10.8785	10.8031	0.9931
5.770	10.8572	10.7825	0.9931
5.775	10.8512	10.7951	0.9948
5.780	10.8185	10.7945	0.9978
5.785	10.8062	10.7778	0.9974
5.790	10.7880	10.7877	1.0000
5.795	10.7912	10.7942	1.0003
5.800	10.7712	10.7761	1.0005
5.805	10.7618	10.7599	0.9998
5.810	10.7571	10.7785	1.0020
5.815	10.7514	10.7752	1.0022
5.820	10.7462	10.7625	1.0015
5.825	10.7372	10.7583	1.0020
5.830	10.7134	10.7517	1.0036
5.835	10.7172	10.7482	1.0029
5.840	10.7052	10.7427	1.0035
5.845	10.6929	10.7489	1.0052
5.850	10.6903	10.7354	1.0042
5.855	10.6793	10.7223	1.0040
5.860	10.6643	10.7201	1.0052
5.865	10.6645	10.7274	1.0059
5.870	10.6580	10.7065	1.0045
5.875	10.6494	10.7052	1.0052
5.880	10.6407	10.6986	1.0054
5.885	10.6489	10.6852	1.0034
5.890	10.6009	10.7060	1.0099
5.895	10.6064	10.6894	1.0078

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Frequency (GHz)	ε_r'	ε_r''	$\tan\delta$
5.900	10.6096	10.6655	1.0053
5.905	10.5891	10.6947	1.0100
5.910	10.5891	10.6394	1.0048
5.915	10.5788	10.6605	1.0077
5.920	10.5731	10.6475	1.0070
5.925	10.5621	10.6375	1.0071
5.930	10.5224	10.6446	1.0116
5.935	10.5167	10.6297	1.0107
5.940	10.5187	10.6333	1.0109
5.945	10.5473	10.6231	1.0072
5.950	10.5053	10.6354	1.0124
5.955	10.5068	10.6301	1.0117
5.960	10.4937	10.6236	1.0124
5.965	10.4980	10.6288	1.0125
5.970	10.4783	10.6367	1.0151
5.975	10.5131	10.5661	1.0050
5.980	10.5528	10.6250	1.0068
5.985	10.4487	10.6213	1.0165
5.990	10.4543	10.5770	1.0117
5.995	10.4568	10.5696	1.0108
6.000	10.4781	10.5696	1.0087

1.2 S_{11} — Smith Chart



Frequency (GHz)	**Re(S_{11})**	**Im(S_{11})**	**$\lvert S_{11}\rvert$ (dB)**
0.001	1.0000	-0.0005	-0.00
0.002	1.0000	-0.0010	0.00
0.003	0.9999	-0.0013	-0.00
0.004	1.0000	-0.0017	-0.00
0.005	0.9999	-0.0022	-0.00
0.006	1.0000	-0.0025	-0.00
0.007	1.0001	-0.0030	0.00
0.008	0.9999	-0.0034	-0.00
0.009	0.9998	-0.0039	-0.00
0.010	0.9999	-0.0043	-0.00
0.011	0.9998	-0.0047	-0.00
0.012	1.0001	-0.0051	0.00
0.013	0.9999	-0.0055	-0.00
0.014	1.0000	-0.0061	0.00
0.015	0.9999	-0.0065	-0.00
0.016	0.9999	-0.0068	-0.00
0.017	0.9999	-0.0072	-0.00
0.018	0.9999	-0.0076	-0.00
0.019	0.9998	-0.0081	-0.00
0.020	0.9999	-0.0085	-0.00
0.021	1.0000	-0.0089	0.00
0.022	0.9998	-0.0093	-0.00
0.023	0.9999	-0.0098	-0.00

(continued on next page)

Frequency (GHz)	**Re**(S_{11})	**Im**(S_{11})	$\lvert S_{11}\rvert$ **(dB)**
0.024	0.9998	-0.0101	-0.00
0.025	0.9998	-0.0106	-0.00
0.026	0.9999	-0.0109	-0.00
0.027	0.9998	-0.0115	-0.00
0.028	0.9997	-0.0119	-0.00
0.029	0.9998	-0.0122	-0.00
0.030	0.9998	-0.0127	-0.00
0.031	0.9998	-0.0131	-0.00
0.032	0.9997	-0.0135	-0.00
0.033	0.9998	-0.0139	-0.00
0.034	0.9996	-0.0143	-0.00
0.035	0.9998	-0.0148	-0.00
0.036	0.9997	-0.0151	-0.00
0.037	0.9996	-0.0156	-0.00
0.038	0.9996	-0.0161	-0.00
0.039	0.9996	-0.0164	-0.00
0.040	0.9996	-0.0168	-0.00
0.041	0.9995	-0.0174	-0.00
0.042	0.9996	-0.0177	-0.00
0.043	0.9996	-0.0180	-0.00
0.044	0.9995	-0.0186	-0.00
0.045	0.9996	-0.0191	-0.00
0.046	0.9994	-0.0195	-0.00
0.047	0.9995	-0.0198	-0.00
0.048	0.9994	-0.0203	-0.00
0.049	0.9995	-0.0207	-0.00
0.050	0.9994	-0.0210	-0.00
0.051	0.9994	-0.0215	-0.00
0.052	0.9994	-0.0220	-0.00
0.053	0.9992	-0.0224	-0.01
0.054	0.9993	-0.0228	-0.00
0.055	0.9993	-0.0231	-0.00
0.056	0.9993	-0.0236	-0.00
0.057	0.9992	-0.0240	-0.00
0.058	0.9993	-0.0245	-0.00
0.059	0.9992	-0.0248	-0.00
0.060	0.9993	-0.0253	-0.00
0.061	0.9991	-0.0257	-0.01
0.062	0.9991	-0.0261	-0.01
0.063	0.9991	-0.0267	-0.01
0.064	0.9990	-0.0270	-0.01
0.065	0.9991	-0.0274	-0.00
0.066	0.9989	-0.0278	-0.01
0.067	0.9990	-0.0283	-0.01
0.068	0.9990	-0.0287	-0.00
0.069	0.9989	-0.0290	-0.01

(continued on next page)

Frequency (GHz)	**Re(S_{11})**	**Im(S_{11})**	**$\lvert S_{11}\rvert$ (dB)**
0.070	0.9988	-0.0294	-0.01
0.071	0.9989	-0.0298	-0.01
0.072	0.9988	-0.0304	-0.01
0.073	0.9988	-0.0307	-0.01
0.074	0.9986	-0.0313	-0.01
0.075	0.9988	-0.0317	-0.01
0.076	0.9987	-0.0320	-0.01
0.077	0.9987	-0.0324	-0.01
0.078	0.9986	-0.0329	-0.01
0.079	0.9986	-0.0333	-0.01
0.080	0.9986	-0.0337	-0.01
0.081	0.9984	-0.0341	-0.01
0.082	0.9985	-0.0346	-0.01
0.083	0.9985	-0.0349	-0.01
0.084	0.9984	-0.0353	-0.01
0.085	0.9983	-0.0357	-0.01
0.086	0.9983	-0.0362	-0.01
0.087	0.9983	-0.0365	-0.01
0.088	0.9983	-0.0371	-0.01
0.089	0.9982	-0.0375	-0.01
0.090	0.9979	-0.0378	-0.01
0.091	0.9981	-0.0384	-0.01
0.092	0.9981	-0.0386	-0.01
0.093	0.9981	-0.0391	-0.01
0.094	0.9979	-0.0395	-0.01
0.095	0.9980	-0.0400	-0.01
0.096	0.9978	-0.0404	-0.01
0.097	0.9979	-0.0408	-0.01
0.098	0.9979	-0.0412	-0.01
0.099	0.9979	-0.0415	-0.01
0.100	0.9977	-0.0419	-0.01
0.105	0.9975	-0.0441	-0.01
0.110	0.9974	-0.0462	-0.01
0.115	0.9970	-0.0483	-0.02
0.120	0.9968	-0.0503	-0.02
0.125	0.9966	-0.0525	-0.02
0.130	0.9963	-0.0544	-0.02
0.135	0.9960	-0.0565	-0.02
0.140	0.9956	-0.0587	-0.02
0.145	0.9954	-0.0606	-0.02
0.150	0.9951	-0.0628	-0.03
0.155	0.9948	-0.0647	-0.03
0.160	0.9944	-0.0669	-0.03
0.165	0.9942	-0.0688	-0.03
0.170	0.9938	-0.0709	-0.03
0.175	0.9936	-0.0730	-0.03

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Frequency (GHz)	**Re(S_{11})**	**Im(S_{11})**	**$\lvert S_{11}\rvert$ (dB)**
0.180	0.9930	-0.0751	-0.04
0.185	0.9927	-0.0770	-0.04
0.190	0.9923	-0.0792	-0.04
0.195	0.9918	-0.0813	-0.04
0.200	0.9915	-0.0833	-0.04
0.205	0.9911	-0.0852	-0.05
0.210	0.9906	-0.0873	-0.05
0.215	0.9901	-0.0894	-0.05
0.220	0.9897	-0.0914	-0.05
0.225	0.9892	-0.0933	-0.06
0.230	0.9888	-0.0954	-0.06
0.235	0.9882	-0.0974	-0.06
0.240	0.9877	-0.0995	-0.06
0.245	0.9871	-0.1015	-0.07
0.250	0.9867	-0.1036	-0.07
0.255	0.9860	-0.1056	-0.07
0.260	0.9855	-0.1075	-0.08
0.265	0.9851	-0.1096	-0.08
0.270	0.9844	-0.1117	-0.08
0.275	0.9839	-0.1135	-0.08
0.280	0.9833	-0.1155	-0.09
0.285	0.9827	-0.1175	-0.09
0.290	0.9820	-0.1195	-0.09
0.295	0.9814	-0.1215	-0.10
0.300	0.9810	-0.1236	-0.10
0.305	0.9803	-0.1255	-0.10
0.310	0.9795	-0.1274	-0.11
0.315	0.9788	-0.1293	-0.11
0.320	0.9782	-0.1313	-0.11
0.325	0.9775	-0.1332	-0.12
0.330	0.9768	-0.1352	-0.12
0.335	0.9762	-0.1371	-0.12
0.340	0.9755	-0.1389	-0.13
0.345	0.9747	-0.1409	-0.13
0.350	0.9739	-0.1429	-0.14
0.355	0.9734	-0.1447	-0.14
0.360	0.9725	-0.1467	-0.14
0.365	0.9719	-0.1485	-0.15
0.370	0.9711	-0.1505	-0.15
0.375	0.9704	-0.1523	-0.16
0.380	0.9695	-0.1542	-0.16
0.385	0.9688	-0.1562	-0.16
0.390	0.9681	-0.1579	-0.17
0.395	0.9672	-0.1599	-0.17
0.400	0.9665	-0.1616	-0.18
0.405	0.9657	-0.1635	-0.18

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Frequency (GHz)	**Re(S_{11})**	**Im(S_{11})**	**$\lvert S_{11}\rvert$ (dB)**
0.410	0.9649	-0.1652	-0.18
0.415	0.9642	-0.1671	-0.19
0.420	0.9633	-0.1688	-0.19
0.425	0.9626	-0.1707	-0.20
0.430	0.9617	-0.1725	-0.20
0.435	0.9609	-0.1744	-0.21
0.440	0.9601	-0.1759	-0.21
0.445	0.9592	-0.1777	-0.21
0.450	0.9585	-0.1795	-0.22
0.455	0.9577	-0.1812	-0.22
0.460	0.9567	-0.1829	-0.23
0.465	0.9559	-0.1847	-0.23
0.470	0.9549	-0.1864	-0.24
0.475	0.9541	-0.1880	-0.24
0.480	0.9533	-0.1898	-0.25
0.485	0.9524	-0.1916	-0.25
0.490	0.9516	-0.1933	-0.26
0.495	0.9508	-0.1948	-0.26
0.500	0.9498	-0.1965	-0.27
0.505	0.9488	-0.1982	-0.27
0.510	0.9481	-0.1998	-0.27
0.515	0.9472	-0.2014	-0.28
0.520	0.9464	-0.2031	-0.28
0.525	0.9455	-0.2047	-0.29
0.530	0.9446	-0.2063	-0.29
0.535	0.9438	-0.2080	-0.30
0.540	0.9429	-0.2095	-0.30
0.545	0.9420	-0.2111	-0.31
0.550	0.9411	-0.2127	-0.31
0.555	0.9402	-0.2143	-0.32
0.560	0.9393	-0.2159	-0.32
0.565	0.9384	-0.2174	-0.33
0.570	0.9375	-0.2189	-0.33
0.575	0.9365	-0.2205	-0.34
0.580	0.9357	-0.2222	-0.34
0.585	0.9348	-0.2236	-0.34
0.590	0.9338	-0.2252	-0.35
0.595	0.9329	-0.2267	-0.35
0.600	0.9319	-0.2282	-0.36
0.605	0.9310	-0.2297	-0.36
0.610	0.9301	-0.2312	-0.37
0.615	0.9291	-0.2327	-0.37
0.620	0.9281	-0.2342	-0.38
0.625	0.9271	-0.2356	-0.39
0.630	0.9263	-0.2371	-0.39
0.635	0.9253	-0.2385	-0.40

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Frequency (GHz)	**Re(S_{11})**	**Im(S_{11})**	**$\lvert S_{11}\rvert$ (dB)**
0.640	0.9243	-0.2401	-0.40
0.645	0.9234	-0.2415	-0.40
0.650	0.9225	-0.2430	-0.41
0.655	0.9215	-0.2443	-0.41
0.660	0.9204	-0.2458	-0.42
0.665	0.9195	-0.2472	-0.43
0.670	0.9185	-0.2486	-0.43
0.675	0.9176	-0.2500	-0.44
0.680	0.9165	-0.2515	-0.44
0.685	0.9155	-0.2526	-0.45
0.690	0.9148	-0.2542	-0.45
0.695	0.9136	-0.2556	-0.46
0.700	0.9125	-0.2568	-0.46
0.705	0.9116	-0.2582	-0.47
0.710	0.9105	-0.2596	-0.47
0.715	0.9095	-0.2609	-0.48
0.720	0.9084	-0.2623	-0.49
0.725	0.9073	-0.2637	-0.49
0.730	0.9063	-0.2651	-0.50
0.735	0.9053	-0.2664	-0.50
0.740	0.9042	-0.2678	-0.51
0.745	0.9032	-0.2691	-0.52
0.750	0.9020	-0.2704	-0.52
0.755	0.9010	-0.2717	-0.53
0.760	0.8999	-0.2728	-0.53
0.765	0.8988	-0.2742	-0.54
0.770	0.8976	-0.2755	-0.55
0.775	0.8965	-0.2768	-0.55
0.780	0.8954	-0.2782	-0.56
0.785	0.8944	-0.2794	-0.56
0.790	0.8931	-0.2808	-0.57
0.795	0.8919	-0.2821	-0.58
0.800	0.8909	-0.2833	-0.59
0.805	0.8898	-0.2845	-0.59
0.810	0.8884	-0.2859	-0.60
0.815	0.8874	-0.2872	-0.61
0.820	0.8861	-0.2883	-0.61
0.825	0.8849	-0.2896	-0.62
0.830	0.8838	-0.2909	-0.63
0.835	0.8827	-0.2920	-0.63
0.840	0.8813	-0.2934	-0.64
0.845	0.8801	-0.2946	-0.65
0.850	0.8789	-0.2958	-0.66
0.855	0.8777	-0.2970	-0.66
0.860	0.8765	-0.2983	-0.67
0.865	0.8752	-0.2995	-0.68

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Frequency (GHz)	**Re(S_{11})**	**Im(S_{11})**	**$\lvert S_{11}\rvert$ (dB)**
0.870	0.8739	-0.3007	-0.68
0.875	0.8727	-0.3017	-0.69
0.880	0.8715	-0.3032	-0.70
0.885	0.8702	-0.3044	-0.71
0.890	0.8689	-0.3055	-0.71
0.895	0.8676	-0.3068	-0.72
0.900	0.8663	-0.3080	-0.73
0.905	0.8650	-0.3091	-0.74
0.910	0.8638	-0.3104	-0.74
0.915	0.8625	-0.3116	-0.75
0.920	0.8611	-0.3126	-0.76
0.925	0.8599	-0.3138	-0.77
0.930	0.8585	-0.3149	-0.78
0.935	0.8573	-0.3161	-0.78
0.940	0.8559	-0.3172	-0.79
0.945	0.8546	-0.3184	-0.80
0.950	0.8533	-0.3197	-0.81
0.955	0.8518	-0.3207	-0.82
0.960	0.8504	-0.3218	-0.83
0.965	0.8492	-0.3230	-0.83
0.970	0.8479	-0.3241	-0.84
0.975	0.8466	-0.3252	-0.85
0.980	0.8451	-0.3264	-0.86
0.985	0.8437	-0.3274	-0.87
0.990	0.8424	-0.3285	-0.87
0.995	0.8410	-0.3295	-0.88
1.000	0.8397	-0.3306	-0.89
1.005	0.8384	-0.3316	-0.90
1.010	0.8370	-0.3326	-0.91
1.015	0.8356	-0.3336	-0.92
1.020	0.8341	-0.3346	-0.93
1.025	0.8328	-0.3356	-0.94
1.030	0.8315	-0.3366	-0.94
1.035	0.8301	-0.3377	-0.95
1.040	0.8288	-0.3386	-0.96
1.045	0.8273	-0.3396	-0.97
1.050	0.8261	-0.3405	-0.98
1.055	0.8247	-0.3416	-0.99
1.060	0.8235	-0.3424	-0.99
1.065	0.8221	-0.3433	-1.00
1.070	0.8208	-0.3442	-1.01
1.075	0.8194	-0.3453	-1.02
1.080	0.8181	-0.3462	-1.03
1.085	0.8168	-0.3471	-1.04
1.090	0.8154	-0.3480	-1.05
1.095	0.8141	-0.3489	-1.05

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Frequency (GHz)	**Re(S_{11})**	**Im(S_{11})**	**$\lvert S_{11}\rvert$ (dB)**
1.100	0.8127	-0.3497	-1.06
1.105	0.8115	-0.3505	-1.07
1.110	0.8102	-0.3513	-1.08
1.115	0.8089	-0.3522	-1.09
1.120	0.8076	-0.3530	-1.10
1.125	0.8062	-0.3538	-1.11
1.130	0.8050	-0.3546	-1.11
1.135	0.8037	-0.3553	-1.12
1.140	0.8026	-0.3561	-1.13
1.145	0.8014	-0.3569	-1.14
1.150	0.8000	-0.3576	-1.15
1.155	0.7988	-0.3583	-1.16
1.160	0.7975	-0.3591	-1.16
1.165	0.7964	-0.3599	-1.17
1.170	0.7952	-0.3606	-1.18
1.175	0.7939	-0.3612	-1.19
1.180	0.7929	-0.3619	-1.19
1.185	0.7917	-0.3626	-1.20
1.190	0.7905	-0.3632	-1.21
1.195	0.7894	-0.3639	-1.22
1.200	0.7882	-0.3645	-1.23
1.205	0.7871	-0.3650	-1.23
1.210	0.7860	-0.3657	-1.24
1.215	0.7847	-0.3662	-1.25
1.220	0.7835	-0.3669	-1.26
1.225	0.7824	-0.3674	-1.27
1.230	0.7813	-0.3680	-1.27
1.235	0.7802	-0.3685	-1.28
1.240	0.7791	-0.3692	-1.29
1.245	0.7781	-0.3696	-1.30
1.250	0.7770	-0.3701	-1.30
1.255	0.7758	-0.3707	-1.31
1.260	0.7747	-0.3712	-1.32
1.265	0.7737	-0.3717	-1.33
1.270	0.7725	-0.3722	-1.34
1.275	0.7715	-0.3727	-1.34
1.280	0.7705	-0.3731	-1.35
1.285	0.7694	-0.3736	-1.36
1.290	0.7684	-0.3742	-1.36
1.295	0.7672	-0.3748	-1.37
1.300	0.7662	-0.3754	-1.38
1.305	0.7650	-0.3759	-1.39
1.310	0.7642	-0.3763	-1.39
1.315	0.7632	-0.3768	-1.40
1.320	0.7622	-0.3772	-1.41
1.325	0.7612	-0.3777	-1.41

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Frequency (GHz)	**Re(S_{11})**	**Im(S_{11})**	**$\lvert S_{11}\rvert$ (dB)**
1.330	0.7601	-0.3779	-1.42
1.335	0.7590	-0.3785	-1.43
1.340	0.7581	-0.3789	-1.44
1.345	0.7570	-0.3793	-1.45
1.350	0.7560	-0.3798	-1.45
1.355	0.7550	-0.3801	-1.46
1.360	0.7540	-0.3806	-1.47
1.365	0.7531	-0.3811	-1.47
1.370	0.7520	-0.3815	-1.48
1.375	0.7509	-0.3818	-1.49
1.380	0.7500	-0.3824	-1.50
1.385	0.7489	-0.3829	-1.50
1.390	0.7476	-0.3832	-1.51
1.395	0.7466	-0.3835	-1.52
1.400	0.7455	-0.3840	-1.53
1.405	0.7445	-0.3845	-1.54
1.410	0.7432	-0.3849	-1.55
1.415	0.7421	-0.3854	-1.55
1.420	0.7410	-0.3858	-1.56
1.425	0.7398	-0.3864	-1.57
1.430	0.7387	-0.3867	-1.58
1.435	0.7377	-0.3874	-1.58
1.440	0.7364	-0.3878	-1.59
1.445	0.7353	-0.3883	-1.60
1.450	0.7342	-0.3886	-1.61
1.455	0.7331	-0.3893	-1.62
1.460	0.7320	-0.3897	-1.63
1.465	0.7308	-0.3902	-1.63
1.470	0.7295	-0.3906	-1.64
1.475	0.7284	-0.3911	-1.65
1.480	0.7272	-0.3916	-1.66
1.485	0.7259	-0.3922	-1.67
1.490	0.7246	-0.3928	-1.68
1.495	0.7234	-0.3932	-1.69
1.500	0.7223	-0.3935	-1.70
1.505	0.7208	-0.3940	-1.71
1.510	0.7196	-0.3945	-1.72
1.515	0.7182	-0.3951	-1.73
1.520	0.7171	-0.3955	-1.74
1.525	0.7156	-0.3960	-1.75
1.530	0.7143	-0.3965	-1.76
1.535	0.7131	-0.3970	-1.76
1.540	0.7118	-0.3975	-1.77
1.545	0.7103	-0.3979	-1.79
1.550	0.7090	-0.3984	-1.80
1.555	0.7076	-0.3989	-1.81

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Frequency (GHz)	**Re**(S_{11})	**Im**(S_{11})	$\lvert S_{11}\rvert$ **(dB)**
1.560	0.7061	-0.3995	-1.82
1.565	0.7047	-0.4000	-1.83
1.570	0.7033	-0.4004	-1.84
1.575	0.7020	-0.4009	-1.85
1.580	0.7006	-0.4014	-1.86
1.585	0.6992	-0.4020	-1.87
1.590	0.6977	-0.4024	-1.88
1.595	0.6964	-0.4030	-1.89
1.600	0.6951	-0.4036	-1.90
1.605	0.6936	-0.4040	-1.91
1.610	0.6923	-0.4044	-1.92
1.615	0.6907	-0.4050	-1.93
1.620	0.6893	-0.4055	-1.94
1.625	0.6880	-0.4059	-1.95
1.630	0.6865	-0.4063	-1.96
1.635	0.6851	-0.4070	-1.97
1.640	0.6836	-0.4073	-1.99
1.645	0.6822	-0.4078	-1.99
1.650	0.6807	-0.4083	-2.01
1.655	0.6792	-0.4088	-2.02
1.660	0.6778	-0.4091	-2.03
1.665	0.6766	-0.4098	-2.04
1.670	0.6749	-0.4101	-2.05
1.675	0.6735	-0.4106	-2.06
1.680	0.6722	-0.4109	-2.07
1.685	0.6708	-0.4113	-2.08
1.690	0.6695	-0.4117	-2.09
1.695	0.6680	-0.4120	-2.10
1.700	0.6666	-0.4123	-2.12
1.705	0.6652	-0.4126	-2.13
1.710	0.6637	-0.4131	-2.14
1.715	0.6624	-0.4135	-2.15
1.720	0.6609	-0.4138	-2.16
1.725	0.6597	-0.4143	-2.17
1.730	0.6581	-0.4147	-2.18
1.735	0.6567	-0.4148	-2.19
1.740	0.6552	-0.4152	-2.21
1.745	0.6539	-0.4155	-2.22
1.750	0.6526	-0.4158	-2.23
1.755	0.6513	-0.4162	-2.24
1.760	0.6500	-0.4164	-2.25
1.765	0.6488	-0.4169	-2.26
1.770	0.6476	-0.4171	-2.27
1.775	0.6465	-0.4175	-2.27
1.780	0.6454	-0.4176	-2.28
1.785	0.6445	-0.4177	-2.29

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Frequency (GHz)	**Re(S_{11})**	**Im(S_{11})**	**$\lvert S_{11}\rvert$ (dB)**
1.790	0.6430	-0.4179	-2.31
1.795	0.6418	-0.4180	-2.32
1.800	0.6406	-0.4182	-2.33
1.805	0.6393	-0.4185	-2.34
1.810	0.6382	-0.4189	-2.35
1.815	0.6371	-0.4191	-2.35
1.820	0.6361	-0.4193	-2.36
1.825	0.6351	-0.4194	-2.37
1.830	0.6340	-0.4197	-2.38
1.835	0.6329	-0.4197	-2.39
1.840	0.6318	-0.4200	-2.40
1.845	0.6308	-0.4200	-2.41
1.850	0.6298	-0.4199	-2.42
1.855	0.6287	-0.4201	-2.43
1.860	0.6277	-0.4201	-2.44
1.865	0.6267	-0.4201	-2.45
1.870	0.6256	-0.4203	-2.46
1.875	0.6246	-0.4205	-2.47
1.880	0.6237	-0.4206	-2.47
1.885	0.6229	-0.4209	-2.48
1.890	0.6221	-0.4210	-2.49
1.895	0.6213	-0.4208	-2.49
1.900	0.6205	-0.4209	-2.50
1.905	0.6195	-0.4210	-2.51
1.910	0.6187	-0.4211	-2.52
1.915	0.6178	-0.4213	-2.53
1.920	0.6170	-0.4214	-2.53
1.925	0.6162	-0.4214	-2.54
1.930	0.6154	-0.4215	-2.55
1.935	0.6146	-0.4217	-2.55
1.940	0.6139	-0.4216	-2.56
1.945	0.6131	-0.4216	-2.57
1.950	0.6125	-0.4218	-2.57
1.955	0.6119	-0.4218	-2.58
1.960	0.6112	-0.4218	-2.58
1.965	0.6104	-0.4219	-2.59
1.970	0.6096	-0.4218	-2.60
1.975	0.6089	-0.4218	-2.61
1.980	0.6083	-0.4217	-2.61
1.985	0.6077	-0.4219	-2.62
1.990	0.6070	-0.4217	-2.63
1.995	0.6062	-0.4217	-2.63
2.000	0.6054	-0.4218	-2.64
2.005	0.6048	-0.4218	-2.65
2.010	0.6042	-0.4219	-2.65
2.015	0.6033	-0.4219	-2.66

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Frequency (GHz)	**Re(S_{11})**	**Im(S_{11})**	**$\lvert S_{11}\rvert$ (dB)**
2.020	0.6027	-0.4219	-2.67
2.025	0.6021	-0.4220	-2.67
2.030	0.6014	-0.4220	-2.68
2.035	0.6007	-0.4219	-2.69
2.040	0.6001	-0.4218	-2.69
2.045	0.5992	-0.4218	-2.70
2.050	0.5986	-0.4220	-2.71
2.055	0.5979	-0.4221	-2.71
2.060	0.5970	-0.4219	-2.72
2.065	0.5964	-0.4220	-2.73
2.070	0.5958	-0.4222	-2.73
2.075	0.5949	-0.4221	-2.74
2.080	0.5943	-0.4221	-2.75
2.085	0.5934	-0.4221	-2.75
2.090	0.5925	-0.4220	-2.76
2.095	0.5917	-0.4221	-2.77
2.100	0.5908	-0.4222	-2.78
2.105	0.5901	-0.4223	-2.79
2.110	0.5891	-0.4223	-2.79
2.115	0.5884	-0.4225	-2.80
2.120	0.5874	-0.4225	-2.81
2.125	0.5864	-0.4227	-2.82
2.130	0.5855	-0.4227	-2.83
2.135	0.5846	-0.4229	-2.83
2.140	0.5836	-0.4231	-2.84
2.145	0.5827	-0.4231	-2.85
2.150	0.5818	-0.4233	-2.86
2.155	0.5807	-0.4234	-2.87
2.160	0.5798	-0.4235	-2.88
2.165	0.5787	-0.4236	-2.89
2.170	0.5777	-0.4237	-2.90
2.175	0.5767	-0.4239	-2.91
2.180	0.5756	-0.4240	-2.92
2.185	0.5746	-0.4242	-2.92
2.190	0.5735	-0.4242	-2.93
2.195	0.5724	-0.4243	-2.94
2.200	0.5714	-0.4247	-2.95
2.205	0.5702	-0.4247	-2.96
2.210	0.5691	-0.4248	-2.97
2.215	0.5679	-0.4251	-2.98
2.220	0.5668	-0.4253	-2.99
2.225	0.5656	-0.4254	-3.00
2.230	0.5646	-0.4254	-3.01
2.235	0.5634	-0.4256	-3.02
2.240	0.5622	-0.4258	-3.03
2.245	0.5610	-0.4260	-3.04

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Frequency (GHz)	**Re(S_{11})**	**Im(S_{11})**	**$\lvert S_{11}\rvert$ (dB)**
2.250	0.5598	-0.4263	-3.05
2.255	0.5587	-0.4265	-3.06
2.260	0.5576	-0.4267	-3.07
2.265	0.5564	-0.4269	-3.08
2.270	0.5551	-0.4271	-3.09
2.275	0.5540	-0.4271	-3.10
2.280	0.5528	-0.4273	-3.11
2.285	0.5517	-0.4275	-3.12
2.290	0.5504	-0.4276	-3.14
2.295	0.5494	-0.4279	-3.14
2.300	0.5482	-0.4281	-3.15
2.305	0.5469	-0.4283	-3.16
2.310	0.5458	-0.4285	-3.17
2.315	0.5445	-0.4287	-3.18
2.320	0.5433	-0.4286	-3.20
2.325	0.5422	-0.4290	-3.21
2.330	0.5410	-0.4291	-3.22
2.335	0.5398	-0.4293	-3.23
2.340	0.5386	-0.4296	-3.24
2.345	0.5374	-0.4297	-3.25
2.350	0.5363	-0.4297	-3.26
2.355	0.5354	-0.4299	-3.27
2.360	0.5342	-0.4300	-3.28
2.365	0.5331	-0.4301	-3.29
2.370	0.5322	-0.4302	-3.29
2.375	0.5310	-0.4305	-3.30
2.380	0.5299	-0.4306	-3.31
2.385	0.5289	-0.4307	-3.32
2.390	0.5279	-0.4308	-3.33
2.395	0.5268	-0.4308	-3.34
2.400	0.5258	-0.4310	-3.35
2.405	0.5249	-0.4310	-3.36
2.410	0.5237	-0.4312	-3.37
2.415	0.5228	-0.4312	-3.38
2.420	0.5219	-0.4313	-3.39
2.425	0.5209	-0.4313	-3.40
2.430	0.5200	-0.4314	-3.41
2.435	0.5191	-0.4314	-3.41
2.440	0.5182	-0.4315	-3.42
2.445	0.5173	-0.4314	-3.43
2.450	0.5166	-0.4315	-3.44
2.455	0.5157	-0.4314	-3.45
2.460	0.5149	-0.4313	-3.46
2.465	0.5141	-0.4313	-3.46
2.470	0.5132	-0.4312	-3.47
2.475	0.5125	-0.4313	-3.48

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Frequency (GHz)	**Re**(S_{11})	**Im**(S_{11})	$\lvert S_{11}\rvert$ **(dB)**
2.480	0.5117	-0.4312	-3.49
2.485	0.5110	-0.4311	-3.50
2.490	0.5102	-0.4310	-3.51
2.495	0.5095	-0.4310	-3.51
2.500	0.5087	-0.4309	-3.52
2.505	0.5080	-0.4309	-3.53
2.510	0.5073	-0.4308	-3.54
2.515	0.5065	-0.4308	-3.54
2.520	0.5058	-0.4307	-3.55
2.525	0.5050	-0.4307	-3.56
2.530	0.5043	-0.4307	-3.57
2.535	0.5035	-0.4306	-3.58
2.540	0.5030	-0.4307	-3.58
2.545	0.5023	-0.4307	-3.59
2.550	0.5017	-0.4306	-3.59
2.555	0.5010	-0.4303	-3.60
2.560	0.5003	-0.4302	-3.61
2.565	0.4999	-0.4298	-3.62
2.570	0.4994	-0.4295	-3.63
2.575	0.4991	-0.4290	-3.63
2.580	0.4990	-0.4286	-3.64
2.585	0.4989	-0.4284	-3.64
2.590	0.4988	-0.4283	-3.64
2.595	0.4986	-0.4283	-3.64
2.600	0.4985	-0.4283	-3.65
2.605	0.4983	-0.4282	-3.65
2.610	0.4981	-0.4283	-3.65
2.615	0.4978	-0.4283	-3.65
2.620	0.4977	-0.4283	-3.65
2.625	0.4975	-0.4281	-3.66
2.630	0.4971	-0.4282	-3.66
2.635	0.4968	-0.4280	-3.67
2.640	0.4965	-0.4278	-3.67
2.645	0.4962	-0.4277	-3.67
2.650	0.4958	-0.4277	-3.68
2.655	0.4956	-0.4276	-3.68
2.660	0.4952	-0.4274	-3.69
2.665	0.4949	-0.4273	-3.69
2.670	0.4945	-0.4272	-3.70
2.675	0.4940	-0.4272	-3.70
2.680	0.4938	-0.4271	-3.70
2.685	0.4934	-0.4272	-3.71
2.690	0.4931	-0.4269	-3.71
2.695	0.4928	-0.4269	-3.71
2.700	0.4925	-0.4268	-3.72
2.705	0.4923	-0.4268	-3.72

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Frequency (GHz)	**Re**(S_{11})	**Im**(S_{11})	$\lvert S_{11}\rvert$ **(dB)**
2.710	0.4920	-0.4266	-3.73
2.715	0.4917	-0.4265	-3.73
2.720	0.4915	-0.4263	-3.73
2.725	0.4912	-0.4261	-3.74
2.730	0.4909	-0.4258	-3.74
2.735	0.4906	-0.4255	-3.75
2.740	0.4902	-0.4251	-3.76
2.745	0.4898	-0.4249	-3.76
2.750	0.4894	-0.4246	-3.77
2.755	0.4890	-0.4242	-3.78
2.760	0.4885	-0.4239	-3.79
2.765	0.4880	-0.4238	-3.79
2.770	0.4874	-0.4233	-3.80
2.775	0.4870	-0.4231	-3.81
2.780	0.4864	-0.4228	-3.82
2.785	0.4859	-0.4226	-3.82
2.790	0.4852	-0.4223	-3.83
2.795	0.4845	-0.4221	-3.84
2.800	0.4841	-0.4220	-3.85
2.805	0.4835	-0.4217	-3.86
2.810	0.4828	-0.4216	-3.86
2.815	0.4822	-0.4214	-3.87
2.820	0.4815	-0.4214	-3.88
2.825	0.4811	-0.4213	-3.88
2.830	0.4804	-0.4211	-3.89
2.835	0.4798	-0.4211	-3.90
2.840	0.4792	-0.4210	-3.91
2.845	0.4785	-0.4208	-3.91
2.850	0.4779	-0.4207	-3.92
2.855	0.4773	-0.4205	-3.93
2.860	0.4768	-0.4205	-3.94
2.865	0.4762	-0.4204	-3.94
2.870	0.4758	-0.4203	-3.95
2.875	0.4756	-0.4200	-3.95
2.880	0.4749	-0.4198	-3.96
2.885	0.4745	-0.4193	-3.97
2.890	0.4738	-0.4192	-3.98
2.895	0.4731	-0.4186	-3.99
2.900	0.4724	-0.4183	-4.00
2.905	0.4716	-0.4180	-4.01
2.910	0.4707	-0.4176	-4.02
2.915	0.4699	-0.4174	-4.03
2.920	0.4690	-0.4174	-4.04
2.925	0.4681	-0.4173	-4.05
2.930	0.4673	-0.4170	-4.06
2.935	0.4664	-0.4170	-4.07

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Frequency (GHz)	**Re(S_{11})**	**Im(S_{11})**	**$\lvert S_{11}\rvert$ (dB)**
2.940	0.4657	-0.4172	-4.08
2.945	0.4649	-0.4171	-4.09
2.950	0.4642	-0.4171	-4.10
2.955	0.4636	-0.4173	-4.10
2.960	0.4630	-0.4174	-4.11
2.965	0.4624	-0.4175	-4.11
2.970	0.4620	-0.4175	-4.11
2.975	0.4617	-0.4177	-4.12
2.980	0.4611	-0.4179	-4.12
2.985	0.4609	-0.4179	-4.12
2.990	0.4608	-0.4178	-4.12
2.995	0.4606	-0.4178	-4.13
3.000	0.4604	-0.4177	-4.13
3.005	0.4585	-0.4172	-4.15
3.010	0.4582	-0.4176	-4.15
3.015	0.4581	-0.4176	-4.15
3.020	0.4578	-0.4176	-4.16
3.025	0.4579	-0.4180	-4.15
3.030	0.4580	-0.4179	-4.15
3.035	0.4581	-0.4175	-4.16
3.040	0.4582	-0.4170	-4.16
3.045	0.4583	-0.4167	-4.16
3.050	0.4583	-0.4162	-4.17
3.055	0.4583	-0.4157	-4.17
3.060	0.4582	-0.4151	-4.18
3.065	0.4577	-0.4143	-4.19
3.070	0.4574	-0.4137	-4.20
3.075	0.4569	-0.4131	-4.21
3.080	0.4565	-0.4123	-4.22
3.085	0.4559	-0.4119	-4.23
3.090	0.4552	-0.4112	-4.24
3.095	0.4546	-0.4109	-4.25
3.100	0.4538	-0.4105	-4.27
3.105	0.4531	-0.4104	-4.27
3.110	0.4524	-0.4101	-4.28
3.115	0.4515	-0.4100	-4.30
3.120	0.4509	-0.4101	-4.30
3.125	0.4503	-0.4103	-4.30
3.130	0.4497	-0.4105	-4.31
3.135	0.4493	-0.4109	-4.31
3.140	0.4489	-0.4111	-4.31
3.145	0.4485	-0.4113	-4.31
3.150	0.4484	-0.4116	-4.31
3.155	0.4483	-0.4119	-4.31
3.160	0.4482	-0.4121	-4.31
3.165	0.4483	-0.4123	-4.31

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Frequency (GHz)	**Re(S_{11})**	**Im(S_{11})**	**$\lvert S_{11}\rvert$ (dB)**
3.170	0.4486	-0.4125	-4.30
3.175	0.4486	-0.4129	-4.30
3.180	0.4491	-0.4128	-4.29
3.185	0.4492	-0.4124	-4.30
3.190	0.4496	-0.4120	-4.30
3.195	0.4498	-0.4118	-4.30
3.200	0.4498	-0.4112	-4.30
3.205	0.4500	-0.4109	-4.30
3.210	0.4498	-0.4100	-4.31
3.215	0.4498	-0.4095	-4.32
3.220	0.4493	-0.4086	-4.33
3.225	0.4486	-0.4082	-4.34
3.230	0.4481	-0.4078	-4.35
3.235	0.4477	-0.4080	-4.36
3.240	0.4469	-0.4080	-4.36
3.245	0.4470	-0.4080	-4.36
3.250	0.4464	-0.4080	-4.37
3.255	0.4459	-0.4081	-4.37
3.260	0.4459	-0.4084	-4.37
3.265	0.4455	-0.4087	-4.37
3.270	0.4455	-0.4085	-4.37
3.275	0.4454	-0.4087	-4.37
3.280	0.4453	-0.4087	-4.37
3.285	0.4451	-0.4089	-4.37
3.290	0.4450	-0.4091	-4.37
3.295	0.4448	-0.4092	-4.37
3.300	0.4448	-0.4092	-4.37
3.305	0.4446	-0.4091	-4.38
3.310	0.4444	-0.4093	-4.38
3.315	0.4445	-0.4095	-4.37
3.320	0.4444	-0.4092	-4.38
3.325	0.4443	-0.4094	-4.38
3.330	0.4443	-0.4093	-4.38
3.335	0.4442	-0.4092	-4.38
3.340	0.4442	-0.4092	-4.38
3.345	0.4443	-0.4089	-4.38
3.350	0.4441	-0.4087	-4.39
3.355	0.4439	-0.4085	-4.39
3.360	0.4440	-0.4081	-4.39
3.365	0.4440	-0.4078	-4.40
3.370	0.4440	-0.4076	-4.40
3.375	0.4434	-0.4071	-4.41
3.380	0.4434	-0.4067	-4.41
3.385	0.4431	-0.4063	-4.42
3.390	0.4425	-0.4059	-4.43
3.395	0.4424	-0.4055	-4.44

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Frequency (GHz)	**Re(S_{11})**	**Im(S_{11})**	**$\lvert S_{11}\rvert$ (dB)**
3.400	0.4417	-0.4050	-4.45
3.405	0.4415	-0.4045	-4.45
3.410	0.4409	-0.4039	-4.47
3.415	0.4403	-0.4034	-4.48
3.420	0.4398	-0.4032	-4.49
3.425	0.4391	-0.4027	-4.50
3.430	0.4386	-0.4024	-4.51
3.435	0.4377	-0.4021	-4.52
3.440	0.4370	-0.4017	-4.53
3.445	0.4361	-0.4016	-4.54
3.450	0.4353	-0.4010	-4.56
3.455	0.4346	-0.4007	-4.57
3.460	0.4335	-0.4007	-4.58
3.465	0.4330	-0.4007	-4.58
3.470	0.4320	-0.4007	-4.59
3.475	0.4312	-0.4006	-4.60
3.480	0.4306	-0.4009	-4.61
3.485	0.4299	-0.4014	-4.61
3.490	0.4294	-0.4014	-4.62
3.495	0.4287	-0.4015	-4.62
3.500	0.4288	-0.4021	-4.61
3.505	0.4285	-0.4024	-4.62
3.510	0.4283	-0.4025	-4.62
3.515	0.4285	-0.4024	-4.62
3.520	0.4285	-0.4023	-4.62
3.525	0.4284	-0.4022	-4.62
3.530	0.4290	-0.4020	-4.61
3.535	0.4293	-0.4014	-4.62
3.540	0.4297	-0.4003	-4.62
3.545	0.4296	-0.3990	-4.64
3.550	0.4296	-0.3976	-4.65
3.555	0.4293	-0.3958	-4.67
3.560	0.4285	-0.3943	-4.70
3.565	0.4272	-0.3928	-4.73
3.570	0.4256	-0.3913	-4.76
3.575	0.4241	-0.3898	-4.79
3.580	0.4215	-0.3885	-4.83
3.585	0.4190	-0.3884	-4.86
3.590	0.4164	-0.3885	-4.89
3.595	0.4141	-0.3895	-4.91
3.600	0.4124	-0.3910	-4.91
3.605	0.4114	-0.3927	-4.90
3.610	0.4111	-0.3941	-4.89
3.615	0.4115	-0.3956	-4.87
3.620	0.4119	-0.3965	-4.86
3.625	0.4127	-0.3978	-4.83

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Frequency (GHz)	**Re(S_{11})**	**Im(S_{11})**	**$\lvert S_{11}\rvert$ (dB)**
3.630	0.4137	-0.3972	-4.83
3.635	0.4146	-0.3968	-4.82
3.640	0.4152	-0.3963	-4.82
3.645	0.4156	-0.3952	-4.83
3.650	0.4157	-0.3946	-4.83
3.655	0.4157	-0.3937	-4.84
3.660	0.4156	-0.3928	-4.85
3.665	0.4151	-0.3920	-4.87
3.670	0.4150	-0.3911	-4.88
3.675	0.4148	-0.3902	-4.89
3.680	0.4145	-0.3893	-4.90
3.685	0.4138	-0.3885	-4.92
3.690	0.4133	-0.3875	-4.94
3.695	0.4125	-0.3867	-4.95
3.700	0.4117	-0.3860	-4.97
3.705	0.4105	-0.3851	-4.99
3.710	0.4092	-0.3845	-5.01
3.715	0.4079	-0.3839	-5.03
3.720	0.4062	-0.3837	-5.06
3.725	0.4047	-0.3838	-5.07
3.730	0.4023	-0.3839	-5.10
3.735	0.4008	-0.3849	-5.10
3.740	0.3994	-0.3865	-5.10
3.745	0.3985	-0.3879	-5.10
3.750	0.3982	-0.3898	-5.08
3.755	0.3983	-0.3916	-5.06
3.760	0.3994	-0.3935	-5.03
3.765	0.4011	-0.3948	-4.99
3.770	0.4032	-0.3949	-4.97
3.775	0.4062	-0.3935	-4.95
3.780	0.4075	-0.3931	-4.94
3.785	0.4086	-0.3900	-4.96
3.790	0.4079	-0.3878	-4.99
3.795	0.4071	-0.3860	-5.02
3.800	0.4057	-0.3849	-5.05
3.805	0.4044	-0.3839	-5.07
3.810	0.4028	-0.3835	-5.10
3.815	0.4014	-0.3831	-5.12
3.820	0.4000	-0.3832	-5.13
3.825	0.3989	-0.3836	-5.14
3.830	0.3979	-0.3837	-5.15
3.835	0.3975	-0.3864	-5.12
3.840	0.3970	-0.3867	-5.13
3.845	0.3969	-0.3867	-5.13
3.850	0.3964	-0.3867	-5.13
3.855	0.3962	-0.3867	-5.14

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Frequency (GHz)	**Re(S_{11})**	**Im(S_{11})**	**$\lvert S_{11}\rvert$ (dB)**
3.860	0.3959	-0.3868	-5.14
3.865	0.3955	-0.3867	-5.14
3.870	0.3953	-0.3869	-5.14
3.875	0.3949	-0.3867	-5.15
3.880	0.3946	-0.3868	-5.15
3.885	0.3943	-0.3866	-5.16
3.890	0.3938	-0.3867	-5.16
3.895	0.3932	-0.3868	-5.17
3.900	0.3928	-0.3868	-5.17
3.905	0.3923	-0.3866	-5.18
3.910	0.3917	-0.3867	-5.19
3.915	0.3911	-0.3869	-5.19
3.920	0.3906	-0.3869	-5.20
3.925	0.3901	-0.3870	-5.20
3.930	0.3897	-0.3871	-5.20
3.935	0.3889	-0.3870	-5.21
3.940	0.3883	-0.3871	-5.22
3.945	0.3876	-0.3871	-5.23
3.950	0.3871	-0.3872	-5.23
3.955	0.3864	-0.3872	-5.24
3.960	0.3857	-0.3872	-5.25
3.965	0.3851	-0.3872	-5.25
3.970	0.3844	-0.3872	-5.26
3.975	0.3836	-0.3872	-5.27
3.980	0.3829	-0.3874	-5.28
3.985	0.3822	-0.3873	-5.29
3.990	0.3815	-0.3872	-5.29
3.995	0.3806	-0.3874	-5.30
4.000	0.3798	-0.3875	-5.31
4.005	0.3791	-0.3874	-5.32
4.010	0.3782	-0.3874	-5.33
4.015	0.3774	-0.3872	-5.34
4.020	0.3765	-0.3873	-5.35
4.025	0.3757	-0.3873	-5.36
4.030	0.3749	-0.3873	-5.37
4.035	0.3740	-0.3872	-5.38
4.040	0.3731	-0.3872	-5.39
4.045	0.3723	-0.3872	-5.40
4.050	0.3714	-0.3870	-5.41
4.055	0.3706	-0.3871	-5.42
4.060	0.3696	-0.3870	-5.43
4.065	0.3687	-0.3871	-5.44
4.070	0.3679	-0.3869	-5.45
4.075	0.3670	-0.3869	-5.46
4.080	0.3661	-0.3869	-5.47
4.085	0.3653	-0.3869	-5.48

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Frequency (GHz)	**Re(S_{11})**	**Im(S_{11})**	**$\lvert S_{11}\rvert$ (dB)**
4.090	0.3643	-0.3868	-5.49
4.095	0.3636	-0.3868	-5.50
4.100	0.3626	-0.3867	-5.51
4.105	0.3615	-0.3866	-5.53
4.110	0.3607	-0.3868	-5.53
4.115	0.3599	-0.3870	-5.54
4.120	0.3589	-0.3869	-5.55
4.125	0.3581	-0.3870	-5.56
4.130	0.3570	-0.3874	-5.57
4.135	0.3558	-0.3873	-5.58
4.140	0.3547	-0.3875	-5.59
4.145	0.3534	-0.3872	-5.61
4.150	0.3524	-0.3875	-5.62
4.155	0.3511	-0.3873	-5.63
4.160	0.3501	-0.3871	-5.65
4.165	0.3492	-0.3867	-5.66
4.170	0.3482	-0.3865	-5.68
4.175	0.3476	-0.3863	-5.68
4.180	0.3469	-0.3862	-5.70
4.185	0.3463	-0.3859	-5.70
4.190	0.3457	-0.3858	-5.71
4.195	0.3450	-0.3857	-5.72
4.200	0.3446	-0.3857	-5.73
4.205	0.3441	-0.3857	-5.73
4.210	0.3435	-0.3856	-5.74
4.215	0.3432	-0.3855	-5.75
4.220	0.3427	-0.3854	-5.75
4.225	0.3422	-0.3853	-5.76
4.230	0.3419	-0.3854	-5.76
4.235	0.3415	-0.3853	-5.77
4.240	0.3410	-0.3852	-5.77
4.245	0.3407	-0.3852	-5.78
4.250	0.3405	-0.3853	-5.78
4.255	0.3401	-0.3852	-5.78
4.260	0.3399	-0.3852	-5.79
4.265	0.3396	-0.3854	-5.79
4.270	0.3396	-0.3854	-5.79
4.275	0.3394	-0.3854	-5.79
4.280	0.3392	-0.3854	-5.79
4.285	0.3391	-0.3855	-5.79
4.290	0.3389	-0.3853	-5.80
4.295	0.3388	-0.3853	-5.80
4.300	0.3386	-0.3853	-5.80
4.305	0.3384	-0.3852	-5.80
4.310	0.3382	-0.3852	-5.81
4.315	0.3380	-0.3852	-5.81

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Frequency (GHz)	**Re(S_{11})**	**Im(S_{11})**	**$\lvert S_{11}\rvert$ (dB)**
4.320	0.3379	-0.3851	-5.81
4.325	0.3377	-0.3851	-5.81
4.330	0.3376	-0.3851	-5.81
4.335	0.3373	-0.3853	-5.81
4.340	0.3372	-0.3855	-5.81
4.345	0.3371	-0.3856	-5.81
4.350	0.3370	-0.3859	-5.81
4.355	0.3369	-0.3860	-5.81
4.360	0.3368	-0.3862	-5.81
4.365	0.3369	-0.3864	-5.80
4.370	0.3370	-0.3867	-5.80
4.375	0.3370	-0.3869	-5.80
4.380	0.3373	-0.3870	-5.79
4.385	0.3377	-0.3875	-5.78
4.390	0.3379	-0.3874	-5.78
4.395	0.3382	-0.3877	-5.77
4.400	0.3384	-0.3877	-5.77
4.405	0.3388	-0.3877	-5.77
4.410	0.3391	-0.3877	-5.76
4.415	0.3395	-0.3876	-5.76
4.420	0.3396	-0.3878	-5.76
4.425	0.3400	-0.3874	-5.76
4.430	0.3401	-0.3874	-5.76
4.435	0.3402	-0.3871	-5.76
4.440	0.3403	-0.3869	-5.76
4.445	0.3404	-0.3866	-5.76
4.450	0.3405	-0.3865	-5.76
4.455	0.3406	-0.3864	-5.76
4.460	0.3406	-0.3863	-5.76
4.465	0.3409	-0.3862	-5.76
4.470	0.3413	-0.3862	-5.76
4.475	0.3418	-0.3863	-5.75
4.480	0.3422	-0.3865	-5.74
4.485	0.3427	-0.3869	-5.73
4.490	0.3433	-0.3873	-5.72
4.495	0.3439	-0.3877	-5.71
4.500	0.3444	-0.3881	-5.70
4.505	0.3449	-0.3883	-5.69
4.510	0.3455	-0.3886	-5.68
4.515	0.3462	-0.3886	-5.67
4.520	0.3467	-0.3887	-5.67
4.525	0.3471	-0.3885	-5.66
4.530	0.3474	-0.3883	-5.66
4.535	0.3479	-0.3882	-5.66
4.540	0.3480	-0.3879	-5.66
4.545	0.3482	-0.3876	-5.66

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Frequency (GHz)	**Re(S_{11})**	**Im(S_{11})**	**$\lvert S_{11}\rvert$ (dB)**
4.550	0.3483	-0.3875	-5.66
4.555	0.3486	-0.3874	-5.66
4.560	0.3486	-0.3873	-5.66
4.565	0.3487	-0.3871	-5.66
4.570	0.3489	-0.3869	-5.66
4.575	0.3488	-0.3865	-5.67
4.580	0.3489	-0.3864	-5.67
4.585	0.3488	-0.3862	-5.67
4.590	0.3490	-0.3858	-5.68
4.595	0.3488	-0.3857	-5.68
4.600	0.3487	-0.3856	-5.68
4.605	0.3486	-0.3853	-5.69
4.610	0.3484	-0.3849	-5.69
4.615	0.3481	-0.3848	-5.70
4.620	0.3479	-0.3845	-5.70
4.625	0.3480	-0.3844	-5.70
4.630	0.3476	-0.3843	-5.71
4.635	0.3475	-0.3842	-5.71
4.640	0.3474	-0.3840	-5.72
4.645	0.3472	-0.3837	-5.72
4.650	0.3471	-0.3835	-5.73
4.655	0.3469	-0.3833	-5.73
4.660	0.3468	-0.3829	-5.74
4.665	0.3470	-0.3827	-5.74
4.670	0.3465	-0.3827	-5.74
4.675	0.3465	-0.3823	-5.75
4.680	0.3466	-0.3821	-5.75
4.685	0.3467	-0.3820	-5.75
4.690	0.3466	-0.3818	-5.75
4.695	0.3467	-0.3816	-5.75
4.700	0.3464	-0.3813	-5.76
4.705	0.3466	-0.3806	-5.77
4.710	0.3465	-0.3808	-5.77
4.715	0.3465	-0.3803	-5.77
4.720	0.3462	-0.3795	-5.79
4.725	0.3461	-0.3794	-5.79
4.730	0.3459	-0.3789	-5.80
4.735	0.3454	-0.3789	-5.80
4.740	0.3453	-0.3781	-5.81
4.745	0.3444	-0.3779	-5.83
4.750	0.3441	-0.3775	-5.83
4.755	0.3431	-0.3772	-5.85
4.760	0.3424	-0.3771	-5.86
4.765	0.3418	-0.3769	-5.87
4.770	0.3409	-0.3769	-5.88
4.775	0.3405	-0.3767	-5.89

(continued on next page)

Frequency (GHz)	**Re(S_{11})**	**Im(S_{11})**	**$\lvert S_{11}\rvert$ (dB)**
4.780	0.3403	-0.3764	-5.89
4.785	0.3399	-0.3768	-5.89
4.790	0.3399	-0.3765	-5.90
4.795	0.3396	-0.3762	-5.90
4.800	0.3398	-0.3759	-5.90
4.805	0.3397	-0.3755	-5.91
4.810	0.3398	-0.3750	-5.92
4.815	0.3398	-0.3743	-5.92
4.820	0.3398	-0.3739	-5.93
4.825	0.3398	-0.3733	-5.94
4.830	0.3397	-0.3727	-5.95
4.835	0.3397	-0.3722	-5.95
4.840	0.3398	-0.3717	-5.96
4.845	0.3395	-0.3712	-5.97
4.850	0.3395	-0.3709	-5.97
4.855	0.3396	-0.3708	-5.97
4.860	0.3395	-0.3703	-5.98
4.865	0.3394	-0.3703	-5.98
4.870	0.3392	-0.3701	-5.99
4.875	0.3393	-0.3701	-5.98
4.880	0.3393	-0.3701	-5.98
4.885	0.3392	-0.3700	-5.99
4.890	0.3392	-0.3702	-5.98
4.895	0.3391	-0.3702	-5.99
4.900	0.3391	-0.3703	-5.98
4.905	0.3392	-0.3705	-5.98
4.910	0.3391	-0.3706	-5.98
4.915	0.3391	-0.3706	-5.98
4.920	0.3390	-0.3708	-5.98
4.925	0.3390	-0.3709	-5.98
4.930	0.3389	-0.3710	-5.98
4.935	0.3388	-0.3712	-5.98
4.940	0.3389	-0.3713	-5.97
4.945	0.3389	-0.3715	-5.97
4.950	0.3388	-0.3716	-5.97
4.955	0.3389	-0.3717	-5.97
4.960	0.3388	-0.3718	-5.97
4.965	0.3387	-0.3719	-5.97
4.970	0.3389	-0.3720	-5.96
4.975	0.3387	-0.3723	-5.96
4.980	0.3386	-0.3723	-5.96
4.985	0.3388	-0.3727	-5.96
4.990	0.3387	-0.3726	-5.96
4.995	0.3387	-0.3729	-5.96
5.000	0.3387	-0.3731	-5.95
5.005	0.3388	-0.3732	-5.95

(continued on next page)

Frequency (GHz)	**Re(S_{11})**	**Im(S_{11})**	**$\lvert S_{11}\rvert$ (dB)**
5.010	0.3387	-0.3733	-5.95
5.015	0.3388	-0.3735	-5.95
5.020	0.3388	-0.3735	-5.95
5.025	0.3391	-0.3737	-5.94
5.030	0.3390	-0.3736	-5.94
5.035	0.3391	-0.3736	-5.94
5.040	0.3393	-0.3736	-5.94
5.045	0.3397	-0.3737	-5.93
5.050	0.3400	-0.3732	-5.94
5.055	0.3401	-0.3733	-5.93
5.060	0.3405	-0.3730	-5.93
5.065	0.3408	-0.3730	-5.93
5.070	0.3410	-0.3726	-5.93
5.075	0.3412	-0.3726	-5.93
5.080	0.3417	-0.3722	-5.93
5.085	0.3417	-0.3721	-5.93
5.090	0.3421	-0.3718	-5.93
5.095	0.3422	-0.3716	-5.93
5.100	0.3423	-0.3713	-5.93
5.105	0.3424	-0.3712	-5.93
5.110	0.3425	-0.3710	-5.94
5.115	0.3427	-0.3708	-5.94
5.120	0.3428	-0.3707	-5.93
5.125	0.3429	-0.3706	-5.94
5.130	0.3431	-0.3705	-5.94
5.135	0.3430	-0.3703	-5.94
5.140	0.3434	-0.3703	-5.93
5.145	0.3436	-0.3702	-5.93
5.150	0.3437	-0.3702	-5.93
5.155	0.3437	-0.3701	-5.93
5.160	0.3438	-0.3700	-5.93
5.165	0.3436	-0.3699	-5.94
5.170	0.3436	-0.3698	-5.94
5.175	0.3435	-0.3697	-5.94
5.180	0.3433	-0.3694	-5.95
5.185	0.3431	-0.3695	-5.95
5.190	0.3428	-0.3692	-5.95
5.195	0.3424	-0.3690	-5.96
5.200	0.3422	-0.3688	-5.97
5.205	0.3419	-0.3686	-5.97
5.210	0.3415	-0.3683	-5.98
5.215	0.3411	-0.3681	-5.99
5.220	0.3407	-0.3679	-6.00
5.225	0.3402	-0.3676	-6.00
5.230	0.3398	-0.3672	-6.02
5.235	0.3394	-0.3668	-6.02

(continued on next page)

Frequency (GHz)	**Re(S_{11})**	**Im(S_{11})**	**$\lvert S_{11}\rvert$ (dB)**
5.240	0.3389	-0.3664	-6.04
5.245	0.3384	-0.3661	-6.05
5.250	0.3378	-0.3657	-6.06
5.255	0.3372	-0.3652	-6.07
5.260	0.3367	-0.3648	-6.08
5.265	0.3363	-0.3646	-6.09
5.270	0.3355	-0.3640	-6.11
5.275	0.3348	-0.3638	-6.12
5.280	0.3341	-0.3634	-6.13
5.285	0.3334	-0.3630	-6.14
5.290	0.3330	-0.3628	-6.15
5.295	0.3319	-0.3623	-6.17
5.300	0.3316	-0.3623	-6.18
5.305	0.3310	-0.3616	-6.19
5.310	0.3302	-0.3617	-6.20
5.315	0.3295	-0.3616	-6.21
5.320	0.3287	-0.3612	-6.23
5.325	0.3285	-0.3610	-6.23
5.330	0.3279	-0.3606	-6.24
5.335	0.3272	-0.3603	-6.25
5.340	0.3267	-0.3602	-6.26
5.345	0.3262	-0.3599	-6.27
5.350	0.3257	-0.3595	-6.28
5.355	0.3250	-0.3590	-6.30
5.360	0.3246	-0.3586	-6.31
5.365	0.3238	-0.3582	-6.32
5.370	0.3232	-0.3577	-6.34
5.375	0.3227	-0.3574	-6.35
5.380	0.3217	-0.3566	-6.37
5.385	0.3210	-0.3562	-6.38
5.390	0.3200	-0.3558	-6.40
5.395	0.3193	-0.3556	-6.41
5.400	0.3185	-0.3553	-6.43
5.405	0.3178	-0.3553	-6.44
5.410	0.3170	-0.3552	-6.45
5.415	0.3164	-0.3547	-6.46
5.420	0.3155	-0.3548	-6.47
5.425	0.3148	-0.3546	-6.48
5.430	0.3141	-0.3543	-6.49
5.435	0.3134	-0.3543	-6.50
5.440	0.3128	-0.3541	-6.51
5.445	0.3122	-0.3540	-6.52
5.450	0.3116	-0.3539	-6.53
5.455	0.3110	-0.3535	-6.54
5.460	0.3107	-0.3537	-6.54
5.465	0.3101	-0.3535	-6.55

(continued on next page)

Frequency (GHz)	**Re(S_{11})**	**Im(S_{11})**	**$\lvert S_{11}\rvert$ (dB)**
5.470	0.3096	-0.3532	-6.56
5.475	0.3089	-0.3529	-6.58
5.480	0.3084	-0.3526	-6.59
5.485	0.3077	-0.3525	-6.60
5.490	0.3070	-0.3522	-6.61
5.495	0.3061	-0.3518	-6.63
5.500	0.3056	-0.3517	-6.63
5.505	0.3049	-0.3516	-6.64
5.510	0.3040	-0.3515	-6.66
5.515	0.3033	-0.3515	-6.66
5.520	0.3026	-0.3515	-6.67
5.525	0.3022	-0.3516	-6.68
5.530	0.3017	-0.3518	-6.68
5.535	0.3011	-0.3518	-6.69
5.540	0.3006	-0.3520	-6.69
5.545	0.3003	-0.3520	-6.69
5.550	0.2998	-0.3521	-6.70
5.555	0.2994	-0.3522	-6.70
5.560	0.2990	-0.3522	-6.71
5.565	0.2986	-0.3523	-6.71
5.570	0.2982	-0.3524	-6.71
5.575	0.2977	-0.3525	-6.72
5.580	0.2973	-0.3525	-6.72
5.585	0.2968	-0.3524	-6.73
5.590	0.2962	-0.3525	-6.74
5.595	0.2956	-0.3527	-6.74
5.600	0.2949	-0.3530	-6.74
5.605	0.2945	-0.3532	-6.75
5.610	0.2938	-0.3535	-6.75
5.615	0.2934	-0.3540	-6.75
5.620	0.2929	-0.3545	-6.75
5.625	0.2926	-0.3550	-6.74
5.630	0.2920	-0.3555	-6.74
5.635	0.2918	-0.3560	-6.74
5.640	0.2915	-0.3564	-6.74
5.645	0.2913	-0.3568	-6.73
5.650	0.2911	-0.3572	-6.73
5.655	0.2910	-0.3576	-6.73
5.660	0.2908	-0.3579	-6.72
5.665	0.2905	-0.3581	-6.72
5.670	0.2903	-0.3588	-6.72
5.675	0.2902	-0.3590	-6.71
5.680	0.2898	-0.3595	-6.71
5.685	0.2896	-0.3599	-6.71
5.690	0.2894	-0.3602	-6.71
5.695	0.2894	-0.3606	-6.70

(continued on next page)

Frequency (GHz)	**Re(S_{11})**	**Im(S_{11})**	**$\lvert S_{11}\rvert$ (dB)**
5.700	0.2892	-0.3609	-6.70
5.705	0.2892	-0.3613	-6.69
5.710	0.2891	-0.3615	-6.69
5.715	0.2890	-0.3618	-6.69
5.720	0.2888	-0.3623	-6.68
5.725	0.2889	-0.3625	-6.68
5.730	0.2888	-0.3628	-6.68
5.735	0.2888	-0.3631	-6.67
5.740	0.2886	-0.3632	-6.67
5.745	0.2887	-0.3634	-6.67
5.750	0.2888	-0.3637	-6.66
5.755	0.2887	-0.3638	-6.66
5.760	0.2886	-0.3640	-6.66
5.765	0.2887	-0.3642	-6.66
5.770	0.2886	-0.3641	-6.66
5.775	0.2887	-0.3643	-6.65
5.780	0.2885	-0.3643	-6.66
5.785	0.2885	-0.3644	-6.65
5.790	0.2885	-0.3644	-6.66
5.795	0.2882	-0.3643	-6.66
5.800	0.2883	-0.3642	-6.66
5.805	0.2882	-0.3640	-6.66
5.810	0.2879	-0.3639	-6.67
5.815	0.2878	-0.3637	-6.67
5.820	0.2875	-0.3635	-6.68
5.825	0.2872	-0.3634	-6.69
5.830	0.2871	-0.3631	-6.69
5.835	0.2868	-0.3629	-6.70
5.840	0.2866	-0.3627	-6.70
5.845	0.2862	-0.3624	-6.71
5.850	0.2859	-0.3620	-6.72
5.855	0.2857	-0.3619	-6.73
5.860	0.2853	-0.3616	-6.73
5.865	0.2850	-0.3611	-6.74
5.870	0.2843	-0.3610	-6.75
5.875	0.2839	-0.3607	-6.76
5.880	0.2835	-0.3604	-6.77
5.885	0.2828	-0.3601	-6.79
5.890	0.2823	-0.3599	-6.79
5.895	0.2820	-0.3596	-6.80
5.900	0.2814	-0.3593	-6.81
5.905	0.2811	-0.3590	-6.82
5.910	0.2807	-0.3592	-6.82
5.915	0.2801	-0.3590	-6.83
5.920	0.2798	-0.3587	-6.84
5.925	0.2794	-0.3582	-6.85

(continued on next page)

Frequency (GHz)	**Re**(S_{11})	**Im**(S_{11})	$\lvert S_{11}\rvert$ **(dB)**
5.930	0.2791	-0.3580	-6.86
5.935	0.2787	-0.3574	-6.87
5.940	0.2782	-0.3573	-6.88
5.945	0.2777	-0.3565	-6.90
5.950	0.2772	-0.3563	-6.91
5.955	0.2769	-0.3559	-6.92
5.960	0.2762	-0.3553	-6.94
5.965	0.2750	-0.3549	-6.96
5.970	0.2745	-0.3545	-6.97
5.975	0.2737	-0.3545	-6.98
5.980	0.2725	-0.3541	-7.00
5.985	0.2718	-0.3538	-7.01
5.990	0.2713	-0.3540	-7.01
5.995	0.2706	-0.3540	-7.02
6.000	0.2700	-0.3538	-7.03

2 Measurement Quality

Reliable band: from 42 MHz (S11 error σ = 1.0e-03); the points below it are kept in the plot (shaded, outside the scale) and in the tables, but are not reliable.
Spike removal: on (isolated point, 2nd difference > 6 x local noise and > S11 error budget, not shared with another sweep); replaced 1 point(s): Short at 4885 MHz ($|\Delta S11|$ = 1.4e-03).
Outliers shared by several sweeps, left alone (instrument artefacts cancelled by the calibration): 100 MHz, 3835 MHz.

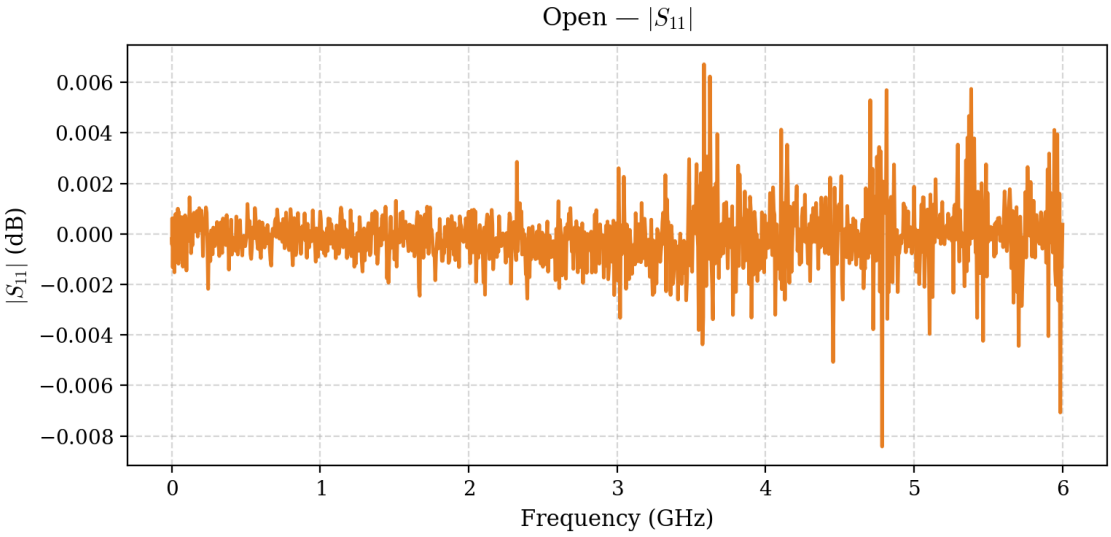
- Unreliable below 42 MHz: 41 point(s) (3 mm probe, S11 error σ = 1.0e-03). There the reference liquids resolve ε_r only to ±116 at 1 MHz … ±2.04 at 41 MHz per σ (usable: $\leq$ ±2); the reference liquids separate -31% at 1 MHz … -10% at 4 MHz unlike the probe model (tolerance ±10%): the S11 there is not the probe's (smoothing, drift or calibration error) and the sample's ε_r drifts by about as much; the sample differs from air by less than 1σ (|S11_dut - S11_open|) at 1 of them. Values kept, but left out of the chart scale and of the summary medians; start the sweep at $\geq$ 42 MHz.

- Gn shows no radiation signature even at the top of the sweep (3 GHz – 6 GHz): mean arg(Gn) = -108° (radiation: -90°, median deviation 20°) and $|Gn| \propto f^{0.4}$ (radiation: f^3). Over the whole sweep it holds noise and the inconsistency between the reference liquids instead (its term is ≈7% of the Ref 1 (Distilled Water) admittance), so the full method's radiation correction is empirical here: compare with the simplified cross-check, especially for samples whose ε_r lies outside the range of the reference liquids.

- 13 frequency point(s) excluded: reference liquids nearly indistinguishable (|S11_ref1-S11_ref2| < 1e-02). Consider starting the sweep above the affected range.

3 Calibration Standard Measurements

3.1 Open



Frequency (GHz)	**Re(S_{11})**	**Im(S_{11})**	**$\lvert S_{11}\rvert$ (dB)**
0.001	1.0000	+0.0001	-0.00
0.002	1.0000	+0.0001	-0.00
0.003	1.0001	+0.0001	0.00
0.004	1.0000	+0.0001	-0.00
0.005	1.0000	+0.0001	-0.00
0.006	1.0000	+0.0001	0.00
0.007	0.9998	+0.0000	-0.00
0.008	1.0000	-0.0001	-0.00
0.009	1.0001	+0.0000	0.00
0.010	0.9999	-0.0000	-0.00
0.011	1.0001	+0.0000	0.00
0.012	0.9999	-0.0001	-0.00
0.013	0.9999	+0.0001	-0.00
0.014	0.9999	+0.0000	-0.00
0.015	1.0000	+0.0001	0.00
0.016	1.0000	+0.0001	0.00
0.017	0.9999	+0.0001	-0.00
0.018	0.9998	-0.0000	-0.00
0.019	1.0000	+0.0001	0.00
0.020	1.0000	+0.0000	-0.00
0.021	1.0000	-0.0000	0.00
0.022	1.0001	-0.0001	0.00
0.023	0.9999	+0.0001	-0.00
0.024	1.0001	+0.0000	0.00
0.025	0.9999	+0.0000	-0.00

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Frequency (GHz)	**Re**(S_{11})	**Im**(S_{11})	$\lvert S_{11}\rvert$ **(dB)**
0.026	1.0001	+0.0002	0.00
0.027	1.0000	+0.0000	-0.00
0.028	1.0001	+0.0001	0.00
0.029	1.0001	+0.0001	0.00
0.030	1.0000	+0.0001	-0.00
0.031	1.0000	-0.0000	-0.00
0.032	1.0001	-0.0001	0.00
0.033	1.0000	-0.0000	-0.00
0.034	0.9999	+0.0001	-0.00
0.035	0.9999	+0.0001	-0.00
0.036	1.0001	-0.0000	0.00
0.037	0.9999	+0.0000	-0.00
0.038	1.0000	+0.0001	-0.00
0.039	0.9999	+0.0001	-0.00
0.040	1.0001	+0.0000	0.00
0.041	1.0001	-0.0000	0.00
0.042	1.0000	+0.0000	-0.00
0.043	1.0000	+0.0001	-0.00
0.044	1.0000	+0.0001	-0.00
0.045	0.9999	+0.0000	-0.00
0.046	1.0000	+0.0000	0.00
0.047	0.9999	-0.0001	-0.00
0.048	1.0001	-0.0000	0.00
0.049	0.9999	+0.0000	-0.00
0.050	1.0000	+0.0000	0.00
0.051	0.9999	+0.0001	-0.00
0.052	1.0000	+0.0000	-0.00
0.053	1.0000	-0.0001	0.00
0.054	1.0001	+0.0000	0.00
0.055	0.9999	+0.0000	-0.00
0.056	1.0000	+0.0001	-0.00
0.057	1.0001	+0.0000	0.00
0.058	1.0000	-0.0000	0.00
0.059	1.0001	+0.0001	0.00
0.060	1.0000	-0.0001	0.00
0.061	1.0000	+0.0001	0.00
0.062	1.0000	+0.0001	-0.00
0.063	1.0000	-0.0000	-0.00
0.064	1.0000	+0.0001	0.00
0.065	1.0000	+0.0000	0.00
0.066	1.0000	+0.0001	-0.00
0.067	1.0000	+0.0001	0.00
0.068	1.0001	-0.0001	0.00
0.069	0.9999	+0.0000	-0.00
0.070	0.9999	-0.0000	-0.00
0.071	1.0000	+0.0000	0.00

(continued on next page)

Frequency (GHz)	**Re**(S_{11})	**Im**(S_{11})	$\lvert S_{11}\rvert$ **(dB)**
0.072	1.0001	+0.0000	0.00
0.073	1.0000	+0.0001	-0.00
0.074	1.0000	+0.0001	-0.00
0.075	0.9999	-0.0000	-0.00
0.076	1.0000	+0.0000	-0.00
0.077	0.9999	+0.0000	-0.00
0.078	1.0000	-0.0000	-0.00
0.079	1.0001	+0.0000	0.00
0.080	1.0000	+0.0001	-0.00
0.081	1.0000	+0.0000	-0.00
0.082	1.0000	+0.0001	0.00
0.083	1.0000	-0.0000	-0.00
0.084	1.0001	+0.0000	0.00
0.085	1.0000	-0.0000	0.00
0.086	1.0001	+0.0000	0.00
0.087	0.9999	+0.0001	-0.00
0.088	1.0000	+0.0001	-0.00
0.089	1.0001	+0.0001	0.00
0.090	1.0001	+0.0001	0.00
0.091	1.0000	+0.0001	-0.00
0.092	1.0000	-0.0000	-0.00
0.093	1.0001	+0.0002	0.00
0.094	1.0000	+0.0001	-0.00
0.095	1.0000	+0.0000	-0.00
0.096	1.0000	+0.0001	0.00
0.097	1.0001	+0.0001	0.00
0.098	1.0000	+0.0000	0.00
0.099	0.9998	-0.0000	-0.00
0.100	1.0000	-0.0000	0.00
0.105	1.0000	+0.0000	0.00
0.110	1.0000	+0.0001	-0.00
0.115	1.0000	+0.0001	-0.00
0.120	1.0002	+0.0001	0.00
0.125	1.0000	-0.0000	0.00
0.130	0.9999	-0.0001	-0.00
0.135	1.0000	-0.0001	-0.00
0.140	0.9999	-0.0000	-0.00
0.145	1.0001	+0.0000	0.00
0.150	1.0001	-0.0000	0.00
0.155	1.0000	+0.0002	0.00
0.160	1.0001	+0.0001	0.00
0.165	1.0000	+0.0000	0.00
0.170	1.0001	+0.0000	0.00
0.175	1.0000	+0.0001	-0.00
0.180	1.0000	-0.0001	0.00
0.185	1.0001	+0.0000	0.00

(continued on next page)

Frequency (GHz)	**Re(S_{11})**	**Im(S_{11})**	**$\lvert S_{11}\rvert$ (dB)**
0.190	1.0000	+0.0001	0.00
0.195	1.0000	+0.0001	0.00
0.200	1.0001	+0.0000	0.00
0.205	1.0001	-0.0001	0.00
0.210	0.9999	+0.0001	-0.00
0.215	1.0000	+0.0000	0.00
0.220	1.0000	-0.0000	0.00
0.225	1.0000	+0.0001	0.00
0.230	1.0001	+0.0001	0.00
0.235	1.0001	-0.0000	0.00
0.240	1.0000	+0.0001	-0.00
0.245	0.9997	+0.0000	-0.00
0.250	1.0000	+0.0002	-0.00
0.255	1.0000	-0.0001	-0.00
0.260	0.9999	+0.0000	-0.00
0.265	1.0000	+0.0000	0.00
0.270	1.0000	+0.0001	0.00
0.275	0.9999	+0.0000	-0.00
0.280	1.0000	+0.0001	0.00
0.285	1.0000	-0.0001	-0.00
0.290	0.9999	+0.0000	-0.00
0.295	1.0000	-0.0000	-0.00
0.300	1.0000	-0.0000	0.00
0.305	1.0001	+0.0001	0.00
0.310	1.0000	+0.0000	-0.00
0.315	1.0000	-0.0000	-0.00
0.320	0.9999	-0.0000	-0.00
0.325	1.0000	+0.0001	0.00
0.330	1.0001	+0.0000	0.00
0.335	0.9999	-0.0001	-0.00
0.340	0.9999	+0.0000	-0.00
0.345	1.0001	-0.0001	0.00
0.350	1.0000	+0.0001	0.00
0.355	1.0000	+0.0001	-0.00
0.360	1.0000	+0.0000	-0.00
0.365	0.9999	-0.0001	-0.00
0.370	1.0001	-0.0000	0.00
0.375	1.0000	+0.0001	-0.00
0.380	1.0000	+0.0000	0.00
0.385	0.9998	+0.0001	-0.00
0.390	1.0000	-0.0000	-0.00
0.395	1.0000	+0.0000	-0.00
0.400	1.0001	+0.0001	0.00
0.405	0.9999	+0.0001	-0.00
0.410	1.0000	-0.0001	-0.00
0.415	0.9999	+0.0001	-0.00

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Frequency (GHz)	**Re(S_{11})**	**Im(S_{11})**	**$\lvert S_{11}\rvert$ (dB)**
0.420	0.9999	-0.0000	-0.00
0.425	0.9999	-0.0001	-0.00
0.430	0.9999	-0.0000	-0.00
0.435	1.0001	+0.0000	0.00
0.440	1.0000	+0.0000	-0.00
0.445	0.9999	+0.0000	-0.00
0.450	1.0000	+0.0000	-0.00
0.455	1.0000	+0.0000	0.00
0.460	1.0001	+0.0000	0.00
0.465	0.9999	+0.0001	-0.00
0.470	0.9999	+0.0002	-0.00
0.475	0.9999	+0.0001	-0.00
0.480	1.0000	-0.0000	-0.00
0.485	1.0000	-0.0000	-0.00
0.490	1.0000	+0.0001	-0.00
0.495	0.9999	+0.0001	-0.00
0.500	1.0000	+0.0000	-0.00
0.505	1.0000	-0.0000	-0.00
0.510	1.0001	+0.0001	0.00
0.515	1.0000	+0.0001	0.00
0.520	1.0001	-0.0001	0.00
0.525	1.0001	+0.0000	0.00
0.530	0.9999	+0.0001	-0.00
0.535	0.9999	+0.0001	-0.00
0.540	0.9999	+0.0000	-0.00
0.545	1.0000	+0.0000	0.00
0.550	1.0000	-0.0000	0.00
0.555	1.0000	-0.0001	-0.00
0.560	1.0001	-0.0000	0.00
0.565	1.0001	+0.0000	0.00
0.570	1.0000	+0.0000	-0.00
0.575	1.0000	-0.0001	-0.00
0.580	0.9999	-0.0000	-0.00
0.585	1.0000	+0.0001	-0.00
0.590	1.0001	-0.0000	0.00
0.595	0.9999	+0.0000	-0.00
0.600	0.9999	-0.0000	-0.00
0.605	1.0000	-0.0000	-0.00
0.610	1.0001	+0.0000	0.00
0.615	0.9999	+0.0001	-0.00
0.620	1.0000	-0.0000	0.00
0.625	0.9999	-0.0000	-0.00
0.630	1.0000	-0.0001	-0.00
0.635	1.0000	+0.0001	-0.00
0.640	1.0000	-0.0000	-0.00
0.645	1.0000	+0.0000	0.00

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Frequency (GHz)	**Re(S_{11})**	**Im(S_{11})**	**$\lvert S_{11}\rvert$ (dB)**
0.650	1.0000	+0.0000	-0.00
0.655	1.0000	+0.0001	-0.00
0.660	0.9999	-0.0000	-0.00
0.665	0.9999	+0.0000	-0.00
0.670	0.9999	+0.0000	-0.00
0.675	1.0000	-0.0000	-0.00
0.680	1.0000	-0.0000	0.00
0.685	1.0000	+0.0001	0.00
0.690	1.0000	-0.0001	-0.00
0.695	1.0001	+0.0000	0.00
0.700	1.0001	+0.0001	0.00
0.705	1.0000	+0.0000	-0.00
0.710	1.0001	-0.0001	0.00
0.715	1.0001	-0.0000	0.00
0.720	1.0000	+0.0001	0.00
0.725	1.0000	+0.0000	-0.00
0.730	1.0000	+0.0000	0.00
0.735	1.0000	+0.0001	0.00
0.740	1.0000	+0.0000	-0.00
0.745	1.0000	-0.0000	-0.00
0.750	1.0001	+0.0001	0.00
0.755	1.0000	-0.0001	0.00
0.760	0.9999	+0.0000	-0.00
0.765	1.0000	+0.0000	0.00
0.770	1.0001	+0.0000	0.00
0.775	1.0000	-0.0001	0.00
0.780	1.0000	-0.0000	0.00
0.785	0.9999	-0.0000	-0.00
0.790	0.9999	+0.0000	-0.00
0.795	1.0000	-0.0000	-0.00
0.800	1.0001	-0.0001	0.00
0.805	1.0001	+0.0000	0.00
0.810	0.9999	+0.0001	-0.00
0.815	0.9999	+0.0000	-0.00
0.820	1.0000	+0.0000	-0.00
0.825	1.0001	+0.0000	0.00
0.830	1.0000	+0.0000	0.00
0.835	0.9999	-0.0000	-0.00
0.840	0.9999	-0.0001	-0.00
0.845	1.0000	-0.0001	0.00
0.850	1.0001	+0.0000	0.00
0.855	1.0000	+0.0000	0.00
0.860	1.0000	+0.0001	0.00
0.865	1.0000	+0.0001	-0.00
0.870	0.9999	+0.0001	-0.00
0.875	1.0000	-0.0000	-0.00

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Frequency (GHz)	**Re(S_{11})**	**Im(S_{11})**	**$\lvert S_{11}\rvert$ (dB)**
0.880	1.0000	+0.0001	0.00
0.885	1.0000	+0.0001	0.00
0.890	1.0000	+0.0001	-0.00
0.895	1.0001	+0.0000	0.00
0.900	1.0000	-0.0000	-0.00
0.905	1.0000	+0.0000	-0.00
0.910	0.9999	+0.0001	-0.00
0.915	1.0000	+0.0001	0.00
0.920	1.0000	+0.0001	-0.00
0.925	0.9999	+0.0001	-0.00
0.930	1.0000	+0.0001	-0.00
0.935	1.0000	+0.0001	0.00
0.940	1.0000	+0.0001	-0.00
0.945	1.0000	-0.0000	-0.00
0.950	1.0001	+0.0000	0.00
0.955	1.0000	+0.0001	0.00
0.960	1.0000	+0.0000	-0.00
0.965	0.9999	-0.0000	-0.00
0.970	0.9999	+0.0001	-0.00
0.975	1.0000	+0.0001	0.00
0.980	1.0000	+0.0001	-0.00
0.985	0.9999	+0.0000	-0.00
0.990	1.0000	+0.0001	-0.00
0.995	1.0000	+0.0001	0.00
1.000	1.0001	-0.0000	0.00
1.005	1.0001	+0.0001	0.00
1.010	0.9999	-0.0000	-0.00
1.015	1.0000	+0.0000	0.00
1.020	1.0000	+0.0001	-0.00
1.025	0.9999	+0.0001	-0.00
1.030	1.0000	+0.0000	0.00
1.035	0.9999	-0.0001	-0.00
1.040	1.0000	+0.0001	-0.00
1.045	1.0000	+0.0001	0.00
1.050	0.9999	+0.0001	-0.00
1.055	1.0001	+0.0001	0.00
1.060	1.0000	+0.0002	-0.00
1.065	1.0001	+0.0000	0.00
1.070	1.0000	+0.0001	-0.00
1.075	1.0000	+0.0001	0.00
1.080	1.0000	+0.0000	0.00
1.085	1.0000	+0.0000	0.00
1.090	0.9999	+0.0001	-0.00
1.095	0.9999	+0.0001	-0.00
1.100	1.0000	+0.0001	0.00
1.105	0.9999	+0.0000	-0.00

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Frequency (GHz)	**Re(S_{11})**	**Im(S_{11})**	**$\lvert S_{11}\rvert$ (dB)**
1.110	1.0000	+0.0002	0.00
1.115	1.0000	+0.0002	-0.00
1.120	1.0000	+0.0000	-0.00
1.125	0.9999	-0.0000	-0.00
1.130	0.9999	+0.0002	-0.00
1.135	1.0001	+0.0000	0.00
1.140	1.0001	-0.0000	0.00
1.145	1.0001	+0.0001	0.00
1.150	1.0000	-0.0000	0.00
1.155	0.9999	+0.0001	-0.00
1.160	1.0001	-0.0000	0.00
1.165	1.0000	+0.0000	0.00
1.170	1.0000	+0.0000	0.00
1.175	1.0000	+0.0001	0.00
1.180	0.9999	-0.0000	-0.00
1.185	1.0000	+0.0001	0.00
1.190	1.0000	+0.0000	0.00
1.195	1.0000	+0.0001	-0.00
1.200	0.9999	+0.0001	-0.00
1.205	1.0001	+0.0000	0.00
1.210	0.9999	-0.0000	-0.00
1.215	1.0000	+0.0000	0.00
1.220	0.9999	+0.0000	-0.00
1.225	1.0000	+0.0000	-0.00
1.230	0.9999	+0.0000	-0.00
1.235	1.0000	+0.0001	-0.00
1.240	0.9999	-0.0001	-0.00
1.245	1.0000	+0.0000	-0.00
1.250	1.0000	+0.0001	0.00
1.255	1.0000	+0.0000	-0.00
1.260	0.9999	-0.0001	-0.00
1.265	1.0001	+0.0000	0.00
1.270	1.0000	+0.0001	0.00
1.275	1.0000	-0.0000	-0.00
1.280	0.9999	+0.0000	-0.00
1.285	0.9998	+0.0000	-0.00
1.290	1.0000	+0.0000	-0.00
1.295	1.0000	+0.0001	0.00
1.300	0.9999	-0.0000	-0.00
1.305	1.0000	-0.0000	0.00
1.310	0.9999	+0.0000	-0.00
1.315	1.0000	+0.0000	-0.00
1.320	1.0000	+0.0001	0.00
1.325	1.0000	-0.0000	0.00
1.330	1.0000	+0.0001	-0.00
1.335	0.9999	+0.0001	-0.00

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Frequency (GHz)	**Re(S_{11})**	**Im(S_{11})**	**$\lvert S_{11}\rvert$ (dB)**
1.340	0.9999	+0.0001	-0.00
1.345	1.0000	+0.0002	0.00
1.350	1.0001	+0.0000	0.00
1.355	1.0001	+0.0001	0.00
1.360	1.0000	-0.0000	0.00
1.365	1.0000	-0.0000	0.00
1.370	1.0001	+0.0001	0.00
1.375	0.9999	+0.0000	-0.00
1.380	1.0001	+0.0001	0.00
1.385	0.9999	+0.0001	-0.00
1.390	0.9999	-0.0000	-0.00
1.395	1.0000	-0.0000	-0.00
1.400	1.0000	-0.0001	-0.00
1.405	1.0000	+0.0002	0.00
1.410	0.9999	+0.0001	-0.00
1.415	1.0001	+0.0001	0.00
1.420	1.0001	-0.0001	0.00
1.425	1.0000	-0.0000	-0.00
1.430	1.0001	+0.0001	0.00
1.435	1.0000	+0.0002	0.00
1.440	1.0001	+0.0002	0.00
1.445	1.0000	+0.0001	0.00
1.450	0.9998	+0.0000	-0.00
1.455	1.0000	+0.0001	-0.00
1.460	0.9998	+0.0001	-0.00
1.465	0.9999	+0.0001	-0.00
1.470	1.0000	+0.0001	-0.00
1.475	1.0001	-0.0001	0.00
1.480	0.9999	+0.0001	-0.00
1.485	1.0000	-0.0000	-0.00
1.490	1.0000	-0.0000	-0.00
1.495	1.0000	+0.0001	0.00
1.500	0.9999	+0.0001	-0.00
1.505	1.0000	+0.0001	-0.00
1.510	1.0002	+0.0001	0.00
1.515	0.9999	+0.0001	-0.00
1.520	1.0000	+0.0001	-0.00
1.525	1.0000	+0.0001	-0.00
1.530	1.0000	+0.0000	0.00
1.535	1.0000	+0.0001	-0.00
1.540	1.0001	-0.0000	0.00
1.545	1.0000	+0.0002	-0.00
1.550	1.0000	+0.0000	0.00
1.555	1.0001	-0.0000	0.00
1.560	0.9999	+0.0000	-0.00
1.565	1.0000	+0.0001	0.00

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Frequency (GHz)	**Re(S_{11})**	**Im(S_{11})**	**$\lvert S_{11}\rvert$ (dB)**
1.570	1.0000	+0.0001	0.00
1.575	1.0001	+0.0001	0.00
1.580	1.0000	+0.0001	-0.00
1.585	1.0000	-0.0000	0.00
1.590	1.0000	+0.0000	-0.00
1.595	0.9999	-0.0000	-0.00
1.600	1.0000	+0.0001	0.00
1.605	1.0000	+0.0000	0.00
1.610	1.0000	+0.0001	0.00
1.615	1.0000	+0.0001	0.00
1.620	0.9999	+0.0001	-0.00
1.625	1.0000	+0.0000	0.00
1.630	1.0000	-0.0000	0.00
1.635	0.9999	+0.0001	-0.00
1.640	0.9999	+0.0001	-0.00
1.645	1.0000	+0.0001	-0.00
1.650	1.0001	-0.0000	0.00
1.655	1.0001	+0.0000	0.00
1.660	1.0000	+0.0002	0.00
1.665	0.9998	+0.0001	-0.00
1.670	0.9997	+0.0001	-0.00
1.675	1.0001	+0.0001	0.00
1.680	1.0000	+0.0000	-0.00
1.685	1.0000	-0.0001	0.00
1.690	1.0001	+0.0000	0.00
1.695	0.9999	-0.0000	-0.00
1.700	1.0000	-0.0001	0.00
1.705	1.0001	+0.0001	0.00
1.710	0.9999	-0.0000	-0.00
1.715	1.0000	-0.0001	-0.00
1.720	1.0001	-0.0000	0.00
1.725	1.0001	+0.0001	0.00
1.730	0.9999	-0.0001	-0.00
1.735	1.0001	+0.0001	0.00
1.740	1.0001	+0.0001	0.00
1.745	1.0000	+0.0002	-0.00
1.750	1.0000	-0.0001	0.00
1.755	1.0000	-0.0001	-0.00
1.760	0.9999	-0.0000	-0.00
1.765	0.9999	-0.0000	-0.00
1.770	0.9999	+0.0002	-0.00
1.775	0.9998	+0.0001	-0.00
1.780	0.9999	-0.0000	-0.00
1.785	0.9999	+0.0001	-0.00
1.790	1.0000	+0.0000	0.00
1.795	0.9999	-0.0001	-0.00

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Frequency (GHz)	**Re(S_{11})**	**Im(S_{11})**	**$\lvert S_{11}\rvert$ (dB)**
1.800	1.0000	-0.0001	-0.00
1.805	1.0001	+0.0001	0.00
1.810	1.0000	-0.0000	-0.00
1.815	1.0000	-0.0000	-0.00
1.820	0.9999	+0.0000	-0.00
1.825	0.9999	-0.0001	-0.00
1.830	0.9999	+0.0000	-0.00
1.835	0.9999	-0.0000	-0.00
1.840	1.0000	+0.0000	0.00
1.845	1.0000	-0.0001	0.00
1.850	1.0000	+0.0001	0.00
1.855	0.9999	-0.0001	-0.00
1.860	1.0001	+0.0001	0.00
1.865	0.9999	+0.0000	-0.00
1.870	0.9999	+0.0001	-0.00
1.875	1.0000	+0.0000	-0.00
1.880	0.9999	-0.0000	-0.00
1.885	1.0001	+0.0000	0.00
1.890	1.0000	-0.0001	0.00
1.895	1.0000	-0.0000	0.00
1.900	0.9999	+0.0001	-0.00
1.905	1.0000	+0.0001	0.00
1.910	0.9999	+0.0000	-0.00
1.915	1.0000	+0.0001	-0.00
1.920	1.0000	+0.0000	-0.00
1.925	1.0000	-0.0000	-0.00
1.930	1.0000	+0.0000	0.00
1.935	0.9998	+0.0000	-0.00
1.940	1.0001	-0.0001	0.00
1.945	1.0000	+0.0001	0.00
1.950	1.0000	+0.0001	0.00
1.955	1.0000	+0.0001	0.00
1.960	1.0000	+0.0000	-0.00
1.965	1.0000	+0.0000	-0.00
1.970	1.0000	-0.0001	-0.00
1.975	1.0000	+0.0000	-0.00
1.980	1.0001	+0.0000	0.00
1.985	1.0001	-0.0001	0.00
1.990	0.9999	-0.0001	-0.00
1.995	1.0000	+0.0001	-0.00
2.000	1.0001	+0.0001	0.00
2.005	1.0000	+0.0000	-0.00
2.010	0.9999	+0.0002	-0.00
2.015	0.9999	+0.0000	-0.00
2.020	1.0001	+0.0001	0.00
2.025	1.0000	+0.0000	-0.00

(continued on next page)

Frequency (GHz)	**Re(S_{11})**	**Im(S_{11})**	**$\lvert S_{11}\rvert$ (dB)**
2.030	0.9999	-0.0001	-0.00
2.035	1.0001	-0.0001	0.00
2.040	1.0000	-0.0000	0.00
2.045	1.0001	+0.0001	0.00
2.050	0.9998	+0.0001	-0.00
2.055	0.9999	+0.0001	-0.00
2.060	1.0000	+0.0001	0.00
2.065	0.9999	+0.0000	-0.00
2.070	1.0000	-0.0001	0.00
2.075	1.0000	+0.0000	0.00
2.080	1.0000	+0.0000	0.00
2.085	0.9999	-0.0001	-0.00
2.090	1.0000	+0.0000	-0.00
2.095	0.9999	-0.0000	-0.00
2.100	0.9998	+0.0000	-0.00
2.105	1.0000	-0.0000	0.00
2.110	0.9997	+0.0000	-0.00
2.115	1.0000	+0.0001	-0.00
2.120	1.0001	+0.0001	0.00
2.125	1.0001	+0.0001	0.00
2.130	1.0001	+0.0000	0.00
2.135	1.0000	+0.0001	-0.00
2.140	0.9999	-0.0001	-0.00
2.145	1.0000	-0.0000	-0.00
2.150	0.9999	-0.0002	-0.00
2.155	1.0001	-0.0001	0.00
2.160	1.0000	+0.0001	0.00
2.165	0.9998	-0.0000	-0.00
2.170	0.9999	+0.0000	-0.00
2.175	0.9999	+0.0001	-0.00
2.180	0.9999	+0.0000	-0.00
2.185	1.0000	+0.0000	-0.00
2.190	0.9999	+0.0001	-0.00
2.195	1.0000	+0.0001	-0.00
2.200	0.9999	+0.0001	-0.00
2.205	1.0001	-0.0000	0.00
2.210	1.0000	+0.0001	-0.00
2.215	1.0000	+0.0001	-0.00
2.220	0.9999	+0.0001	-0.00
2.225	1.0000	+0.0002	-0.00
2.230	1.0000	+0.0001	0.00
2.235	1.0000	-0.0001	-0.00
2.240	1.0000	-0.0000	-0.00
2.245	0.9999	+0.0001	-0.00
2.250	0.9999	-0.0000	-0.00
2.255	1.0000	+0.0001	0.00

(continued on next page)

Frequency (GHz)	**Re(S_{11})**	**Im(S_{11})**	**$\lvert S_{11}\rvert$ (dB)**
2.260	0.9999	+0.0002	-0.00
2.265	0.9999	+0.0001	-0.00
2.270	1.0000	-0.0000	0.00
2.275	1.0001	+0.0001	0.00
2.280	1.0001	+0.0000	0.00
2.285	0.9998	+0.0001	-0.00
2.290	0.9999	+0.0000	-0.00
2.295	1.0000	-0.0001	0.00
2.300	1.0000	+0.0000	-0.00
2.305	1.0001	+0.0001	0.00
2.310	1.0000	-0.0001	-0.00
2.315	0.9999	-0.0001	-0.00
2.320	0.9999	+0.0000	-0.00
2.325	1.0003	-0.0001	0.00
2.330	0.9999	+0.0001	-0.00
2.335	1.0000	+0.0001	0.00
2.340	1.0000	-0.0000	-0.00
2.345	1.0001	+0.0002	0.00
2.350	1.0000	+0.0001	0.00
2.355	1.0001	-0.0001	0.00
2.360	0.9999	+0.0000	-0.00
2.365	1.0000	+0.0002	0.00
2.370	0.9999	+0.0001	-0.00
2.375	1.0000	+0.0000	-0.00
2.380	0.9999	+0.0000	-0.00
2.385	0.9998	+0.0002	-0.00
2.390	0.9998	+0.0001	-0.00
2.395	0.9997	+0.0002	-0.00
2.400	1.0000	+0.0001	-0.00
2.405	1.0000	+0.0001	0.00
2.410	0.9998	-0.0001	-0.00
2.415	1.0000	+0.0001	0.00
2.420	0.9999	+0.0001	-0.00
2.425	1.0000	+0.0001	-0.00
2.430	1.0000	+0.0002	0.00
2.435	1.0000	-0.0000	-0.00
2.440	0.9999	+0.0001	-0.00
2.445	0.9999	-0.0001	-0.00
2.450	0.9999	-0.0001	-0.00
2.455	0.9999	-0.0001	-0.00
2.460	1.0000	-0.0001	0.00
2.465	0.9999	-0.0000	-0.00
2.470	1.0000	+0.0001	-0.00
2.475	1.0000	+0.0001	0.00
2.480	1.0000	+0.0001	-0.00
2.485	0.9999	-0.0000	-0.00

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Frequency (GHz)	**Re(S_{11})**	**Im(S_{11})**	**$\lvert S_{11}\rvert$ (dB)**
2.490	0.9999	+0.0000	-0.00
2.495	1.0000	+0.0000	0.00
2.500	0.9998	+0.0000	-0.00
2.505	1.0000	-0.0000	-0.00
2.510	0.9999	-0.0000	-0.00
2.515	1.0000	+0.0000	-0.00
2.520	0.9999	+0.0000	-0.00
2.525	1.0000	+0.0002	-0.00
2.530	1.0001	+0.0001	0.00
2.535	0.9998	+0.0000	-0.00
2.540	0.9999	+0.0001	-0.00
2.545	1.0000	-0.0000	-0.00
2.550	1.0000	+0.0001	-0.00
2.555	0.9999	-0.0000	-0.00
2.560	0.9999	+0.0001	-0.00
2.565	0.9999	+0.0000	-0.00
2.570	1.0000	+0.0000	-0.00
2.575	0.9999	-0.0001	-0.00
2.580	0.9999	+0.0000	-0.00
2.585	0.9998	+0.0002	-0.00
2.590	1.0000	+0.0000	0.00
2.595	1.0000	-0.0001	-0.00
2.600	1.0000	-0.0000	-0.00
2.605	1.0001	+0.0002	0.00
2.610	0.9998	-0.0001	-0.00
2.615	1.0001	-0.0002	0.00
2.620	1.0000	-0.0000	0.00
2.625	1.0000	+0.0000	-0.00
2.630	0.9998	-0.0001	-0.00
2.635	1.0000	+0.0001	-0.00
2.640	0.9999	+0.0001	-0.00
2.645	0.9999	-0.0000	-0.00
2.650	1.0000	+0.0002	0.00
2.655	0.9998	+0.0001	-0.00
2.660	1.0000	+0.0001	0.00
2.665	1.0000	-0.0000	0.00
2.670	1.0000	+0.0002	-0.00
2.675	1.0001	-0.0001	0.00
2.680	1.0000	-0.0000	0.00
2.685	1.0001	+0.0001	0.00
2.690	1.0000	+0.0003	0.00
2.695	0.9999	+0.0001	-0.00
2.700	1.0000	+0.0000	-0.00
2.705	0.9998	+0.0000	-0.00
2.710	1.0000	+0.0001	-0.00
2.715	1.0000	-0.0002	0.00

(continued on next page)

Frequency (GHz)	**Re(S_{11})**	**Im(S_{11})**	**$\lvert S_{11}\rvert$ (dB)**
2.720	0.9999	+0.0000	-0.00
2.725	1.0001	+0.0002	0.00
2.730	1.0000	+0.0003	0.00
2.735	0.9998	+0.0003	-0.00
2.740	1.0000	-0.0000	0.00
2.745	1.0000	+0.0000	-0.00
2.750	0.9998	+0.0002	-0.00
2.755	0.9999	+0.0002	-0.00
2.760	0.9999	+0.0002	-0.00
2.765	1.0000	+0.0001	0.00
2.770	0.9998	+0.0001	-0.00
2.775	1.0000	+0.0001	-0.00
2.780	1.0000	+0.0001	0.00
2.785	0.9999	+0.0001	-0.00
2.790	1.0000	+0.0001	-0.00
2.795	0.9998	+0.0000	-0.00
2.800	1.0000	+0.0001	0.00
2.805	1.0000	-0.0000	-0.00
2.810	1.0000	+0.0002	-0.00
2.815	1.0000	+0.0000	-0.00
2.820	0.9999	+0.0000	-0.00
2.825	1.0000	+0.0002	0.00
2.830	0.9998	+0.0002	-0.00
2.835	0.9999	+0.0001	-0.00
2.840	0.9999	+0.0003	-0.00
2.845	0.9999	+0.0000	-0.00
2.850	1.0001	+0.0001	0.00
2.855	0.9998	+0.0002	-0.00
2.860	1.0001	+0.0000	0.00
2.865	0.9999	+0.0003	-0.00
2.870	1.0000	+0.0001	-0.00
2.875	1.0000	+0.0001	-0.00
2.880	0.9999	+0.0002	-0.00
2.885	1.0000	-0.0001	0.00
2.890	1.0000	+0.0004	-0.00
2.895	0.9999	+0.0002	-0.00
2.900	1.0000	+0.0001	-0.00
2.905	1.0000	+0.0002	-0.00
2.910	0.9998	+0.0001	-0.00
2.915	1.0000	+0.0001	-0.00
2.920	0.9999	+0.0001	-0.00
2.925	1.0000	-0.0001	0.00
2.930	0.9999	+0.0002	-0.00
2.935	1.0001	+0.0001	0.00
2.940	0.9998	+0.0001	-0.00
2.945	0.9998	+0.0000	-0.00

(continued on next page)

Frequency (GHz)	**Re(S_{11})**	**Im(S_{11})**	**$\lvert S_{11}\rvert$ (dB)**
2.950	1.0000	+0.0000	0.00
2.955	1.0000	-0.0001	-0.00
2.960	0.9999	+0.0003	-0.00
2.965	0.9999	+0.0000	-0.00
2.970	0.9999	+0.0001	-0.00
2.975	1.0001	+0.0001	0.00
2.980	0.9997	-0.0002	-0.00
2.985	0.9999	+0.0002	-0.00
2.990	0.9998	+0.0002	-0.00
2.995	1.0000	-0.0000	0.00
3.000	0.9999	+0.0002	-0.00
3.005	0.9998	+0.0003	-0.00
3.010	1.0003	-0.0002	0.00
3.015	1.0000	+0.0002	-0.00
3.020	0.9996	+0.0004	-0.00
3.025	1.0000	+0.0001	0.00
3.030	0.9999	+0.0002	-0.00
3.035	0.9999	-0.0000	-0.00
3.040	0.9999	+0.0002	-0.00
3.045	1.0003	+0.0000	0.00
3.050	1.0001	+0.0001	0.00
3.055	1.0000	+0.0000	0.00
3.060	0.9997	+0.0001	-0.00
3.065	1.0000	-0.0003	0.00
3.070	0.9999	+0.0003	-0.00
3.075	0.9998	+0.0001	-0.00
3.080	0.9998	+0.0001	-0.00
3.085	1.0000	-0.0000	0.00
3.090	1.0000	+0.0001	-0.00
3.095	0.9999	+0.0001	-0.00
3.100	0.9998	-0.0001	-0.00
3.105	0.9999	-0.0001	-0.00
3.110	0.9999	-0.0001	-0.00
3.115	1.0000	-0.0000	-0.00
3.120	0.9999	+0.0001	-0.00
3.125	1.0001	+0.0001	0.00
3.130	1.0000	+0.0001	0.00
3.135	1.0000	+0.0000	-0.00
3.140	0.9999	+0.0000	-0.00
3.145	1.0000	+0.0001	0.00
3.150	0.9999	+0.0001	-0.00
3.155	0.9999	-0.0000	-0.00
3.160	0.9999	-0.0000	-0.00
3.165	1.0001	+0.0003	0.00
3.170	0.9999	-0.0000	-0.00
3.175	0.9999	+0.0002	-0.00

(continued on next page)

Frequency (GHz)	**Re(S_{11})**	**Im(S_{11})**	$\lvert S_{11}\rvert$ **(dB)**
3.180	1.0000	-0.0001	-0.00
3.185	1.0001	-0.0001	0.00
3.190	1.0000	+0.0002	0.00
3.195	0.9999	+0.0001	-0.00
3.200	0.9999	+0.0002	-0.00
3.205	1.0000	+0.0000	-0.00
3.210	1.0000	+0.0002	-0.00
3.215	1.0000	+0.0000	-0.00
3.220	0.9998	+0.0001	-0.00
3.225	0.9999	+0.0000	-0.00
3.230	1.0001	-0.0002	0.00
3.235	0.9998	-0.0000	-0.00
3.240	0.9997	+0.0001	-0.00
3.245	1.0000	+0.0000	-0.00
3.250	0.9997	+0.0003	-0.00
3.255	0.9999	+0.0002	-0.00
3.260	0.9999	-0.0001	-0.00
3.265	0.9998	-0.0001	-0.00
3.270	1.0000	+0.0000	0.00
3.275	1.0000	+0.0001	0.00
3.280	1.0000	-0.0001	-0.00
3.285	0.9999	+0.0001	-0.00
3.290	0.9999	+0.0001	-0.00
3.295	0.9999	+0.0001	-0.00
3.300	1.0000	-0.0003	-0.00
3.305	1.0001	+0.0001	0.00
3.310	1.0001	+0.0002	0.00
3.315	1.0000	+0.0000	-0.00
3.320	1.0000	+0.0001	-0.00
3.325	1.0003	-0.0000	0.00
3.330	0.9997	-0.0001	-0.00
3.335	1.0002	+0.0001	0.00
3.340	1.0000	-0.0000	0.00
3.345	1.0001	-0.0001	0.00
3.350	0.9999	+0.0002	-0.00
3.355	1.0000	-0.0001	0.00
3.360	0.9997	+0.0003	-0.00
3.365	1.0000	+0.0003	0.00
3.370	1.0000	+0.0001	-0.00
3.375	0.9998	+0.0001	-0.00
3.380	0.9998	+0.0001	-0.00
3.385	0.9999	+0.0001	-0.00
3.390	1.0000	+0.0001	-0.00
3.395	0.9999	-0.0000	-0.00
3.400	0.9999	+0.0003	-0.00
3.405	0.9999	+0.0002	-0.00

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Frequency (GHz)	**Re(S_{11})**	**Im(S_{11})**	**$\lvert S_{11}\rvert$ (dB)**
3.410	0.9997	+0.0002	-0.00
3.415	0.9999	+0.0004	-0.00
3.420	0.9999	+0.0003	-0.00
3.425	1.0000	+0.0001	0.00
3.430	0.9999	+0.0001	-0.00
3.435	0.9998	+0.0002	-0.00
3.440	1.0000	+0.0002	-0.00
3.445	0.9999	+0.0003	-0.00
3.450	1.0000	+0.0003	-0.00
3.455	0.9998	+0.0003	-0.00
3.460	1.0000	+0.0003	-0.00
3.465	0.9997	+0.0002	-0.00
3.470	0.9998	+0.0003	-0.00
3.475	0.9999	+0.0000	-0.00
3.480	1.0001	+0.0002	0.00
3.485	1.0003	-0.0000	0.00
3.490	1.0000	+0.0002	-0.00
3.495	0.9999	-0.0000	-0.00
3.500	1.0001	+0.0001	0.00
3.505	0.9998	+0.0000	-0.00
3.510	1.0000	+0.0000	0.00
3.515	1.0001	+0.0002	0.00
3.520	1.0000	+0.0003	0.00
3.525	1.0000	+0.0005	0.00
3.530	1.0001	+0.0000	0.00
3.535	1.0003	+0.0001	0.00
3.540	1.0000	+0.0006	0.00
3.545	0.9999	+0.0000	-0.00
3.550	0.9996	+0.0004	-0.00
3.555	1.0002	-0.0002	0.00
3.560	1.0000	+0.0001	-0.00
3.565	1.0003	-0.0004	0.00
3.570	1.0000	+0.0003	-0.00
3.575	0.9995	+0.0005	-0.00
3.580	0.9996	+0.0006	-0.00
3.585	1.0008	-0.0003	0.01
3.590	1.0000	-0.0000	-0.00
3.595	1.0003	-0.0002	0.00
3.600	0.9998	+0.0003	-0.00
3.605	1.0004	+0.0001	0.00
3.610	1.0003	+0.0003	0.00
3.615	1.0000	+0.0001	0.00
3.620	0.9999	+0.0000	-0.00
3.625	1.0007	-0.0009	0.01
3.630	1.0002	+0.0005	0.00
3.635	0.9999	+0.0001	-0.00

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Frequency (GHz)	**Re(S_{11})**	**Im(S_{11})**	**$\lvert S_{11}\rvert$ (dB)**
3.640	0.9998	+0.0000	-0.00
3.645	0.9996	+0.0001	-0.00
3.650	1.0001	+0.0005	0.00
3.655	1.0002	+0.0001	0.00
3.660	0.9999	+0.0003	-0.00
3.665	1.0002	+0.0001	0.00
3.670	0.9999	+0.0004	-0.00
3.675	1.0005	+0.0004	0.00
3.680	1.0000	+0.0002	0.00
3.685	0.9998	+0.0003	-0.00
3.690	1.0000	+0.0001	0.00
3.695	0.9999	+0.0000	-0.00
3.700	0.9999	+0.0001	-0.00
3.705	1.0000	-0.0000	0.00
3.710	1.0000	+0.0001	0.00
3.715	0.9999	+0.0000	-0.00
3.720	1.0001	-0.0001	0.00
3.725	0.9999	+0.0001	-0.00
3.730	1.0000	+0.0001	0.00
3.735	1.0001	+0.0002	0.00
3.740	1.0000	+0.0002	-0.00
3.745	0.9998	-0.0000	-0.00
3.750	1.0000	+0.0001	0.00
3.755	1.0001	+0.0000	0.00
3.760	1.0000	-0.0000	0.00
3.765	0.9999	+0.0003	-0.00
3.770	1.0000	+0.0002	0.00
3.775	1.0001	+0.0001	0.00
3.780	0.9996	+0.0011	-0.00
3.785	0.9999	+0.0005	-0.00
3.790	0.9998	+0.0000	-0.00
3.795	1.0000	-0.0001	0.00
3.800	0.9997	-0.0000	-0.00
3.805	0.9999	-0.0003	-0.00
3.810	1.0000	+0.0001	-0.00
3.815	1.0003	-0.0000	0.00
3.820	0.9999	+0.0001	-0.00
3.825	1.0003	+0.0002	0.00
3.830	1.0001	+0.0001	0.00
3.835	1.0000	+0.0001	0.00
3.840	1.0001	-0.0001	0.00
3.845	1.0000	+0.0001	-0.00
3.850	0.9998	-0.0001	-0.00
3.855	0.9999	+0.0003	-0.00
3.860	1.0002	+0.0002	0.00
3.865	0.9999	+0.0005	-0.00

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Frequency (GHz)	**Re(S_{11})**	**Im(S_{11})**	$\lvert S_{11}\rvert$ **(dB)**
3.870	1.0001	+0.0000	0.00
3.875	0.9999	+0.0001	-0.00
3.880	1.0000	-0.0000	0.00
3.885	0.9999	+0.0001	-0.00
3.890	0.9998	-0.0002	-0.00
3.895	1.0002	-0.0003	0.00
3.900	1.0001	+0.0002	0.00
3.905	0.9996	+0.0002	-0.00
3.910	0.9999	+0.0003	-0.00
3.915	1.0000	+0.0000	-0.00
3.920	0.9998	+0.0001	-0.00
3.925	1.0001	+0.0001	0.00
3.930	1.0001	-0.0000	0.00
3.935	1.0000	+0.0001	-0.00
3.940	1.0000	-0.0000	-0.00
3.945	0.9998	-0.0002	-0.00
3.950	1.0000	+0.0001	-0.00
3.955	0.9998	+0.0001	-0.00
3.960	1.0000	+0.0001	-0.00
3.965	0.9999	-0.0001	-0.00
3.970	0.9998	+0.0002	-0.00
3.975	1.0001	+0.0001	0.00
3.980	1.0000	+0.0001	0.00
3.985	1.0000	+0.0001	0.00
3.990	0.9998	+0.0000	-0.00
3.995	1.0001	+0.0001	0.00
4.000	0.9999	+0.0001	-0.00
4.005	1.0000	+0.0002	0.00
4.010	1.0000	+0.0002	0.00
4.015	0.9999	+0.0001	-0.00
4.020	0.9999	+0.0002	-0.00
4.025	1.0000	-0.0001	-0.00
4.030	1.0002	+0.0003	0.00
4.035	1.0000	+0.0002	-0.00
4.040	1.0000	+0.0001	0.00
4.045	0.9999	+0.0002	-0.00
4.050	1.0002	-0.0001	0.00
4.055	1.0001	+0.0001	0.00
4.060	1.0001	+0.0002	0.00
4.065	0.9996	+0.0004	-0.00
4.070	0.9999	+0.0003	-0.00
4.075	1.0000	+0.0002	0.00
4.080	0.9999	+0.0001	-0.00
4.085	0.9998	+0.0003	-0.00
4.090	0.9999	+0.0000	-0.00
4.095	1.0000	+0.0003	-0.00

(continued on next page)

Frequency (GHz)	**Re(S_{11})**	**Im(S_{11})**	**$\lvert S_{11}\rvert$ (dB)**
4.100	0.9998	+0.0001	-0.00
4.105	1.0005	+0.0003	0.00
4.110	0.9997	-0.0001	-0.00
4.115	1.0001	+0.0002	0.00
4.120	0.9998	+0.0003	-0.00
4.125	1.0001	-0.0002	0.00
4.130	0.9997	-0.0002	-0.00
4.135	1.0001	+0.0002	0.00
4.140	1.0002	+0.0003	0.00
4.145	1.0004	-0.0006	0.00
4.150	1.0002	+0.0004	0.00
4.155	0.9999	+0.0003	-0.00
4.160	0.9998	-0.0000	-0.00
4.165	1.0001	+0.0001	0.00
4.170	1.0000	+0.0002	-0.00
4.175	0.9998	+0.0002	-0.00
4.180	1.0000	+0.0000	-0.00
4.185	1.0001	+0.0000	0.00
4.190	1.0001	+0.0001	0.00
4.195	1.0001	+0.0001	0.00
4.200	1.0000	+0.0001	0.00
4.205	1.0001	-0.0003	0.00
4.210	1.0002	+0.0002	0.00
4.215	0.9999	+0.0003	-0.00
4.220	1.0000	+0.0001	0.00
4.225	1.0000	+0.0000	-0.00
4.230	0.9999	-0.0000	-0.00
4.235	1.0000	+0.0003	-0.00
4.240	0.9999	+0.0001	-0.00
4.245	1.0001	-0.0003	0.00
4.250	1.0001	+0.0001	0.00
4.255	1.0000	-0.0003	-0.00
4.260	1.0000	+0.0001	-0.00
4.265	1.0002	-0.0001	0.00
4.270	1.0000	+0.0000	-0.00
4.275	1.0000	+0.0001	0.00
4.280	1.0000	+0.0001	-0.00
4.285	0.9999	+0.0001	-0.00
4.290	1.0000	+0.0001	-0.00
4.295	1.0000	+0.0002	0.00
4.300	0.9998	+0.0000	-0.00
4.305	1.0000	+0.0001	0.00
4.310	1.0001	+0.0001	0.00
4.315	1.0000	+0.0001	-0.00
4.320	1.0001	+0.0001	0.00
4.325	1.0000	+0.0002	0.00

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Frequency (GHz)	**Re(S_{11})**	**Im(S_{11})**	**$\lvert S_{11}\rvert$ (dB)**
4.330	1.0001	+0.0000	0.00
4.335	1.0000	+0.0001	-0.00
4.340	1.0001	+0.0000	0.00
4.345	1.0000	-0.0001	0.00
4.350	1.0000	-0.0001	-0.00
4.355	0.9998	+0.0000	-0.00
4.360	1.0001	+0.0001	0.00
4.365	1.0000	+0.0002	0.00
4.370	0.9999	-0.0000	-0.00
4.375	1.0002	+0.0003	0.00
4.380	1.0001	+0.0000	0.00
4.385	1.0000	+0.0002	0.00
4.390	0.9999	-0.0001	-0.00
4.395	1.0001	+0.0001	0.00
4.400	0.9999	+0.0001	-0.00
4.405	1.0002	-0.0001	0.00
4.410	1.0000	-0.0001	-0.00
4.415	1.0001	-0.0000	0.00
4.420	0.9999	+0.0003	-0.00
4.425	0.9999	+0.0001	-0.00
4.430	1.0000	+0.0001	0.00
4.435	1.0003	+0.0001	0.00
4.440	0.9999	-0.0002	-0.00
4.445	1.0000	+0.0003	0.00
4.450	0.9997	+0.0001	-0.00
4.455	0.9994	+0.0003	-0.01
4.460	0.9998	-0.0000	-0.00
4.465	1.0002	+0.0003	0.00
4.470	1.0000	-0.0002	-0.00
4.475	1.0001	+0.0000	0.00
4.480	0.9999	-0.0000	-0.00
4.485	1.0000	+0.0004	-0.00
4.490	1.0001	+0.0001	0.00
4.495	1.0001	-0.0003	0.00
4.500	1.0000	-0.0001	0.00
4.505	1.0001	+0.0003	0.00
4.510	1.0003	+0.0004	0.00
4.515	0.9998	+0.0002	-0.00
4.520	0.9997	-0.0002	-0.00
4.525	1.0000	-0.0003	0.00
4.530	0.9999	-0.0000	-0.00
4.535	1.0000	-0.0002	0.00
4.540	1.0000	-0.0001	0.00
4.545	0.9999	+0.0000	-0.00
4.550	1.0001	+0.0000	0.00
4.555	1.0000	+0.0001	0.00

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Frequency (GHz)	**Re(S_{11})**	**Im(S_{11})**	**$\lvert S_{11}\rvert$ (dB)**
4.560	0.9999	+0.0000	-0.00
4.565	0.9999	+0.0001	-0.00
4.570	1.0000	-0.0001	0.00
4.575	1.0000	+0.0001	0.00
4.580	0.9999	+0.0000	-0.00
4.585	0.9999	+0.0001	-0.00
4.590	1.0001	+0.0001	0.00
4.595	1.0000	+0.0000	-0.00
4.600	1.0000	+0.0002	-0.00
4.605	1.0001	+0.0001	0.00
4.610	0.9999	+0.0002	-0.00
4.615	1.0000	+0.0001	-0.00
4.620	1.0000	+0.0001	0.00
4.625	0.9999	-0.0002	-0.00
4.630	1.0000	+0.0001	-0.00
4.635	1.0000	+0.0000	-0.00
4.640	1.0000	+0.0002	-0.00
4.645	0.9998	+0.0003	-0.00
4.650	1.0000	+0.0001	0.00
4.655	1.0001	+0.0003	0.00
4.660	0.9998	-0.0002	-0.00
4.665	0.9999	-0.0002	-0.00
4.670	0.9998	+0.0004	-0.00
4.675	1.0002	+0.0001	0.00
4.680	1.0000	+0.0003	-0.00
4.685	1.0001	+0.0001	0.00
4.690	0.9999	+0.0002	-0.00
4.695	1.0001	+0.0000	0.00
4.700	1.0000	+0.0002	-0.00
4.705	1.0006	-0.0003	0.01
4.710	0.9999	+0.0007	-0.00
4.715	1.0000	+0.0001	0.00
4.720	1.0003	-0.0000	0.00
4.725	0.9996	+0.0003	-0.00
4.730	1.0000	-0.0000	-0.00
4.735	1.0001	+0.0000	0.00
4.740	1.0002	-0.0001	0.00
4.745	0.9999	+0.0004	-0.00
4.750	1.0004	+0.0001	0.00
4.755	1.0000	-0.0002	-0.00
4.760	1.0003	-0.0001	0.00
4.765	1.0004	-0.0002	0.00
4.770	0.9997	-0.0004	-0.00
4.775	1.0004	+0.0002	0.00
4.780	1.0001	-0.0001	0.00
4.785	0.9990	-0.0002	-0.01

(continued on next page)

Frequency (GHz)	**Re(S_{11})**	**Im(S_{11})**	**$\lvert S_{11}\rvert$ (dB)**
4.790	1.0002	-0.0001	0.00
4.795	1.0001	+0.0001	0.00
4.800	1.0002	-0.0002	0.00
4.805	1.0001	-0.0003	0.00
4.810	1.0001	-0.0002	0.00
4.815	1.0007	-0.0002	0.01
4.820	0.9996	+0.0003	-0.00
4.825	1.0000	+0.0000	-0.00
4.830	0.9997	-0.0000	-0.00
4.835	1.0000	+0.0001	0.00
4.840	1.0001	-0.0001	0.00
4.845	1.0002	-0.0002	0.00
4.850	1.0000	+0.0002	-0.00
4.855	0.9998	+0.0001	-0.00
4.860	1.0000	-0.0001	0.00
4.865	1.0003	+0.0003	0.00
4.870	1.0000	+0.0002	0.00
4.875	1.0000	+0.0001	0.00
4.880	0.9999	-0.0001	-0.00
4.885	1.0000	+0.0001	0.00
4.890	0.9997	-0.0007	-0.00
4.895	0.9999	+0.0003	-0.00
4.900	0.9999	-0.0001	-0.00
4.905	0.9999	-0.0001	-0.00
4.910	1.0000	+0.0002	-0.00
4.915	1.0000	+0.0001	-0.00
4.920	1.0000	+0.0000	-0.00
4.925	1.0000	+0.0000	-0.00
4.930	1.0001	+0.0001	0.00
4.935	1.0000	+0.0001	0.00
4.940	1.0001	-0.0001	0.00
4.945	0.9999	+0.0001	-0.00
4.950	1.0000	-0.0000	0.00
4.955	1.0000	-0.0000	-0.00
4.960	1.0001	-0.0001	0.00
4.965	0.9999	+0.0000	-0.00
4.970	1.0000	+0.0000	-0.00
4.975	1.0000	-0.0001	0.00
4.980	1.0000	+0.0001	0.00
4.985	0.9999	-0.0001	-0.00
4.990	1.0000	-0.0001	-0.00
4.995	1.0000	+0.0002	-0.00
5.000	1.0002	+0.0002	0.00
5.005	1.0000	+0.0001	-0.00
5.010	1.0000	+0.0001	0.00
5.015	0.9998	-0.0000	-0.00

(continued on next page)

Frequency (GHz)	**Re(S_{11})**	**Im(S_{11})**	**$\lvert S_{11}\rvert$ (dB)**
5.020	1.0000	+0.0001	-0.00
5.025	0.9999	+0.0001	-0.00
5.030	1.0000	+0.0003	0.00
5.035	0.9998	+0.0001	-0.00
5.040	0.9999	+0.0001	-0.00
5.045	1.0000	-0.0001	-0.00
5.050	1.0001	+0.0001	0.00
5.055	0.9999	-0.0001	-0.00
5.060	1.0002	+0.0001	0.00
5.065	1.0002	+0.0003	0.00
5.070	0.9998	+0.0001	-0.00
5.075	0.9999	-0.0001	-0.00
5.080	1.0000	+0.0001	0.00
5.085	1.0000	+0.0003	0.00
5.090	1.0001	+0.0001	0.00
5.095	1.0000	-0.0003	-0.00
5.100	0.9999	-0.0002	-0.00
5.105	0.9995	+0.0000	-0.00
5.110	1.0002	-0.0000	0.00
5.115	0.9999	+0.0001	-0.00
5.120	1.0001	+0.0001	0.00
5.125	0.9997	+0.0000	-0.00
5.130	1.0001	-0.0000	0.00
5.135	1.0000	+0.0004	-0.00
5.140	1.0000	+0.0002	-0.00
5.145	1.0002	+0.0001	0.00
5.150	1.0000	+0.0001	0.00
5.155	1.0001	+0.0000	0.00
5.160	1.0001	-0.0000	0.00
5.165	1.0000	+0.0002	-0.00
5.170	1.0000	+0.0000	0.00
5.175	1.0001	+0.0001	0.00
5.180	1.0000	+0.0001	-0.00
5.185	1.0001	+0.0002	0.00
5.190	1.0001	+0.0001	0.00
5.195	1.0000	+0.0001	-0.00
5.200	1.0000	+0.0001	-0.00
5.205	1.0000	+0.0000	-0.00
5.210	1.0000	+0.0000	0.00
5.215	1.0000	-0.0000	-0.00
5.220	1.0000	+0.0001	0.00
5.225	1.0000	+0.0001	-0.00
5.230	1.0001	+0.0001	0.00
5.235	1.0000	+0.0000	-0.00
5.240	1.0001	+0.0001	0.00
5.245	1.0000	+0.0001	0.00

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Frequency (GHz)	**Re(S_{11})**	**Im(S_{11})**	$\lvert S_{11}\rvert$ **(dB)**
5.250	1.0000	+0.0000	0.00
5.255	1.0000	+0.0002	-0.00
5.260	1.0001	-0.0002	0.00
5.265	0.9997	-0.0001	-0.00
5.270	1.0001	+0.0002	0.00
5.275	1.0001	+0.0002	0.00
5.280	1.0000	+0.0001	-0.00
5.285	1.0000	+0.0001	-0.00
5.290	1.0002	+0.0000	0.00
5.295	1.0004	+0.0000	0.00
5.300	1.0000	+0.0001	0.00
5.305	0.9999	-0.0003	-0.00
5.310	1.0001	+0.0003	0.00
5.315	1.0000	-0.0001	-0.00
5.320	1.0000	+0.0000	0.00
5.325	1.0000	+0.0001	-0.00
5.330	0.9999	+0.0000	-0.00
5.335	1.0000	+0.0002	-0.00
5.340	0.9998	+0.0002	-0.00
5.345	1.0001	+0.0002	0.00
5.350	1.0003	-0.0002	0.00
5.355	0.9998	+0.0003	-0.00
5.360	1.0004	+0.0001	0.00
5.365	0.9999	-0.0005	-0.00
5.370	1.0005	+0.0002	0.00
5.375	1.0000	+0.0002	-0.00
5.380	1.0002	-0.0002	0.00
5.385	1.0007	+0.0003	0.01
5.390	1.0000	+0.0004	-0.00
5.395	1.0000	-0.0002	0.00
5.400	1.0001	+0.0003	0.00
5.405	1.0004	-0.0001	0.00
5.410	1.0001	+0.0003	0.00
5.415	0.9999	-0.0005	-0.00
5.420	1.0000	+0.0002	-0.00
5.425	0.9996	-0.0004	-0.00
5.430	1.0002	-0.0001	0.00
5.435	1.0000	-0.0001	0.00
5.440	0.9999	+0.0003	-0.00
5.445	0.9998	-0.0001	-0.00
5.450	1.0001	+0.0003	0.00
5.455	1.0002	-0.0003	0.00
5.460	1.0000	-0.0000	0.00
5.465	0.9995	-0.0002	-0.00
5.470	1.0001	+0.0002	0.00
5.475	1.0000	-0.0000	0.00

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Frequency (GHz)	**Re(S_{11})**	**Im(S_{11})**	**$\lvert S_{11}\rvert$ (dB)**
5.480	1.0001	+0.0001	0.00
5.485	1.0003	-0.0001	0.00
5.490	0.9998	+0.0003	-0.00
5.495	1.0002	-0.0003	0.00
5.500	1.0000	+0.0001	-0.00
5.505	1.0000	+0.0000	0.00
5.510	1.0000	-0.0001	-0.00
5.515	1.0000	-0.0001	0.00
5.520	1.0000	+0.0001	-0.00
5.525	1.0000	+0.0000	0.00
5.530	1.0000	+0.0000	0.00
5.535	1.0000	-0.0002	0.00
5.540	0.9999	+0.0001	-0.00
5.545	1.0000	+0.0001	0.00
5.550	0.9999	+0.0000	-0.00
5.555	1.0001	+0.0001	0.00
5.560	1.0000	+0.0000	-0.00
5.565	1.0001	-0.0000	0.00
5.570	1.0001	+0.0001	0.00
5.575	1.0001	+0.0000	0.00
5.580	1.0000	-0.0000	-0.00
5.585	1.0000	+0.0000	0.00
5.590	1.0000	+0.0000	0.00
5.595	1.0000	-0.0000	0.00
5.600	0.9999	+0.0000	-0.00
5.605	1.0001	-0.0001	0.00
5.610	1.0002	+0.0001	0.00
5.615	1.0001	+0.0002	0.00
5.620	1.0000	+0.0000	-0.00
5.625	0.9999	+0.0001	-0.00
5.630	1.0000	+0.0001	-0.00
5.635	1.0001	-0.0001	0.00
5.640	1.0001	-0.0000	0.00
5.645	1.0002	+0.0002	0.00
5.650	1.0001	-0.0001	0.00
5.655	0.9998	-0.0000	-0.00
5.660	0.9999	+0.0002	-0.00
5.665	1.0001	-0.0003	0.00
5.670	0.9997	+0.0004	-0.00
5.675	0.9999	+0.0001	-0.00
5.680	0.9999	+0.0003	-0.00
5.685	1.0000	-0.0003	-0.00
5.690	1.0001	-0.0003	0.00
5.695	0.9997	-0.0002	-0.00
5.700	1.0000	-0.0000	0.00
5.705	0.9995	+0.0003	-0.00

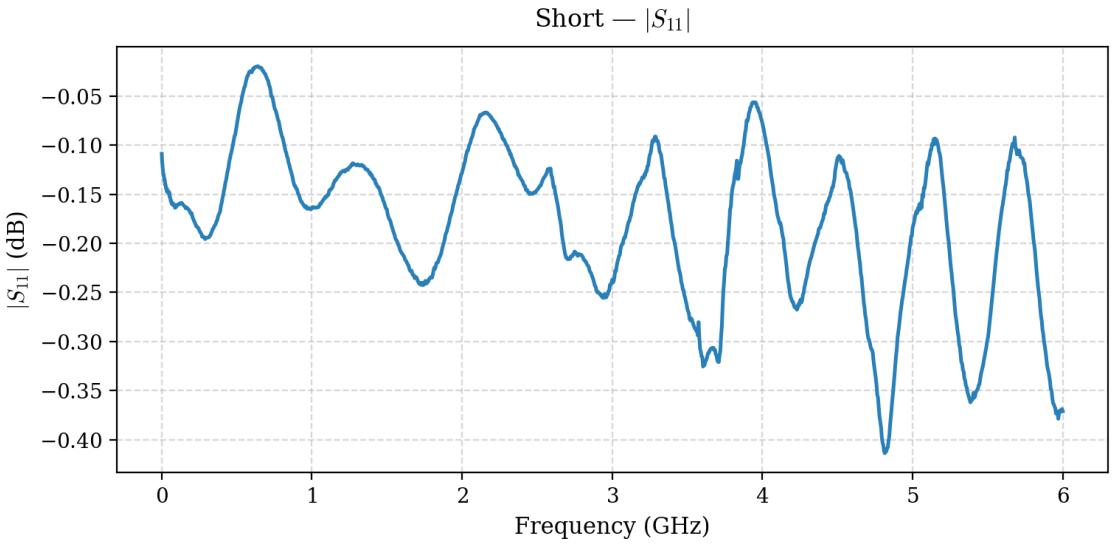
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Frequency (GHz)	**Re(S_{11})**	**Im(S_{11})**	**$\lvert S_{11}\rvert$ (dB)**
5.710	1.0000	-0.0001	0.00
5.715	1.0001	+0.0004	0.00
5.720	0.9999	-0.0001	-0.00
5.725	0.9997	-0.0001	-0.00
5.730	0.9998	-0.0001	-0.00
5.735	0.9999	+0.0001	-0.00
5.740	1.0000	-0.0000	-0.00
5.745	0.9999	-0.0000	-0.00
5.750	1.0002	-0.0000	0.00
5.755	0.9999	-0.0001	-0.00
5.760	1.0001	+0.0003	0.00
5.765	1.0003	-0.0001	0.00
5.770	1.0002	-0.0000	0.00
5.775	1.0001	-0.0000	0.00
5.780	1.0000	-0.0000	-0.00
5.785	1.0002	+0.0001	0.00
5.790	1.0000	+0.0001	0.00
5.795	0.9999	+0.0000	-0.00
5.800	1.0000	-0.0000	0.00
5.805	1.0002	+0.0002	0.00
5.810	1.0000	-0.0001	-0.00
5.815	0.9999	-0.0001	-0.00
5.820	1.0001	+0.0000	0.00
5.825	1.0001	-0.0001	0.00
5.830	1.0000	+0.0001	0.00
5.835	1.0000	+0.0001	-0.00
5.840	1.0000	+0.0001	-0.00
5.845	1.0000	-0.0001	-0.00
5.850	1.0000	+0.0000	0.00
5.855	1.0001	+0.0001	0.00
5.860	1.0000	+0.0001	-0.00
5.865	0.9999	-0.0000	-0.00
5.870	1.0000	+0.0002	-0.00
5.875	1.0000	+0.0000	-0.00
5.880	1.0001	+0.0001	0.00
5.885	1.0001	-0.0000	0.00
5.890	0.9999	+0.0002	-0.00
5.895	0.9999	+0.0000	-0.00
5.900	1.0003	+0.0002	0.00
5.905	0.9995	+0.0001	-0.00
5.910	1.0004	+0.0003	0.00
5.915	1.0000	+0.0001	-0.00
5.920	1.0002	-0.0001	0.00
5.925	1.0000	+0.0001	-0.00
5.930	1.0000	+0.0004	0.00
5.935	0.9999	+0.0004	-0.00

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Frequency (GHz)	Re(S_{11})	Im(S_{11})	$\lvert S_{11}\rvert$ (dB)
5.940	1.0000	+0.0002	-0.00
5.945	1.0005	+0.0001	0.00
5.950	0.9998	-0.0001	-0.00
5.955	0.9998	+0.0003	-0.00
5.960	0.9999	+0.0002	-0.00
5.965	1.0005	+0.0002	0.00
5.970	0.9997	+0.0002	-0.00
5.975	0.9997	-0.0000	-0.00
5.980	1.0002	-0.0005	0.00
5.985	0.9992	+0.0004	-0.01
5.990	0.9999	+0.0003	-0.00
5.995	0.9998	+0.0000	-0.00
6.000	1.0000	-0.0004	0.00

3.2 Short



Frequency (GHz)	Re(S_{11})	Im(S_{11})	$\lvert S_{11}\rvert$ (dB)
0.001	-0.9875	-0.0025	-0.11
0.002	-0.9871	-0.0006	-0.11
0.003	-0.9868	+0.0001	-0.12
0.004	-0.9865	+0.0003	-0.12
0.005	-0.9864	+0.0007	-0.12
0.006	-0.9861	+0.0008	-0.12
0.007	-0.9859	+0.0010	-0.12
0.008	-0.9858	+0.0011	-0.12
0.009	-0.9857	+0.0012	-0.13

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Frequency (GHz)	**Re**(S_{11})	**Im**(S_{11})	$\lvert S_{11}\rvert$ **(dB)**
0.010	-0.9855	+0.0013	-0.13
0.011	-0.9854	+0.0008	-0.13
0.012	-0.9851	+0.0010	-0.13
0.013	-0.9851	+0.0014	-0.13
0.014	-0.9851	+0.0013	-0.13
0.015	-0.9849	+0.0016	-0.13
0.016	-0.9848	+0.0014	-0.13
0.017	-0.9846	+0.0015	-0.13
0.018	-0.9847	+0.0015	-0.13
0.019	-0.9845	+0.0015	-0.14
0.020	-0.9844	+0.0014	-0.14
0.021	-0.9845	+0.0014	-0.14
0.022	-0.9844	+0.0012	-0.14
0.023	-0.9841	+0.0007	-0.14
0.024	-0.9840	+0.0008	-0.14
0.025	-0.9841	+0.0014	-0.14
0.026	-0.9839	+0.0014	-0.14
0.027	-0.9838	+0.0015	-0.14
0.028	-0.9838	+0.0014	-0.14
0.029	-0.9836	+0.0013	-0.14
0.030	-0.9836	+0.0012	-0.14
0.031	-0.9835	+0.0013	-0.14
0.032	-0.9835	+0.0013	-0.14
0.033	-0.9835	+0.0013	-0.14
0.034	-0.9835	+0.0012	-0.14
0.035	-0.9832	+0.0011	-0.15
0.036	-0.9833	+0.0012	-0.15
0.037	-0.9834	+0.0010	-0.14
0.038	-0.9834	+0.0010	-0.15
0.039	-0.9832	+0.0008	-0.15
0.040	-0.9832	+0.0010	-0.15
0.041	-0.9831	+0.0008	-0.15
0.042	-0.9832	+0.0008	-0.15
0.043	-0.9830	+0.0008	-0.15
0.044	-0.9832	+0.0007	-0.15
0.045	-0.9832	+0.0007	-0.15
0.046	-0.9830	+0.0008	-0.15
0.047	-0.9830	+0.0006	-0.15
0.048	-0.9830	+0.0004	-0.15
0.049	-0.9830	+0.0004	-0.15
0.050	-0.9831	+0.0004	-0.15
0.051	-0.9829	+0.0005	-0.15
0.052	-0.9829	+0.0004	-0.15
0.053	-0.9828	+0.0003	-0.15
0.054	-0.9826	+0.0003	-0.15
0.055	-0.9826	+0.0002	-0.15

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Frequency (GHz)	**Re(S_{11})**	**Im(S_{11})**	**$\lvert S_{11}\rvert$ (dB)**
0.056	-0.9826	+0.0002	-0.15
0.057	-0.9825	+0.0002	-0.15
0.058	-0.9824	+0.0001	-0.15
0.059	-0.9823	+0.0001	-0.16
0.060	-0.9822	+0.0001	-0.16
0.061	-0.9821	+0.0000	-0.16
0.062	-0.9822	+0.0001	-0.16
0.063	-0.9821	+0.0000	-0.16
0.064	-0.9820	-0.0001	-0.16
0.065	-0.9819	-0.0001	-0.16
0.066	-0.9820	-0.0002	-0.16
0.067	-0.9820	-0.0003	-0.16
0.068	-0.9818	-0.0003	-0.16
0.069	-0.9818	-0.0004	-0.16
0.070	-0.9818	-0.0004	-0.16
0.071	-0.9817	-0.0005	-0.16
0.072	-0.9818	-0.0007	-0.16
0.073	-0.9817	-0.0007	-0.16
0.074	-0.9817	-0.0007	-0.16
0.075	-0.9817	-0.0009	-0.16
0.076	-0.9818	-0.0010	-0.16
0.077	-0.9817	-0.0010	-0.16
0.078	-0.9818	-0.0010	-0.16
0.079	-0.9816	-0.0012	-0.16
0.080	-0.9817	-0.0011	-0.16
0.081	-0.9816	-0.0012	-0.16
0.082	-0.9817	-0.0013	-0.16
0.083	-0.9816	-0.0013	-0.16
0.084	-0.9816	-0.0014	-0.16
0.085	-0.9816	-0.0014	-0.16
0.086	-0.9816	-0.0015	-0.16
0.087	-0.9814	-0.0015	-0.16
0.088	-0.9814	-0.0015	-0.16
0.089	-0.9813	-0.0016	-0.16
0.090	-0.9813	-0.0016	-0.16
0.091	-0.9814	-0.0016	-0.16
0.092	-0.9815	-0.0016	-0.16
0.093	-0.9814	-0.0016	-0.16
0.094	-0.9814	-0.0018	-0.16
0.095	-0.9815	-0.0017	-0.16
0.096	-0.9814	-0.0018	-0.16
0.097	-0.9816	-0.0018	-0.16
0.098	-0.9816	-0.0019	-0.16
0.099	-0.9816	-0.0019	-0.16
0.100	-0.9816	-0.0020	-0.16
0.105	-0.9816	-0.0020	-0.16

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Frequency (GHz)	**Re**(S_{11})	**Im**(S_{11})	$\lvert S_{11}\rvert$ **(dB)**
0.110	-0.9817	-0.0021	-0.16
0.115	-0.9817	-0.0022	-0.16
0.120	-0.9817	-0.0023	-0.16
0.125	-0.9818	-0.0023	-0.16
0.130	-0.9818	-0.0022	-0.16
0.135	-0.9819	-0.0021	-0.16
0.140	-0.9817	-0.0020	-0.16
0.145	-0.9816	-0.0021	-0.16
0.150	-0.9815	-0.0021	-0.16
0.155	-0.9814	-0.0021	-0.16
0.160	-0.9813	-0.0022	-0.16
0.165	-0.9814	-0.0022	-0.16
0.170	-0.9813	-0.0023	-0.16
0.175	-0.9812	-0.0023	-0.17
0.180	-0.9811	-0.0021	-0.17
0.185	-0.9810	-0.0022	-0.17
0.190	-0.9808	-0.0023	-0.17
0.195	-0.9806	-0.0025	-0.17
0.200	-0.9807	-0.0022	-0.17
0.205	-0.9803	-0.0024	-0.17
0.210	-0.9800	-0.0024	-0.18
0.215	-0.9798	-0.0027	-0.18
0.220	-0.9797	-0.0028	-0.18
0.225	-0.9795	-0.0030	-0.18
0.230	-0.9792	-0.0033	-0.18
0.235	-0.9792	-0.0033	-0.18
0.240	-0.9790	-0.0034	-0.18
0.245	-0.9790	-0.0036	-0.18
0.250	-0.9787	-0.0038	-0.19
0.255	-0.9785	-0.0041	-0.19
0.260	-0.9784	-0.0044	-0.19
0.265	-0.9783	-0.0046	-0.19
0.270	-0.9780	-0.0050	-0.19
0.275	-0.9780	-0.0053	-0.19
0.280	-0.9780	-0.0057	-0.19
0.285	-0.9778	-0.0060	-0.19
0.290	-0.9777	-0.0061	-0.20
0.295	-0.9778	-0.0066	-0.19
0.300	-0.9778	-0.0071	-0.19
0.305	-0.9779	-0.0076	-0.19
0.310	-0.9779	-0.0082	-0.19
0.315	-0.9779	-0.0090	-0.19
0.320	-0.9780	-0.0096	-0.19
0.325	-0.9781	-0.0100	-0.19
0.330	-0.9783	-0.0105	-0.19
0.335	-0.9785	-0.0111	-0.19

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Frequency (GHz)	**Re(S_{11})**	**Im(S_{11})**	**$\lvert S_{11}\rvert$ (dB)**
0.340	-0.9788	-0.0116	-0.19
0.345	-0.9789	-0.0123	-0.18
0.350	-0.9791	-0.0129	-0.18
0.355	-0.9792	-0.0135	-0.18
0.360	-0.9793	-0.0141	-0.18
0.365	-0.9796	-0.0148	-0.18
0.370	-0.9797	-0.0154	-0.18
0.375	-0.9801	-0.0162	-0.17
0.380	-0.9804	-0.0167	-0.17
0.385	-0.9806	-0.0174	-0.17
0.390	-0.9807	-0.0180	-0.17
0.395	-0.9814	-0.0187	-0.16
0.400	-0.9817	-0.0196	-0.16
0.405	-0.9822	-0.0204	-0.15
0.410	-0.9825	-0.0213	-0.15
0.415	-0.9830	-0.0222	-0.15
0.420	-0.9834	-0.0229	-0.14
0.425	-0.9839	-0.0236	-0.14
0.430	-0.9842	-0.0246	-0.14
0.435	-0.9848	-0.0253	-0.13
0.440	-0.9852	-0.0261	-0.13
0.445	-0.9856	-0.0269	-0.12
0.450	-0.9862	-0.0276	-0.12
0.455	-0.9866	-0.0285	-0.11
0.460	-0.9870	-0.0291	-0.11
0.465	-0.9873	-0.0299	-0.11
0.470	-0.9877	-0.0306	-0.10
0.475	-0.9881	-0.0314	-0.10
0.480	-0.9886	-0.0321	-0.10
0.485	-0.9890	-0.0330	-0.09
0.490	-0.9892	-0.0337	-0.09
0.495	-0.9897	-0.0343	-0.09
0.500	-0.9902	-0.0350	-0.08
0.505	-0.9907	-0.0357	-0.08
0.510	-0.9912	-0.0364	-0.07
0.515	-0.9917	-0.0368	-0.07
0.520	-0.9922	-0.0373	-0.06
0.525	-0.9925	-0.0382	-0.06
0.530	-0.9928	-0.0387	-0.06
0.535	-0.9933	-0.0392	-0.05
0.540	-0.9937	-0.0396	-0.05
0.545	-0.9940	-0.0401	-0.05
0.550	-0.9944	-0.0405	-0.04
0.555	-0.9945	-0.0410	-0.04
0.560	-0.9950	-0.0414	-0.04
0.565	-0.9953	-0.0418	-0.03

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Frequency (GHz)	**Re(S_{11})**	**Im(S_{11})**	**$\lvert S_{11}\rvert$ (dB)**
0.570	-0.9956	-0.0423	-0.03
0.575	-0.9957	-0.0426	-0.03
0.580	-0.9960	-0.0429	-0.03
0.585	-0.9962	-0.0432	-0.03
0.590	-0.9961	-0.0435	-0.03
0.595	-0.9962	-0.0436	-0.02
0.600	-0.9961	-0.0438	-0.03
0.605	-0.9962	-0.0439	-0.02
0.610	-0.9965	-0.0442	-0.02
0.615	-0.9965	-0.0442	-0.02
0.620	-0.9966	-0.0441	-0.02
0.625	-0.9967	-0.0443	-0.02
0.630	-0.9967	-0.0441	-0.02
0.635	-0.9967	-0.0441	-0.02
0.640	-0.9967	-0.0441	-0.02
0.645	-0.9967	-0.0441	-0.02
0.650	-0.9966	-0.0440	-0.02
0.655	-0.9966	-0.0438	-0.02
0.660	-0.9966	-0.0436	-0.02
0.665	-0.9966	-0.0435	-0.02
0.670	-0.9963	-0.0432	-0.02
0.675	-0.9963	-0.0430	-0.02
0.680	-0.9961	-0.0428	-0.03
0.685	-0.9960	-0.0424	-0.03
0.690	-0.9959	-0.0424	-0.03
0.695	-0.9957	-0.0421	-0.03
0.700	-0.9955	-0.0417	-0.03
0.705	-0.9953	-0.0413	-0.03
0.710	-0.9952	-0.0408	-0.03
0.715	-0.9947	-0.0404	-0.04
0.720	-0.9945	-0.0399	-0.04
0.725	-0.9942	-0.0394	-0.04
0.730	-0.9935	-0.0389	-0.05
0.735	-0.9934	-0.0385	-0.05
0.740	-0.9932	-0.0380	-0.05
0.745	-0.9928	-0.0375	-0.06
0.750	-0.9923	-0.0368	-0.06
0.755	-0.9920	-0.0362	-0.06
0.760	-0.9919	-0.0357	-0.07
0.765	-0.9916	-0.0349	-0.07
0.770	-0.9913	-0.0342	-0.07
0.775	-0.9909	-0.0335	-0.07
0.780	-0.9908	-0.0329	-0.08
0.785	-0.9903	-0.0321	-0.08
0.790	-0.9898	-0.0312	-0.08
0.795	-0.9895	-0.0307	-0.09

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Frequency (GHz)	**Re**(S_{11})	**Im**(S_{11})	$\lvert S_{11}\rvert$ **(dB)**
0.800	-0.9892	-0.0301	-0.09
0.805	-0.9890	-0.0292	-0.09
0.810	-0.9887	-0.0285	-0.10
0.815	-0.9883	-0.0279	-0.10
0.820	-0.9879	-0.0273	-0.10
0.825	-0.9875	-0.0264	-0.11
0.830	-0.9873	-0.0258	-0.11
0.835	-0.9869	-0.0250	-0.11
0.840	-0.9868	-0.0243	-0.11
0.845	-0.9862	-0.0236	-0.12
0.850	-0.9860	-0.0229	-0.12
0.855	-0.9856	-0.0223	-0.12
0.860	-0.9853	-0.0215	-0.13
0.865	-0.9850	-0.0208	-0.13
0.870	-0.9848	-0.0201	-0.13
0.875	-0.9843	-0.0195	-0.14
0.880	-0.9840	-0.0188	-0.14
0.885	-0.9837	-0.0184	-0.14
0.890	-0.9837	-0.0178	-0.14
0.895	-0.9834	-0.0173	-0.14
0.900	-0.9834	-0.0166	-0.14
0.905	-0.9831	-0.0162	-0.15
0.910	-0.9828	-0.0155	-0.15
0.915	-0.9826	-0.0152	-0.15
0.920	-0.9823	-0.0147	-0.15
0.925	-0.9821	-0.0141	-0.16
0.930	-0.9819	-0.0137	-0.16
0.935	-0.9819	-0.0133	-0.16
0.940	-0.9816	-0.0129	-0.16
0.945	-0.9816	-0.0125	-0.16
0.950	-0.9814	-0.0121	-0.16
0.955	-0.9813	-0.0120	-0.16
0.960	-0.9814	-0.0119	-0.16
0.965	-0.9813	-0.0117	-0.16
0.970	-0.9811	-0.0115	-0.16
0.975	-0.9812	-0.0112	-0.16
0.980	-0.9812	-0.0111	-0.16
0.985	-0.9811	-0.0108	-0.16
0.990	-0.9811	-0.0110	-0.17
0.995	-0.9811	-0.0109	-0.17
1.000	-0.9812	-0.0110	-0.16
1.005	-0.9813	-0.0108	-0.16
1.010	-0.9813	-0.0108	-0.16
1.015	-0.9812	-0.0111	-0.16
1.020	-0.9813	-0.0114	-0.16
1.025	-0.9813	-0.0114	-0.16

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Frequency (GHz)	**Re(S_{11})**	**Im(S_{11})**	**$\lvert S_{11}\rvert$ (dB)**
1.030	-0.9812	-0.0116	-0.16
1.035	-0.9813	-0.0118	-0.16
1.040	-0.9813	-0.0121	-0.16
1.045	-0.9813	-0.0124	-0.16
1.050	-0.9815	-0.0131	-0.16
1.055	-0.9817	-0.0134	-0.16
1.060	-0.9816	-0.0138	-0.16
1.065	-0.9818	-0.0143	-0.16
1.070	-0.9819	-0.0147	-0.16
1.075	-0.9821	-0.0150	-0.16
1.080	-0.9822	-0.0156	-0.15
1.085	-0.9822	-0.0163	-0.15
1.090	-0.9825	-0.0169	-0.15
1.095	-0.9824	-0.0173	-0.15
1.100	-0.9825	-0.0177	-0.15
1.105	-0.9826	-0.0185	-0.15
1.110	-0.9826	-0.0192	-0.15
1.115	-0.9830	-0.0197	-0.15
1.120	-0.9832	-0.0203	-0.15
1.125	-0.9831	-0.0210	-0.15
1.130	-0.9833	-0.0218	-0.14
1.135	-0.9833	-0.0226	-0.14
1.140	-0.9835	-0.0233	-0.14
1.145	-0.9838	-0.0241	-0.14
1.150	-0.9840	-0.0247	-0.14
1.155	-0.9842	-0.0254	-0.14
1.160	-0.9841	-0.0262	-0.14
1.165	-0.9843	-0.0267	-0.13
1.170	-0.9844	-0.0275	-0.13
1.175	-0.9845	-0.0282	-0.13
1.180	-0.9845	-0.0290	-0.13
1.185	-0.9848	-0.0297	-0.13
1.190	-0.9848	-0.0305	-0.13
1.195	-0.9849	-0.0310	-0.13
1.200	-0.9849	-0.0315	-0.13
1.205	-0.9850	-0.0324	-0.13
1.210	-0.9850	-0.0330	-0.13
1.215	-0.9849	-0.0337	-0.13
1.220	-0.9849	-0.0342	-0.13
1.225	-0.9850	-0.0348	-0.13
1.230	-0.9850	-0.0354	-0.13
1.235	-0.9851	-0.0358	-0.12
1.240	-0.9850	-0.0362	-0.13
1.245	-0.9852	-0.0366	-0.12
1.250	-0.9852	-0.0371	-0.12
1.255	-0.9853	-0.0376	-0.12

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Frequency (GHz)	**Re(S_{11})**	**Im(S_{11})**	**$\lvert S_{11}\rvert$ (dB)**
1.260	-0.9855	-0.0379	-0.12
1.265	-0.9854	-0.0382	-0.12
1.270	-0.9857	-0.0386	-0.12
1.275	-0.9857	-0.0386	-0.12
1.280	-0.9856	-0.0388	-0.12
1.285	-0.9855	-0.0389	-0.12
1.290	-0.9855	-0.0390	-0.12
1.295	-0.9855	-0.0388	-0.12
1.300	-0.9855	-0.0386	-0.12
1.305	-0.9856	-0.0385	-0.12
1.310	-0.9855	-0.0387	-0.12
1.315	-0.9855	-0.0385	-0.12
1.320	-0.9854	-0.0382	-0.12
1.325	-0.9855	-0.0383	-0.12
1.330	-0.9855	-0.0380	-0.12
1.335	-0.9855	-0.0376	-0.12
1.340	-0.9854	-0.0372	-0.12
1.345	-0.9854	-0.0368	-0.12
1.350	-0.9853	-0.0364	-0.12
1.355	-0.9852	-0.0358	-0.12
1.360	-0.9851	-0.0351	-0.12
1.365	-0.9851	-0.0344	-0.12
1.370	-0.9851	-0.0338	-0.12
1.375	-0.9850	-0.0331	-0.13
1.380	-0.9849	-0.0324	-0.13
1.385	-0.9847	-0.0316	-0.13
1.390	-0.9849	-0.0307	-0.13
1.395	-0.9846	-0.0297	-0.13
1.400	-0.9846	-0.0286	-0.13
1.405	-0.9844	-0.0277	-0.13
1.410	-0.9843	-0.0267	-0.13
1.415	-0.9843	-0.0258	-0.13
1.420	-0.9842	-0.0246	-0.14
1.425	-0.9838	-0.0235	-0.14
1.430	-0.9838	-0.0221	-0.14
1.435	-0.9836	-0.0209	-0.14
1.440	-0.9835	-0.0198	-0.14
1.445	-0.9835	-0.0185	-0.14
1.450	-0.9833	-0.0173	-0.14
1.455	-0.9831	-0.0160	-0.15
1.460	-0.9827	-0.0146	-0.15
1.465	-0.9826	-0.0135	-0.15
1.470	-0.9824	-0.0121	-0.15
1.475	-0.9821	-0.0106	-0.16
1.480	-0.9820	-0.0092	-0.16
1.485	-0.9817	-0.0078	-0.16

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Frequency (GHz)	**Re(S_{11})**	**Im(S_{11})**	**$\lvert S_{11}\rvert$ (dB)**
1.490	-0.9814	-0.0068	-0.16
1.495	-0.9812	-0.0051	-0.16
1.500	-0.9811	-0.0037	-0.17
1.505	-0.9806	-0.0023	-0.17
1.510	-0.9803	-0.0010	-0.17
1.515	-0.9800	+0.0003	-0.18
1.520	-0.9799	+0.0015	-0.18
1.525	-0.9797	+0.0030	-0.18
1.530	-0.9793	+0.0043	-0.18
1.535	-0.9790	+0.0058	-0.18
1.540	-0.9787	+0.0071	-0.19
1.545	-0.9785	+0.0083	-0.19
1.550	-0.9782	+0.0095	-0.19
1.555	-0.9778	+0.0107	-0.19
1.560	-0.9776	+0.0118	-0.20
1.565	-0.9775	+0.0129	-0.20
1.570	-0.9773	+0.0139	-0.20
1.575	-0.9770	+0.0149	-0.20
1.580	-0.9768	+0.0157	-0.20
1.585	-0.9765	+0.0167	-0.21
1.590	-0.9762	+0.0175	-0.21
1.595	-0.9761	+0.0185	-0.21
1.600	-0.9759	+0.0191	-0.21
1.605	-0.9758	+0.0199	-0.21
1.610	-0.9756	+0.0207	-0.21
1.615	-0.9755	+0.0215	-0.21
1.620	-0.9752	+0.0223	-0.22
1.625	-0.9750	+0.0230	-0.22
1.630	-0.9749	+0.0237	-0.22
1.635	-0.9747	+0.0242	-0.22
1.640	-0.9745	+0.0247	-0.22
1.645	-0.9743	+0.0252	-0.22
1.650	-0.9740	+0.0254	-0.23
1.655	-0.9737	+0.0257	-0.23
1.660	-0.9737	+0.0261	-0.23
1.665	-0.9737	+0.0263	-0.23
1.670	-0.9733	+0.0263	-0.23
1.675	-0.9732	+0.0263	-0.23
1.680	-0.9732	+0.0265	-0.23
1.685	-0.9729	+0.0265	-0.24
1.690	-0.9730	+0.0262	-0.23
1.695	-0.9730	+0.0259	-0.23
1.700	-0.9729	+0.0257	-0.24
1.705	-0.9726	+0.0254	-0.24
1.710	-0.9725	+0.0252	-0.24
1.715	-0.9723	+0.0249	-0.24

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Frequency (GHz)	**Re(S_{11})**	**Im(S_{11})**	**$\lvert S_{11}\rvert$ (dB)**
1.720	-0.9724	+0.0243	-0.24
1.725	-0.9722	+0.0237	-0.24
1.730	-0.9724	+0.0231	-0.24
1.735	-0.9725	+0.0226	-0.24
1.740	-0.9722	+0.0218	-0.24
1.745	-0.9725	+0.0209	-0.24
1.750	-0.9724	+0.0198	-0.24
1.755	-0.9728	+0.0190	-0.24
1.760	-0.9725	+0.0180	-0.24
1.765	-0.9725	+0.0169	-0.24
1.770	-0.9728	+0.0162	-0.24
1.775	-0.9727	+0.0151	-0.24
1.780	-0.9730	+0.0139	-0.24
1.785	-0.9731	+0.0131	-0.24
1.790	-0.9733	+0.0121	-0.23
1.795	-0.9733	+0.0108	-0.23
1.800	-0.9733	+0.0094	-0.24
1.805	-0.9736	+0.0082	-0.23
1.810	-0.9740	+0.0072	-0.23
1.815	-0.9744	+0.0061	-0.22
1.820	-0.9743	+0.0048	-0.23
1.825	-0.9746	+0.0038	-0.22
1.830	-0.9750	+0.0023	-0.22
1.835	-0.9750	+0.0010	-0.22
1.840	-0.9751	-0.0004	-0.22
1.845	-0.9754	-0.0015	-0.22
1.850	-0.9756	-0.0029	-0.21
1.855	-0.9758	-0.0041	-0.21
1.860	-0.9760	-0.0054	-0.21
1.865	-0.9763	-0.0066	-0.21
1.870	-0.9767	-0.0079	-0.20
1.875	-0.9770	-0.0091	-0.20
1.880	-0.9773	-0.0104	-0.20
1.885	-0.9777	-0.0115	-0.20
1.890	-0.9781	-0.0129	-0.19
1.895	-0.9783	-0.0140	-0.19
1.900	-0.9785	-0.0151	-0.19
1.905	-0.9789	-0.0162	-0.18
1.910	-0.9792	-0.0172	-0.18
1.915	-0.9795	-0.0182	-0.18
1.920	-0.9800	-0.0193	-0.17
1.925	-0.9803	-0.0203	-0.17
1.930	-0.9805	-0.0213	-0.17
1.935	-0.9808	-0.0221	-0.17
1.940	-0.9813	-0.0227	-0.16
1.945	-0.9816	-0.0238	-0.16

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Frequency (GHz)	**Re(S_{11})**	**Im(S_{11})**	**$\lvert S_{11}\rvert$ (dB)**
1.950	-0.9819	-0.0244	-0.16
1.955	-0.9823	-0.0251	-0.15
1.960	-0.9825	-0.0257	-0.15
1.965	-0.9829	-0.0263	-0.15
1.970	-0.9831	-0.0270	-0.14
1.975	-0.9835	-0.0273	-0.14
1.980	-0.9837	-0.0278	-0.14
1.985	-0.9841	-0.0283	-0.14
1.990	-0.9844	-0.0286	-0.13
1.995	-0.9848	-0.0289	-0.13
2.000	-0.9849	-0.0289	-0.13
2.005	-0.9853	-0.0289	-0.12
2.010	-0.9856	-0.0289	-0.12
2.015	-0.9860	-0.0290	-0.12
2.020	-0.9862	-0.0288	-0.12
2.025	-0.9868	-0.0285	-0.11
2.030	-0.9871	-0.0282	-0.11
2.035	-0.9875	-0.0281	-0.11
2.040	-0.9875	-0.0277	-0.11
2.045	-0.9880	-0.0273	-0.10
2.050	-0.9882	-0.0267	-0.10
2.055	-0.9884	-0.0263	-0.10
2.060	-0.9888	-0.0258	-0.09
2.065	-0.9891	-0.0250	-0.09
2.070	-0.9893	-0.0244	-0.09
2.075	-0.9897	-0.0235	-0.09
2.080	-0.9901	-0.0228	-0.08
2.085	-0.9903	-0.0220	-0.08
2.090	-0.9905	-0.0213	-0.08
2.095	-0.9906	-0.0204	-0.08
2.100	-0.9909	-0.0195	-0.08
2.105	-0.9912	-0.0184	-0.08
2.110	-0.9914	-0.0174	-0.07
2.115	-0.9916	-0.0163	-0.07
2.120	-0.9919	-0.0151	-0.07
2.125	-0.9919	-0.0139	-0.07
2.130	-0.9920	-0.0130	-0.07
2.135	-0.9920	-0.0116	-0.07
2.140	-0.9922	-0.0104	-0.07
2.145	-0.9922	-0.0092	-0.07
2.150	-0.9922	-0.0079	-0.07
2.155	-0.9923	-0.0068	-0.07
2.160	-0.9923	-0.0056	-0.07
2.165	-0.9923	-0.0043	-0.07
2.170	-0.9923	-0.0031	-0.07
2.175	-0.9921	-0.0020	-0.07

(continued on next page)

Frequency (GHz)	**Re(S_{11})**	**Im(S_{11})**	**$\lvert S_{11}\rvert$ (dB)**
2.180	-0.9921	-0.0006	-0.07
2.185	-0.9920	+0.0006	-0.07
2.190	-0.9918	+0.0018	-0.07
2.195	-0.9918	+0.0030	-0.07
2.200	-0.9916	+0.0043	-0.07
2.205	-0.9915	+0.0055	-0.07
2.210	-0.9913	+0.0066	-0.08
2.215	-0.9913	+0.0077	-0.08
2.220	-0.9912	+0.0088	-0.08
2.225	-0.9909	+0.0095	-0.08
2.230	-0.9908	+0.0106	-0.08
2.235	-0.9906	+0.0115	-0.08
2.240	-0.9904	+0.0124	-0.08
2.245	-0.9903	+0.0132	-0.08
2.250	-0.9900	+0.0139	-0.09
2.255	-0.9899	+0.0147	-0.09
2.260	-0.9898	+0.0155	-0.09
2.265	-0.9895	+0.0162	-0.09
2.270	-0.9892	+0.0167	-0.09
2.275	-0.9890	+0.0173	-0.09
2.280	-0.9890	+0.0177	-0.09
2.285	-0.9888	+0.0183	-0.10
2.290	-0.9886	+0.0185	-0.10
2.295	-0.9884	+0.0190	-0.10
2.300	-0.9883	+0.0194	-0.10
2.305	-0.9881	+0.0197	-0.10
2.310	-0.9878	+0.0198	-0.11
2.315	-0.9877	+0.0200	-0.11
2.320	-0.9873	+0.0199	-0.11
2.325	-0.9873	+0.0200	-0.11
2.330	-0.9869	+0.0200	-0.11
2.335	-0.9868	+0.0199	-0.11
2.340	-0.9865	+0.0200	-0.12
2.345	-0.9863	+0.0198	-0.12
2.350	-0.9860	+0.0195	-0.12
2.355	-0.9859	+0.0189	-0.12
2.360	-0.9857	+0.0183	-0.12
2.365	-0.9856	+0.0176	-0.12
2.370	-0.9854	+0.0170	-0.13
2.375	-0.9849	+0.0161	-0.13
2.380	-0.9849	+0.0155	-0.13
2.385	-0.9848	+0.0145	-0.13
2.390	-0.9846	+0.0137	-0.13
2.395	-0.9845	+0.0126	-0.14
2.400	-0.9844	+0.0118	-0.14
2.405	-0.9842	+0.0105	-0.14

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Frequency (GHz)	**Re(S_{11})**	**Im(S_{11})**	**$\lvert S_{11}\rvert$ (dB)**
2.410	-0.9839	+0.0094	-0.14
2.415	-0.9838	+0.0081	-0.14
2.420	-0.9838	+0.0069	-0.14
2.425	-0.9833	+0.0056	-0.15
2.430	-0.9834	+0.0043	-0.15
2.435	-0.9833	+0.0028	-0.15
2.440	-0.9831	+0.0012	-0.15
2.445	-0.9830	-0.0003	-0.15
2.450	-0.9829	-0.0018	-0.15
2.455	-0.9830	-0.0035	-0.15
2.460	-0.9829	-0.0052	-0.15
2.465	-0.9829	-0.0068	-0.15
2.470	-0.9829	-0.0084	-0.15
2.475	-0.9830	-0.0100	-0.15
2.480	-0.9830	-0.0115	-0.15
2.485	-0.9830	-0.0133	-0.15
2.490	-0.9830	-0.0151	-0.15
2.495	-0.9831	-0.0167	-0.15
2.500	-0.9832	-0.0183	-0.15
2.505	-0.9833	-0.0201	-0.14
2.510	-0.9834	-0.0218	-0.14
2.515	-0.9834	-0.0234	-0.14
2.520	-0.9835	-0.0251	-0.14
2.525	-0.9835	-0.0266	-0.14
2.530	-0.9838	-0.0282	-0.14
2.535	-0.9840	-0.0297	-0.14
2.540	-0.9841	-0.0311	-0.13
2.545	-0.9842	-0.0327	-0.13
2.550	-0.9843	-0.0341	-0.13
2.555	-0.9844	-0.0355	-0.13
2.560	-0.9845	-0.0369	-0.13
2.565	-0.9846	-0.0383	-0.13
2.570	-0.9849	-0.0394	-0.12
2.575	-0.9850	-0.0403	-0.12
2.580	-0.9849	-0.0411	-0.12
2.585	-0.9850	-0.0417	-0.12
2.590	-0.9849	-0.0423	-0.12
2.595	-0.9841	-0.0433	-0.13
2.600	-0.9839	-0.0439	-0.13
2.605	-0.9833	-0.0451	-0.14
2.610	-0.9826	-0.0460	-0.14
2.615	-0.9827	-0.0470	-0.14
2.620	-0.9819	-0.0482	-0.15
2.625	-0.9815	-0.0495	-0.15
2.630	-0.9809	-0.0506	-0.16
2.635	-0.9805	-0.0518	-0.16

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Frequency (GHz)	**Re(S_{11})**	**Im(S_{11})**	**$\lvert S_{11}\rvert$ (dB)**
2.640	-0.9799	-0.0528	-0.16
2.645	-0.9793	-0.0539	-0.17
2.650	-0.9786	-0.0547	-0.17
2.655	-0.9782	-0.0556	-0.18
2.660	-0.9775	-0.0567	-0.18
2.665	-0.9766	-0.0576	-0.19
2.670	-0.9759	-0.0586	-0.20
2.675	-0.9753	-0.0599	-0.20
2.680	-0.9747	-0.0610	-0.21
2.685	-0.9741	-0.0622	-0.21
2.690	-0.9737	-0.0634	-0.21
2.695	-0.9734	-0.0647	-0.22
2.700	-0.9734	-0.0656	-0.21
2.705	-0.9731	-0.0666	-0.22
2.710	-0.9731	-0.0676	-0.22
2.715	-0.9731	-0.0682	-0.22
2.720	-0.9730	-0.0690	-0.22
2.725	-0.9733	-0.0690	-0.21
2.730	-0.9733	-0.0693	-0.21
2.735	-0.9733	-0.0696	-0.21
2.740	-0.9734	-0.0697	-0.21
2.745	-0.9736	-0.0697	-0.21
2.750	-0.9738	-0.0698	-0.21
2.755	-0.9737	-0.0699	-0.21
2.760	-0.9735	-0.0698	-0.21
2.765	-0.9735	-0.0696	-0.21
2.770	-0.9733	-0.0695	-0.21
2.775	-0.9735	-0.0694	-0.21
2.780	-0.9735	-0.0692	-0.21
2.785	-0.9734	-0.0694	-0.21
2.790	-0.9734	-0.0692	-0.21
2.795	-0.9735	-0.0690	-0.21
2.800	-0.9734	-0.0689	-0.21
2.805	-0.9732	-0.0689	-0.21
2.810	-0.9730	-0.0684	-0.22
2.815	-0.9729	-0.0685	-0.22
2.820	-0.9728	-0.0683	-0.22
2.825	-0.9728	-0.0682	-0.22
2.830	-0.9725	-0.0678	-0.22
2.835	-0.9724	-0.0675	-0.22
2.840	-0.9723	-0.0672	-0.22
2.845	-0.9722	-0.0670	-0.22
2.850	-0.9718	-0.0667	-0.23
2.855	-0.9717	-0.0664	-0.23
2.860	-0.9715	-0.0663	-0.23
2.865	-0.9713	-0.0661	-0.23

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Frequency (GHz)	**Re(S_{11})**	**Im(S_{11})**	**$\lvert S_{11}\rvert$ (dB)**
2.870	-0.9711	-0.0658	-0.23
2.875	-0.9708	-0.0658	-0.24
2.880	-0.9705	-0.0657	-0.24
2.885	-0.9702	-0.0654	-0.24
2.890	-0.9701	-0.0653	-0.24
2.895	-0.9697	-0.0651	-0.25
2.900	-0.9695	-0.0650	-0.25
2.905	-0.9695	-0.0649	-0.25
2.910	-0.9694	-0.0649	-0.25
2.915	-0.9693	-0.0648	-0.25
2.920	-0.9692	-0.0648	-0.25
2.925	-0.9692	-0.0650	-0.25
2.930	-0.9690	-0.0653	-0.25
2.935	-0.9689	-0.0654	-0.25
2.940	-0.9688	-0.0657	-0.26
2.945	-0.9688	-0.0658	-0.26
2.950	-0.9692	-0.0661	-0.25
2.955	-0.9689	-0.0665	-0.25
2.960	-0.9687	-0.0671	-0.26
2.965	-0.9689	-0.0675	-0.25
2.970	-0.9690	-0.0678	-0.25
2.975	-0.9694	-0.0683	-0.25
2.980	-0.9694	-0.0688	-0.25
2.985	-0.9700	-0.0692	-0.24
2.990	-0.9700	-0.0699	-0.24
2.995	-0.9701	-0.0702	-0.24
3.000	-0.9705	-0.0711	-0.24
3.005	-0.9701	-0.0713	-0.24
3.010	-0.9702	-0.0721	-0.24
3.015	-0.9706	-0.0728	-0.23
3.020	-0.9713	-0.0733	-0.23
3.025	-0.9712	-0.0742	-0.23
3.030	-0.9718	-0.0752	-0.22
3.035	-0.9721	-0.0761	-0.22
3.040	-0.9723	-0.0774	-0.22
3.045	-0.9726	-0.0783	-0.21
3.050	-0.9726	-0.0794	-0.21
3.055	-0.9732	-0.0802	-0.21
3.060	-0.9735	-0.0817	-0.20
3.065	-0.9736	-0.0828	-0.20
3.070	-0.9736	-0.0837	-0.20
3.075	-0.9739	-0.0850	-0.20
3.080	-0.9741	-0.0861	-0.19
3.085	-0.9740	-0.0872	-0.19
3.090	-0.9743	-0.0879	-0.19
3.095	-0.9745	-0.0889	-0.19

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Frequency (GHz)	**Re(S_{11})**	**Im(S_{11})**	**$\lvert S_{11}\rvert$ (dB)**
3.100	-0.9748	-0.0900	-0.18
3.105	-0.9752	-0.0908	-0.18
3.110	-0.9754	-0.0917	-0.18
3.115	-0.9754	-0.0926	-0.18
3.120	-0.9758	-0.0934	-0.17
3.125	-0.9759	-0.0942	-0.17
3.130	-0.9764	-0.0950	-0.17
3.135	-0.9768	-0.0959	-0.16
3.140	-0.9771	-0.0970	-0.16
3.145	-0.9773	-0.0976	-0.16
3.150	-0.9776	-0.0983	-0.15
3.155	-0.9779	-0.0990	-0.15
3.160	-0.9780	-0.0996	-0.15
3.165	-0.9783	-0.1001	-0.15
3.170	-0.9786	-0.1007	-0.14
3.175	-0.9786	-0.1010	-0.14
3.180	-0.9789	-0.1017	-0.14
3.185	-0.9788	-0.1019	-0.14
3.190	-0.9790	-0.1024	-0.14
3.195	-0.9791	-0.1029	-0.14
3.200	-0.9793	-0.1036	-0.13
3.205	-0.9793	-0.1037	-0.13
3.210	-0.9792	-0.1040	-0.13
3.215	-0.9796	-0.1048	-0.13
3.220	-0.9795	-0.1050	-0.13
3.225	-0.9799	-0.1053	-0.13
3.230	-0.9798	-0.1063	-0.13
3.235	-0.9804	-0.1061	-0.12
3.240	-0.9805	-0.1063	-0.12
3.245	-0.9812	-0.1063	-0.11
3.250	-0.9816	-0.1067	-0.11
3.255	-0.9818	-0.1066	-0.11
3.260	-0.9823	-0.1064	-0.10
3.265	-0.9830	-0.1067	-0.10
3.270	-0.9834	-0.1062	-0.10
3.275	-0.9832	-0.1061	-0.10
3.280	-0.9836	-0.1056	-0.09
3.285	-0.9839	-0.1058	-0.09
3.290	-0.9839	-0.1052	-0.09
3.295	-0.9835	-0.1048	-0.10
3.300	-0.9836	-0.1044	-0.10
3.305	-0.9834	-0.1041	-0.10
3.310	-0.9831	-0.1035	-0.10
3.315	-0.9824	-0.1031	-0.11
3.320	-0.9824	-0.1018	-0.11
3.325	-0.9822	-0.1016	-0.11

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Frequency (GHz)	**Re(S_{11})**	**Im(S_{11})**	**$\lvert S_{11}\rvert$ (dB)**
3.330	-0.9817	-0.1005	-0.12
3.335	-0.9811	-0.0998	-0.12
3.340	-0.9807	-0.0985	-0.13
3.345	-0.9799	-0.0977	-0.13
3.350	-0.9799	-0.0964	-0.13
3.355	-0.9793	-0.0955	-0.14
3.360	-0.9789	-0.0943	-0.15
3.365	-0.9785	-0.0928	-0.15
3.370	-0.9779	-0.0915	-0.16
3.375	-0.9776	-0.0903	-0.16
3.380	-0.9770	-0.0893	-0.17
3.385	-0.9766	-0.0882	-0.17
3.390	-0.9760	-0.0866	-0.18
3.395	-0.9756	-0.0855	-0.18
3.400	-0.9752	-0.0841	-0.19
3.405	-0.9745	-0.0829	-0.19
3.410	-0.9743	-0.0817	-0.20
3.415	-0.9741	-0.0803	-0.20
3.420	-0.9737	-0.0793	-0.20
3.425	-0.9732	-0.0780	-0.21
3.430	-0.9727	-0.0768	-0.21
3.435	-0.9723	-0.0755	-0.22
3.440	-0.9718	-0.0744	-0.22
3.445	-0.9714	-0.0733	-0.23
3.450	-0.9712	-0.0722	-0.23
3.455	-0.9709	-0.0708	-0.23
3.460	-0.9703	-0.0697	-0.24
3.465	-0.9701	-0.0690	-0.24
3.470	-0.9696	-0.0675	-0.25
3.475	-0.9696	-0.0666	-0.25
3.480	-0.9691	-0.0657	-0.25
3.485	-0.9689	-0.0647	-0.26
3.490	-0.9685	-0.0638	-0.26
3.495	-0.9683	-0.0631	-0.26
3.500	-0.9678	-0.0619	-0.27
3.505	-0.9675	-0.0613	-0.27
3.510	-0.9674	-0.0603	-0.27
3.515	-0.9668	-0.0595	-0.28
3.520	-0.9667	-0.0587	-0.28
3.525	-0.9670	-0.0572	-0.28
3.530	-0.9670	-0.0555	-0.28
3.535	-0.9667	-0.0556	-0.28
3.540	-0.9664	-0.0554	-0.28
3.545	-0.9662	-0.0550	-0.28
3.550	-0.9660	-0.0546	-0.29
3.555	-0.9659	-0.0544	-0.29

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Frequency (GHz)	**Re**(S_{11})	**Im**(S_{11})	$\lvert S_{11}\rvert$ **(dB)**
3.560	-0.9657	-0.0541	-0.29
3.565	-0.9659	-0.0539	-0.29
3.570	-0.9653	-0.0535	-0.29
3.575	-0.9667	-0.0544	-0.28
3.580	-0.9650	-0.0537	-0.30
3.585	-0.9637	-0.0538	-0.31
3.590	-0.9634	-0.0542	-0.31
3.595	-0.9629	-0.0539	-0.32
3.600	-0.9628	-0.0536	-0.32
3.605	-0.9618	-0.0529	-0.33
3.610	-0.9619	-0.0523	-0.32
3.615	-0.9620	-0.0523	-0.32
3.620	-0.9623	-0.0524	-0.32
3.625	-0.9625	-0.0527	-0.32
3.630	-0.9628	-0.0526	-0.32
3.635	-0.9631	-0.0527	-0.31
3.640	-0.9634	-0.0526	-0.31
3.645	-0.9633	-0.0532	-0.31
3.650	-0.9636	-0.0533	-0.31
3.655	-0.9637	-0.0535	-0.31
3.660	-0.9637	-0.0535	-0.31
3.665	-0.9638	-0.0536	-0.31
3.670	-0.9637	-0.0540	-0.31
3.675	-0.9638	-0.0543	-0.31
3.680	-0.9635	-0.0546	-0.31
3.685	-0.9636	-0.0542	-0.31
3.690	-0.9631	-0.0539	-0.31
3.695	-0.9629	-0.0532	-0.31
3.700	-0.9625	-0.0526	-0.32
3.705	-0.9624	-0.0512	-0.32
3.710	-0.9624	-0.0501	-0.32
3.715	-0.9633	-0.0484	-0.31
3.720	-0.9639	-0.0465	-0.31
3.725	-0.9657	-0.0450	-0.29
3.730	-0.9668	-0.0434	-0.28
3.735	-0.9690	-0.0432	-0.27
3.740	-0.9709	-0.0423	-0.25
3.745	-0.9719	-0.0431	-0.24
3.750	-0.9734	-0.0430	-0.23
3.755	-0.9736	-0.0432	-0.22
3.760	-0.9748	-0.0433	-0.21
3.765	-0.9751	-0.0428	-0.21
3.770	-0.9763	-0.0423	-0.20
3.775	-0.9775	-0.0422	-0.19
3.780	-0.9788	-0.0424	-0.18
3.785	-0.9802	-0.0424	-0.17

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Frequency (GHz)	**Re(S_{11})**	**Im(S_{11})**	**$\lvert S_{11}\rvert$ (dB)**
3.790	-0.9806	-0.0422	-0.16
3.795	-0.9819	-0.0420	-0.15
3.800	-0.9826	-0.0420	-0.14
3.805	-0.9830	-0.0421	-0.14
3.810	-0.9834	-0.0415	-0.14
3.815	-0.9841	-0.0413	-0.13
3.820	-0.9846	-0.0407	-0.13
3.825	-0.9850	-0.0407	-0.12
3.830	-0.9859	-0.0400	-0.12
3.835	-0.9839	-0.0388	-0.13
3.840	-0.9839	-0.0374	-0.13
3.845	-0.9851	-0.0363	-0.12
3.850	-0.9858	-0.0356	-0.12
3.855	-0.9859	-0.0343	-0.12
3.860	-0.9865	-0.0334	-0.11
3.865	-0.9869	-0.0319	-0.11
3.870	-0.9876	-0.0307	-0.10
3.875	-0.9881	-0.0293	-0.10
3.880	-0.9888	-0.0279	-0.09
3.885	-0.9896	-0.0266	-0.09
3.890	-0.9901	-0.0252	-0.08
3.895	-0.9911	-0.0240	-0.07
3.900	-0.9910	-0.0224	-0.08
3.905	-0.9914	-0.0206	-0.07
3.910	-0.9921	-0.0191	-0.07
3.915	-0.9923	-0.0175	-0.07
3.920	-0.9928	-0.0158	-0.06
3.925	-0.9929	-0.0140	-0.06
3.930	-0.9933	-0.0131	-0.06
3.935	-0.9934	-0.0115	-0.06
3.940	-0.9934	-0.0099	-0.06
3.945	-0.9935	-0.0083	-0.06
3.950	-0.9934	-0.0067	-0.06
3.955	-0.9935	-0.0050	-0.06
3.960	-0.9933	-0.0035	-0.06
3.965	-0.9929	-0.0020	-0.06
3.970	-0.9929	-0.0007	-0.06
3.975	-0.9925	+0.0011	-0.07
3.980	-0.9923	+0.0027	-0.07
3.985	-0.9921	+0.0042	-0.07
3.990	-0.9917	+0.0055	-0.07
3.995	-0.9914	+0.0071	-0.07
4.000	-0.9910	+0.0087	-0.08
4.005	-0.9906	+0.0099	-0.08
4.010	-0.9902	+0.0113	-0.08
4.015	-0.9896	+0.0125	-0.09

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Frequency (GHz)	**Re(S_{11})**	**Im(S_{11})**	**$\lvert S_{11}\rvert$ (dB)**
4.020	-0.9892	+0.0137	-0.09
4.025	-0.9886	+0.0146	-0.10
4.030	-0.9883	+0.0159	-0.10
4.035	-0.9878	+0.0169	-0.11
4.040	-0.9872	+0.0181	-0.11
4.045	-0.9867	+0.0191	-0.11
4.050	-0.9860	+0.0198	-0.12
4.055	-0.9853	+0.0206	-0.13
4.060	-0.9846	+0.0213	-0.13
4.065	-0.9841	+0.0216	-0.14
4.070	-0.9836	+0.0221	-0.14
4.075	-0.9829	+0.0227	-0.15
4.080	-0.9823	+0.0229	-0.15
4.085	-0.9819	+0.0230	-0.16
4.090	-0.9814	+0.0231	-0.16
4.095	-0.9807	+0.0232	-0.17
4.100	-0.9802	+0.0233	-0.17
4.105	-0.9799	+0.0234	-0.17
4.110	-0.9795	+0.0232	-0.18
4.115	-0.9792	+0.0232	-0.18
4.120	-0.9793	+0.0232	-0.18
4.125	-0.9789	+0.0232	-0.18
4.130	-0.9783	+0.0229	-0.19
4.135	-0.9778	+0.0232	-0.19
4.140	-0.9775	+0.0232	-0.20
4.145	-0.9764	+0.0230	-0.20
4.150	-0.9759	+0.0226	-0.21
4.155	-0.9752	+0.0223	-0.22
4.160	-0.9745	+0.0217	-0.22
4.165	-0.9737	+0.0203	-0.23
4.170	-0.9728	+0.0193	-0.24
4.175	-0.9724	+0.0176	-0.24
4.180	-0.9718	+0.0165	-0.25
4.185	-0.9712	+0.0151	-0.25
4.190	-0.9709	+0.0134	-0.26
4.195	-0.9707	+0.0117	-0.26
4.200	-0.9705	+0.0100	-0.26
4.205	-0.9705	+0.0082	-0.26
4.210	-0.9701	+0.0062	-0.26
4.215	-0.9699	+0.0041	-0.27
4.220	-0.9699	+0.0021	-0.27
4.225	-0.9700	+0.0000	-0.26
4.230	-0.9697	-0.0021	-0.27
4.235	-0.9699	-0.0045	-0.27
4.240	-0.9703	-0.0068	-0.26
4.245	-0.9701	-0.0093	-0.26

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Frequency (GHz)	**Re(S_{11})**	**Im(S_{11})**	**$\lvert S_{11}\rvert$ (dB)**
4.250	-0.9706	-0.0112	-0.26
4.255	-0.9707	-0.0137	-0.26
4.260	-0.9704	-0.0158	-0.26
4.265	-0.9706	-0.0181	-0.26
4.270	-0.9712	-0.0203	-0.25
4.275	-0.9713	-0.0228	-0.25
4.280	-0.9719	-0.0252	-0.24
4.285	-0.9722	-0.0276	-0.24
4.290	-0.9726	-0.0300	-0.24
4.295	-0.9728	-0.0322	-0.23
4.300	-0.9732	-0.0346	-0.23
4.305	-0.9737	-0.0370	-0.23
4.310	-0.9739	-0.0394	-0.22
4.315	-0.9742	-0.0416	-0.22
4.320	-0.9744	-0.0439	-0.22
4.325	-0.9745	-0.0463	-0.21
4.330	-0.9748	-0.0487	-0.21
4.335	-0.9750	-0.0509	-0.21
4.340	-0.9752	-0.0530	-0.21
4.345	-0.9754	-0.0552	-0.20
4.350	-0.9756	-0.0575	-0.20
4.355	-0.9757	-0.0594	-0.20
4.360	-0.9758	-0.0619	-0.19
4.365	-0.9760	-0.0640	-0.19
4.370	-0.9761	-0.0660	-0.19
4.375	-0.9765	-0.0681	-0.19
4.380	-0.9765	-0.0700	-0.18
4.385	-0.9764	-0.0720	-0.18
4.390	-0.9765	-0.0740	-0.18
4.395	-0.9767	-0.0757	-0.18
4.400	-0.9768	-0.0772	-0.18
4.405	-0.9769	-0.0789	-0.17
4.410	-0.9773	-0.0810	-0.17
4.415	-0.9771	-0.0826	-0.17
4.420	-0.9775	-0.0841	-0.17
4.425	-0.9779	-0.0854	-0.16
4.430	-0.9778	-0.0874	-0.16
4.435	-0.9780	-0.0887	-0.16
4.440	-0.9780	-0.0899	-0.16
4.445	-0.9782	-0.0918	-0.15
4.450	-0.9782	-0.0927	-0.15
4.455	-0.9785	-0.0945	-0.15
4.460	-0.9785	-0.0960	-0.15
4.465	-0.9788	-0.0978	-0.14
4.470	-0.9786	-0.0994	-0.14
4.475	-0.9790	-0.1015	-0.14

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Frequency (GHz)	**Re**(S_{11})	**Im**(S_{11})	$\lvert S_{11}\rvert$ **(dB)**
4.480	-0.9792	-0.1033	-0.13
4.485	-0.9800	-0.1049	-0.13
4.490	-0.9804	-0.1059	-0.12
4.495	-0.9807	-0.1074	-0.12
4.500	-0.9811	-0.1082	-0.11
4.505	-0.9811	-0.1090	-0.11
4.510	-0.9812	-0.1093	-0.11
4.515	-0.9809	-0.1096	-0.11
4.520	-0.9810	-0.1101	-0.11
4.525	-0.9804	-0.1102	-0.12
4.530	-0.9803	-0.1100	-0.12
4.535	-0.9807	-0.1099	-0.12
4.540	-0.9802	-0.1096	-0.12
4.545	-0.9798	-0.1089	-0.12
4.550	-0.9794	-0.1092	-0.13
4.555	-0.9794	-0.1089	-0.13
4.560	-0.9793	-0.1085	-0.13
4.565	-0.9789	-0.1081	-0.13
4.570	-0.9787	-0.1076	-0.13
4.575	-0.9785	-0.1072	-0.14
4.580	-0.9777	-0.1066	-0.14
4.585	-0.9777	-0.1060	-0.15
4.590	-0.9772	-0.1056	-0.15
4.595	-0.9767	-0.1051	-0.15
4.600	-0.9762	-0.1045	-0.16
4.605	-0.9756	-0.1038	-0.17
4.610	-0.9751	-0.1035	-0.17
4.615	-0.9744	-0.1029	-0.18
4.620	-0.9738	-0.1024	-0.18
4.625	-0.9734	-0.1019	-0.19
4.630	-0.9725	-0.1013	-0.20
4.635	-0.9717	-0.1009	-0.20
4.640	-0.9711	-0.1005	-0.21
4.645	-0.9704	-0.0999	-0.22
4.650	-0.9699	-0.0993	-0.22
4.655	-0.9688	-0.0989	-0.23
4.660	-0.9683	-0.0987	-0.23
4.665	-0.9674	-0.0981	-0.24
4.670	-0.9666	-0.0982	-0.25
4.675	-0.9655	-0.0981	-0.26
4.680	-0.9649	-0.0979	-0.27
4.685	-0.9640	-0.0981	-0.27
4.690	-0.9636	-0.0982	-0.28
4.695	-0.9626	-0.0986	-0.29
4.700	-0.9619	-0.0986	-0.29
4.705	-0.9614	-0.0983	-0.30

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Frequency (GHz)	**Re(S_{11})**	**Im(S_{11})**	**$\lvert S_{11}\rvert$ (dB)**
4.710	-0.9612	-0.0988	-0.30
4.715	-0.9608	-0.0990	-0.30
4.720	-0.9605	-0.0984	-0.31
4.725	-0.9601	-0.0984	-0.31
4.730	-0.9601	-0.0978	-0.31
4.735	-0.9598	-0.0975	-0.31
4.740	-0.9590	-0.0968	-0.32
4.745	-0.9584	-0.0960	-0.33
4.750	-0.9578	-0.0954	-0.33
4.755	-0.9572	-0.0945	-0.34
4.760	-0.9565	-0.0943	-0.34
4.765	-0.9554	-0.0936	-0.35
4.770	-0.9551	-0.0935	-0.36
4.775	-0.9540	-0.0926	-0.37
4.780	-0.9530	-0.0933	-0.38
4.785	-0.9524	-0.0937	-0.38
4.790	-0.9518	-0.0935	-0.39
4.795	-0.9506	-0.0942	-0.40
4.800	-0.9499	-0.0951	-0.40
4.805	-0.9495	-0.0962	-0.41
4.810	-0.9487	-0.0977	-0.41
4.815	-0.9483	-0.0990	-0.41
4.820	-0.9483	-0.1004	-0.41
4.825	-0.9483	-0.1011	-0.41
4.830	-0.9484	-0.1033	-0.41
4.835	-0.9484	-0.1046	-0.41
4.840	-0.9490	-0.1057	-0.40
4.845	-0.9494	-0.1069	-0.40
4.850	-0.9505	-0.1081	-0.39
4.855	-0.9512	-0.1098	-0.38
4.860	-0.9520	-0.1108	-0.37
4.865	-0.9530	-0.1116	-0.36
4.870	-0.9537	-0.1125	-0.35
4.875	-0.9546	-0.1138	-0.34
4.880	-0.9555	-0.1146	-0.33
4.885	-0.9564	-0.1168	-0.32
4.890	-0.9574	-0.1162	-0.31
4.895	-0.9581	-0.1169	-0.31
4.900	-0.9593	-0.1178	-0.30
4.905	-0.9599	-0.1183	-0.29
4.910	-0.9605	-0.1188	-0.28
4.915	-0.9612	-0.1192	-0.28
4.920	-0.9620	-0.1197	-0.27
4.925	-0.9625	-0.1205	-0.26
4.930	-0.9631	-0.1210	-0.26
4.935	-0.9635	-0.1215	-0.25

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Frequency (GHz)	**Re(S_{11})**	**Im(S_{11})**	**$\lvert S_{11}\rvert$ (dB)**
4.940	-0.9641	-0.1220	-0.25
4.945	-0.9646	-0.1226	-0.24
4.950	-0.9652	-0.1233	-0.24
4.955	-0.9656	-0.1238	-0.23
4.960	-0.9663	-0.1240	-0.23
4.965	-0.9667	-0.1248	-0.22
4.970	-0.9673	-0.1253	-0.22
4.975	-0.9677	-0.1256	-0.21
4.980	-0.9686	-0.1261	-0.20
4.985	-0.9689	-0.1268	-0.20
4.990	-0.9698	-0.1266	-0.19
4.995	-0.9698	-0.1271	-0.19
5.000	-0.9707	-0.1270	-0.18
5.005	-0.9711	-0.1270	-0.18
5.010	-0.9714	-0.1270	-0.18
5.015	-0.9719	-0.1274	-0.17
5.020	-0.9723	-0.1267	-0.17
5.025	-0.9726	-0.1274	-0.17
5.030	-0.9725	-0.1266	-0.17
5.035	-0.9730	-0.1270	-0.16
5.040	-0.9730	-0.1257	-0.17
5.045	-0.9735	-0.1268	-0.16
5.050	-0.9730	-0.1260	-0.17
5.055	-0.9737	-0.1267	-0.16
5.060	-0.9732	-0.1262	-0.16
5.065	-0.9747	-0.1258	-0.15
5.070	-0.9744	-0.1265	-0.15
5.075	-0.9756	-0.1257	-0.14
5.080	-0.9754	-0.1261	-0.14
5.085	-0.9762	-0.1252	-0.14
5.090	-0.9770	-0.1255	-0.13
5.095	-0.9780	-0.1245	-0.12
5.100	-0.9787	-0.1246	-0.12
5.105	-0.9785	-0.1234	-0.12
5.110	-0.9794	-0.1224	-0.11
5.115	-0.9799	-0.1210	-0.11
5.120	-0.9806	-0.1200	-0.11
5.125	-0.9810	-0.1182	-0.10
5.130	-0.9818	-0.1168	-0.10
5.135	-0.9819	-0.1149	-0.10
5.140	-0.9828	-0.1131	-0.09
5.145	-0.9824	-0.1106	-0.10
5.150	-0.9833	-0.1089	-0.09
5.155	-0.9833	-0.1067	-0.10
5.160	-0.9837	-0.1042	-0.09
5.165	-0.9835	-0.1020	-0.10

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Frequency (GHz)	**Re(S_{11})**	**Im(S_{11})**	**$\lvert S_{11}\rvert$ (dB)**
5.170	-0.9835	-0.0994	-0.10
5.175	-0.9831	-0.0970	-0.11
5.180	-0.9831	-0.0943	-0.11
5.185	-0.9828	-0.0918	-0.11
5.190	-0.9822	-0.0891	-0.12
5.195	-0.9815	-0.0867	-0.13
5.200	-0.9808	-0.0840	-0.14
5.205	-0.9804	-0.0815	-0.14
5.210	-0.9797	-0.0790	-0.15
5.215	-0.9790	-0.0765	-0.16
5.220	-0.9781	-0.0740	-0.17
5.225	-0.9773	-0.0716	-0.18
5.230	-0.9765	-0.0692	-0.18
5.235	-0.9758	-0.0670	-0.19
5.240	-0.9753	-0.0647	-0.20
5.245	-0.9745	-0.0625	-0.21
5.250	-0.9735	-0.0605	-0.22
5.255	-0.9727	-0.0583	-0.22
5.260	-0.9721	-0.0563	-0.23
5.265	-0.9713	-0.0542	-0.24
5.270	-0.9708	-0.0526	-0.24
5.275	-0.9699	-0.0508	-0.25
5.280	-0.9690	-0.0489	-0.26
5.285	-0.9680	-0.0473	-0.27
5.290	-0.9675	-0.0456	-0.28
5.295	-0.9669	-0.0443	-0.28
5.300	-0.9662	-0.0425	-0.29
5.305	-0.9658	-0.0410	-0.29
5.310	-0.9649	-0.0392	-0.30
5.315	-0.9646	-0.0379	-0.31
5.320	-0.9638	-0.0367	-0.31
5.325	-0.9631	-0.0350	-0.32
5.330	-0.9629	-0.0342	-0.32
5.335	-0.9623	-0.0326	-0.33
5.340	-0.9621	-0.0316	-0.33
5.345	-0.9614	-0.0308	-0.34
5.350	-0.9612	-0.0289	-0.34
5.355	-0.9606	-0.0282	-0.34
5.360	-0.9605	-0.0270	-0.35
5.365	-0.9600	-0.0262	-0.35
5.370	-0.9597	-0.0251	-0.35
5.375	-0.9594	-0.0242	-0.36
5.380	-0.9592	-0.0229	-0.36
5.385	-0.9589	-0.0221	-0.36
5.390	-0.9591	-0.0211	-0.36
5.395	-0.9592	-0.0206	-0.36

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Frequency (GHz)	**Re(S_{11})**	**Im(S_{11})**	**$\lvert S_{11}\rvert$ (dB)**
5.400	-0.9594	-0.0197	-0.36
5.405	-0.9601	-0.0189	-0.35
5.410	-0.9596	-0.0181	-0.36
5.415	-0.9596	-0.0178	-0.36
5.420	-0.9601	-0.0172	-0.35
5.425	-0.9603	-0.0164	-0.35
5.430	-0.9605	-0.0159	-0.35
5.435	-0.9606	-0.0157	-0.35
5.440	-0.9610	-0.0154	-0.34
5.445	-0.9611	-0.0151	-0.34
5.450	-0.9617	-0.0152	-0.34
5.455	-0.9622	-0.0153	-0.33
5.460	-0.9626	-0.0149	-0.33
5.465	-0.9629	-0.0152	-0.33
5.470	-0.9633	-0.0156	-0.32
5.475	-0.9639	-0.0155	-0.32
5.480	-0.9643	-0.0161	-0.31
5.485	-0.9648	-0.0161	-0.31
5.490	-0.9652	-0.0165	-0.31
5.495	-0.9658	-0.0167	-0.30
5.500	-0.9662	-0.0170	-0.30
5.505	-0.9667	-0.0174	-0.29
5.510	-0.9676	-0.0177	-0.29
5.515	-0.9683	-0.0180	-0.28
5.520	-0.9692	-0.0186	-0.27
5.525	-0.9699	-0.0194	-0.26
5.530	-0.9708	-0.0195	-0.26
5.535	-0.9717	-0.0202	-0.25
5.540	-0.9726	-0.0209	-0.24
5.545	-0.9735	-0.0217	-0.23
5.550	-0.9743	-0.0223	-0.22
5.555	-0.9752	-0.0231	-0.22
5.560	-0.9760	-0.0237	-0.21
5.565	-0.9767	-0.0244	-0.20
5.570	-0.9775	-0.0251	-0.19
5.575	-0.9783	-0.0259	-0.19
5.580	-0.9789	-0.0264	-0.18
5.585	-0.9796	-0.0270	-0.18
5.590	-0.9803	-0.0277	-0.17
5.595	-0.9808	-0.0282	-0.16
5.600	-0.9812	-0.0284	-0.16
5.605	-0.9818	-0.0289	-0.16
5.610	-0.9824	-0.0294	-0.15
5.615	-0.9831	-0.0297	-0.14
5.620	-0.9835	-0.0298	-0.14
5.625	-0.9842	-0.0301	-0.13

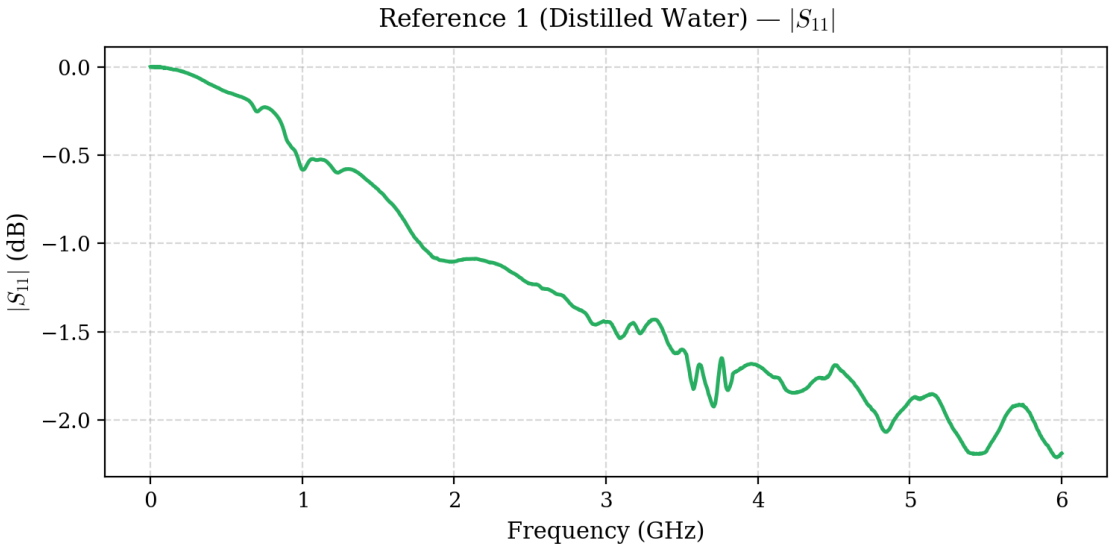
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Frequency (GHz)	**Re**(S_{11})	**Im**(S_{11})	$\lvert S_{11}\rvert$ **(dB)**
5.630	-0.9852	-0.0302	-0.13
5.635	-0.9852	-0.0303	-0.13
5.640	-0.9861	-0.0305	-0.12
5.645	-0.9862	-0.0309	-0.12
5.650	-0.9867	-0.0315	-0.11
5.655	-0.9873	-0.0316	-0.11
5.660	-0.9878	-0.0317	-0.10
5.665	-0.9882	-0.0323	-0.10
5.670	-0.9882	-0.0318	-0.10
5.675	-0.9882	-0.0321	-0.10
5.680	-0.9889	-0.0320	-0.09
5.685	-0.9878	-0.0321	-0.10
5.690	-0.9875	-0.0324	-0.10
5.695	-0.9870	-0.0322	-0.11
5.700	-0.9874	-0.0323	-0.11
5.705	-0.9872	-0.0322	-0.11
5.710	-0.9874	-0.0323	-0.11
5.715	-0.9868	-0.0324	-0.11
5.720	-0.9866	-0.0325	-0.11
5.725	-0.9866	-0.0326	-0.11
5.730	-0.9865	-0.0320	-0.11
5.735	-0.9864	-0.0327	-0.11
5.740	-0.9859	-0.0319	-0.12
5.745	-0.9859	-0.0321	-0.12
5.750	-0.9850	-0.0314	-0.13
5.755	-0.9844	-0.0313	-0.13
5.760	-0.9840	-0.0311	-0.14
5.765	-0.9834	-0.0307	-0.14
5.770	-0.9825	-0.0305	-0.15
5.775	-0.9821	-0.0299	-0.15
5.780	-0.9816	-0.0295	-0.16
5.785	-0.9807	-0.0286	-0.17
5.790	-0.9799	-0.0287	-0.17
5.795	-0.9791	-0.0280	-0.18
5.800	-0.9782	-0.0275	-0.19
5.805	-0.9773	-0.0271	-0.20
5.810	-0.9767	-0.0265	-0.20
5.815	-0.9760	-0.0257	-0.21
5.820	-0.9749	-0.0255	-0.22
5.825	-0.9738	-0.0249	-0.23
5.830	-0.9729	-0.0242	-0.24
5.835	-0.9720	-0.0237	-0.24
5.840	-0.9713	-0.0233	-0.25
5.845	-0.9707	-0.0227	-0.26
5.850	-0.9696	-0.0221	-0.27
5.855	-0.9687	-0.0217	-0.27

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Frequency (GHz)	**Re**(S_{11})	**Im**(S_{11})	$\lvert S_{11}\rvert$ **(dB)**
5.860	-0.9681	-0.0212	-0.28
5.865	-0.9672	-0.0207	-0.29
5.870	-0.9667	-0.0202	-0.29
5.875	-0.9658	-0.0198	-0.30
5.880	-0.9653	-0.0195	-0.30
5.885	-0.9647	-0.0191	-0.31
5.890	-0.9641	-0.0188	-0.32
5.895	-0.9632	-0.0188	-0.32
5.900	-0.9628	-0.0186	-0.33
5.905	-0.9624	-0.0186	-0.33
5.910	-0.9620	-0.0185	-0.34
5.915	-0.9614	-0.0185	-0.34
5.920	-0.9605	-0.0185	-0.35
5.925	-0.9604	-0.0183	-0.35
5.930	-0.9595	-0.0188	-0.36
5.935	-0.9589	-0.0187	-0.36
5.940	-0.9588	-0.0188	-0.36
5.945	-0.9585	-0.0192	-0.37
5.950	-0.9583	-0.0195	-0.37
5.955	-0.9577	-0.0196	-0.37
5.960	-0.9579	-0.0197	-0.37
5.965	-0.9578	-0.0201	-0.37
5.970	-0.9571	-0.0209	-0.38
5.975	-0.9577	-0.0213	-0.37
5.980	-0.9580	-0.0225	-0.37
5.985	-0.9579	-0.0230	-0.37
5.990	-0.9579	-0.0243	-0.37
5.995	-0.9581	-0.0253	-0.37
6.000	-0.9578	-0.0261	-0.37

3.3 Reference 1 (Distilled Water)



Frequency (GHz)	**Re(S_{11})**	**Im(S_{11})**	**$\lvert S_{11}\rvert$ (dB)**
0.001	1.0000	-0.0011	0.00
0.002	1.0001	-0.0019	0.00
0.003	1.0000	-0.0032	0.00
0.004	0.9999	-0.0040	-0.00
0.005	1.0000	-0.0051	0.00
0.006	0.9999	-0.0060	-0.00
0.007	0.9998	-0.0070	-0.00
0.008	0.9999	-0.0082	-0.00
0.009	0.9999	-0.0090	-0.00
0.010	0.9998	-0.0101	-0.00
0.011	1.0000	-0.0110	0.00
0.012	0.9998	-0.0120	-0.00
0.013	0.9998	-0.0129	-0.00
0.014	0.9999	-0.0141	-0.00
0.015	0.9997	-0.0150	-0.00
0.016	0.9998	-0.0160	-0.00
0.017	0.9997	-0.0170	-0.00
0.018	0.9997	-0.0182	-0.00
0.019	0.9997	-0.0191	-0.00
0.020	0.9997	-0.0201	-0.00
0.021	0.9996	-0.0210	-0.00
0.022	0.9996	-0.0221	-0.00
0.023	0.9998	-0.0232	0.00
0.024	0.9995	-0.0242	-0.00
0.025	0.9995	-0.0252	-0.00
0.026	0.9995	-0.0260	-0.00

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Frequency (GHz)	**Re(S_{11})**	**Im(S_{11})**	**$\lvert S_{11}\rvert$ (dB)**
0.027	0.9995	-0.0270	-0.00
0.028	0.9994	-0.0281	-0.00
0.029	0.9994	-0.0292	-0.00
0.030	0.9995	-0.0301	-0.00
0.031	0.9994	-0.0310	-0.00
0.032	0.9994	-0.0320	-0.00
0.033	0.9993	-0.0330	-0.00
0.034	0.9992	-0.0340	-0.00
0.035	0.9992	-0.0349	-0.00
0.036	0.9993	-0.0360	-0.00
0.037	0.9992	-0.0370	-0.00
0.038	0.9990	-0.0380	-0.00
0.039	0.9989	-0.0390	-0.00
0.040	0.9991	-0.0401	-0.00
0.041	0.9989	-0.0409	-0.00
0.042	0.9989	-0.0419	-0.00
0.043	0.9989	-0.0428	-0.00
0.044	0.9988	-0.0440	-0.00
0.045	0.9987	-0.0450	-0.00
0.046	0.9986	-0.0461	-0.00
0.047	0.9986	-0.0471	-0.00
0.048	0.9986	-0.0480	-0.00
0.049	0.9985	-0.0491	-0.00
0.050	0.9985	-0.0500	-0.00
0.051	0.9985	-0.0510	-0.00
0.052	0.9985	-0.0520	-0.00
0.053	0.9983	-0.0530	-0.00
0.054	0.9982	-0.0540	-0.00
0.055	0.9981	-0.0550	-0.00
0.056	0.9982	-0.0560	-0.00
0.057	0.9980	-0.0569	-0.00
0.058	0.9981	-0.0580	-0.00
0.059	0.9979	-0.0591	-0.00
0.060	0.9979	-0.0600	-0.00
0.061	0.9979	-0.0611	-0.00
0.062	0.9977	-0.0620	-0.00
0.063	0.9977	-0.0630	-0.00
0.064	0.9976	-0.0639	-0.00
0.065	0.9976	-0.0650	-0.00
0.066	0.9975	-0.0660	-0.00
0.067	0.9973	-0.0670	-0.00
0.068	0.9974	-0.0679	-0.00
0.069	0.9974	-0.0690	-0.00
0.070	0.9972	-0.0699	-0.00
0.071	0.9971	-0.0709	-0.00
0.072	0.9970	-0.0718	-0.00

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Frequency (GHz)	**Re(S_{11})**	**Im(S_{11})**	**$\lvert S_{11}\rvert$ (dB)**
0.073	0.9968	-0.0729	-0.00
0.074	0.9969	-0.0739	-0.00
0.075	0.9967	-0.0750	-0.00
0.076	0.9967	-0.0758	-0.00
0.077	0.9965	-0.0769	-0.00
0.078	0.9965	-0.0779	-0.00
0.079	0.9963	-0.0788	-0.01
0.080	0.9963	-0.0798	-0.00
0.081	0.9962	-0.0809	-0.00
0.082	0.9961	-0.0818	-0.00
0.083	0.9961	-0.0829	-0.00
0.084	0.9959	-0.0838	-0.00
0.085	0.9958	-0.0848	-0.01
0.086	0.9957	-0.0858	-0.01
0.087	0.9956	-0.0868	-0.01
0.088	0.9955	-0.0877	-0.01
0.089	0.9955	-0.0887	-0.00
0.090	0.9952	-0.0897	-0.01
0.091	0.9952	-0.0907	-0.01
0.092	0.9949	-0.0917	-0.01
0.093	0.9949	-0.0928	-0.01
0.094	0.9948	-0.0938	-0.01
0.095	0.9946	-0.0947	-0.01
0.096	0.9946	-0.0957	-0.01
0.097	0.9945	-0.0967	-0.01
0.098	0.9944	-0.0977	-0.01
0.099	0.9943	-0.0986	-0.01
0.100	0.9942	-0.0996	-0.01
0.105	0.9937	-0.1046	-0.01
0.110	0.9931	-0.1094	-0.01
0.115	0.9925	-0.1144	-0.01
0.120	0.9917	-0.1193	-0.01
0.125	0.9910	-0.1242	-0.01
0.130	0.9902	-0.1292	-0.01
0.135	0.9895	-0.1341	-0.01
0.140	0.9888	-0.1389	-0.01
0.145	0.9879	-0.1440	-0.01
0.150	0.9871	-0.1487	-0.02
0.155	0.9862	-0.1536	-0.02
0.160	0.9855	-0.1583	-0.02
0.165	0.9847	-0.1633	-0.02
0.170	0.9837	-0.1681	-0.02
0.175	0.9829	-0.1730	-0.02
0.180	0.9818	-0.1779	-0.02
0.185	0.9808	-0.1827	-0.02
0.190	0.9797	-0.1876	-0.02

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Frequency (GHz)	**Re(S_{11})**	**Im(S_{11})**	**$\lvert S_{11}\rvert$ (dB)**
0.195	0.9786	-0.1924	-0.02
0.200	0.9776	-0.1973	-0.02
0.205	0.9763	-0.2020	-0.03
0.210	0.9753	-0.2070	-0.03
0.215	0.9740	-0.2117	-0.03
0.220	0.9730	-0.2166	-0.03
0.225	0.9715	-0.2213	-0.03
0.230	0.9703	-0.2261	-0.03
0.235	0.9689	-0.2308	-0.03
0.240	0.9677	-0.2356	-0.04
0.245	0.9663	-0.2405	-0.04
0.250	0.9650	-0.2453	-0.04
0.255	0.9634	-0.2499	-0.04
0.260	0.9621	-0.2546	-0.04
0.265	0.9606	-0.2594	-0.04
0.270	0.9591	-0.2641	-0.05
0.275	0.9576	-0.2688	-0.05
0.280	0.9561	-0.2735	-0.05
0.285	0.9544	-0.2781	-0.05
0.290	0.9529	-0.2828	-0.05
0.295	0.9513	-0.2874	-0.05
0.300	0.9496	-0.2920	-0.06
0.305	0.9480	-0.2968	-0.06
0.310	0.9463	-0.3013	-0.06
0.315	0.9445	-0.3058	-0.06
0.320	0.9427	-0.3104	-0.07
0.325	0.9410	-0.3149	-0.07
0.330	0.9393	-0.3194	-0.07
0.335	0.9374	-0.3239	-0.07
0.340	0.9355	-0.3286	-0.07
0.345	0.9337	-0.3328	-0.08
0.350	0.9319	-0.3373	-0.08
0.355	0.9299	-0.3418	-0.08
0.360	0.9280	-0.3461	-0.08
0.365	0.9260	-0.3507	-0.09
0.370	0.9240	-0.3551	-0.09
0.375	0.9221	-0.3595	-0.09
0.380	0.9200	-0.3638	-0.09
0.385	0.9179	-0.3681	-0.10
0.390	0.9160	-0.3725	-0.10
0.395	0.9139	-0.3767	-0.10
0.400	0.9121	-0.3810	-0.10
0.405	0.9100	-0.3853	-0.10
0.410	0.9079	-0.3895	-0.11
0.415	0.9058	-0.3936	-0.11
0.420	0.9037	-0.3978	-0.11

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Frequency (GHz)	**Re(S_{11})**	**Im(S_{11})**	**$\lvert S_{11}\rvert$ (dB)**
0.425	0.9018	-0.4019	-0.11
0.430	0.8996	-0.4060	-0.11
0.435	0.8974	-0.4102	-0.12
0.440	0.8953	-0.4142	-0.12
0.445	0.8932	-0.4183	-0.12
0.450	0.8912	-0.4225	-0.12
0.455	0.8888	-0.4265	-0.12
0.460	0.8866	-0.4304	-0.13
0.465	0.8843	-0.4344	-0.13
0.470	0.8821	-0.4384	-0.13
0.475	0.8798	-0.4422	-0.13
0.480	0.8776	-0.4463	-0.14
0.485	0.8754	-0.4501	-0.14
0.490	0.8732	-0.4541	-0.14
0.495	0.8710	-0.4579	-0.14
0.500	0.8686	-0.4618	-0.14
0.505	0.8664	-0.4656	-0.14
0.510	0.8641	-0.4694	-0.15
0.515	0.8618	-0.4732	-0.15
0.520	0.8596	-0.4770	-0.15
0.525	0.8574	-0.4809	-0.15
0.530	0.8552	-0.4844	-0.15
0.535	0.8529	-0.4881	-0.15
0.540	0.8506	-0.4919	-0.15
0.545	0.8482	-0.4956	-0.15
0.550	0.8457	-0.4993	-0.16
0.555	0.8434	-0.5030	-0.16
0.560	0.8409	-0.5067	-0.16
0.565	0.8386	-0.5103	-0.16
0.570	0.8361	-0.5139	-0.16
0.575	0.8337	-0.5176	-0.16
0.580	0.8313	-0.5212	-0.17
0.585	0.8288	-0.5246	-0.17
0.590	0.8264	-0.5282	-0.17
0.595	0.8241	-0.5317	-0.17
0.600	0.8216	-0.5352	-0.17
0.605	0.8190	-0.5387	-0.17
0.610	0.8164	-0.5422	-0.18
0.615	0.8138	-0.5457	-0.18
0.620	0.8113	-0.5492	-0.18
0.625	0.8086	-0.5526	-0.18
0.630	0.8059	-0.5561	-0.18
0.635	0.8033	-0.5595	-0.18
0.640	0.8006	-0.5628	-0.19
0.645	0.7980	-0.5662	-0.19
0.650	0.7952	-0.5694	-0.19

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Frequency (GHz)	**Re(S_{11})**	**Im(S_{11})**	**$\lvert S_{11}\rvert$ (dB)**
0.655	0.7922	-0.5726	-0.20
0.660	0.7893	-0.5758	-0.20
0.665	0.7864	-0.5789	-0.21
0.670	0.7832	-0.5819	-0.21
0.675	0.7802	-0.5847	-0.22
0.680	0.7773	-0.5872	-0.23
0.685	0.7742	-0.5895	-0.24
0.690	0.7716	-0.5917	-0.24
0.695	0.7690	-0.5937	-0.25
0.700	0.7669	-0.5961	-0.25
0.705	0.7648	-0.5986	-0.25
0.710	0.7630	-0.6012	-0.25
0.715	0.7612	-0.6043	-0.25
0.720	0.7593	-0.6073	-0.24
0.725	0.7571	-0.6107	-0.24
0.730	0.7549	-0.6142	-0.24
0.735	0.7525	-0.6175	-0.23
0.740	0.7500	-0.6210	-0.23
0.745	0.7474	-0.6244	-0.23
0.750	0.7447	-0.6278	-0.23
0.755	0.7420	-0.6310	-0.23
0.760	0.7390	-0.6341	-0.23
0.765	0.7361	-0.6376	-0.23
0.770	0.7331	-0.6409	-0.23
0.775	0.7298	-0.6440	-0.23
0.780	0.7268	-0.6473	-0.24
0.785	0.7235	-0.6506	-0.24
0.790	0.7203	-0.6536	-0.24
0.795	0.7170	-0.6567	-0.24
0.800	0.7138	-0.6596	-0.25
0.805	0.7104	-0.6626	-0.25
0.810	0.7070	-0.6656	-0.26
0.815	0.7036	-0.6685	-0.26
0.820	0.7000	-0.6715	-0.26
0.825	0.6966	-0.6743	-0.27
0.830	0.6929	-0.6772	-0.27
0.835	0.6894	-0.6800	-0.28
0.840	0.6856	-0.6827	-0.29
0.845	0.6821	-0.6854	-0.29
0.850	0.6782	-0.6881	-0.30
0.855	0.6742	-0.6907	-0.31
0.860	0.6702	-0.6932	-0.32
0.865	0.6662	-0.6956	-0.33
0.870	0.6621	-0.6978	-0.34
0.875	0.6579	-0.6999	-0.35
0.880	0.6537	-0.7017	-0.36

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Frequency (GHz)	**Re(S_{11})**	**Im(S_{11})**	**$\lvert S_{11}\rvert$ (dB)**
0.885	0.6496	-0.7031	-0.38
0.890	0.6459	-0.7045	-0.39
0.895	0.6423	-0.7056	-0.41
0.900	0.6390	-0.7069	-0.42
0.905	0.6362	-0.7084	-0.43
0.910	0.6332	-0.7101	-0.43
0.915	0.6302	-0.7121	-0.44
0.920	0.6270	-0.7140	-0.44
0.925	0.6238	-0.7157	-0.45
0.930	0.6208	-0.7176	-0.46
0.935	0.6177	-0.7195	-0.46
0.940	0.6146	-0.7217	-0.46
0.945	0.6113	-0.7239	-0.47
0.950	0.6077	-0.7263	-0.47
0.955	0.6039	-0.7283	-0.48
0.960	0.6001	-0.7300	-0.49
0.965	0.5964	-0.7315	-0.50
0.970	0.5926	-0.7330	-0.51
0.975	0.5889	-0.7340	-0.53
0.980	0.5855	-0.7348	-0.54
0.985	0.5823	-0.7353	-0.56
0.990	0.5798	-0.7359	-0.57
0.995	0.5771	-0.7364	-0.58
1.000	0.5753	-0.7372	-0.58
1.005	0.5737	-0.7384	-0.58
1.010	0.5719	-0.7398	-0.58
1.015	0.5702	-0.7416	-0.58
1.020	0.5686	-0.7437	-0.57
1.025	0.5668	-0.7461	-0.57
1.030	0.5648	-0.7488	-0.56
1.035	0.5625	-0.7515	-0.55
1.040	0.5602	-0.7544	-0.54
1.045	0.5575	-0.7572	-0.53
1.050	0.5547	-0.7599	-0.53
1.055	0.5518	-0.7626	-0.53
1.060	0.5488	-0.7651	-0.52
1.065	0.5454	-0.7675	-0.52
1.070	0.5422	-0.7698	-0.52
1.075	0.5390	-0.7720	-0.52
1.080	0.5358	-0.7740	-0.53
1.085	0.5327	-0.7759	-0.53
1.090	0.5297	-0.7777	-0.53
1.095	0.5268	-0.7798	-0.53
1.100	0.5237	-0.7817	-0.53
1.105	0.5209	-0.7838	-0.53
1.110	0.5179	-0.7858	-0.53

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Frequency (GHz)	**Re(S_{11})**	**Im(S_{11})**	**$\lvert S_{11}\rvert$ (dB)**
1.115	0.5149	-0.7879	-0.53
1.120	0.5119	-0.7901	-0.52
1.125	0.5087	-0.7921	-0.53
1.130	0.5054	-0.7941	-0.53
1.135	0.5023	-0.7958	-0.53
1.140	0.4992	-0.7978	-0.53
1.145	0.4959	-0.7997	-0.53
1.150	0.4925	-0.8015	-0.53
1.155	0.4893	-0.8032	-0.53
1.160	0.4861	-0.8047	-0.54
1.165	0.4828	-0.8062	-0.54
1.170	0.4794	-0.8076	-0.55
1.175	0.4764	-0.8089	-0.55
1.180	0.4732	-0.8102	-0.55
1.185	0.4702	-0.8114	-0.56
1.190	0.4673	-0.8123	-0.56
1.195	0.4645	-0.8132	-0.57
1.200	0.4618	-0.8141	-0.57
1.205	0.4590	-0.8149	-0.58
1.210	0.4563	-0.8157	-0.59
1.215	0.4539	-0.8165	-0.59
1.220	0.4515	-0.8172	-0.60
1.225	0.4492	-0.8183	-0.60
1.230	0.4471	-0.8194	-0.60
1.235	0.4448	-0.8204	-0.60
1.240	0.4428	-0.8217	-0.60
1.245	0.4407	-0.8230	-0.60
1.250	0.4385	-0.8246	-0.59
1.255	0.4362	-0.8261	-0.59
1.260	0.4339	-0.8275	-0.59
1.265	0.4316	-0.8290	-0.59
1.270	0.4292	-0.8306	-0.58
1.275	0.4265	-0.8321	-0.58
1.280	0.4241	-0.8337	-0.58
1.285	0.4212	-0.8350	-0.58
1.290	0.4184	-0.8367	-0.58
1.295	0.4156	-0.8381	-0.58
1.300	0.4129	-0.8394	-0.58
1.305	0.4102	-0.8408	-0.58
1.310	0.4076	-0.8420	-0.58
1.315	0.4048	-0.8433	-0.58
1.320	0.4020	-0.8445	-0.58
1.325	0.3993	-0.8458	-0.58
1.330	0.3966	-0.8468	-0.58
1.335	0.3937	-0.8480	-0.58
1.340	0.3909	-0.8491	-0.59

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Frequency (GHz)	**Re(S_{11})**	**Im(S_{11})**	**$\lvert S_{11}\rvert$ (dB)**
1.345	0.3880	-0.8500	-0.59
1.350	0.3853	-0.8512	-0.59
1.355	0.3823	-0.8521	-0.59
1.360	0.3794	-0.8531	-0.60
1.365	0.3764	-0.8541	-0.60
1.370	0.3736	-0.8551	-0.60
1.375	0.3707	-0.8559	-0.60
1.380	0.3677	-0.8567	-0.61
1.385	0.3648	-0.8577	-0.61
1.390	0.3618	-0.8586	-0.61
1.395	0.3587	-0.8594	-0.62
1.400	0.3557	-0.8603	-0.62
1.405	0.3525	-0.8611	-0.63
1.410	0.3494	-0.8619	-0.63
1.415	0.3463	-0.8627	-0.63
1.420	0.3431	-0.8635	-0.64
1.425	0.3399	-0.8644	-0.64
1.430	0.3368	-0.8652	-0.64
1.435	0.3335	-0.8661	-0.65
1.440	0.3304	-0.8669	-0.65
1.445	0.3273	-0.8677	-0.65
1.450	0.3240	-0.8684	-0.66
1.455	0.3208	-0.8694	-0.66
1.460	0.3175	-0.8701	-0.67
1.465	0.3143	-0.8707	-0.67
1.470	0.3111	-0.8716	-0.67
1.475	0.3077	-0.8723	-0.68
1.480	0.3044	-0.8731	-0.68
1.485	0.3008	-0.8738	-0.69
1.490	0.2975	-0.8747	-0.69
1.495	0.2940	-0.8754	-0.69
1.500	0.2908	-0.8761	-0.70
1.505	0.2872	-0.8766	-0.70
1.510	0.2836	-0.8773	-0.71
1.515	0.2801	-0.8780	-0.71
1.520	0.2767	-0.8787	-0.71
1.525	0.2731	-0.8795	-0.72
1.530	0.2695	-0.8801	-0.72
1.535	0.2658	-0.8808	-0.72
1.540	0.2621	-0.8813	-0.73
1.545	0.2584	-0.8817	-0.74
1.550	0.2549	-0.8823	-0.74
1.555	0.2509	-0.8827	-0.75
1.560	0.2472	-0.8832	-0.75
1.565	0.2434	-0.8838	-0.76
1.570	0.2396	-0.8844	-0.76

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Frequency (GHz)	**Re(S_{11})**	**Im(S_{11})**	**$\lvert S_{11}\rvert$ (dB)**
1.575	0.2358	-0.8848	-0.77
1.580	0.2323	-0.8853	-0.77
1.585	0.2284	-0.8861	-0.77
1.590	0.2247	-0.8865	-0.78
1.595	0.2209	-0.8870	-0.78
1.600	0.2173	-0.8874	-0.78
1.605	0.2136	-0.8877	-0.79
1.610	0.2099	-0.8879	-0.80
1.615	0.2061	-0.8883	-0.80
1.620	0.2023	-0.8887	-0.81
1.625	0.1985	-0.8890	-0.81
1.630	0.1947	-0.8892	-0.82
1.635	0.1909	-0.8894	-0.82
1.640	0.1869	-0.8893	-0.83
1.645	0.1832	-0.8893	-0.84
1.650	0.1793	-0.8895	-0.84
1.655	0.1756	-0.8898	-0.85
1.660	0.1720	-0.8896	-0.86
1.665	0.1684	-0.8896	-0.86
1.670	0.1648	-0.8898	-0.87
1.675	0.1612	-0.8896	-0.88
1.680	0.1577	-0.8896	-0.88
1.685	0.1541	-0.8893	-0.89
1.690	0.1507	-0.8891	-0.90
1.695	0.1476	-0.8890	-0.90
1.700	0.1441	-0.8887	-0.91
1.705	0.1408	-0.8886	-0.92
1.710	0.1374	-0.8883	-0.93
1.715	0.1340	-0.8880	-0.93
1.720	0.1310	-0.8878	-0.94
1.725	0.1276	-0.8876	-0.95
1.730	0.1247	-0.8875	-0.95
1.735	0.1215	-0.8870	-0.96
1.740	0.1185	-0.8868	-0.97
1.745	0.1157	-0.8867	-0.97
1.750	0.1128	-0.8866	-0.98
1.755	0.1101	-0.8865	-0.98
1.760	0.1074	-0.8861	-0.99
1.765	0.1048	-0.8860	-0.99
1.770	0.1023	-0.8860	-0.99
1.775	0.0999	-0.8858	-1.00
1.780	0.0976	-0.8855	-1.00
1.785	0.0950	-0.8850	-1.01
1.790	0.0923	-0.8846	-1.02
1.795	0.0899	-0.8843	-1.02
1.800	0.0875	-0.8841	-1.03

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Frequency (GHz)	**Re(S_{11})**	**Im(S_{11})**	**$\lvert S_{11}\rvert$ (dB)**
1.805	0.0852	-0.8839	-1.03
1.810	0.0829	-0.8838	-1.03
1.815	0.0808	-0.8836	-1.04
1.820	0.0785	-0.8834	-1.04
1.825	0.0766	-0.8831	-1.05
1.830	0.0747	-0.8830	-1.05
1.835	0.0725	-0.8826	-1.06
1.840	0.0706	-0.8824	-1.06
1.845	0.0688	-0.8822	-1.06
1.850	0.0667	-0.8815	-1.07
1.855	0.0648	-0.8815	-1.07
1.860	0.0628	-0.8809	-1.08
1.865	0.0608	-0.8811	-1.08
1.870	0.0590	-0.8809	-1.08
1.875	0.0569	-0.8809	-1.08
1.880	0.0551	-0.8809	-1.08
1.885	0.0535	-0.8812	-1.08
1.890	0.0522	-0.8810	-1.09
1.895	0.0504	-0.8807	-1.09
1.900	0.0486	-0.8804	-1.09
1.905	0.0470	-0.8803	-1.09
1.910	0.0452	-0.8804	-1.09
1.915	0.0434	-0.8804	-1.10
1.920	0.0419	-0.8805	-1.10
1.925	0.0403	-0.8804	-1.10
1.930	0.0389	-0.8805	-1.10
1.935	0.0374	-0.8804	-1.10
1.940	0.0359	-0.8804	-1.10
1.945	0.0345	-0.8804	-1.10
1.950	0.0330	-0.8803	-1.10
1.955	0.0316	-0.8803	-1.10
1.960	0.0302	-0.8802	-1.10
1.965	0.0288	-0.8803	-1.10
1.970	0.0274	-0.8801	-1.10
1.975	0.0262	-0.8803	-1.10
1.980	0.0248	-0.8804	-1.10
1.985	0.0235	-0.8802	-1.10
1.990	0.0220	-0.8804	-1.10
1.995	0.0206	-0.8805	-1.10
2.000	0.0190	-0.8805	-1.10
2.005	0.0175	-0.8807	-1.10
2.010	0.0159	-0.8808	-1.10
2.015	0.0145	-0.8809	-1.10
2.020	0.0130	-0.8810	-1.10
2.025	0.0112	-0.8813	-1.10
2.030	0.0095	-0.8815	-1.09

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Frequency (GHz)	**Re(S_{11})**	**Im(S_{11})**	**$\lvert S_{11}\rvert$ (dB)**
2.035	0.0079	-0.8815	-1.10
2.040	0.0061	-0.8816	-1.09
2.045	0.0044	-0.8817	-1.09
2.050	0.0026	-0.8817	-1.09
2.055	0.0007	-0.8820	-1.09
2.060	-0.0012	-0.8819	-1.09
2.065	-0.0031	-0.8821	-1.09
2.070	-0.0052	-0.8821	-1.09
2.075	-0.0073	-0.8821	-1.09
2.080	-0.0095	-0.8822	-1.09
2.085	-0.0117	-0.8821	-1.09
2.090	-0.0140	-0.8821	-1.09
2.095	-0.0163	-0.8821	-1.09
2.100	-0.0188	-0.8820	-1.09
2.105	-0.0212	-0.8820	-1.09
2.110	-0.0237	-0.8820	-1.09
2.115	-0.0261	-0.8819	-1.09
2.120	-0.0286	-0.8818	-1.09
2.125	-0.0312	-0.8818	-1.09
2.130	-0.0341	-0.8816	-1.09
2.135	-0.0366	-0.8814	-1.09
2.140	-0.0393	-0.8815	-1.09
2.145	-0.0419	-0.8814	-1.09
2.150	-0.0445	-0.8812	-1.09
2.155	-0.0473	-0.8809	-1.09
2.160	-0.0501	-0.8806	-1.09
2.165	-0.0529	-0.8804	-1.09
2.170	-0.0558	-0.8800	-1.09
2.175	-0.0585	-0.8798	-1.09
2.180	-0.0613	-0.8795	-1.09
2.185	-0.0641	-0.8791	-1.10
2.190	-0.0671	-0.8790	-1.10
2.195	-0.0700	-0.8785	-1.10
2.200	-0.0729	-0.8783	-1.10
2.205	-0.0759	-0.8778	-1.10
2.210	-0.0788	-0.8775	-1.10
2.215	-0.0817	-0.8770	-1.10
2.220	-0.0846	-0.8766	-1.10
2.225	-0.0877	-0.8761	-1.11
2.230	-0.0906	-0.8756	-1.11
2.235	-0.0936	-0.8752	-1.11
2.240	-0.0967	-0.8750	-1.11
2.245	-0.0995	-0.8745	-1.11
2.250	-0.1025	-0.8742	-1.11
2.255	-0.1056	-0.8737	-1.11
2.260	-0.1083	-0.8733	-1.11

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Frequency (GHz)	**Re(S_{11})**	**Im(S_{11})**	**$\lvert S_{11}\rvert$ (dB)**
2.265	-0.1114	-0.8728	-1.11
2.270	-0.1141	-0.8723	-1.11
2.275	-0.1170	-0.8716	-1.12
2.280	-0.1199	-0.8711	-1.12
2.285	-0.1228	-0.8707	-1.12
2.290	-0.1255	-0.8700	-1.12
2.295	-0.1283	-0.8695	-1.12
2.300	-0.1312	-0.8689	-1.12
2.305	-0.1339	-0.8684	-1.12
2.310	-0.1369	-0.8677	-1.13
2.315	-0.1394	-0.8669	-1.13
2.320	-0.1423	-0.8662	-1.13
2.325	-0.1449	-0.8656	-1.13
2.330	-0.1474	-0.8649	-1.14
2.335	-0.1503	-0.8643	-1.14
2.340	-0.1526	-0.8635	-1.14
2.345	-0.1553	-0.8627	-1.14
2.350	-0.1577	-0.8620	-1.15
2.355	-0.1601	-0.8612	-1.15
2.360	-0.1625	-0.8603	-1.15
2.365	-0.1646	-0.8598	-1.16
2.370	-0.1667	-0.8589	-1.16
2.375	-0.1690	-0.8584	-1.16
2.380	-0.1711	-0.8576	-1.16
2.385	-0.1732	-0.8570	-1.17
2.390	-0.1754	-0.8564	-1.17
2.395	-0.1773	-0.8557	-1.17
2.400	-0.1792	-0.8551	-1.17
2.405	-0.1813	-0.8544	-1.18
2.410	-0.1831	-0.8536	-1.18
2.415	-0.1849	-0.8531	-1.18
2.420	-0.1866	-0.8523	-1.18
2.425	-0.1882	-0.8517	-1.19
2.430	-0.1899	-0.8508	-1.19
2.435	-0.1914	-0.8503	-1.19
2.440	-0.1929	-0.8498	-1.20
2.445	-0.1945	-0.8490	-1.20
2.450	-0.1958	-0.8483	-1.20
2.455	-0.1969	-0.8476	-1.21
2.460	-0.1982	-0.8471	-1.21
2.465	-0.1995	-0.8464	-1.21
2.470	-0.2007	-0.8459	-1.22
2.475	-0.2018	-0.8453	-1.22
2.480	-0.2033	-0.8448	-1.22
2.485	-0.2044	-0.8443	-1.22
2.490	-0.2055	-0.8436	-1.23

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Frequency (GHz)	**Re(S_{11})**	**Im(S_{11})**	**$\lvert S_{11}\rvert$ (dB)**
2.495	-0.2066	-0.8434	-1.23
2.500	-0.2080	-0.8429	-1.23
2.505	-0.2090	-0.8426	-1.23
2.510	-0.2101	-0.8422	-1.23
2.515	-0.2111	-0.8417	-1.23
2.520	-0.2123	-0.8415	-1.23
2.525	-0.2133	-0.8411	-1.23
2.530	-0.2144	-0.8408	-1.23
2.535	-0.2155	-0.8405	-1.23
2.540	-0.2166	-0.8404	-1.23
2.545	-0.2174	-0.8400	-1.23
2.550	-0.2183	-0.8396	-1.23
2.555	-0.2192	-0.8392	-1.24
2.560	-0.2200	-0.8386	-1.24
2.565	-0.2204	-0.8382	-1.24
2.570	-0.2209	-0.8375	-1.25
2.575	-0.2213	-0.8371	-1.25
2.580	-0.2216	-0.8365	-1.26
2.585	-0.2221	-0.8364	-1.26
2.590	-0.2225	-0.8361	-1.26
2.595	-0.2229	-0.8361	-1.26
2.600	-0.2234	-0.8358	-1.26
2.605	-0.2241	-0.8356	-1.26
2.610	-0.2246	-0.8354	-1.26
2.615	-0.2253	-0.8353	-1.26
2.620	-0.2258	-0.8350	-1.26
2.625	-0.2263	-0.8346	-1.26
2.630	-0.2268	-0.8342	-1.26
2.635	-0.2273	-0.8339	-1.27
2.640	-0.2280	-0.8335	-1.27
2.645	-0.2285	-0.8331	-1.27
2.650	-0.2291	-0.8328	-1.27
2.655	-0.2297	-0.8323	-1.28
2.660	-0.2303	-0.8317	-1.28
2.665	-0.2309	-0.8314	-1.28
2.670	-0.2317	-0.8307	-1.29
2.675	-0.2321	-0.8305	-1.29
2.680	-0.2327	-0.8302	-1.29
2.685	-0.2330	-0.8300	-1.29
2.690	-0.2336	-0.8298	-1.29
2.695	-0.2341	-0.8296	-1.29
2.700	-0.2344	-0.8294	-1.29
2.705	-0.2349	-0.8292	-1.29
2.710	-0.2350	-0.8290	-1.29
2.715	-0.2357	-0.8284	-1.30
2.720	-0.2362	-0.8282	-1.30

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Frequency (GHz)	**Re(S_{11})**	**Im(S_{11})**	**$\lvert S_{11}\rvert$ (dB)**
2.725	-0.2369	-0.8276	-1.30
2.730	-0.2374	-0.8268	-1.31
2.735	-0.2383	-0.8261	-1.31
2.740	-0.2391	-0.8253	-1.32
2.745	-0.2402	-0.8247	-1.32
2.750	-0.2409	-0.8239	-1.33
2.755	-0.2419	-0.8230	-1.33
2.760	-0.2432	-0.8221	-1.34
2.765	-0.2444	-0.8213	-1.34
2.770	-0.2456	-0.8205	-1.35
2.775	-0.2467	-0.8196	-1.35
2.780	-0.2481	-0.8189	-1.35
2.785	-0.2493	-0.8181	-1.36
2.790	-0.2508	-0.8173	-1.36
2.795	-0.2522	-0.8168	-1.36
2.800	-0.2533	-0.8161	-1.37
2.805	-0.2550	-0.8155	-1.37
2.810	-0.2563	-0.8148	-1.37
2.815	-0.2576	-0.8143	-1.37
2.820	-0.2590	-0.8136	-1.37
2.825	-0.2603	-0.8127	-1.38
2.830	-0.2617	-0.8123	-1.38
2.835	-0.2630	-0.8116	-1.38
2.840	-0.2644	-0.8112	-1.38
2.845	-0.2657	-0.8103	-1.38
2.850	-0.2670	-0.8096	-1.39
2.855	-0.2682	-0.8089	-1.39
2.860	-0.2694	-0.8082	-1.39
2.865	-0.2705	-0.8074	-1.40
2.870	-0.2715	-0.8066	-1.40
2.875	-0.2724	-0.8057	-1.41
2.880	-0.2733	-0.8046	-1.41
2.885	-0.2745	-0.8035	-1.42
2.890	-0.2758	-0.8022	-1.43
2.895	-0.2773	-0.8009	-1.44
2.900	-0.2786	-0.7996	-1.44
2.905	-0.2803	-0.7985	-1.45
2.910	-0.2817	-0.7975	-1.46
2.915	-0.2836	-0.7966	-1.46
2.920	-0.2853	-0.7958	-1.46
2.925	-0.2870	-0.7952	-1.46
2.930	-0.2887	-0.7944	-1.46
2.935	-0.2900	-0.7940	-1.46
2.940	-0.2914	-0.7937	-1.46
2.945	-0.2929	-0.7934	-1.46
2.950	-0.2940	-0.7931	-1.45

(continued on next page)

Frequency (GHz)	**Re**(S_{11})	**Im**(S_{11})	$\lvert S_{11}\rvert$ **(dB)**
2.955	-0.2953	-0.7929	-1.45
2.960	-0.2960	-0.7928	-1.45
2.965	-0.2971	-0.7926	-1.45
2.970	-0.2978	-0.7925	-1.45
2.975	-0.2985	-0.7925	-1.44
2.980	-0.2992	-0.7925	-1.44
2.985	-0.2998	-0.7925	-1.44
2.990	-0.3000	-0.7921	-1.44
2.995	-0.3003	-0.7916	-1.45
3.000	-0.3008	-0.7915	-1.45
3.005	-0.3044	-0.7903	-1.44
3.010	-0.3045	-0.7902	-1.44
3.015	-0.3047	-0.7900	-1.44
3.020	-0.3049	-0.7899	-1.45
3.025	-0.3046	-0.7900	-1.45
3.030	-0.3048	-0.7895	-1.45
3.035	-0.3048	-0.7890	-1.45
3.040	-0.3047	-0.7885	-1.46
3.045	-0.3047	-0.7874	-1.47
3.050	-0.3048	-0.7868	-1.48
3.055	-0.3049	-0.7859	-1.48
3.060	-0.3048	-0.7849	-1.49
3.065	-0.3054	-0.7837	-1.50
3.070	-0.3057	-0.7827	-1.51
3.075	-0.3065	-0.7819	-1.52
3.080	-0.3070	-0.7808	-1.53
3.085	-0.3079	-0.7797	-1.53
3.090	-0.3092	-0.7787	-1.54
3.095	-0.3105	-0.7783	-1.54
3.100	-0.3117	-0.7779	-1.53
3.105	-0.3131	-0.7779	-1.53
3.110	-0.3143	-0.7777	-1.53
3.115	-0.3156	-0.7774	-1.52
3.120	-0.3168	-0.7774	-1.52
3.125	-0.3180	-0.7778	-1.51
3.130	-0.3188	-0.7780	-1.51
3.135	-0.3196	-0.7788	-1.50
3.140	-0.3202	-0.7797	-1.48
3.145	-0.3210	-0.7800	-1.48
3.150	-0.3213	-0.7808	-1.47
3.155	-0.3217	-0.7814	-1.46
3.160	-0.3218	-0.7817	-1.46
3.165	-0.3219	-0.7820	-1.46
3.170	-0.3220	-0.7822	-1.45
3.175	-0.3220	-0.7825	-1.45
3.180	-0.3214	-0.7829	-1.45

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Frequency (GHz)	**Re(S_{11})**	**Im(S_{11})**	**$\lvert S_{11}\rvert$ (dB)**
3.185	-0.3216	-0.7824	-1.45
3.190	-0.3213	-0.7819	-1.46
3.195	-0.3210	-0.7813	-1.47
3.200	-0.3215	-0.7803	-1.47
3.205	-0.3214	-0.7792	-1.48
3.210	-0.3218	-0.7784	-1.49
3.215	-0.3221	-0.7772	-1.50
3.220	-0.3232	-0.7760	-1.51
3.225	-0.3243	-0.7753	-1.51
3.230	-0.3258	-0.7749	-1.51
3.235	-0.3269	-0.7750	-1.50
3.240	-0.3281	-0.7750	-1.50
3.245	-0.3291	-0.7754	-1.49
3.250	-0.3305	-0.7752	-1.49
3.255	-0.3315	-0.7758	-1.48
3.260	-0.3325	-0.7760	-1.47
3.265	-0.3338	-0.7761	-1.46
3.270	-0.3347	-0.7762	-1.46
3.275	-0.3355	-0.7765	-1.45
3.280	-0.3364	-0.7764	-1.45
3.285	-0.3374	-0.7770	-1.44
3.290	-0.3383	-0.7764	-1.44
3.295	-0.3394	-0.7765	-1.44
3.300	-0.3401	-0.7767	-1.43
3.305	-0.3404	-0.7766	-1.43
3.310	-0.3414	-0.7763	-1.43
3.315	-0.3421	-0.7760	-1.43
3.320	-0.3425	-0.7759	-1.43
3.325	-0.3434	-0.7754	-1.43
3.330	-0.3437	-0.7752	-1.43
3.335	-0.3442	-0.7747	-1.43
3.340	-0.3448	-0.7740	-1.44
3.345	-0.3457	-0.7729	-1.45
3.350	-0.3462	-0.7719	-1.45
3.355	-0.3467	-0.7710	-1.46
3.360	-0.3474	-0.7698	-1.47
3.365	-0.3478	-0.7688	-1.48
3.370	-0.3490	-0.7674	-1.48
3.375	-0.3494	-0.7660	-1.49
3.380	-0.3506	-0.7644	-1.50
3.385	-0.3515	-0.7628	-1.52
3.390	-0.3526	-0.7614	-1.52
3.395	-0.3537	-0.7595	-1.54
3.400	-0.3549	-0.7577	-1.55
3.405	-0.3560	-0.7560	-1.56
3.410	-0.3576	-0.7542	-1.57

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Frequency (GHz)	**Re**(S_{11})	**Im**(S_{11})	$\lvert S_{11}\rvert$ **(dB)**
3.415	-0.3591	-0.7527	-1.58
3.420	-0.3607	-0.7515	-1.58
3.425	-0.3621	-0.7497	-1.59
3.430	-0.3637	-0.7481	-1.60
3.435	-0.3654	-0.7468	-1.60
3.440	-0.3671	-0.7452	-1.61
3.445	-0.3693	-0.7434	-1.62
3.450	-0.3713	-0.7422	-1.62
3.455	-0.3729	-0.7410	-1.62
3.460	-0.3748	-0.7402	-1.62
3.465	-0.3768	-0.7396	-1.62
3.470	-0.3784	-0.7383	-1.62
3.475	-0.3801	-0.7377	-1.62
3.480	-0.3815	-0.7372	-1.62
3.485	-0.3836	-0.7370	-1.61
3.490	-0.3849	-0.7370	-1.60
3.495	-0.3859	-0.7365	-1.60
3.500	-0.3866	-0.7365	-1.60
3.505	-0.3870	-0.7360	-1.60
3.510	-0.3877	-0.7355	-1.60
3.515	-0.3879	-0.7348	-1.61
3.520	-0.3877	-0.7343	-1.61
3.525	-0.3879	-0.7332	-1.62
3.530	-0.3875	-0.7328	-1.63
3.535	-0.3867	-0.7303	-1.66
3.540	-0.3866	-0.7279	-1.68
3.545	-0.3866	-0.7253	-1.70
3.550	-0.3873	-0.7224	-1.73
3.555	-0.3877	-0.7190	-1.76
3.560	-0.3894	-0.7159	-1.78
3.565	-0.3909	-0.7134	-1.79
3.570	-0.3936	-0.7102	-1.81
3.575	-0.3960	-0.7071	-1.83
3.580	-0.4000	-0.7054	-1.82
3.585	-0.4046	-0.7048	-1.80
3.590	-0.4077	-0.7051	-1.78
3.595	-0.4106	-0.7063	-1.76
3.600	-0.4126	-0.7081	-1.73
3.605	-0.4139	-0.7100	-1.70
3.610	-0.4137	-0.7118	-1.69
3.615	-0.4130	-0.7127	-1.68
3.620	-0.4117	-0.7129	-1.69
3.625	-0.4102	-0.7136	-1.69
3.630	-0.4091	-0.7128	-1.70
3.635	-0.4074	-0.7115	-1.73
3.640	-0.4067	-0.7098	-1.74

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Frequency (GHz)	**Re(S_{11})**	**Im(S_{11})**	**$\lvert S_{11}\rvert$ (dB)**
3.645	-0.4064	-0.7082	-1.76
3.650	-0.4063	-0.7063	-1.78
3.655	-0.4066	-0.7043	-1.80
3.660	-0.4070	-0.7029	-1.81
3.665	-0.4075	-0.7013	-1.82
3.670	-0.4080	-0.6994	-1.83
3.675	-0.4083	-0.6971	-1.85
3.680	-0.4090	-0.6956	-1.86
3.685	-0.4095	-0.6939	-1.88
3.690	-0.4106	-0.6917	-1.89
3.695	-0.4123	-0.6893	-1.90
3.700	-0.4140	-0.6870	-1.92
3.705	-0.4166	-0.6848	-1.92
3.710	-0.4195	-0.6827	-1.92
3.715	-0.4231	-0.6814	-1.92
3.720	-0.4272	-0.6800	-1.91
3.725	-0.4314	-0.6805	-1.88
3.730	-0.4369	-0.6803	-1.85
3.735	-0.4404	-0.6828	-1.80
3.740	-0.4438	-0.6855	-1.76
3.745	-0.4451	-0.6885	-1.73
3.750	-0.4457	-0.6924	-1.69
3.755	-0.4450	-0.6956	-1.66
3.760	-0.4432	-0.6982	-1.65
3.765	-0.4403	-0.6999	-1.65
3.770	-0.4369	-0.6998	-1.67
3.775	-0.4339	-0.6980	-1.70
3.780	-0.4311	-0.6954	-1.74
3.785	-0.4307	-0.6915	-1.78
3.790	-0.4317	-0.6874	-1.81
3.795	-0.4343	-0.6844	-1.82
3.800	-0.4370	-0.6820	-1.83
3.805	-0.4394	-0.6804	-1.83
3.810	-0.4425	-0.6794	-1.82
3.815	-0.4452	-0.6787	-1.81
3.820	-0.4479	-0.6784	-1.80
3.825	-0.4503	-0.6782	-1.79
3.830	-0.4526	-0.6788	-1.77
3.835	-0.4534	-0.6816	-1.74
3.840	-0.4549	-0.6811	-1.73
3.845	-0.4559	-0.6808	-1.73
3.850	-0.4572	-0.6803	-1.73
3.855	-0.4583	-0.6795	-1.73
3.860	-0.4599	-0.6791	-1.72
3.865	-0.4613	-0.6782	-1.72
3.870	-0.4627	-0.6775	-1.72

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Frequency (GHz)	**Re(S_{11})**	**Im(S_{11})**	**$\lvert S_{11}\rvert$ (dB)**
3.875	-0.4642	-0.6768	-1.72
3.880	-0.4657	-0.6761	-1.71
3.885	-0.4677	-0.6755	-1.71
3.890	-0.4691	-0.6742	-1.71
3.895	-0.4708	-0.6734	-1.71
3.900	-0.4726	-0.6725	-1.70
3.905	-0.4743	-0.6716	-1.70
3.910	-0.4761	-0.6708	-1.70
3.915	-0.4782	-0.6698	-1.69
3.920	-0.4798	-0.6687	-1.69
3.925	-0.4818	-0.6675	-1.69
3.930	-0.4834	-0.6667	-1.69
3.935	-0.4855	-0.6652	-1.69
3.940	-0.4869	-0.6644	-1.68
3.945	-0.4889	-0.6630	-1.68
3.950	-0.4906	-0.6620	-1.68
3.955	-0.4925	-0.6606	-1.68
3.960	-0.4942	-0.6593	-1.68
3.965	-0.4959	-0.6579	-1.68
3.970	-0.4975	-0.6566	-1.68
3.975	-0.4991	-0.6552	-1.69
3.980	-0.5012	-0.6536	-1.68
3.985	-0.5028	-0.6523	-1.69
3.990	-0.5046	-0.6508	-1.69
3.995	-0.5061	-0.6491	-1.69
4.000	-0.5079	-0.6477	-1.69
4.005	-0.5095	-0.6459	-1.70
4.010	-0.5114	-0.6442	-1.70
4.015	-0.5127	-0.6430	-1.70
4.020	-0.5145	-0.6412	-1.70
4.025	-0.5158	-0.6397	-1.71
4.030	-0.5174	-0.6383	-1.71
4.035	-0.5191	-0.6364	-1.71
4.040	-0.5205	-0.6347	-1.71
4.045	-0.5219	-0.6331	-1.72
4.050	-0.5233	-0.6314	-1.72
4.055	-0.5247	-0.6298	-1.73
4.060	-0.5259	-0.6282	-1.73
4.065	-0.5273	-0.6268	-1.73
4.070	-0.5285	-0.6251	-1.74
4.075	-0.5296	-0.6238	-1.74
4.080	-0.5309	-0.6225	-1.74
4.085	-0.5322	-0.6210	-1.75
4.090	-0.5331	-0.6198	-1.75
4.095	-0.5340	-0.6185	-1.75
4.100	-0.5353	-0.6170	-1.76

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Frequency (GHz)	**Re(S_{11})**	**Im(S_{11})**	**$\lvert S_{11}\rvert$ (dB)**
4.105	-0.5364	-0.6159	-1.76
4.110	-0.5375	-0.6148	-1.76
4.115	-0.5389	-0.6138	-1.76
4.120	-0.5399	-0.6128	-1.76
4.125	-0.5416	-0.6113	-1.76
4.130	-0.5426	-0.6103	-1.76
4.135	-0.5438	-0.6089	-1.76
4.140	-0.5450	-0.6077	-1.76
4.145	-0.5458	-0.6060	-1.77
4.150	-0.5468	-0.6043	-1.78
4.155	-0.5475	-0.6029	-1.78
4.160	-0.5480	-0.6016	-1.79
4.165	-0.5480	-0.6004	-1.80
4.170	-0.5481	-0.5994	-1.81
4.175	-0.5476	-0.5984	-1.82
4.180	-0.5479	-0.5979	-1.82
4.185	-0.5474	-0.5971	-1.83
4.190	-0.5476	-0.5965	-1.83
4.195	-0.5473	-0.5966	-1.83
4.200	-0.5471	-0.5963	-1.84
4.205	-0.5469	-0.5962	-1.84
4.210	-0.5468	-0.5961	-1.84
4.215	-0.5468	-0.5960	-1.84
4.220	-0.5464	-0.5962	-1.84
4.225	-0.5463	-0.5960	-1.85
4.230	-0.5460	-0.5964	-1.85
4.235	-0.5459	-0.5966	-1.84
4.240	-0.5456	-0.5967	-1.85
4.245	-0.5452	-0.5971	-1.85
4.250	-0.5450	-0.5976	-1.84
4.255	-0.5447	-0.5980	-1.84
4.260	-0.5442	-0.5982	-1.84
4.265	-0.5438	-0.5988	-1.84
4.270	-0.5437	-0.5992	-1.84
4.275	-0.5431	-0.5996	-1.84
4.280	-0.5431	-0.6000	-1.84
4.285	-0.5425	-0.6007	-1.84
4.290	-0.5423	-0.6010	-1.84
4.295	-0.5418	-0.6015	-1.83
4.300	-0.5418	-0.6022	-1.83
4.305	-0.5416	-0.6026	-1.83
4.310	-0.5413	-0.6032	-1.83
4.315	-0.5410	-0.6037	-1.82
4.320	-0.5406	-0.6043	-1.82
4.325	-0.5405	-0.6051	-1.82
4.330	-0.5401	-0.6056	-1.81

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Frequency (GHz)	**Re(S_{11})**	**Im(S_{11})**	**$\lvert S_{11}\rvert$ (dB)**
4.335	-0.5400	-0.6064	-1.81
4.340	-0.5399	-0.6072	-1.80
4.345	-0.5396	-0.6082	-1.80
4.350	-0.5395	-0.6089	-1.79
4.355	-0.5390	-0.6098	-1.79
4.360	-0.5386	-0.6107	-1.78
4.365	-0.5381	-0.6117	-1.78
4.370	-0.5378	-0.6124	-1.78
4.375	-0.5370	-0.6135	-1.77
4.380	-0.5367	-0.6143	-1.77
4.385	-0.5362	-0.6152	-1.77
4.390	-0.5353	-0.6161	-1.76
4.395	-0.5347	-0.6168	-1.76
4.400	-0.5343	-0.6175	-1.76
4.405	-0.5337	-0.6180	-1.76
4.410	-0.5330	-0.6186	-1.76
4.415	-0.5322	-0.6190	-1.76
4.420	-0.5321	-0.6192	-1.76
4.425	-0.5315	-0.6193	-1.77
4.430	-0.5311	-0.6200	-1.76
4.435	-0.5308	-0.6199	-1.77
4.440	-0.5307	-0.6200	-1.77
4.445	-0.5304	-0.6206	-1.76
4.450	-0.5304	-0.6205	-1.76
4.455	-0.5306	-0.6214	-1.75
4.460	-0.5297	-0.6218	-1.76
4.465	-0.5295	-0.6227	-1.75
4.470	-0.5292	-0.6239	-1.74
4.475	-0.5289	-0.6248	-1.74
4.480	-0.5285	-0.6263	-1.73
4.485	-0.5285	-0.6275	-1.72
4.490	-0.5279	-0.6293	-1.71
4.495	-0.5279	-0.6306	-1.70
4.500	-0.5274	-0.6315	-1.69
4.505	-0.5270	-0.6326	-1.69
4.510	-0.5266	-0.6327	-1.69
4.515	-0.5259	-0.6331	-1.69
4.520	-0.5255	-0.6337	-1.69
4.525	-0.5246	-0.6335	-1.70
4.530	-0.5248	-0.6328	-1.70
4.535	-0.5244	-0.6322	-1.71
4.540	-0.5246	-0.6318	-1.71
4.545	-0.5245	-0.6307	-1.72
4.550	-0.5249	-0.6304	-1.72
4.555	-0.5251	-0.6297	-1.72
4.560	-0.5255	-0.6293	-1.73

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Frequency (GHz)	**Re(S_{11})**	**Im(S_{11})**	**$\lvert S_{11}\rvert$ (dB)**
4.565	-0.5255	-0.6286	-1.73
4.570	-0.5258	-0.6277	-1.74
4.575	-0.5263	-0.6268	-1.74
4.580	-0.5264	-0.6261	-1.75
4.585	-0.5269	-0.6250	-1.75
4.590	-0.5273	-0.6245	-1.75
4.595	-0.5275	-0.6237	-1.76
4.600	-0.5282	-0.6227	-1.76
4.605	-0.5283	-0.6218	-1.77
4.610	-0.5289	-0.6209	-1.77
4.615	-0.5295	-0.6196	-1.78
4.620	-0.5299	-0.6189	-1.78
4.625	-0.5305	-0.6178	-1.78
4.630	-0.5307	-0.6172	-1.79
4.635	-0.5309	-0.6161	-1.80
4.640	-0.5309	-0.6151	-1.80
4.645	-0.5312	-0.6143	-1.81
4.650	-0.5314	-0.6136	-1.81
4.655	-0.5318	-0.6126	-1.82
4.660	-0.5317	-0.6118	-1.82
4.665	-0.5317	-0.6109	-1.83
4.670	-0.5317	-0.6102	-1.84
4.675	-0.5315	-0.6095	-1.84
4.680	-0.5313	-0.6086	-1.85
4.685	-0.5309	-0.6083	-1.86
4.690	-0.5307	-0.6076	-1.87
4.695	-0.5305	-0.6068	-1.87
4.700	-0.5301	-0.6063	-1.88
4.705	-0.5299	-0.6054	-1.89
4.710	-0.5300	-0.6048	-1.89
4.715	-0.5299	-0.6039	-1.90
4.720	-0.5299	-0.6028	-1.91
4.725	-0.5303	-0.6021	-1.91
4.730	-0.5310	-0.6007	-1.92
4.735	-0.5316	-0.6000	-1.92
4.740	-0.5325	-0.5987	-1.92
4.745	-0.5329	-0.5975	-1.93
4.750	-0.5337	-0.5957	-1.94
4.755	-0.5348	-0.5945	-1.94
4.760	-0.5354	-0.5933	-1.95
4.765	-0.5361	-0.5924	-1.95
4.770	-0.5363	-0.5916	-1.95
4.775	-0.5361	-0.5908	-1.96
4.780	-0.5363	-0.5898	-1.97
4.785	-0.5361	-0.5891	-1.98
4.790	-0.5351	-0.5887	-1.99

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Frequency (GHz)	**Re(S_{11})**	**Im(S_{11})**	**$\lvert S_{11}\rvert$ (dB)**
4.795	-0.5345	-0.5880	-2.00
4.800	-0.5337	-0.5874	-2.01
4.805	-0.5329	-0.5868	-2.02
4.810	-0.5320	-0.5858	-2.03
4.815	-0.5312	-0.5858	-2.04
4.820	-0.5301	-0.5853	-2.05
4.825	-0.5298	-0.5852	-2.05
4.830	-0.5291	-0.5850	-2.06
4.835	-0.5287	-0.5849	-2.07
4.840	-0.5284	-0.5849	-2.07
4.845	-0.5285	-0.5848	-2.07
4.850	-0.5283	-0.5851	-2.07
4.855	-0.5284	-0.5856	-2.06
4.860	-0.5286	-0.5860	-2.06
4.865	-0.5292	-0.5859	-2.05
4.870	-0.5293	-0.5867	-2.05
4.875	-0.5298	-0.5874	-2.04
4.880	-0.5302	-0.5880	-2.03
4.885	-0.5308	-0.5885	-2.02
4.890	-0.5315	-0.5892	-2.01
4.895	-0.5319	-0.5897	-2.00
4.900	-0.5325	-0.5899	-2.00
4.905	-0.5329	-0.5904	-1.99
4.910	-0.5335	-0.5903	-1.99
4.915	-0.5341	-0.5908	-1.98
4.920	-0.5348	-0.5910	-1.97
4.925	-0.5352	-0.5911	-1.97
4.930	-0.5356	-0.5913	-1.96
4.935	-0.5361	-0.5916	-1.96
4.940	-0.5366	-0.5917	-1.95
4.945	-0.5370	-0.5920	-1.95
4.950	-0.5371	-0.5922	-1.94
4.955	-0.5377	-0.5923	-1.94
4.960	-0.5381	-0.5925	-1.93
4.965	-0.5384	-0.5928	-1.93
4.970	-0.5387	-0.5931	-1.92
4.975	-0.5392	-0.5933	-1.92
4.980	-0.5397	-0.5936	-1.91
4.985	-0.5399	-0.5939	-1.91
4.990	-0.5405	-0.5944	-1.90
4.995	-0.5408	-0.5946	-1.90
5.000	-0.5414	-0.5949	-1.89
5.005	-0.5418	-0.5951	-1.89
5.010	-0.5425	-0.5951	-1.88
5.015	-0.5426	-0.5953	-1.88
5.020	-0.5431	-0.5954	-1.87

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Frequency (GHz)	**Re(S_{11})**	**Im(S_{11})**	**$\lvert S_{11}\rvert$ (dB)**
5.025	-0.5432	-0.5956	-1.87
5.030	-0.5437	-0.5950	-1.87
5.035	-0.5438	-0.5955	-1.87
5.040	-0.5439	-0.5948	-1.87
5.045	-0.5438	-0.5952	-1.87
5.050	-0.5436	-0.5945	-1.88
5.055	-0.5436	-0.5949	-1.88
5.060	-0.5433	-0.5943	-1.88
5.065	-0.5438	-0.5945	-1.88
5.070	-0.5434	-0.5941	-1.88
5.075	-0.5444	-0.5940	-1.88
5.080	-0.5440	-0.5941	-1.88
5.085	-0.5452	-0.5936	-1.87
5.090	-0.5452	-0.5935	-1.87
5.095	-0.5464	-0.5933	-1.87
5.100	-0.5468	-0.5929	-1.87
5.105	-0.5477	-0.5922	-1.87
5.110	-0.5484	-0.5921	-1.86
5.115	-0.5493	-0.5913	-1.86
5.120	-0.5500	-0.5909	-1.86
5.125	-0.5512	-0.5900	-1.86
5.130	-0.5520	-0.5898	-1.85
5.135	-0.5527	-0.5888	-1.86
5.140	-0.5538	-0.5879	-1.86
5.145	-0.5552	-0.5870	-1.85
5.150	-0.5559	-0.5862	-1.85
5.155	-0.5573	-0.5846	-1.85
5.160	-0.5585	-0.5832	-1.86
5.165	-0.5595	-0.5817	-1.86
5.170	-0.5609	-0.5804	-1.86
5.175	-0.5623	-0.5785	-1.87
5.180	-0.5632	-0.5766	-1.87
5.185	-0.5643	-0.5752	-1.88
5.190	-0.5655	-0.5727	-1.89
5.195	-0.5665	-0.5710	-1.89
5.200	-0.5676	-0.5685	-1.90
5.205	-0.5687	-0.5665	-1.91
5.210	-0.5695	-0.5645	-1.92
5.215	-0.5706	-0.5623	-1.93
5.220	-0.5715	-0.5600	-1.94
5.225	-0.5728	-0.5574	-1.95
5.230	-0.5736	-0.5553	-1.96
5.235	-0.5745	-0.5531	-1.97
5.240	-0.5755	-0.5509	-1.97
5.245	-0.5764	-0.5489	-1.98
5.250	-0.5772	-0.5466	-1.99

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Frequency (GHz)	**Re(S_{11})**	**Im(S_{11})**	**$\lvert S_{11}\rvert$ (dB)**
5.255	-0.5784	-0.5443	-2.00
5.260	-0.5792	-0.5421	-2.01
5.265	-0.5803	-0.5401	-2.02
5.270	-0.5814	-0.5381	-2.02
5.275	-0.5822	-0.5360	-2.03
5.280	-0.5829	-0.5341	-2.04
5.285	-0.5837	-0.5320	-2.05
5.290	-0.5848	-0.5303	-2.05
5.295	-0.5856	-0.5283	-2.06
5.300	-0.5863	-0.5267	-2.07
5.305	-0.5874	-0.5250	-2.07
5.310	-0.5878	-0.5233	-2.08
5.315	-0.5886	-0.5217	-2.09
5.320	-0.5894	-0.5198	-2.09
5.325	-0.5900	-0.5182	-2.10
5.330	-0.5905	-0.5167	-2.11
5.335	-0.5910	-0.5150	-2.11
5.340	-0.5915	-0.5133	-2.12
5.345	-0.5925	-0.5116	-2.13
5.350	-0.5926	-0.5101	-2.14
5.355	-0.5933	-0.5083	-2.14
5.360	-0.5937	-0.5067	-2.15
5.365	-0.5946	-0.5051	-2.16
5.370	-0.5950	-0.5034	-2.16
5.375	-0.5959	-0.5014	-2.17
5.380	-0.5967	-0.5001	-2.17
5.385	-0.5972	-0.4983	-2.18
5.390	-0.5981	-0.4971	-2.18
5.395	-0.5988	-0.4959	-2.19
5.400	-0.5999	-0.4944	-2.19
5.405	-0.6007	-0.4932	-2.19
5.410	-0.6017	-0.4920	-2.19
5.415	-0.6026	-0.4909	-2.19
5.420	-0.6033	-0.4895	-2.19
5.425	-0.6046	-0.4884	-2.19
5.430	-0.6051	-0.4874	-2.19
5.435	-0.6057	-0.4866	-2.19
5.440	-0.6066	-0.4858	-2.19
5.445	-0.6072	-0.4848	-2.19
5.450	-0.6079	-0.4839	-2.19
5.455	-0.6084	-0.4831	-2.19
5.460	-0.6093	-0.4825	-2.19
5.465	-0.6098	-0.4816	-2.19
5.470	-0.6104	-0.4811	-2.19
5.475	-0.6109	-0.4803	-2.19
5.480	-0.6117	-0.4797	-2.19

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Frequency (GHz)	**Re(S_{11})**	**Im(S_{11})**	**$\lvert S_{11}\rvert$ (dB)**
5.485	-0.6125	-0.4788	-2.19
5.490	-0.6131	-0.4782	-2.19
5.495	-0.6139	-0.4775	-2.18
5.500	-0.6147	-0.4769	-2.18
5.505	-0.6154	-0.4764	-2.18
5.510	-0.6168	-0.4762	-2.17
5.515	-0.6177	-0.4761	-2.16
5.520	-0.6186	-0.4760	-2.15
5.525	-0.6193	-0.4760	-2.15
5.530	-0.6203	-0.4760	-2.14
5.535	-0.6210	-0.4761	-2.13
5.540	-0.6217	-0.4762	-2.12
5.545	-0.6221	-0.4763	-2.12
5.550	-0.6229	-0.4765	-2.11
5.555	-0.6236	-0.4766	-2.10
5.560	-0.6243	-0.4768	-2.10
5.565	-0.6250	-0.4768	-2.09
5.570	-0.6259	-0.4768	-2.08
5.575	-0.6266	-0.4770	-2.08
5.580	-0.6273	-0.4771	-2.07
5.585	-0.6281	-0.4770	-2.06
5.590	-0.6290	-0.4771	-2.05
5.595	-0.6299	-0.4771	-2.04
5.600	-0.6309	-0.4772	-2.04
5.605	-0.6319	-0.4774	-2.03
5.610	-0.6330	-0.4775	-2.02
5.615	-0.6337	-0.4779	-2.01
5.620	-0.6348	-0.4780	-2.00
5.625	-0.6356	-0.4785	-1.99
5.630	-0.6364	-0.4787	-1.98
5.635	-0.6370	-0.4789	-1.97
5.640	-0.6376	-0.4791	-1.96
5.645	-0.6382	-0.4794	-1.96
5.650	-0.6389	-0.4794	-1.95
5.655	-0.6393	-0.4795	-1.95
5.660	-0.6399	-0.4795	-1.94
5.665	-0.6404	-0.4797	-1.94
5.670	-0.6410	-0.4798	-1.93
5.675	-0.6415	-0.4797	-1.93
5.680	-0.6424	-0.4796	-1.92
5.685	-0.6421	-0.4794	-1.92
5.690	-0.6425	-0.4791	-1.92
5.695	-0.6430	-0.4791	-1.92
5.700	-0.6430	-0.4789	-1.92
5.705	-0.6436	-0.4786	-1.92
5.710	-0.6435	-0.4784	-1.92

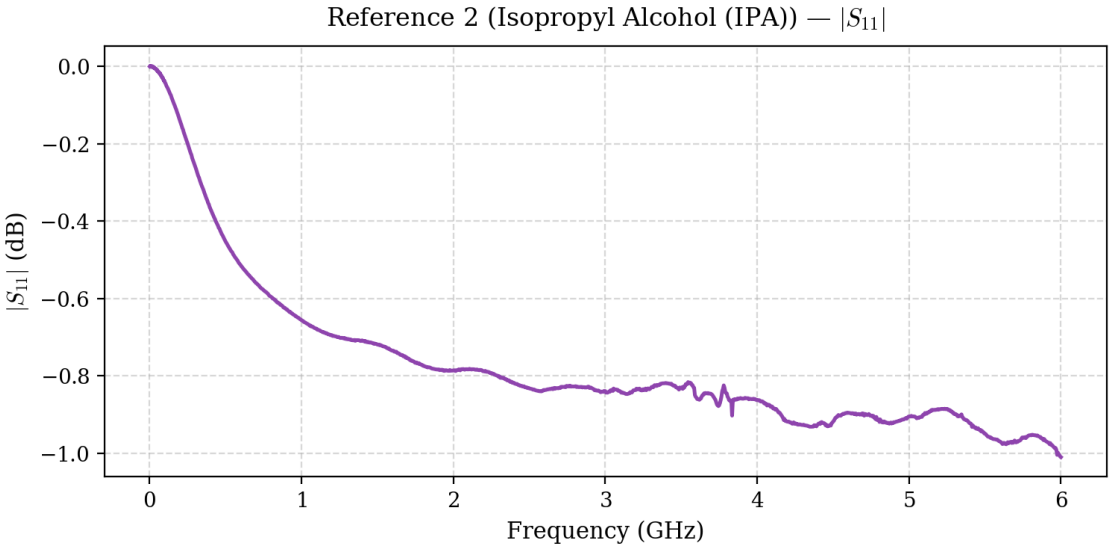
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Frequency (GHz)	**Re(S_{11})**	**Im(S_{11})**	**$\lvert S_{11}\rvert$ (dB)**
5.715	-0.6439	-0.4784	-1.92
5.720	-0.6442	-0.4784	-1.91
5.725	-0.6441	-0.4781	-1.92
5.730	-0.6445	-0.4775	-1.92
5.735	-0.6446	-0.4774	-1.91
5.740	-0.6447	-0.4767	-1.92
5.745	-0.6453	-0.4768	-1.91
5.750	-0.6450	-0.4758	-1.92
5.755	-0.6450	-0.4754	-1.92
5.760	-0.6448	-0.4748	-1.93
5.765	-0.6454	-0.4741	-1.93
5.770	-0.6450	-0.4735	-1.94
5.775	-0.6453	-0.4721	-1.94
5.780	-0.6448	-0.4717	-1.95
5.785	-0.6452	-0.4705	-1.95
5.790	-0.6452	-0.4698	-1.96
5.795	-0.6450	-0.4685	-1.97
5.800	-0.6448	-0.4671	-1.98
5.805	-0.6448	-0.4661	-1.99
5.810	-0.6443	-0.4649	-2.00
5.815	-0.6444	-0.4637	-2.00
5.820	-0.6445	-0.4623	-2.01
5.825	-0.6443	-0.4613	-2.02
5.830	-0.6442	-0.4598	-2.03
5.835	-0.6441	-0.4584	-2.04
5.840	-0.6438	-0.4569	-2.05
5.845	-0.6441	-0.4555	-2.06
5.850	-0.6441	-0.4544	-2.07
5.855	-0.6438	-0.4530	-2.08
5.860	-0.6440	-0.4514	-2.09
5.865	-0.6441	-0.4504	-2.09
5.870	-0.6442	-0.4487	-2.10
5.875	-0.6443	-0.4477	-2.11
5.880	-0.6446	-0.4463	-2.11
5.885	-0.6447	-0.4450	-2.12
5.890	-0.6447	-0.4439	-2.13
5.895	-0.6447	-0.4429	-2.13
5.900	-0.6449	-0.4419	-2.14
5.905	-0.6451	-0.4410	-2.14
5.910	-0.6452	-0.4404	-2.15
5.915	-0.6449	-0.4393	-2.15
5.920	-0.6449	-0.4381	-2.16
5.925	-0.6450	-0.4373	-2.17
5.930	-0.6447	-0.4361	-2.18
5.935	-0.6446	-0.4351	-2.18
5.940	-0.6448	-0.4341	-2.19

(continued on next page)

Frequency (GHz)	Re(S_{11})	Im(S_{11})	$\lvert S_{11}\rvert$ (dB)
5.945	-0.6444	-0.4334	-2.20
5.950	-0.6445	-0.4322	-2.20
5.955	-0.6448	-0.4312	-2.21
5.960	-0.6451	-0.4303	-2.21
5.965	-0.6454	-0.4296	-2.21
5.970	-0.6460	-0.4286	-2.21
5.975	-0.6468	-0.4279	-2.21
5.980	-0.6469	-0.4282	-2.20
5.985	-0.6475	-0.4276	-2.20
5.990	-0.6480	-0.4277	-2.20
5.995	-0.6486	-0.4279	-2.19
6.000	-0.6488	-0.4281	-2.19

3.4 Reference 2 (Isopropyl Alcohol (IPA))



Frequency (GHz)	Re(S_{11})	Im(S_{11})	$\lvert S_{11}\rvert$ (dB)
0.001	1.0000	-0.0006	-0.00
0.002	1.0000	-0.0009	0.00
0.003	1.0002	-0.0011	0.00
0.004	1.0000	-0.0013	-0.00
0.005	1.0001	-0.0016	0.00
0.006	1.0000	-0.0018	-0.00
0.007	0.9999	-0.0020	-0.00
0.008	1.0000	-0.0023	0.00
0.009	1.0000	-0.0026	0.00
0.010	1.0000	-0.0028	0.00

(continued on next page)

Frequency (GHz)	**Re(S_{11})**	**Im(S_{11})**	**$\lvert S_{11}\rvert$ (dB)**
0.011	1.0000	-0.0031	-0.00
0.012	0.9999	-0.0033	-0.00
0.013	0.9999	-0.0037	-0.00
0.014	0.9999	-0.0038	-0.00
0.015	0.9999	-0.0040	-0.00
0.016	0.9999	-0.0042	-0.00
0.017	1.0001	-0.0046	0.00
0.018	0.9999	-0.0048	-0.00
0.019	0.9999	-0.0050	-0.00
0.020	0.9998	-0.0053	-0.00
0.021	0.9998	-0.0055	-0.00
0.022	0.9999	-0.0058	-0.00
0.023	0.9998	-0.0060	-0.00
0.024	0.9998	-0.0062	-0.00
0.025	0.9997	-0.0064	-0.00
0.026	0.9996	-0.0067	-0.00
0.027	0.9997	-0.0071	-0.00
0.028	0.9996	-0.0072	-0.00
0.029	0.9997	-0.0075	-0.00
0.030	0.9996	-0.0077	-0.00
0.031	0.9995	-0.0079	-0.00
0.032	0.9994	-0.0081	-0.00
0.033	0.9995	-0.0085	-0.00
0.034	0.9994	-0.0087	-0.00
0.035	0.9993	-0.0089	-0.01
0.036	0.9994	-0.0091	-0.01
0.037	0.9994	-0.0094	-0.01
0.038	0.9993	-0.0095	-0.01
0.039	0.9992	-0.0099	-0.01
0.040	0.9992	-0.0102	-0.01
0.041	0.9991	-0.0103	-0.01
0.042	0.9991	-0.0106	-0.01
0.043	0.9991	-0.0107	-0.01
0.044	0.9991	-0.0110	-0.01
0.045	0.9991	-0.0112	-0.01
0.046	0.9990	-0.0117	-0.01
0.047	0.9989	-0.0119	-0.01
0.048	0.9987	-0.0120	-0.01
0.049	0.9988	-0.0122	-0.01
0.050	0.9986	-0.0125	-0.01
0.051	0.9988	-0.0128	-0.01
0.052	0.9986	-0.0130	-0.01
0.053	0.9985	-0.0132	-0.01
0.054	0.9985	-0.0135	-0.01
0.055	0.9985	-0.0137	-0.01
0.056	0.9983	-0.0139	-0.01

(continued on next page)

Frequency (GHz)	**Re(S_{11})**	**Im(S_{11})**	**$\lvert S_{11}\rvert$ (dB)**
0.057	0.9984	-0.0141	-0.01
0.058	0.9983	-0.0145	-0.01
0.059	0.9982	-0.0146	-0.01
0.060	0.9982	-0.0148	-0.01
0.061	0.9982	-0.0151	-0.01
0.062	0.9981	-0.0154	-0.02
0.063	0.9979	-0.0154	-0.02
0.064	0.9979	-0.0157	-0.02
0.065	0.9980	-0.0161	-0.02
0.066	0.9979	-0.0161	-0.02
0.067	0.9977	-0.0164	-0.02
0.068	0.9978	-0.0168	-0.02
0.069	0.9976	-0.0169	-0.02
0.070	0.9975	-0.0172	-0.02
0.071	0.9976	-0.0173	-0.02
0.072	0.9974	-0.0175	-0.02
0.073	0.9973	-0.0177	-0.02
0.074	0.9972	-0.0180	-0.02
0.075	0.9970	-0.0183	-0.02
0.076	0.9971	-0.0185	-0.02
0.077	0.9971	-0.0187	-0.02
0.078	0.9970	-0.0189	-0.02
0.079	0.9969	-0.0192	-0.03
0.080	0.9968	-0.0194	-0.03
0.081	0.9966	-0.0195	-0.03
0.082	0.9966	-0.0198	-0.03
0.083	0.9966	-0.0200	-0.03
0.084	0.9966	-0.0203	-0.03
0.085	0.9965	-0.0206	-0.03
0.086	0.9964	-0.0208	-0.03
0.087	0.9962	-0.0209	-0.03
0.088	0.9962	-0.0211	-0.03
0.089	0.9961	-0.0213	-0.03
0.090	0.9959	-0.0216	-0.03
0.091	0.9959	-0.0219	-0.03
0.092	0.9957	-0.0219	-0.04
0.093	0.9956	-0.0223	-0.04
0.094	0.9956	-0.0224	-0.04
0.095	0.9955	-0.0225	-0.04
0.096	0.9954	-0.0230	-0.04
0.097	0.9954	-0.0231	-0.04
0.098	0.9953	-0.0233	-0.04
0.099	0.9952	-0.0235	-0.04
0.100	0.9950	-0.0237	-0.04
0.105	0.9946	-0.0248	-0.04
0.110	0.9939	-0.0257	-0.05

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Frequency (GHz)	**Re(S_{11})**	**Im(S_{11})**	**$\lvert S_{11}\rvert$ (dB)**
0.115	0.9935	-0.0268	-0.05
0.120	0.9931	-0.0278	-0.06
0.125	0.9925	-0.0287	-0.06
0.130	0.9919	-0.0297	-0.07
0.135	0.9914	-0.0305	-0.07
0.140	0.9910	-0.0315	-0.07
0.145	0.9902	-0.0325	-0.08
0.150	0.9896	-0.0334	-0.09
0.155	0.9890	-0.0341	-0.09
0.160	0.9883	-0.0351	-0.10
0.165	0.9878	-0.0358	-0.10
0.170	0.9871	-0.0366	-0.11
0.175	0.9866	-0.0374	-0.11
0.180	0.9858	-0.0382	-0.12
0.185	0.9852	-0.0388	-0.12
0.190	0.9844	-0.0397	-0.13
0.195	0.9837	-0.0405	-0.14
0.200	0.9831	-0.0412	-0.14
0.205	0.9824	-0.0418	-0.15
0.210	0.9817	-0.0424	-0.15
0.215	0.9809	-0.0433	-0.16
0.220	0.9804	-0.0438	-0.16
0.225	0.9795	-0.0444	-0.17
0.230	0.9788	-0.0449	-0.18
0.235	0.9782	-0.0455	-0.18
0.240	0.9775	-0.0461	-0.19
0.245	0.9768	-0.0468	-0.19
0.250	0.9762	-0.0473	-0.20
0.255	0.9754	-0.0477	-0.21
0.260	0.9746	-0.0482	-0.21
0.265	0.9740	-0.0487	-0.22
0.270	0.9733	-0.0493	-0.22
0.275	0.9728	-0.0496	-0.23
0.280	0.9718	-0.0502	-0.24
0.285	0.9712	-0.0505	-0.24
0.290	0.9705	-0.0509	-0.25
0.295	0.9699	-0.0514	-0.25
0.300	0.9692	-0.0517	-0.26
0.305	0.9686	-0.0521	-0.26
0.310	0.9679	-0.0525	-0.27
0.315	0.9672	-0.0527	-0.28
0.320	0.9665	-0.0530	-0.28
0.325	0.9660	-0.0535	-0.29
0.330	0.9652	-0.0538	-0.29
0.335	0.9645	-0.0541	-0.30
0.340	0.9639	-0.0543	-0.31

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Frequency (GHz)	**Re(S_{11})**	**Im(S_{11})**	**$\lvert S_{11}\rvert$ (dB)**
0.345	0.9634	-0.0545	-0.31
0.350	0.9627	-0.0548	-0.32
0.355	0.9622	-0.0552	-0.32
0.360	0.9615	-0.0554	-0.33
0.365	0.9609	-0.0556	-0.33
0.370	0.9603	-0.0558	-0.34
0.375	0.9597	-0.0560	-0.34
0.380	0.9592	-0.0563	-0.35
0.385	0.9585	-0.0564	-0.35
0.390	0.9581	-0.0566	-0.36
0.395	0.9573	-0.0569	-0.36
0.400	0.9568	-0.0570	-0.37
0.405	0.9564	-0.0572	-0.37
0.410	0.9557	-0.0572	-0.38
0.415	0.9553	-0.0574	-0.38
0.420	0.9548	-0.0576	-0.39
0.425	0.9543	-0.0577	-0.39
0.430	0.9538	-0.0579	-0.39
0.435	0.9532	-0.0580	-0.40
0.440	0.9528	-0.0581	-0.40
0.445	0.9524	-0.0583	-0.41
0.450	0.9519	-0.0584	-0.41
0.455	0.9513	-0.0584	-0.42
0.460	0.9510	-0.0585	-0.42
0.465	0.9504	-0.0586	-0.42
0.470	0.9500	-0.0587	-0.43
0.475	0.9495	-0.0589	-0.43
0.480	0.9491	-0.0589	-0.44
0.485	0.9487	-0.0590	-0.44
0.490	0.9482	-0.0590	-0.44
0.495	0.9478	-0.0592	-0.45
0.500	0.9474	-0.0592	-0.45
0.505	0.9470	-0.0592	-0.46
0.510	0.9466	-0.0592	-0.46
0.515	0.9462	-0.0594	-0.46
0.520	0.9458	-0.0594	-0.47
0.525	0.9455	-0.0594	-0.47
0.530	0.9451	-0.0595	-0.47
0.535	0.9448	-0.0596	-0.48
0.540	0.9445	-0.0596	-0.48
0.545	0.9442	-0.0597	-0.48
0.550	0.9437	-0.0597	-0.49
0.555	0.9434	-0.0598	-0.49
0.560	0.9430	-0.0599	-0.49
0.565	0.9427	-0.0599	-0.49
0.570	0.9424	-0.0600	-0.50

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Frequency (GHz)	**Re(S_{11})**	**Im(S_{11})**	**$\lvert S_{11}\rvert$ (dB)**
0.575	0.9422	-0.0599	-0.50
0.580	0.9419	-0.0600	-0.50
0.585	0.9414	-0.0599	-0.51
0.590	0.9412	-0.0600	-0.51
0.595	0.9409	-0.0601	-0.51
0.600	0.9406	-0.0601	-0.51
0.605	0.9403	-0.0602	-0.52
0.610	0.9400	-0.0602	-0.52
0.615	0.9398	-0.0602	-0.52
0.620	0.9395	-0.0602	-0.52
0.625	0.9392	-0.0603	-0.53
0.630	0.9390	-0.0602	-0.53
0.635	0.9388	-0.0603	-0.53
0.640	0.9385	-0.0604	-0.53
0.645	0.9382	-0.0604	-0.54
0.650	0.9380	-0.0605	-0.54
0.655	0.9378	-0.0604	-0.54
0.660	0.9375	-0.0604	-0.54
0.665	0.9373	-0.0604	-0.54
0.670	0.9370	-0.0605	-0.55
0.675	0.9368	-0.0605	-0.55
0.680	0.9366	-0.0607	-0.55
0.685	0.9363	-0.0606	-0.55
0.690	0.9361	-0.0606	-0.56
0.695	0.9358	-0.0606	-0.56
0.700	0.9356	-0.0606	-0.56
0.705	0.9355	-0.0607	-0.56
0.710	0.9352	-0.0608	-0.56
0.715	0.9350	-0.0608	-0.57
0.720	0.9349	-0.0608	-0.57
0.725	0.9345	-0.0608	-0.57
0.730	0.9343	-0.0607	-0.57
0.735	0.9341	-0.0609	-0.57
0.740	0.9340	-0.0608	-0.58
0.745	0.9338	-0.0609	-0.58
0.750	0.9336	-0.0610	-0.58
0.755	0.9334	-0.0610	-0.58
0.760	0.9332	-0.0610	-0.58
0.765	0.9331	-0.0610	-0.58
0.770	0.9329	-0.0611	-0.58
0.775	0.9327	-0.0610	-0.59
0.780	0.9325	-0.0612	-0.59
0.785	0.9322	-0.0612	-0.59
0.790	0.9320	-0.0611	-0.59
0.795	0.9318	-0.0613	-0.59
0.800	0.9317	-0.0613	-0.60

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Frequency (GHz)	**Re**(S_{11})	**Im**(S_{11})	$\lvert S_{11}\rvert$ **(dB)**
0.805	0.9315	-0.0614	-0.60
0.810	0.9312	-0.0614	-0.60
0.815	0.9312	-0.0614	-0.60
0.820	0.9310	-0.0615	-0.60
0.825	0.9307	-0.0615	-0.60
0.830	0.9307	-0.0615	-0.60
0.835	0.9303	-0.0617	-0.61
0.840	0.9301	-0.0617	-0.61
0.845	0.9301	-0.0616	-0.61
0.850	0.9299	-0.0617	-0.61
0.855	0.9298	-0.0619	-0.61
0.860	0.9295	-0.0618	-0.62
0.865	0.9294	-0.0621	-0.62
0.870	0.9293	-0.0621	-0.62
0.875	0.9291	-0.0620	-0.62
0.880	0.9287	-0.0620	-0.62
0.885	0.9286	-0.0621	-0.62
0.890	0.9286	-0.0622	-0.62
0.895	0.9283	-0.0623	-0.63
0.900	0.9282	-0.0621	-0.63
0.905	0.9281	-0.0623	-0.63
0.910	0.9278	-0.0624	-0.63
0.915	0.9277	-0.0625	-0.63
0.920	0.9274	-0.0624	-0.63
0.925	0.9275	-0.0625	-0.63
0.930	0.9273	-0.0626	-0.64
0.935	0.9270	-0.0627	-0.64
0.940	0.9269	-0.0627	-0.64
0.945	0.9267	-0.0628	-0.64
0.950	0.9266	-0.0627	-0.64
0.955	0.9265	-0.0628	-0.64
0.960	0.9262	-0.0629	-0.65
0.965	0.9262	-0.0630	-0.65
0.970	0.9260	-0.0629	-0.65
0.975	0.9258	-0.0630	-0.65
0.980	0.9258	-0.0630	-0.65
0.985	0.9256	-0.0631	-0.65
0.990	0.9254	-0.0632	-0.65
0.995	0.9253	-0.0633	-0.65
1.000	0.9251	-0.0633	-0.66
1.005	0.9250	-0.0634	-0.66
1.010	0.9249	-0.0636	-0.66
1.015	0.9246	-0.0636	-0.66
1.020	0.9245	-0.0636	-0.66
1.025	0.9244	-0.0636	-0.66
1.030	0.9243	-0.0637	-0.66

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Frequency (GHz)	**Re**(S_{11})	**Im**(S_{11})	$\lvert S_{11}\rvert$ **(dB)**
1.035	0.9241	-0.0638	-0.66
1.040	0.9240	-0.0638	-0.67
1.045	0.9239	-0.0639	-0.67
1.050	0.9237	-0.0639	-0.67
1.055	0.9235	-0.0640	-0.67
1.060	0.9235	-0.0640	-0.67
1.065	0.9233	-0.0640	-0.67
1.070	0.9233	-0.0642	-0.67
1.075	0.9232	-0.0643	-0.67
1.080	0.9230	-0.0643	-0.67
1.085	0.9229	-0.0643	-0.68
1.090	0.9228	-0.0644	-0.68
1.095	0.9226	-0.0645	-0.68
1.100	0.9226	-0.0646	-0.68
1.105	0.9223	-0.0647	-0.68
1.110	0.9223	-0.0647	-0.68
1.115	0.9222	-0.0647	-0.68
1.120	0.9222	-0.0648	-0.68
1.125	0.9220	-0.0648	-0.68
1.130	0.9219	-0.0649	-0.68
1.135	0.9218	-0.0651	-0.69
1.140	0.9216	-0.0650	-0.69
1.145	0.9217	-0.0651	-0.69
1.150	0.9216	-0.0650	-0.69
1.155	0.9215	-0.0652	-0.69
1.160	0.9213	-0.0653	-0.69
1.165	0.9214	-0.0654	-0.69
1.170	0.9212	-0.0654	-0.69
1.175	0.9210	-0.0655	-0.69
1.180	0.9210	-0.0657	-0.69
1.185	0.9210	-0.0655	-0.69
1.190	0.9208	-0.0656	-0.69
1.195	0.9208	-0.0657	-0.69
1.200	0.9208	-0.0657	-0.69
1.205	0.9206	-0.0658	-0.70
1.210	0.9206	-0.0658	-0.70
1.215	0.9205	-0.0659	-0.70
1.220	0.9205	-0.0658	-0.70
1.225	0.9204	-0.0659	-0.70
1.230	0.9204	-0.0660	-0.70
1.235	0.9202	-0.0661	-0.70
1.240	0.9202	-0.0661	-0.70
1.245	0.9203	-0.0661	-0.70
1.250	0.9201	-0.0663	-0.70
1.255	0.9200	-0.0663	-0.70
1.260	0.9200	-0.0663	-0.70

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Frequency (GHz)	**Re(S_{11})**	**Im(S_{11})**	**$\lvert S_{11}\rvert$ (dB)**
1.265	0.9200	-0.0664	-0.70
1.270	0.9199	-0.0664	-0.70
1.275	0.9199	-0.0665	-0.70
1.280	0.9198	-0.0665	-0.70
1.285	0.9199	-0.0666	-0.70
1.290	0.9197	-0.0666	-0.70
1.295	0.9197	-0.0667	-0.70
1.300	0.9196	-0.0669	-0.71
1.305	0.9197	-0.0669	-0.70
1.310	0.9196	-0.0669	-0.71
1.315	0.9196	-0.0669	-0.71
1.320	0.9194	-0.0670	-0.71
1.325	0.9195	-0.0671	-0.71
1.330	0.9193	-0.0671	-0.71
1.335	0.9193	-0.0672	-0.71
1.340	0.9193	-0.0671	-0.71
1.345	0.9192	-0.0672	-0.71
1.350	0.9193	-0.0674	-0.71
1.355	0.9191	-0.0673	-0.71
1.360	0.9193	-0.0674	-0.71
1.365	0.9193	-0.0675	-0.71
1.370	0.9193	-0.0675	-0.71
1.375	0.9193	-0.0675	-0.71
1.380	0.9192	-0.0676	-0.71
1.385	0.9191	-0.0677	-0.71
1.390	0.9190	-0.0679	-0.71
1.395	0.9192	-0.0680	-0.71
1.400	0.9192	-0.0680	-0.71
1.405	0.9190	-0.0680	-0.71
1.410	0.9190	-0.0681	-0.71
1.415	0.9189	-0.0681	-0.71
1.420	0.9190	-0.0683	-0.71
1.425	0.9187	-0.0682	-0.71
1.430	0.9187	-0.0684	-0.71
1.435	0.9187	-0.0683	-0.71
1.440	0.9187	-0.0685	-0.71
1.445	0.9186	-0.0687	-0.71
1.450	0.9185	-0.0687	-0.71
1.455	0.9186	-0.0689	-0.71
1.460	0.9185	-0.0691	-0.71
1.465	0.9184	-0.0691	-0.71
1.470	0.9183	-0.0691	-0.72
1.475	0.9182	-0.0692	-0.72
1.480	0.9182	-0.0692	-0.72
1.485	0.9181	-0.0695	-0.72
1.490	0.9181	-0.0695	-0.72

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Frequency (GHz)	**Re(S_{11})**	**Im(S_{11})**	**$\lvert S_{11}\rvert$ (dB)**
1.495	0.9180	-0.0696	-0.72
1.500	0.9179	-0.0698	-0.72
1.505	0.9179	-0.0699	-0.72
1.510	0.9178	-0.0699	-0.72
1.515	0.9178	-0.0701	-0.72
1.520	0.9177	-0.0703	-0.72
1.525	0.9176	-0.0704	-0.72
1.530	0.9175	-0.0704	-0.72
1.535	0.9174	-0.0705	-0.72
1.540	0.9173	-0.0707	-0.72
1.545	0.9172	-0.0709	-0.72
1.550	0.9172	-0.0709	-0.72
1.555	0.9171	-0.0710	-0.73
1.560	0.9169	-0.0712	-0.73
1.565	0.9168	-0.0714	-0.73
1.570	0.9166	-0.0714	-0.73
1.575	0.9167	-0.0716	-0.73
1.580	0.9165	-0.0715	-0.73
1.585	0.9163	-0.0718	-0.73
1.590	0.9162	-0.0719	-0.73
1.595	0.9161	-0.0721	-0.73
1.600	0.9161	-0.0721	-0.73
1.605	0.9160	-0.0723	-0.74
1.610	0.9158	-0.0723	-0.74
1.615	0.9157	-0.0725	-0.74
1.620	0.9157	-0.0727	-0.74
1.625	0.9157	-0.0729	-0.74
1.630	0.9153	-0.0730	-0.74
1.635	0.9153	-0.0731	-0.74
1.640	0.9152	-0.0731	-0.74
1.645	0.9150	-0.0734	-0.74
1.650	0.9148	-0.0736	-0.75
1.655	0.9148	-0.0736	-0.75
1.660	0.9147	-0.0738	-0.75
1.665	0.9147	-0.0740	-0.75
1.670	0.9143	-0.0741	-0.75
1.675	0.9143	-0.0743	-0.75
1.680	0.9143	-0.0743	-0.75
1.685	0.9139	-0.0744	-0.75
1.690	0.9141	-0.0746	-0.75
1.695	0.9138	-0.0746	-0.75
1.700	0.9137	-0.0748	-0.75
1.705	0.9137	-0.0749	-0.76
1.710	0.9134	-0.0750	-0.76
1.715	0.9132	-0.0750	-0.76
1.720	0.9132	-0.0754	-0.76

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Frequency (GHz)	**Re(S_{11})**	**Im(S_{11})**	**$\lvert S_{11}\rvert$ (dB)**
1.725	0.9130	-0.0753	-0.76
1.730	0.9130	-0.0754	-0.76
1.735	0.9129	-0.0756	-0.76
1.740	0.9128	-0.0757	-0.76
1.745	0.9125	-0.0760	-0.77
1.750	0.9124	-0.0759	-0.77
1.755	0.9125	-0.0761	-0.77
1.760	0.9123	-0.0762	-0.77
1.765	0.9123	-0.0763	-0.77
1.770	0.9121	-0.0764	-0.77
1.775	0.9122	-0.0766	-0.77
1.780	0.9120	-0.0766	-0.77
1.785	0.9119	-0.0768	-0.77
1.790	0.9118	-0.0768	-0.77
1.795	0.9117	-0.0769	-0.77
1.800	0.9115	-0.0770	-0.77
1.805	0.9115	-0.0772	-0.77
1.810	0.9113	-0.0772	-0.78
1.815	0.9113	-0.0774	-0.78
1.820	0.9112	-0.0774	-0.78
1.825	0.9111	-0.0776	-0.78
1.830	0.9110	-0.0777	-0.78
1.835	0.9110	-0.0779	-0.78
1.840	0.9110	-0.0780	-0.78
1.845	0.9109	-0.0781	-0.78
1.850	0.9108	-0.0780	-0.78
1.855	0.9107	-0.0782	-0.78
1.860	0.9106	-0.0780	-0.78
1.865	0.9106	-0.0781	-0.78
1.870	0.9106	-0.0782	-0.78
1.875	0.9105	-0.0784	-0.78
1.880	0.9104	-0.0786	-0.78
1.885	0.9104	-0.0787	-0.78
1.890	0.9104	-0.0788	-0.78
1.895	0.9103	-0.0789	-0.78
1.900	0.9103	-0.0790	-0.78
1.905	0.9103	-0.0790	-0.78
1.910	0.9101	-0.0791	-0.79
1.915	0.9102	-0.0793	-0.78
1.920	0.9101	-0.0793	-0.79
1.925	0.9102	-0.0795	-0.78
1.930	0.9100	-0.0794	-0.79
1.935	0.9100	-0.0796	-0.79
1.940	0.9099	-0.0797	-0.79
1.945	0.9100	-0.0799	-0.79
1.950	0.9100	-0.0799	-0.79

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Frequency (GHz)	**Re(S_{11})**	**Im(S_{11})**	**$\lvert S_{11}\rvert$ (dB)**
1.955	0.9100	-0.0800	-0.79
1.960	0.9099	-0.0800	-0.79
1.965	0.9098	-0.0802	-0.79
1.970	0.9099	-0.0803	-0.79
1.975	0.9100	-0.0804	-0.79
1.980	0.9099	-0.0804	-0.79
1.985	0.9099	-0.0804	-0.79
1.990	0.9098	-0.0806	-0.79
1.995	0.9099	-0.0806	-0.79
2.000	0.9099	-0.0808	-0.79
2.005	0.9099	-0.0807	-0.79
2.010	0.9099	-0.0809	-0.79
2.015	0.9097	-0.0810	-0.79
2.020	0.9099	-0.0811	-0.79
2.025	0.9100	-0.0812	-0.79
2.030	0.9099	-0.0812	-0.79
2.035	0.9101	-0.0814	-0.78
2.040	0.9101	-0.0814	-0.78
2.045	0.9099	-0.0816	-0.79
2.050	0.9101	-0.0816	-0.78
2.055	0.9102	-0.0816	-0.78
2.060	0.9101	-0.0819	-0.78
2.065	0.9102	-0.0818	-0.78
2.070	0.9100	-0.0821	-0.78
2.075	0.9100	-0.0821	-0.78
2.080	0.9101	-0.0822	-0.78
2.085	0.9101	-0.0824	-0.78
2.090	0.9101	-0.0824	-0.78
2.095	0.9100	-0.0825	-0.78
2.100	0.9100	-0.0828	-0.78
2.105	0.9102	-0.0829	-0.78
2.110	0.9100	-0.0828	-0.78
2.115	0.9101	-0.0830	-0.78
2.120	0.9099	-0.0831	-0.78
2.125	0.9100	-0.0833	-0.78
2.130	0.9100	-0.0835	-0.78
2.135	0.9100	-0.0835	-0.78
2.140	0.9099	-0.0837	-0.78
2.145	0.9098	-0.0839	-0.78
2.150	0.9098	-0.0840	-0.78
2.155	0.9099	-0.0842	-0.78
2.160	0.9097	-0.0843	-0.78
2.165	0.9097	-0.0844	-0.78
2.170	0.9096	-0.0846	-0.79
2.175	0.9097	-0.0847	-0.78
2.180	0.9096	-0.0849	-0.79

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Frequency (GHz)	**Re(S_{11})**	**Im(S_{11})**	**$\lvert S_{11}\rvert$ (dB)**
2.185	0.9096	-0.0850	-0.79
2.190	0.9094	-0.0852	-0.79
2.195	0.9093	-0.0855	-0.79
2.200	0.9092	-0.0856	-0.79
2.205	0.9092	-0.0857	-0.79
2.210	0.9093	-0.0859	-0.79
2.215	0.9091	-0.0860	-0.79
2.220	0.9090	-0.0862	-0.79
2.225	0.9090	-0.0864	-0.79
2.230	0.9089	-0.0866	-0.79
2.235	0.9089	-0.0869	-0.79
2.240	0.9087	-0.0869	-0.79
2.245	0.9086	-0.0872	-0.79
2.250	0.9084	-0.0874	-0.79
2.255	0.9085	-0.0874	-0.79
2.260	0.9084	-0.0877	-0.79
2.265	0.9083	-0.0879	-0.79
2.270	0.9081	-0.0880	-0.80
2.275	0.9080	-0.0882	-0.80
2.280	0.9079	-0.0884	-0.80
2.285	0.9078	-0.0886	-0.80
2.290	0.9077	-0.0888	-0.80
2.295	0.9077	-0.0888	-0.80
2.300	0.9073	-0.0889	-0.80
2.305	0.9075	-0.0894	-0.80
2.310	0.9071	-0.0893	-0.81
2.315	0.9072	-0.0895	-0.80
2.320	0.9070	-0.0898	-0.81
2.325	0.9069	-0.0901	-0.81
2.330	0.9070	-0.0903	-0.80
2.335	0.9068	-0.0904	-0.81
2.340	0.9066	-0.0904	-0.81
2.345	0.9066	-0.0908	-0.81
2.350	0.9065	-0.0910	-0.81
2.355	0.9062	-0.0912	-0.81
2.360	0.9062	-0.0913	-0.81
2.365	0.9062	-0.0914	-0.81
2.370	0.9059	-0.0916	-0.81
2.375	0.9057	-0.0916	-0.82
2.380	0.9058	-0.0920	-0.81
2.385	0.9055	-0.0921	-0.82
2.390	0.9056	-0.0923	-0.82
2.395	0.9054	-0.0926	-0.82
2.400	0.9053	-0.0929	-0.82
2.405	0.9051	-0.0929	-0.82
2.410	0.9050	-0.0931	-0.82

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Frequency (GHz)	**Re(S_{11})**	**Im(S_{11})**	**$\lvert S_{11}\rvert$ (dB)**
2.415	0.9050	-0.0933	-0.82
2.420	0.9050	-0.0935	-0.82
2.425	0.9047	-0.0936	-0.82
2.430	0.9047	-0.0936	-0.82
2.435	0.9046	-0.0938	-0.82
2.440	0.9044	-0.0940	-0.83
2.445	0.9044	-0.0942	-0.83
2.450	0.9044	-0.0943	-0.83
2.455	0.9042	-0.0945	-0.83
2.460	0.9042	-0.0946	-0.83
2.465	0.9042	-0.0947	-0.83
2.470	0.9040	-0.0948	-0.83
2.475	0.9040	-0.0949	-0.83
2.480	0.9039	-0.0951	-0.83
2.485	0.9039	-0.0952	-0.83
2.490	0.9037	-0.0953	-0.83
2.495	0.9037	-0.0955	-0.83
2.500	0.9036	-0.0954	-0.83
2.505	0.9035	-0.0957	-0.83
2.510	0.9034	-0.0958	-0.83
2.515	0.9034	-0.0959	-0.83
2.520	0.9033	-0.0961	-0.83
2.525	0.9031	-0.0962	-0.84
2.530	0.9032	-0.0964	-0.84
2.535	0.9031	-0.0964	-0.84
2.540	0.9030	-0.0965	-0.84
2.545	0.9029	-0.0967	-0.84
2.550	0.9028	-0.0968	-0.84
2.555	0.9028	-0.0968	-0.84
2.560	0.9028	-0.0970	-0.84
2.565	0.9027	-0.0970	-0.84
2.570	0.9027	-0.0970	-0.84
2.575	0.9027	-0.0971	-0.84
2.580	0.9026	-0.0971	-0.84
2.585	0.9028	-0.0972	-0.84
2.590	0.9028	-0.0972	-0.84
2.595	0.9029	-0.0975	-0.84
2.600	0.9029	-0.0976	-0.84
2.605	0.9029	-0.0976	-0.84
2.610	0.9029	-0.0977	-0.84
2.615	0.9030	-0.0979	-0.84
2.620	0.9031	-0.0981	-0.83
2.625	0.9030	-0.0982	-0.83
2.630	0.9031	-0.0984	-0.83
2.635	0.9032	-0.0981	-0.83
2.640	0.9031	-0.0985	-0.83

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Frequency (GHz)	**Re(S_{11})**	**Im(S_{11})**	**$\lvert S_{11}\rvert$ (dB)**
2.645	0.9031	-0.0985	-0.83
2.650	0.9032	-0.0987	-0.83
2.655	0.9030	-0.0987	-0.83
2.660	0.9032	-0.0990	-0.83
2.665	0.9033	-0.0989	-0.83
2.670	0.9031	-0.0988	-0.83
2.675	0.9032	-0.0993	-0.83
2.680	0.9032	-0.0993	-0.83
2.685	0.9032	-0.0995	-0.83
2.690	0.9032	-0.0995	-0.83
2.695	0.9032	-0.0997	-0.83
2.700	0.9033	-0.0999	-0.83
2.705	0.9032	-0.0999	-0.83
2.710	0.9035	-0.1001	-0.83
2.715	0.9035	-0.1002	-0.83
2.720	0.9035	-0.1003	-0.83
2.725	0.9034	-0.1003	-0.83
2.730	0.9034	-0.1004	-0.83
2.735	0.9035	-0.1005	-0.83
2.740	0.9034	-0.1005	-0.83
2.745	0.9037	-0.1008	-0.83
2.750	0.9036	-0.1007	-0.83
2.755	0.9036	-0.1008	-0.83
2.760	0.9037	-0.1008	-0.83
2.765	0.9035	-0.1010	-0.83
2.770	0.9037	-0.1010	-0.83
2.775	0.9034	-0.1011	-0.83
2.780	0.9035	-0.1011	-0.83
2.785	0.9035	-0.1013	-0.83
2.790	0.9035	-0.1012	-0.83
2.795	0.9035	-0.1015	-0.83
2.800	0.9035	-0.1015	-0.83
2.805	0.9035	-0.1017	-0.83
2.810	0.9033	-0.1018	-0.83
2.815	0.9034	-0.1020	-0.83
2.820	0.9033	-0.1020	-0.83
2.825	0.9032	-0.1020	-0.83
2.830	0.9032	-0.1021	-0.83
2.835	0.9033	-0.1022	-0.83
2.840	0.9032	-0.1024	-0.83
2.845	0.9032	-0.1027	-0.83
2.850	0.9031	-0.1026	-0.83
2.855	0.9031	-0.1027	-0.83
2.860	0.9031	-0.1030	-0.83
2.865	0.9030	-0.1031	-0.83
2.870	0.9030	-0.1033	-0.83

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Frequency (GHz)	**Re(S_{11})**	**Im(S_{11})**	**$\lvert S_{11}\rvert$ (dB)**
2.875	0.9030	-0.1033	-0.83
2.880	0.9030	-0.1034	-0.83
2.885	0.9031	-0.1036	-0.83
2.890	0.9032	-0.1036	-0.83
2.895	0.9028	-0.1036	-0.83
2.900	0.9028	-0.1037	-0.83
2.905	0.9029	-0.1038	-0.83
2.910	0.9027	-0.1039	-0.83
2.915	0.9026	-0.1041	-0.83
2.920	0.9025	-0.1041	-0.83
2.925	0.9023	-0.1042	-0.84
2.930	0.9021	-0.1042	-0.84
2.935	0.9022	-0.1045	-0.84
2.940	0.9020	-0.1045	-0.84
2.945	0.9019	-0.1049	-0.84
2.950	0.9019	-0.1051	-0.84
2.955	0.9020	-0.1049	-0.84
2.960	0.9018	-0.1057	-0.84
2.965	0.9016	-0.1053	-0.84
2.970	0.9014	-0.1056	-0.84
2.975	0.9016	-0.1058	-0.84
2.980	0.9015	-0.1060	-0.84
2.985	0.9015	-0.1060	-0.84
2.990	0.9017	-0.1063	-0.84
2.995	0.9016	-0.1063	-0.84
3.000	0.9016	-0.1066	-0.84
3.005	0.9012	-0.1064	-0.84
3.010	0.9012	-0.1069	-0.84
3.015	0.9011	-0.1068	-0.84
3.020	0.9013	-0.1071	-0.84
3.025	0.9014	-0.1073	-0.84
3.030	0.9014	-0.1071	-0.84
3.035	0.9018	-0.1073	-0.84
3.040	0.9017	-0.1073	-0.84
3.045	0.9019	-0.1075	-0.84
3.050	0.9015	-0.1076	-0.84
3.055	0.9022	-0.1075	-0.83
3.060	0.9019	-0.1074	-0.84
3.065	0.9020	-0.1072	-0.84
3.070	0.9020	-0.1072	-0.84
3.075	0.9019	-0.1071	-0.84
3.080	0.9019	-0.1071	-0.84
3.085	0.9016	-0.1072	-0.84
3.090	0.9017	-0.1070	-0.84
3.095	0.9016	-0.1070	-0.84
3.100	0.9014	-0.1070	-0.84

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Frequency (GHz)	**Re(S_{11})**	**Im(S_{11})**	**$\lvert S_{11}\rvert$ (dB)**
3.105	0.9011	-0.1072	-0.84
3.110	0.9012	-0.1073	-0.84
3.115	0.9010	-0.1073	-0.84
3.120	0.9010	-0.1074	-0.84
3.125	0.9009	-0.1076	-0.84
3.130	0.9008	-0.1077	-0.85
3.135	0.9006	-0.1081	-0.85
3.140	0.9007	-0.1081	-0.85
3.145	0.9006	-0.1084	-0.85
3.150	0.9006	-0.1085	-0.85
3.155	0.9007	-0.1088	-0.85
3.160	0.9007	-0.1088	-0.85
3.165	0.9010	-0.1090	-0.84
3.170	0.9009	-0.1090	-0.84
3.175	0.9009	-0.1092	-0.84
3.180	0.9012	-0.1095	-0.84
3.185	0.9014	-0.1098	-0.84
3.190	0.9015	-0.1093	-0.84
3.195	0.9015	-0.1096	-0.84
3.200	0.9015	-0.1097	-0.84
3.205	0.9019	-0.1097	-0.83
3.210	0.9018	-0.1095	-0.83
3.215	0.9018	-0.1094	-0.83
3.220	0.9019	-0.1090	-0.83
3.225	0.9019	-0.1095	-0.83
3.230	0.9015	-0.1094	-0.84
3.235	0.9016	-0.1096	-0.84
3.240	0.9014	-0.1098	-0.84
3.245	0.9015	-0.1097	-0.84
3.250	0.9017	-0.1103	-0.83
3.255	0.9014	-0.1103	-0.84
3.260	0.9016	-0.1104	-0.83
3.265	0.9018	-0.1106	-0.83
3.270	0.9014	-0.1106	-0.84
3.275	0.9016	-0.1108	-0.83
3.280	0.9018	-0.1109	-0.83
3.285	0.9017	-0.1116	-0.83
3.290	0.9015	-0.1111	-0.83
3.295	0.9016	-0.1115	-0.83
3.300	0.9019	-0.1115	-0.83
3.305	0.9021	-0.1116	-0.83
3.310	0.9021	-0.1117	-0.83
3.315	0.9020	-0.1120	-0.83
3.320	0.9023	-0.1120	-0.83
3.325	0.9026	-0.1121	-0.82
3.330	0.9023	-0.1125	-0.83

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Frequency (GHz)	**Re(S_{11})**	**Im(S_{11})**	**$\lvert S_{11}\rvert$ (dB)**
3.335	0.9023	-0.1126	-0.83
3.340	0.9025	-0.1127	-0.82
3.345	0.9024	-0.1128	-0.82
3.350	0.9024	-0.1130	-0.82
3.355	0.9026	-0.1129	-0.82
3.360	0.9026	-0.1131	-0.82
3.365	0.9028	-0.1131	-0.82
3.370	0.9028	-0.1135	-0.82
3.375	0.9029	-0.1131	-0.82
3.380	0.9029	-0.1132	-0.82
3.385	0.9029	-0.1135	-0.82
3.390	0.9029	-0.1135	-0.82
3.395	0.9029	-0.1136	-0.82
3.400	0.9029	-0.1138	-0.82
3.405	0.9028	-0.1137	-0.82
3.410	0.9029	-0.1137	-0.82
3.415	0.9028	-0.1139	-0.82
3.420	0.9027	-0.1138	-0.82
3.425	0.9027	-0.1138	-0.82
3.430	0.9027	-0.1139	-0.82
3.435	0.9027	-0.1140	-0.82
3.440	0.9024	-0.1139	-0.82
3.445	0.9023	-0.1143	-0.82
3.450	0.9020	-0.1147	-0.83
3.455	0.9021	-0.1145	-0.83
3.460	0.9020	-0.1148	-0.83
3.465	0.9016	-0.1149	-0.83
3.470	0.9016	-0.1151	-0.83
3.475	0.9014	-0.1152	-0.83
3.480	0.9012	-0.1153	-0.83
3.485	0.9009	-0.1154	-0.84
3.490	0.9014	-0.1162	-0.83
3.495	0.9013	-0.1166	-0.83
3.500	0.9011	-0.1166	-0.83
3.505	0.9014	-0.1165	-0.83
3.510	0.9010	-0.1167	-0.83
3.515	0.9015	-0.1171	-0.83
3.520	0.9012	-0.1172	-0.83
3.525	0.9016	-0.1181	-0.83
3.530	0.9021	-0.1180	-0.82
3.535	0.9020	-0.1180	-0.82
3.540	0.9022	-0.1179	-0.82
3.545	0.9027	-0.1177	-0.82
3.550	0.9023	-0.1175	-0.82
3.555	0.9026	-0.1175	-0.82
3.560	0.9026	-0.1169	-0.82

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Frequency (GHz)	**Re(S_{11})**	**Im(S_{11})**	**$\lvert S_{11}\rvert$ (dB)**
3.565	0.9025	-0.1167	-0.82
3.570	0.9020	-0.1167	-0.82
3.575	0.9020	-0.1163	-0.82
3.580	0.9016	-0.1155	-0.83
3.585	0.9014	-0.1153	-0.83
3.590	0.8994	-0.1153	-0.85
3.595	0.8989	-0.1157	-0.85
3.600	0.8986	-0.1162	-0.86
3.605	0.8982	-0.1166	-0.86
3.610	0.8981	-0.1171	-0.86
3.615	0.8979	-0.1183	-0.86
3.620	0.8977	-0.1189	-0.86
3.625	0.8980	-0.1191	-0.86
3.630	0.8979	-0.1198	-0.86
3.635	0.8985	-0.1198	-0.85
3.640	0.8989	-0.1196	-0.85
3.645	0.8989	-0.1197	-0.85
3.650	0.8993	-0.1197	-0.85
3.655	0.8992	-0.1199	-0.85
3.660	0.8994	-0.1197	-0.84
3.665	0.8994	-0.1197	-0.84
3.670	0.8993	-0.1196	-0.85
3.675	0.8995	-0.1195	-0.84
3.680	0.8993	-0.1194	-0.85
3.685	0.8993	-0.1193	-0.85
3.690	0.8993	-0.1194	-0.85
3.695	0.8993	-0.1194	-0.85
3.700	0.8992	-0.1190	-0.85
3.705	0.8990	-0.1189	-0.85
3.710	0.8987	-0.1190	-0.85
3.715	0.8984	-0.1187	-0.86
3.720	0.8980	-0.1185	-0.86
3.725	0.8977	-0.1188	-0.86
3.730	0.8970	-0.1189	-0.87
3.735	0.8964	-0.1194	-0.87
3.740	0.8960	-0.1198	-0.88
3.745	0.8958	-0.1207	-0.88
3.750	0.8958	-0.1217	-0.88
3.755	0.8963	-0.1227	-0.87
3.760	0.8970	-0.1235	-0.86
3.765	0.8980	-0.1239	-0.85
3.770	0.8995	-0.1235	-0.84
3.775	0.9001	-0.1229	-0.83
3.780	0.9011	-0.1229	-0.82
3.785	0.9007	-0.1210	-0.83
3.790	0.8996	-0.1202	-0.84

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Frequency (GHz)	**Re(S_{11})**	**Im(S_{11})**	**$\lvert S_{11}\rvert$ (dB)**
3.795	0.8994	-0.1202	-0.84
3.800	0.8993	-0.1202	-0.85
3.805	0.8984	-0.1200	-0.85
3.810	0.8980	-0.1203	-0.86
3.815	0.8973	-0.1207	-0.86
3.820	0.8974	-0.1206	-0.86
3.825	0.8970	-0.1210	-0.87
3.830	0.8970	-0.1214	-0.87
3.835	0.8927	-0.1242	-0.90
3.840	0.8971	-0.1227	-0.86
3.845	0.8973	-0.1228	-0.86
3.850	0.8970	-0.1230	-0.86
3.855	0.8973	-0.1231	-0.86
3.860	0.8972	-0.1233	-0.86
3.865	0.8971	-0.1236	-0.86
3.870	0.8974	-0.1236	-0.86
3.875	0.8972	-0.1239	-0.86
3.880	0.8973	-0.1238	-0.86
3.885	0.8973	-0.1242	-0.86
3.890	0.8974	-0.1244	-0.86
3.895	0.8973	-0.1244	-0.86
3.900	0.8973	-0.1245	-0.86
3.905	0.8972	-0.1249	-0.86
3.910	0.8972	-0.1247	-0.86
3.915	0.8972	-0.1252	-0.86
3.920	0.8973	-0.1254	-0.86
3.925	0.8972	-0.1256	-0.86
3.930	0.8972	-0.1258	-0.86
3.935	0.8971	-0.1258	-0.86
3.940	0.8971	-0.1263	-0.86
3.945	0.8968	-0.1265	-0.86
3.950	0.8968	-0.1267	-0.86
3.955	0.8969	-0.1269	-0.86
3.960	0.8968	-0.1271	-0.86
3.965	0.8967	-0.1273	-0.86
3.970	0.8967	-0.1275	-0.86
3.975	0.8968	-0.1279	-0.86
3.980	0.8966	-0.1280	-0.86
3.985	0.8966	-0.1282	-0.86
3.990	0.8964	-0.1284	-0.86
3.995	0.8963	-0.1288	-0.86
4.000	0.8962	-0.1290	-0.86
4.005	0.8962	-0.1293	-0.86
4.010	0.8961	-0.1297	-0.86
4.015	0.8961	-0.1298	-0.86
4.020	0.8958	-0.1300	-0.87

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Frequency (GHz)	**Re(S_{11})**	**Im(S_{11})**	**$\lvert S_{11}\rvert$ (dB)**
4.025	0.8956	-0.1301	-0.87
4.030	0.8957	-0.1306	-0.87
4.035	0.8956	-0.1309	-0.87
4.040	0.8954	-0.1310	-0.87
4.045	0.8950	-0.1311	-0.87
4.050	0.8950	-0.1317	-0.87
4.055	0.8947	-0.1319	-0.87
4.060	0.8947	-0.1323	-0.87
4.065	0.8942	-0.1324	-0.88
4.070	0.8944	-0.1326	-0.87
4.075	0.8942	-0.1330	-0.88
4.080	0.8943	-0.1334	-0.88
4.085	0.8939	-0.1334	-0.88
4.090	0.8937	-0.1338	-0.88
4.095	0.8938	-0.1342	-0.88
4.100	0.8936	-0.1343	-0.88
4.105	0.8931	-0.1346	-0.88
4.110	0.8930	-0.1350	-0.88
4.115	0.8930	-0.1353	-0.88
4.120	0.8926	-0.1355	-0.89
4.125	0.8928	-0.1358	-0.89
4.130	0.8925	-0.1362	-0.89
4.135	0.8921	-0.1366	-0.89
4.140	0.8917	-0.1365	-0.89
4.145	0.8917	-0.1369	-0.89
4.150	0.8914	-0.1375	-0.90
4.155	0.8910	-0.1378	-0.90
4.160	0.8907	-0.1378	-0.90
4.165	0.8904	-0.1383	-0.91
4.170	0.8900	-0.1384	-0.91
4.175	0.8894	-0.1382	-0.91
4.180	0.8897	-0.1385	-0.91
4.185	0.8891	-0.1387	-0.92
4.190	0.8890	-0.1390	-0.92
4.195	0.8889	-0.1392	-0.92
4.200	0.8888	-0.1390	-0.92
4.205	0.8888	-0.1395	-0.92
4.210	0.8887	-0.1399	-0.92
4.215	0.8886	-0.1394	-0.92
4.220	0.8885	-0.1399	-0.92
4.225	0.8884	-0.1395	-0.92
4.230	0.8881	-0.1401	-0.92
4.235	0.8881	-0.1402	-0.92
4.240	0.8884	-0.1405	-0.92
4.245	0.8882	-0.1406	-0.92
4.250	0.8880	-0.1407	-0.92

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Frequency (GHz)	**Re(S_{11})**	**Im(S_{11})**	**$\lvert S_{11}\rvert$ (dB)**
4.255	0.8881	-0.1409	-0.92
4.260	0.8878	-0.1411	-0.93
4.265	0.8879	-0.1409	-0.92
4.270	0.8877	-0.1413	-0.93
4.275	0.8877	-0.1414	-0.93
4.280	0.8877	-0.1415	-0.93
4.285	0.8876	-0.1417	-0.93
4.290	0.8876	-0.1418	-0.93
4.295	0.8875	-0.1418	-0.93
4.300	0.8875	-0.1420	-0.93
4.305	0.8875	-0.1421	-0.93
4.310	0.8874	-0.1422	-0.93
4.315	0.8873	-0.1423	-0.93
4.320	0.8873	-0.1423	-0.93
4.325	0.8872	-0.1424	-0.93
4.330	0.8872	-0.1425	-0.93
4.335	0.8871	-0.1426	-0.93
4.340	0.8870	-0.1427	-0.93
4.345	0.8869	-0.1430	-0.93
4.350	0.8868	-0.1431	-0.93
4.355	0.8868	-0.1432	-0.93
4.360	0.8868	-0.1434	-0.93
4.365	0.8868	-0.1437	-0.93
4.370	0.8868	-0.1436	-0.93
4.375	0.8871	-0.1440	-0.93
4.380	0.8871	-0.1441	-0.93
4.385	0.8869	-0.1445	-0.93
4.390	0.8867	-0.1444	-0.93
4.395	0.8870	-0.1447	-0.93
4.400	0.8872	-0.1448	-0.93
4.405	0.8874	-0.1453	-0.92
4.410	0.8874	-0.1451	-0.92
4.415	0.8873	-0.1450	-0.92
4.420	0.8877	-0.1450	-0.92
4.425	0.8878	-0.1449	-0.92
4.430	0.8875	-0.1451	-0.92
4.435	0.8874	-0.1454	-0.92
4.440	0.8874	-0.1454	-0.92
4.445	0.8873	-0.1455	-0.92
4.450	0.8868	-0.1454	-0.93
4.455	0.8870	-0.1457	-0.93
4.460	0.8866	-0.1456	-0.93
4.465	0.8866	-0.1453	-0.93
4.470	0.8868	-0.1454	-0.93
4.475	0.8868	-0.1450	-0.93
4.480	0.8870	-0.1447	-0.93

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Frequency (GHz)	**Re(S_{11})**	**Im(S_{11})**	**$\lvert S_{11}\rvert$ (dB)**
4.485	0.8868	-0.1451	-0.93
4.490	0.8874	-0.1452	-0.92
4.495	0.8876	-0.1454	-0.92
4.500	0.8878	-0.1456	-0.92
4.505	0.8880	-0.1456	-0.92
4.510	0.8884	-0.1461	-0.91
4.515	0.8887	-0.1461	-0.91
4.520	0.8888	-0.1463	-0.91
4.525	0.8891	-0.1461	-0.90
4.530	0.8891	-0.1464	-0.91
4.535	0.8895	-0.1465	-0.90
4.540	0.8893	-0.1462	-0.90
4.545	0.8893	-0.1464	-0.90
4.550	0.8893	-0.1463	-0.90
4.555	0.8895	-0.1465	-0.90
4.560	0.8896	-0.1465	-0.90
4.565	0.8898	-0.1465	-0.90
4.570	0.8897	-0.1467	-0.90
4.575	0.8898	-0.1466	-0.90
4.580	0.8899	-0.1466	-0.90
4.585	0.8899	-0.1468	-0.90
4.590	0.8899	-0.1468	-0.90
4.595	0.8900	-0.1470	-0.90
4.600	0.8900	-0.1469	-0.90
4.605	0.8899	-0.1472	-0.90
4.610	0.8899	-0.1471	-0.90
4.615	0.8898	-0.1474	-0.90
4.620	0.8897	-0.1473	-0.90
4.625	0.8898	-0.1477	-0.90
4.630	0.8897	-0.1476	-0.90
4.635	0.8895	-0.1478	-0.90
4.640	0.8896	-0.1480	-0.90
4.645	0.8894	-0.1478	-0.90
4.650	0.8896	-0.1481	-0.90
4.655	0.8893	-0.1481	-0.90
4.660	0.8894	-0.1482	-0.90
4.665	0.8894	-0.1484	-0.90
4.670	0.8894	-0.1487	-0.90
4.675	0.8892	-0.1488	-0.90
4.680	0.8892	-0.1486	-0.90
4.685	0.8892	-0.1487	-0.90
4.690	0.8895	-0.1487	-0.90
4.695	0.8891	-0.1488	-0.90
4.700	0.8894	-0.1490	-0.90
4.705	0.8888	-0.1496	-0.90
4.710	0.8892	-0.1495	-0.90

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Frequency (GHz)	**Re(S_{11})**	**Im(S_{11})**	**$\lvert S_{11}\rvert$ (dB)**
4.715	0.8892	-0.1494	-0.90
4.720	0.8890	-0.1496	-0.90
4.725	0.8890	-0.1493	-0.90
4.730	0.8890	-0.1493	-0.90
4.735	0.8893	-0.1492	-0.90
4.740	0.8891	-0.1493	-0.90
4.745	0.8887	-0.1495	-0.90
4.750	0.8888	-0.1494	-0.90
4.755	0.8888	-0.1497	-0.90
4.760	0.8884	-0.1500	-0.91
4.765	0.8887	-0.1497	-0.90
4.770	0.8882	-0.1501	-0.91
4.775	0.8878	-0.1507	-0.91
4.780	0.8876	-0.1506	-0.91
4.785	0.8882	-0.1509	-0.91
4.790	0.8874	-0.1508	-0.91
4.795	0.8874	-0.1511	-0.91
4.800	0.8874	-0.1512	-0.91
4.805	0.8868	-0.1510	-0.92
4.810	0.8875	-0.1513	-0.91
4.815	0.8870	-0.1509	-0.92
4.820	0.8867	-0.1511	-0.92
4.825	0.8870	-0.1508	-0.92
4.830	0.8866	-0.1512	-0.92
4.835	0.8870	-0.1510	-0.92
4.840	0.8869	-0.1508	-0.92
4.845	0.8867	-0.1507	-0.92
4.850	0.8867	-0.1508	-0.92
4.855	0.8867	-0.1503	-0.92
4.860	0.8866	-0.1505	-0.92
4.865	0.8868	-0.1499	-0.92
4.870	0.8865	-0.1503	-0.92
4.875	0.8867	-0.1503	-0.92
4.880	0.8868	-0.1504	-0.92
4.885	0.8865	-0.1504	-0.92
4.890	0.8867	-0.1505	-0.92
4.895	0.8866	-0.1504	-0.92
4.900	0.8868	-0.1505	-0.92
4.905	0.8868	-0.1505	-0.92
4.910	0.8868	-0.1507	-0.92
4.915	0.8869	-0.1508	-0.92
4.920	0.8870	-0.1509	-0.92
4.925	0.8870	-0.1510	-0.92
4.930	0.8870	-0.1512	-0.92
4.935	0.8870	-0.1513	-0.92
4.940	0.8872	-0.1514	-0.92

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Frequency (GHz)	**Re(S_{11})**	**Im(S_{11})**	**$\lvert S_{11}\rvert$ (dB)**
4.945	0.8871	-0.1516	-0.92
4.950	0.8872	-0.1517	-0.91
4.955	0.8874	-0.1518	-0.91
4.960	0.8873	-0.1520	-0.91
4.965	0.8873	-0.1520	-0.91
4.970	0.8875	-0.1522	-0.91
4.975	0.8876	-0.1522	-0.91
4.980	0.8876	-0.1526	-0.91
4.985	0.8874	-0.1528	-0.91
4.990	0.8877	-0.1529	-0.91
4.995	0.8878	-0.1533	-0.91
5.000	0.8879	-0.1534	-0.90
5.005	0.8879	-0.1537	-0.90
5.010	0.8878	-0.1540	-0.91
5.015	0.8879	-0.1542	-0.90
5.020	0.8877	-0.1543	-0.91
5.025	0.8877	-0.1549	-0.90
5.030	0.8876	-0.1551	-0.91
5.035	0.8876	-0.1549	-0.90
5.040	0.8875	-0.1552	-0.91
5.045	0.8873	-0.1551	-0.91
5.050	0.8871	-0.1551	-0.91
5.055	0.8872	-0.1547	-0.91
5.060	0.8874	-0.1548	-0.91
5.065	0.8873	-0.1544	-0.91
5.070	0.8875	-0.1545	-0.91
5.075	0.8876	-0.1545	-0.91
5.080	0.8876	-0.1545	-0.91
5.085	0.8876	-0.1543	-0.91
5.090	0.8881	-0.1543	-0.90
5.095	0.8882	-0.1543	-0.90
5.100	0.8883	-0.1542	-0.90
5.105	0.8887	-0.1542	-0.90
5.110	0.8886	-0.1544	-0.90
5.115	0.8884	-0.1545	-0.90
5.120	0.8888	-0.1545	-0.89
5.125	0.8887	-0.1546	-0.90
5.130	0.8889	-0.1546	-0.89
5.135	0.8889	-0.1546	-0.89
5.140	0.8892	-0.1547	-0.89
5.145	0.8891	-0.1550	-0.89
5.150	0.8894	-0.1550	-0.89
5.155	0.8893	-0.1552	-0.89
5.160	0.8893	-0.1553	-0.89
5.165	0.8894	-0.1554	-0.89
5.170	0.8893	-0.1555	-0.89

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Frequency (GHz)	**Re(S_{11})**	**Im(S_{11})**	**$\lvert S_{11}\rvert$ (dB)**
5.175	0.8893	-0.1557	-0.89
5.180	0.8893	-0.1556	-0.89
5.185	0.8894	-0.1558	-0.89
5.190	0.8895	-0.1558	-0.89
5.195	0.8893	-0.1559	-0.89
5.200	0.8894	-0.1559	-0.89
5.205	0.8894	-0.1560	-0.89
5.210	0.8894	-0.1562	-0.89
5.215	0.8895	-0.1563	-0.89
5.220	0.8894	-0.1564	-0.89
5.225	0.8894	-0.1566	-0.89
5.230	0.8893	-0.1566	-0.89
5.235	0.8894	-0.1569	-0.88
5.240	0.8892	-0.1571	-0.89
5.245	0.8893	-0.1570	-0.89
5.250	0.8893	-0.1573	-0.89
5.255	0.8892	-0.1575	-0.89
5.260	0.8889	-0.1574	-0.89
5.265	0.8890	-0.1574	-0.89
5.270	0.8888	-0.1577	-0.89
5.275	0.8885	-0.1579	-0.89
5.280	0.8885	-0.1580	-0.89
5.285	0.8882	-0.1580	-0.89
5.290	0.8881	-0.1585	-0.89
5.295	0.8878	-0.1583	-0.90
5.300	0.8877	-0.1587	-0.90
5.305	0.8875	-0.1588	-0.90
5.310	0.8874	-0.1593	-0.90
5.315	0.8873	-0.1593	-0.90
5.320	0.8868	-0.1596	-0.90
5.325	0.8870	-0.1595	-0.90
5.330	0.8868	-0.1598	-0.90
5.335	0.8868	-0.1602	-0.90
5.340	0.8868	-0.1603	-0.90
5.345	0.8872	-0.1606	-0.90
5.350	0.8866	-0.1604	-0.91
5.355	0.8863	-0.1606	-0.91
5.360	0.8859	-0.1610	-0.91
5.365	0.8858	-0.1607	-0.91
5.370	0.8857	-0.1612	-0.91
5.375	0.8855	-0.1609	-0.91
5.380	0.8855	-0.1612	-0.91
5.385	0.8853	-0.1615	-0.92
5.390	0.8851	-0.1614	-0.92
5.395	0.8847	-0.1616	-0.92
5.400	0.8846	-0.1617	-0.92

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Frequency (GHz)	**Re(S_{11})**	**Im(S_{11})**	**$\lvert S_{11}\rvert$ (dB)**
5.405	0.8844	-0.1623	-0.92
5.410	0.8842	-0.1622	-0.93
5.415	0.8834	-0.1628	-0.93
5.420	0.8838	-0.1628	-0.93
5.425	0.8834	-0.1633	-0.93
5.430	0.8833	-0.1630	-0.93
5.435	0.8830	-0.1633	-0.93
5.440	0.8831	-0.1636	-0.93
5.445	0.8827	-0.1635	-0.94
5.450	0.8827	-0.1640	-0.94
5.455	0.8828	-0.1641	-0.94
5.460	0.8824	-0.1642	-0.94
5.465	0.8823	-0.1641	-0.94
5.470	0.8822	-0.1647	-0.94
5.475	0.8818	-0.1646	-0.94
5.480	0.8819	-0.1648	-0.94
5.485	0.8816	-0.1649	-0.94
5.490	0.8815	-0.1652	-0.95
5.495	0.8812	-0.1652	-0.95
5.500	0.8811	-0.1653	-0.95
5.505	0.8809	-0.1654	-0.95
5.510	0.8806	-0.1656	-0.95
5.515	0.8804	-0.1657	-0.96
5.520	0.8801	-0.1660	-0.96
5.525	0.8801	-0.1659	-0.96
5.530	0.8800	-0.1662	-0.96
5.535	0.8798	-0.1663	-0.96
5.540	0.8798	-0.1665	-0.96
5.545	0.8795	-0.1668	-0.96
5.550	0.8794	-0.1668	-0.96
5.555	0.8793	-0.1669	-0.96
5.560	0.8792	-0.1671	-0.96
5.565	0.8792	-0.1671	-0.96
5.570	0.8792	-0.1672	-0.96
5.575	0.8791	-0.1673	-0.97
5.580	0.8791	-0.1674	-0.96
5.585	0.8789	-0.1676	-0.97
5.590	0.8787	-0.1676	-0.97
5.595	0.8785	-0.1679	-0.97
5.600	0.8784	-0.1679	-0.97
5.605	0.8782	-0.1681	-0.97
5.610	0.8780	-0.1685	-0.97
5.615	0.8776	-0.1687	-0.98
5.620	0.8776	-0.1691	-0.98
5.625	0.8778	-0.1693	-0.97
5.630	0.8775	-0.1697	-0.98

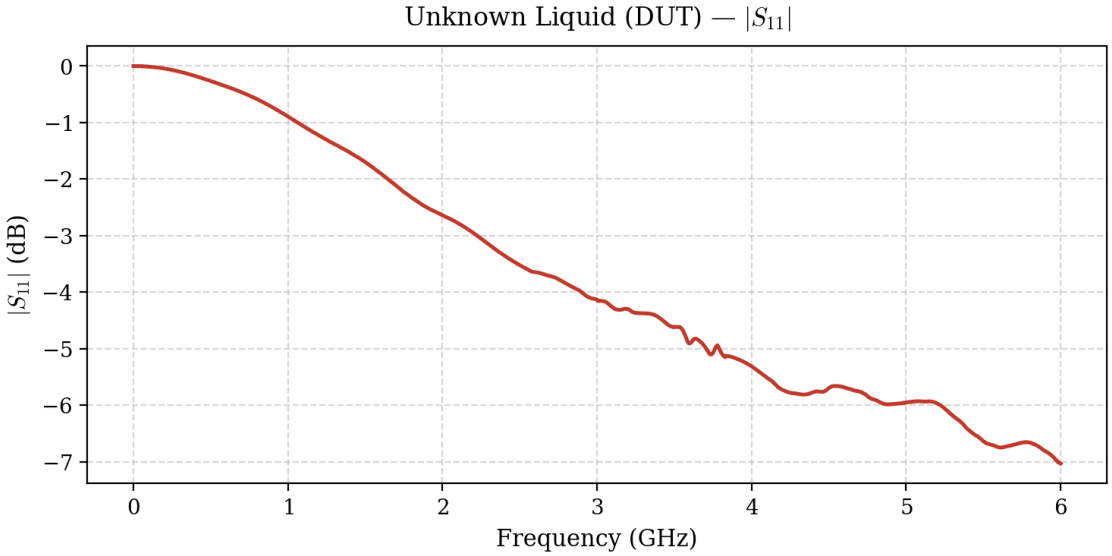
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Frequency (GHz)	**Re(S_{11})**	**Im(S_{11})**	**$\lvert S_{11}\rvert$ (dB)**
5.635	0.8775	-0.1700	-0.98
5.640	0.8773	-0.1702	-0.98
5.645	0.8774	-0.1706	-0.98
5.650	0.8777	-0.1709	-0.97
5.655	0.8773	-0.1712	-0.97
5.660	0.8776	-0.1713	-0.97
5.665	0.8776	-0.1712	-0.97
5.670	0.8775	-0.1721	-0.97
5.675	0.8773	-0.1720	-0.97
5.680	0.8775	-0.1724	-0.97
5.685	0.8772	-0.1726	-0.97
5.690	0.8773	-0.1732	-0.97
5.695	0.8777	-0.1734	-0.97
5.700	0.8772	-0.1737	-0.97
5.705	0.8770	-0.1741	-0.97
5.710	0.8773	-0.1740	-0.97
5.715	0.8771	-0.1742	-0.97
5.720	0.8773	-0.1747	-0.97
5.725	0.8772	-0.1752	-0.97
5.730	0.8772	-0.1754	-0.97
5.735	0.8774	-0.1754	-0.97
5.740	0.8777	-0.1759	-0.96
5.745	0.8780	-0.1759	-0.96
5.750	0.8779	-0.1765	-0.96
5.755	0.8778	-0.1764	-0.96
5.760	0.8777	-0.1768	-0.96
5.765	0.8779	-0.1768	-0.96
5.770	0.8776	-0.1770	-0.96
5.775	0.8779	-0.1771	-0.96
5.780	0.8779	-0.1776	-0.96
5.785	0.8781	-0.1779	-0.95
5.790	0.8781	-0.1781	-0.95
5.795	0.8781	-0.1781	-0.95
5.800	0.8781	-0.1786	-0.95
5.805	0.8780	-0.1785	-0.95
5.810	0.8781	-0.1788	-0.95
5.815	0.8781	-0.1789	-0.95
5.820	0.8780	-0.1790	-0.95
5.825	0.8779	-0.1793	-0.95
5.830	0.8779	-0.1795	-0.95
5.835	0.8778	-0.1795	-0.95
5.840	0.8777	-0.1797	-0.95
5.845	0.8777	-0.1801	-0.95
5.850	0.8775	-0.1800	-0.96
5.855	0.8774	-0.1802	-0.96
5.860	0.8772	-0.1805	-0.96

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Frequency (GHz)	Re(S_{11})	Im(S_{11})	$\lvert S_{11}\rvert$ (dB)
5.865	0.8772	-0.1805	-0.96
5.870	0.8768	-0.1807	-0.96
5.875	0.8767	-0.1809	-0.96
5.880	0.8766	-0.1811	-0.96
5.885	0.8762	-0.1812	-0.97
5.890	0.8764	-0.1817	-0.96
5.895	0.8760	-0.1818	-0.97
5.900	0.8758	-0.1818	-0.97
5.905	0.8757	-0.1820	-0.97
5.910	0.8754	-0.1823	-0.97
5.915	0.8753	-0.1825	-0.97
5.920	0.8751	-0.1828	-0.97
5.925	0.8747	-0.1829	-0.98
5.930	0.8749	-0.1833	-0.97
5.935	0.8744	-0.1833	-0.98
5.940	0.8744	-0.1835	-0.98
5.945	0.8743	-0.1831	-0.98
5.950	0.8739	-0.1838	-0.98
5.955	0.8737	-0.1835	-0.99
5.960	0.8734	-0.1838	-0.99
5.965	0.8735	-0.1840	-0.99
5.970	0.8729	-0.1842	-0.99
5.975	0.8716	-0.1840	-1.00
5.980	0.8722	-0.1839	-1.00
5.985	0.8715	-0.1848	-1.00
5.990	0.8713	-0.1851	-1.01
5.995	0.8709	-0.1854	-1.01
6.000	0.8707	-0.1854	-1.01

3.5 Unknown Liquid (DUT)



Frequency (GHz)	**Re(S_{11})**	**Im(S_{11})**	**$\lvert S_{11}\rvert$ (dB)**
0.001	1.0000	-0.0005	-0.00
0.002	1.0000	-0.0010	0.00
0.003	0.9999	-0.0013	-0.00
0.004	1.0000	-0.0017	-0.00
0.005	0.9999	-0.0022	-0.00
0.006	1.0000	-0.0025	-0.00
0.007	1.0001	-0.0030	0.00
0.008	0.9999	-0.0034	-0.00
0.009	0.9998	-0.0039	-0.00
0.010	0.9999	-0.0043	-0.00
0.011	0.9998	-0.0047	-0.00
0.012	1.0001	-0.0051	0.00
0.013	0.9999	-0.0055	-0.00
0.014	1.0000	-0.0061	0.00
0.015	0.9999	-0.0065	-0.00
0.016	0.9999	-0.0068	-0.00
0.017	0.9999	-0.0072	-0.00
0.018	0.9999	-0.0076	-0.00
0.019	0.9998	-0.0081	-0.00
0.020	0.9999	-0.0085	-0.00
0.021	1.0000	-0.0089	0.00
0.022	0.9998	-0.0093	-0.00
0.023	0.9999	-0.0098	-0.00
0.024	0.9998	-0.0101	-0.00
0.025	0.9998	-0.0106	-0.00
0.026	0.9999	-0.0109	-0.00
0.027	0.9998	-0.0115	-0.00
0.028	0.9997	-0.0119	-0.00
0.029	0.9998	-0.0122	-0.00
0.030	0.9998	-0.0127	-0.00
0.031	0.9998	-0.0131	-0.00
0.032	0.9997	-0.0135	-0.00
0.033	0.9998	-0.0139	-0.00
0.034	0.9996	-0.0143	-0.00
0.035	0.9998	-0.0148	-0.00
0.036	0.9997	-0.0151	-0.00
0.037	0.9996	-0.0156	-0.00
0.038	0.9996	-0.0161	-0.00
0.039	0.9996	-0.0164	-0.00
0.040	0.9996	-0.0168	-0.00
0.041	0.9995	-0.0174	-0.00
0.042	0.9996	-0.0177	-0.00
0.043	0.9996	-0.0180	-0.00
0.044	0.9995	-0.0186	-0.00
0.045	0.9996	-0.0191	-0.00
0.046	0.9994	-0.0195	-0.00

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Frequency (GHz)	**Re**(S_{11})	**Im**(S_{11})	$\lvert S_{11}\rvert$ **(dB)**
0.047	0.9995	-0.0198	-0.00
0.048	0.9994	-0.0203	-0.00
0.049	0.9995	-0.0207	-0.00
0.050	0.9994	-0.0210	-0.00
0.051	0.9994	-0.0215	-0.00
0.052	0.9994	-0.0220	-0.00
0.053	0.9992	-0.0224	-0.01
0.054	0.9993	-0.0228	-0.00
0.055	0.9993	-0.0231	-0.00
0.056	0.9993	-0.0236	-0.00
0.057	0.9992	-0.0240	-0.00
0.058	0.9993	-0.0245	-0.00
0.059	0.9992	-0.0248	-0.00
0.060	0.9993	-0.0253	-0.00
0.061	0.9991	-0.0257	-0.01
0.062	0.9991	-0.0261	-0.01
0.063	0.9991	-0.0267	-0.01
0.064	0.9990	-0.0270	-0.01
0.065	0.9991	-0.0274	-0.00
0.066	0.9989	-0.0278	-0.01
0.067	0.9990	-0.0283	-0.01
0.068	0.9990	-0.0287	-0.00
0.069	0.9989	-0.0290	-0.01
0.070	0.9988	-0.0294	-0.01
0.071	0.9989	-0.0298	-0.01
0.072	0.9988	-0.0304	-0.01
0.073	0.9988	-0.0307	-0.01
0.074	0.9986	-0.0313	-0.01
0.075	0.9988	-0.0317	-0.01
0.076	0.9987	-0.0320	-0.01
0.077	0.9987	-0.0324	-0.01
0.078	0.9986	-0.0329	-0.01
0.079	0.9986	-0.0333	-0.01
0.080	0.9986	-0.0337	-0.01
0.081	0.9984	-0.0341	-0.01
0.082	0.9985	-0.0346	-0.01
0.083	0.9985	-0.0349	-0.01
0.084	0.9984	-0.0353	-0.01
0.085	0.9983	-0.0357	-0.01
0.086	0.9983	-0.0362	-0.01
0.087	0.9983	-0.0365	-0.01
0.088	0.9983	-0.0371	-0.01
0.089	0.9982	-0.0375	-0.01
0.090	0.9979	-0.0378	-0.01
0.091	0.9981	-0.0384	-0.01
0.092	0.9981	-0.0386	-0.01

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Frequency (GHz)	**Re(S_{11})**	**Im(S_{11})**	**$\lvert S_{11}\rvert$ (dB)**
0.093	0.9981	-0.0391	-0.01
0.094	0.9979	-0.0395	-0.01
0.095	0.9980	-0.0400	-0.01
0.096	0.9978	-0.0404	-0.01
0.097	0.9979	-0.0408	-0.01
0.098	0.9979	-0.0412	-0.01
0.099	0.9979	-0.0415	-0.01
0.100	0.9977	-0.0419	-0.01
0.105	0.9975	-0.0441	-0.01
0.110	0.9974	-0.0462	-0.01
0.115	0.9970	-0.0483	-0.02
0.120	0.9968	-0.0503	-0.02
0.125	0.9966	-0.0525	-0.02
0.130	0.9963	-0.0544	-0.02
0.135	0.9960	-0.0565	-0.02
0.140	0.9956	-0.0587	-0.02
0.145	0.9954	-0.0606	-0.02
0.150	0.9951	-0.0628	-0.03
0.155	0.9948	-0.0647	-0.03
0.160	0.9944	-0.0669	-0.03
0.165	0.9942	-0.0688	-0.03
0.170	0.9938	-0.0709	-0.03
0.175	0.9936	-0.0730	-0.03
0.180	0.9930	-0.0751	-0.04
0.185	0.9927	-0.0770	-0.04
0.190	0.9923	-0.0792	-0.04
0.195	0.9918	-0.0813	-0.04
0.200	0.9915	-0.0833	-0.04
0.205	0.9911	-0.0852	-0.05
0.210	0.9906	-0.0873	-0.05
0.215	0.9901	-0.0894	-0.05
0.220	0.9897	-0.0914	-0.05
0.225	0.9892	-0.0933	-0.06
0.230	0.9888	-0.0954	-0.06
0.235	0.9882	-0.0974	-0.06
0.240	0.9877	-0.0995	-0.06
0.245	0.9871	-0.1015	-0.07
0.250	0.9867	-0.1036	-0.07
0.255	0.9860	-0.1056	-0.07
0.260	0.9855	-0.1075	-0.08
0.265	0.9851	-0.1096	-0.08
0.270	0.9844	-0.1117	-0.08
0.275	0.9839	-0.1135	-0.08
0.280	0.9833	-0.1155	-0.09
0.285	0.9827	-0.1175	-0.09
0.290	0.9820	-0.1195	-0.09

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Frequency (GHz)	**Re(S_{11})**	**Im(S_{11})**	**$\lvert S_{11}\rvert$ (dB)**
0.295	0.9814	-0.1215	-0.10
0.300	0.9810	-0.1236	-0.10
0.305	0.9803	-0.1255	-0.10
0.310	0.9795	-0.1274	-0.11
0.315	0.9788	-0.1293	-0.11
0.320	0.9782	-0.1313	-0.11
0.325	0.9775	-0.1332	-0.12
0.330	0.9768	-0.1352	-0.12
0.335	0.9762	-0.1371	-0.12
0.340	0.9755	-0.1389	-0.13
0.345	0.9747	-0.1409	-0.13
0.350	0.9739	-0.1429	-0.14
0.355	0.9734	-0.1447	-0.14
0.360	0.9725	-0.1467	-0.14
0.365	0.9719	-0.1485	-0.15
0.370	0.9711	-0.1505	-0.15
0.375	0.9704	-0.1523	-0.16
0.380	0.9695	-0.1542	-0.16
0.385	0.9688	-0.1562	-0.16
0.390	0.9681	-0.1579	-0.17
0.395	0.9672	-0.1599	-0.17
0.400	0.9665	-0.1616	-0.18
0.405	0.9657	-0.1635	-0.18
0.410	0.9649	-0.1652	-0.18
0.415	0.9642	-0.1671	-0.19
0.420	0.9633	-0.1688	-0.19
0.425	0.9626	-0.1707	-0.20
0.430	0.9617	-0.1725	-0.20
0.435	0.9609	-0.1744	-0.21
0.440	0.9601	-0.1759	-0.21
0.445	0.9592	-0.1777	-0.21
0.450	0.9585	-0.1795	-0.22
0.455	0.9577	-0.1812	-0.22
0.460	0.9567	-0.1829	-0.23
0.465	0.9559	-0.1847	-0.23
0.470	0.9549	-0.1864	-0.24
0.475	0.9541	-0.1880	-0.24
0.480	0.9533	-0.1898	-0.25
0.485	0.9524	-0.1916	-0.25
0.490	0.9516	-0.1933	-0.26
0.495	0.9508	-0.1948	-0.26
0.500	0.9498	-0.1965	-0.27
0.505	0.9488	-0.1982	-0.27
0.510	0.9481	-0.1998	-0.27
0.515	0.9472	-0.2014	-0.28
0.520	0.9464	-0.2031	-0.28

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Frequency (GHz)	**Re(S_{11})**	**Im(S_{11})**	**$\lvert S_{11}\rvert$ (dB)**
0.525	0.9455	-0.2047	-0.29
0.530	0.9446	-0.2063	-0.29
0.535	0.9438	-0.2080	-0.30
0.540	0.9429	-0.2095	-0.30
0.545	0.9420	-0.2111	-0.31
0.550	0.9411	-0.2127	-0.31
0.555	0.9402	-0.2143	-0.32
0.560	0.9393	-0.2159	-0.32
0.565	0.9384	-0.2174	-0.33
0.570	0.9375	-0.2189	-0.33
0.575	0.9365	-0.2205	-0.34
0.580	0.9357	-0.2222	-0.34
0.585	0.9348	-0.2236	-0.34
0.590	0.9338	-0.2252	-0.35
0.595	0.9329	-0.2267	-0.35
0.600	0.9319	-0.2282	-0.36
0.605	0.9310	-0.2297	-0.36
0.610	0.9301	-0.2312	-0.37
0.615	0.9291	-0.2327	-0.37
0.620	0.9281	-0.2342	-0.38
0.625	0.9271	-0.2356	-0.39
0.630	0.9263	-0.2371	-0.39
0.635	0.9253	-0.2385	-0.40
0.640	0.9243	-0.2401	-0.40
0.645	0.9234	-0.2415	-0.40
0.650	0.9225	-0.2430	-0.41
0.655	0.9215	-0.2443	-0.41
0.660	0.9204	-0.2458	-0.42
0.665	0.9195	-0.2472	-0.43
0.670	0.9185	-0.2486	-0.43
0.675	0.9176	-0.2500	-0.44
0.680	0.9165	-0.2515	-0.44
0.685	0.9155	-0.2526	-0.45
0.690	0.9148	-0.2542	-0.45
0.695	0.9136	-0.2556	-0.46
0.700	0.9125	-0.2568	-0.46
0.705	0.9116	-0.2582	-0.47
0.710	0.9105	-0.2596	-0.47
0.715	0.9095	-0.2609	-0.48
0.720	0.9084	-0.2623	-0.49
0.725	0.9073	-0.2637	-0.49
0.730	0.9063	-0.2651	-0.50
0.735	0.9053	-0.2664	-0.50
0.740	0.9042	-0.2678	-0.51
0.745	0.9032	-0.2691	-0.52
0.750	0.9020	-0.2704	-0.52

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Frequency (GHz)	**Re(S_{11})**	**Im(S_{11})**	**$\lvert S_{11}\rvert$ (dB)**
0.755	0.9010	-0.2717	-0.53
0.760	0.8999	-0.2728	-0.53
0.765	0.8988	-0.2742	-0.54
0.770	0.8976	-0.2755	-0.55
0.775	0.8965	-0.2768	-0.55
0.780	0.8954	-0.2782	-0.56
0.785	0.8944	-0.2794	-0.56
0.790	0.8931	-0.2808	-0.57
0.795	0.8919	-0.2821	-0.58
0.800	0.8909	-0.2833	-0.59
0.805	0.8898	-0.2845	-0.59
0.810	0.8884	-0.2859	-0.60
0.815	0.8874	-0.2872	-0.61
0.820	0.8861	-0.2883	-0.61
0.825	0.8849	-0.2896	-0.62
0.830	0.8838	-0.2909	-0.63
0.835	0.8827	-0.2920	-0.63
0.840	0.8813	-0.2934	-0.64
0.845	0.8801	-0.2946	-0.65
0.850	0.8789	-0.2958	-0.66
0.855	0.8777	-0.2970	-0.66
0.860	0.8765	-0.2983	-0.67
0.865	0.8752	-0.2995	-0.68
0.870	0.8739	-0.3007	-0.68
0.875	0.8727	-0.3017	-0.69
0.880	0.8715	-0.3032	-0.70
0.885	0.8702	-0.3044	-0.71
0.890	0.8689	-0.3055	-0.71
0.895	0.8676	-0.3068	-0.72
0.900	0.8663	-0.3080	-0.73
0.905	0.8650	-0.3091	-0.74
0.910	0.8638	-0.3104	-0.74
0.915	0.8625	-0.3116	-0.75
0.920	0.8611	-0.3126	-0.76
0.925	0.8599	-0.3138	-0.77
0.930	0.8585	-0.3149	-0.78
0.935	0.8573	-0.3161	-0.78
0.940	0.8559	-0.3172	-0.79
0.945	0.8546	-0.3184	-0.80
0.950	0.8533	-0.3197	-0.81
0.955	0.8518	-0.3207	-0.82
0.960	0.8504	-0.3218	-0.83
0.965	0.8492	-0.3230	-0.83
0.970	0.8479	-0.3241	-0.84
0.975	0.8466	-0.3252	-0.85
0.980	0.8451	-0.3264	-0.86

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Frequency (GHz)	**Re(S_{11})**	**Im(S_{11})**	**$\lvert S_{11}\rvert$ (dB)**
0.985	0.8437	-0.3274	-0.87
0.990	0.8424	-0.3285	-0.87
0.995	0.8410	-0.3295	-0.88
1.000	0.8397	-0.3306	-0.89
1.005	0.8384	-0.3316	-0.90
1.010	0.8370	-0.3326	-0.91
1.015	0.8356	-0.3336	-0.92
1.020	0.8341	-0.3346	-0.93
1.025	0.8328	-0.3356	-0.94
1.030	0.8315	-0.3366	-0.94
1.035	0.8301	-0.3377	-0.95
1.040	0.8288	-0.3386	-0.96
1.045	0.8273	-0.3396	-0.97
1.050	0.8261	-0.3405	-0.98
1.055	0.8247	-0.3416	-0.99
1.060	0.8235	-0.3424	-0.99
1.065	0.8221	-0.3433	-1.00
1.070	0.8208	-0.3442	-1.01
1.075	0.8194	-0.3453	-1.02
1.080	0.8181	-0.3462	-1.03
1.085	0.8168	-0.3471	-1.04
1.090	0.8154	-0.3480	-1.05
1.095	0.8141	-0.3489	-1.05
1.100	0.8127	-0.3497	-1.06
1.105	0.8115	-0.3505	-1.07
1.110	0.8102	-0.3513	-1.08
1.115	0.8089	-0.3522	-1.09
1.120	0.8076	-0.3530	-1.10
1.125	0.8062	-0.3538	-1.11
1.130	0.8050	-0.3546	-1.11
1.135	0.8037	-0.3553	-1.12
1.140	0.8026	-0.3561	-1.13
1.145	0.8014	-0.3569	-1.14
1.150	0.8000	-0.3576	-1.15
1.155	0.7988	-0.3583	-1.16
1.160	0.7975	-0.3591	-1.16
1.165	0.7964	-0.3599	-1.17
1.170	0.7952	-0.3606	-1.18
1.175	0.7939	-0.3612	-1.19
1.180	0.7929	-0.3619	-1.19
1.185	0.7917	-0.3626	-1.20
1.190	0.7905	-0.3632	-1.21
1.195	0.7894	-0.3639	-1.22
1.200	0.7882	-0.3645	-1.23
1.205	0.7871	-0.3650	-1.23
1.210	0.7860	-0.3657	-1.24

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Frequency (GHz)	**Re(S_{11})**	**Im(S_{11})**	**$\lvert S_{11}\rvert$ (dB)**
1.215	0.7847	-0.3662	-1.25
1.220	0.7835	-0.3669	-1.26
1.225	0.7824	-0.3674	-1.27
1.230	0.7813	-0.3680	-1.27
1.235	0.7802	-0.3685	-1.28
1.240	0.7791	-0.3692	-1.29
1.245	0.7781	-0.3696	-1.30
1.250	0.7770	-0.3701	-1.30
1.255	0.7758	-0.3707	-1.31
1.260	0.7747	-0.3712	-1.32
1.265	0.7737	-0.3717	-1.33
1.270	0.7725	-0.3722	-1.34
1.275	0.7715	-0.3727	-1.34
1.280	0.7705	-0.3731	-1.35
1.285	0.7694	-0.3736	-1.36
1.290	0.7684	-0.3742	-1.36
1.295	0.7672	-0.3748	-1.37
1.300	0.7662	-0.3754	-1.38
1.305	0.7650	-0.3759	-1.39
1.310	0.7642	-0.3763	-1.39
1.315	0.7632	-0.3768	-1.40
1.320	0.7622	-0.3772	-1.41
1.325	0.7612	-0.3777	-1.41
1.330	0.7601	-0.3779	-1.42
1.335	0.7590	-0.3785	-1.43
1.340	0.7581	-0.3789	-1.44
1.345	0.7570	-0.3793	-1.45
1.350	0.7560	-0.3798	-1.45
1.355	0.7550	-0.3801	-1.46
1.360	0.7540	-0.3806	-1.47
1.365	0.7531	-0.3811	-1.47
1.370	0.7520	-0.3815	-1.48
1.375	0.7509	-0.3818	-1.49
1.380	0.7500	-0.3824	-1.50
1.385	0.7489	-0.3829	-1.50
1.390	0.7476	-0.3832	-1.51
1.395	0.7466	-0.3835	-1.52
1.400	0.7455	-0.3840	-1.53
1.405	0.7445	-0.3845	-1.54
1.410	0.7432	-0.3849	-1.55
1.415	0.7421	-0.3854	-1.55
1.420	0.7410	-0.3858	-1.56
1.425	0.7398	-0.3864	-1.57
1.430	0.7387	-0.3867	-1.58
1.435	0.7377	-0.3874	-1.58
1.440	0.7364	-0.3878	-1.59

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Frequency (GHz)	**Re(S_{11})**	**Im(S_{11})**	**$\lvert S_{11}\rvert$ (dB)**
1.445	0.7353	-0.3883	-1.60
1.450	0.7342	-0.3886	-1.61
1.455	0.7331	-0.3893	-1.62
1.460	0.7320	-0.3897	-1.63
1.465	0.7308	-0.3902	-1.63
1.470	0.7295	-0.3906	-1.64
1.475	0.7284	-0.3911	-1.65
1.480	0.7272	-0.3916	-1.66
1.485	0.7259	-0.3922	-1.67
1.490	0.7246	-0.3928	-1.68
1.495	0.7234	-0.3932	-1.69
1.500	0.7223	-0.3935	-1.70
1.505	0.7208	-0.3940	-1.71
1.510	0.7196	-0.3945	-1.72
1.515	0.7182	-0.3951	-1.73
1.520	0.7171	-0.3955	-1.74
1.525	0.7156	-0.3960	-1.75
1.530	0.7143	-0.3965	-1.76
1.535	0.7131	-0.3970	-1.76
1.540	0.7118	-0.3975	-1.77
1.545	0.7103	-0.3979	-1.79
1.550	0.7090	-0.3984	-1.80
1.555	0.7076	-0.3989	-1.81
1.560	0.7061	-0.3995	-1.82
1.565	0.7047	-0.4000	-1.83
1.570	0.7033	-0.4004	-1.84
1.575	0.7020	-0.4009	-1.85
1.580	0.7006	-0.4014	-1.86
1.585	0.6992	-0.4020	-1.87
1.590	0.6977	-0.4024	-1.88
1.595	0.6964	-0.4030	-1.89
1.600	0.6951	-0.4036	-1.90
1.605	0.6936	-0.4040	-1.91
1.610	0.6923	-0.4044	-1.92
1.615	0.6907	-0.4050	-1.93
1.620	0.6893	-0.4055	-1.94
1.625	0.6880	-0.4059	-1.95
1.630	0.6865	-0.4063	-1.96
1.635	0.6851	-0.4070	-1.97
1.640	0.6836	-0.4073	-1.99
1.645	0.6822	-0.4078	-1.99
1.650	0.6807	-0.4083	-2.01
1.655	0.6792	-0.4088	-2.02
1.660	0.6778	-0.4091	-2.03
1.665	0.6766	-0.4098	-2.04
1.670	0.6749	-0.4101	-2.05

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Frequency (GHz)	**Re(S_{11})**	**Im(S_{11})**	**$\lvert S_{11}\rvert$ (dB)**
1.675	0.6735	-0.4106	-2.06
1.680	0.6722	-0.4109	-2.07
1.685	0.6708	-0.4113	-2.08
1.690	0.6695	-0.4117	-2.09
1.695	0.6680	-0.4120	-2.10
1.700	0.6666	-0.4123	-2.12
1.705	0.6652	-0.4126	-2.13
1.710	0.6637	-0.4131	-2.14
1.715	0.6624	-0.4135	-2.15
1.720	0.6609	-0.4138	-2.16
1.725	0.6597	-0.4143	-2.17
1.730	0.6581	-0.4147	-2.18
1.735	0.6567	-0.4148	-2.19
1.740	0.6552	-0.4152	-2.21
1.745	0.6539	-0.4155	-2.22
1.750	0.6526	-0.4158	-2.23
1.755	0.6513	-0.4162	-2.24
1.760	0.6500	-0.4164	-2.25
1.765	0.6488	-0.4169	-2.26
1.770	0.6476	-0.4171	-2.27
1.775	0.6465	-0.4175	-2.27
1.780	0.6454	-0.4176	-2.28
1.785	0.6445	-0.4177	-2.29
1.790	0.6430	-0.4179	-2.31
1.795	0.6418	-0.4180	-2.32
1.800	0.6406	-0.4182	-2.33
1.805	0.6393	-0.4185	-2.34
1.810	0.6382	-0.4189	-2.35
1.815	0.6371	-0.4191	-2.35
1.820	0.6361	-0.4193	-2.36
1.825	0.6351	-0.4194	-2.37
1.830	0.6340	-0.4197	-2.38
1.835	0.6329	-0.4197	-2.39
1.840	0.6318	-0.4200	-2.40
1.845	0.6308	-0.4200	-2.41
1.850	0.6298	-0.4199	-2.42
1.855	0.6287	-0.4201	-2.43
1.860	0.6277	-0.4201	-2.44
1.865	0.6267	-0.4201	-2.45
1.870	0.6256	-0.4203	-2.46
1.875	0.6246	-0.4205	-2.47
1.880	0.6237	-0.4206	-2.47
1.885	0.6229	-0.4209	-2.48
1.890	0.6221	-0.4210	-2.49
1.895	0.6213	-0.4208	-2.49
1.900	0.6205	-0.4209	-2.50

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Frequency (GHz)	**Re(S_{11})**	**Im(S_{11})**	**$\lvert S_{11}\rvert$ (dB)**
1.905	0.6195	-0.4210	-2.51
1.910	0.6187	-0.4211	-2.52
1.915	0.6178	-0.4213	-2.53
1.920	0.6170	-0.4214	-2.53
1.925	0.6162	-0.4214	-2.54
1.930	0.6154	-0.4215	-2.55
1.935	0.6146	-0.4217	-2.55
1.940	0.6139	-0.4216	-2.56
1.945	0.6131	-0.4216	-2.57
1.950	0.6125	-0.4218	-2.57
1.955	0.6119	-0.4218	-2.58
1.960	0.6112	-0.4218	-2.58
1.965	0.6104	-0.4219	-2.59
1.970	0.6096	-0.4218	-2.60
1.975	0.6089	-0.4218	-2.61
1.980	0.6083	-0.4217	-2.61
1.985	0.6077	-0.4219	-2.62
1.990	0.6070	-0.4217	-2.63
1.995	0.6062	-0.4217	-2.63
2.000	0.6054	-0.4218	-2.64
2.005	0.6048	-0.4218	-2.65
2.010	0.6042	-0.4219	-2.65
2.015	0.6033	-0.4219	-2.66
2.020	0.6027	-0.4219	-2.67
2.025	0.6021	-0.4220	-2.67
2.030	0.6014	-0.4220	-2.68
2.035	0.6007	-0.4219	-2.69
2.040	0.6001	-0.4218	-2.69
2.045	0.5992	-0.4218	-2.70
2.050	0.5986	-0.4220	-2.71
2.055	0.5979	-0.4221	-2.71
2.060	0.5970	-0.4219	-2.72
2.065	0.5964	-0.4220	-2.73
2.070	0.5958	-0.4222	-2.73
2.075	0.5949	-0.4221	-2.74
2.080	0.5943	-0.4221	-2.75
2.085	0.5934	-0.4221	-2.75
2.090	0.5925	-0.4220	-2.76
2.095	0.5917	-0.4221	-2.77
2.100	0.5908	-0.4222	-2.78
2.105	0.5901	-0.4223	-2.79
2.110	0.5891	-0.4223	-2.79
2.115	0.5884	-0.4225	-2.80
2.120	0.5874	-0.4225	-2.81
2.125	0.5864	-0.4227	-2.82
2.130	0.5855	-0.4227	-2.83

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Frequency (GHz)	**Re(S_{11})**	**Im(S_{11})**	**$\lvert S_{11}\rvert$ (dB)**
2.135	0.5846	-0.4229	-2.83
2.140	0.5836	-0.4231	-2.84
2.145	0.5827	-0.4231	-2.85
2.150	0.5818	-0.4233	-2.86
2.155	0.5807	-0.4234	-2.87
2.160	0.5798	-0.4235	-2.88
2.165	0.5787	-0.4236	-2.89
2.170	0.5777	-0.4237	-2.90
2.175	0.5767	-0.4239	-2.91
2.180	0.5756	-0.4240	-2.92
2.185	0.5746	-0.4242	-2.92
2.190	0.5735	-0.4242	-2.93
2.195	0.5724	-0.4243	-2.94
2.200	0.5714	-0.4247	-2.95
2.205	0.5702	-0.4247	-2.96
2.210	0.5691	-0.4248	-2.97
2.215	0.5679	-0.4251	-2.98
2.220	0.5668	-0.4253	-2.99
2.225	0.5656	-0.4254	-3.00
2.230	0.5646	-0.4254	-3.01
2.235	0.5634	-0.4256	-3.02
2.240	0.5622	-0.4258	-3.03
2.245	0.5610	-0.4260	-3.04
2.250	0.5598	-0.4263	-3.05
2.255	0.5587	-0.4265	-3.06
2.260	0.5576	-0.4267	-3.07
2.265	0.5564	-0.4269	-3.08
2.270	0.5551	-0.4271	-3.09
2.275	0.5540	-0.4271	-3.10
2.280	0.5528	-0.4273	-3.11
2.285	0.5517	-0.4275	-3.12
2.290	0.5504	-0.4276	-3.14
2.295	0.5494	-0.4279	-3.14
2.300	0.5482	-0.4281	-3.15
2.305	0.5469	-0.4283	-3.16
2.310	0.5458	-0.4285	-3.17
2.315	0.5445	-0.4287	-3.18
2.320	0.5433	-0.4286	-3.20
2.325	0.5422	-0.4290	-3.21
2.330	0.5410	-0.4291	-3.22
2.335	0.5398	-0.4293	-3.23
2.340	0.5386	-0.4296	-3.24
2.345	0.5374	-0.4297	-3.25
2.350	0.5363	-0.4297	-3.26
2.355	0.5354	-0.4299	-3.27
2.360	0.5342	-0.4300	-3.28

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Frequency (GHz)	**Re(S_{11})**	**Im(S_{11})**	**$\lvert S_{11}\rvert$ (dB)**
2.365	0.5331	-0.4301	-3.29
2.370	0.5322	-0.4302	-3.29
2.375	0.5310	-0.4305	-3.30
2.380	0.5299	-0.4306	-3.31
2.385	0.5289	-0.4307	-3.32
2.390	0.5279	-0.4308	-3.33
2.395	0.5268	-0.4308	-3.34
2.400	0.5258	-0.4310	-3.35
2.405	0.5249	-0.4310	-3.36
2.410	0.5237	-0.4312	-3.37
2.415	0.5228	-0.4312	-3.38
2.420	0.5219	-0.4313	-3.39
2.425	0.5209	-0.4313	-3.40
2.430	0.5200	-0.4314	-3.41
2.435	0.5191	-0.4314	-3.41
2.440	0.5182	-0.4315	-3.42
2.445	0.5173	-0.4314	-3.43
2.450	0.5166	-0.4315	-3.44
2.455	0.5157	-0.4314	-3.45
2.460	0.5149	-0.4313	-3.46
2.465	0.5141	-0.4313	-3.46
2.470	0.5132	-0.4312	-3.47
2.475	0.5125	-0.4313	-3.48
2.480	0.5117	-0.4312	-3.49
2.485	0.5110	-0.4311	-3.50
2.490	0.5102	-0.4310	-3.51
2.495	0.5095	-0.4310	-3.51
2.500	0.5087	-0.4309	-3.52
2.505	0.5080	-0.4309	-3.53
2.510	0.5073	-0.4308	-3.54
2.515	0.5065	-0.4308	-3.54
2.520	0.5058	-0.4307	-3.55
2.525	0.5050	-0.4307	-3.56
2.530	0.5043	-0.4307	-3.57
2.535	0.5035	-0.4306	-3.58
2.540	0.5030	-0.4307	-3.58
2.545	0.5023	-0.4307	-3.59
2.550	0.5017	-0.4306	-3.59
2.555	0.5010	-0.4303	-3.60
2.560	0.5003	-0.4302	-3.61
2.565	0.4999	-0.4298	-3.62
2.570	0.4994	-0.4295	-3.63
2.575	0.4991	-0.4290	-3.63
2.580	0.4990	-0.4286	-3.64
2.585	0.4989	-0.4284	-3.64
2.590	0.4988	-0.4283	-3.64

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Frequency (GHz)	**Re(S_{11})**	**Im(S_{11})**	**$\lvert S_{11}\rvert$ (dB)**
2.595	0.4986	-0.4283	-3.64
2.600	0.4985	-0.4283	-3.65
2.605	0.4983	-0.4282	-3.65
2.610	0.4981	-0.4283	-3.65
2.615	0.4978	-0.4283	-3.65
2.620	0.4977	-0.4283	-3.65
2.625	0.4975	-0.4281	-3.66
2.630	0.4971	-0.4282	-3.66
2.635	0.4968	-0.4280	-3.67
2.640	0.4965	-0.4278	-3.67
2.645	0.4962	-0.4277	-3.67
2.650	0.4958	-0.4277	-3.68
2.655	0.4956	-0.4276	-3.68
2.660	0.4952	-0.4274	-3.69
2.665	0.4949	-0.4273	-3.69
2.670	0.4945	-0.4272	-3.70
2.675	0.4940	-0.4272	-3.70
2.680	0.4938	-0.4271	-3.70
2.685	0.4934	-0.4272	-3.71
2.690	0.4931	-0.4269	-3.71
2.695	0.4928	-0.4269	-3.71
2.700	0.4925	-0.4268	-3.72
2.705	0.4923	-0.4268	-3.72
2.710	0.4920	-0.4266	-3.73
2.715	0.4917	-0.4265	-3.73
2.720	0.4915	-0.4263	-3.73
2.725	0.4912	-0.4261	-3.74
2.730	0.4909	-0.4258	-3.74
2.735	0.4906	-0.4255	-3.75
2.740	0.4902	-0.4251	-3.76
2.745	0.4898	-0.4249	-3.76
2.750	0.4894	-0.4246	-3.77
2.755	0.4890	-0.4242	-3.78
2.760	0.4885	-0.4239	-3.79
2.765	0.4880	-0.4238	-3.79
2.770	0.4874	-0.4233	-3.80
2.775	0.4870	-0.4231	-3.81
2.780	0.4864	-0.4228	-3.82
2.785	0.4859	-0.4226	-3.82
2.790	0.4852	-0.4223	-3.83
2.795	0.4845	-0.4221	-3.84
2.800	0.4841	-0.4220	-3.85
2.805	0.4835	-0.4217	-3.86
2.810	0.4828	-0.4216	-3.86
2.815	0.4822	-0.4214	-3.87
2.820	0.4815	-0.4214	-3.88

(continued on next page)

Frequency (GHz)	**Re**(S_{11})	**Im**(S_{11})	$\lvert S_{11}\rvert$ **(dB)**
2.825	0.4811	-0.4213	-3.88
2.830	0.4804	-0.4211	-3.89
2.835	0.4798	-0.4211	-3.90
2.840	0.4792	-0.4210	-3.91
2.845	0.4785	-0.4208	-3.91
2.850	0.4779	-0.4207	-3.92
2.855	0.4773	-0.4205	-3.93
2.860	0.4768	-0.4205	-3.94
2.865	0.4762	-0.4204	-3.94
2.870	0.4758	-0.4203	-3.95
2.875	0.4756	-0.4200	-3.95
2.880	0.4749	-0.4198	-3.96
2.885	0.4745	-0.4193	-3.97
2.890	0.4738	-0.4192	-3.98
2.895	0.4731	-0.4186	-3.99
2.900	0.4724	-0.4183	-4.00
2.905	0.4716	-0.4180	-4.01
2.910	0.4707	-0.4176	-4.02
2.915	0.4699	-0.4174	-4.03
2.920	0.4690	-0.4174	-4.04
2.925	0.4681	-0.4173	-4.05
2.930	0.4673	-0.4170	-4.06
2.935	0.4664	-0.4170	-4.07
2.940	0.4657	-0.4172	-4.08
2.945	0.4649	-0.4171	-4.09
2.950	0.4642	-0.4171	-4.10
2.955	0.4636	-0.4173	-4.10
2.960	0.4630	-0.4174	-4.11
2.965	0.4624	-0.4175	-4.11
2.970	0.4620	-0.4175	-4.11
2.975	0.4617	-0.4177	-4.12
2.980	0.4611	-0.4179	-4.12
2.985	0.4609	-0.4179	-4.12
2.990	0.4608	-0.4178	-4.12
2.995	0.4606	-0.4178	-4.13
3.000	0.4604	-0.4177	-4.13
3.005	0.4585	-0.4172	-4.15
3.010	0.4582	-0.4176	-4.15
3.015	0.4581	-0.4176	-4.15
3.020	0.4578	-0.4176	-4.16
3.025	0.4579	-0.4180	-4.15
3.030	0.4580	-0.4179	-4.15
3.035	0.4581	-0.4175	-4.16
3.040	0.4582	-0.4170	-4.16
3.045	0.4583	-0.4167	-4.16
3.050	0.4583	-0.4162	-4.17

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Frequency (GHz)	**Re(S_{11})**	**Im(S_{11})**	**$\lvert S_{11}\rvert$ (dB)**
3.055	0.4583	-0.4157	-4.17
3.060	0.4582	-0.4151	-4.18
3.065	0.4577	-0.4143	-4.19
3.070	0.4574	-0.4137	-4.20
3.075	0.4569	-0.4131	-4.21
3.080	0.4565	-0.4123	-4.22
3.085	0.4559	-0.4119	-4.23
3.090	0.4552	-0.4112	-4.24
3.095	0.4546	-0.4109	-4.25
3.100	0.4538	-0.4105	-4.27
3.105	0.4531	-0.4104	-4.27
3.110	0.4524	-0.4101	-4.28
3.115	0.4515	-0.4100	-4.30
3.120	0.4509	-0.4101	-4.30
3.125	0.4503	-0.4103	-4.30
3.130	0.4497	-0.4105	-4.31
3.135	0.4493	-0.4109	-4.31
3.140	0.4489	-0.4111	-4.31
3.145	0.4485	-0.4113	-4.31
3.150	0.4484	-0.4116	-4.31
3.155	0.4483	-0.4119	-4.31
3.160	0.4482	-0.4121	-4.31
3.165	0.4483	-0.4123	-4.31
3.170	0.4486	-0.4125	-4.30
3.175	0.4486	-0.4129	-4.30
3.180	0.4491	-0.4128	-4.29
3.185	0.4492	-0.4124	-4.30
3.190	0.4496	-0.4120	-4.30
3.195	0.4498	-0.4118	-4.30
3.200	0.4498	-0.4112	-4.30
3.205	0.4500	-0.4109	-4.30
3.210	0.4498	-0.4100	-4.31
3.215	0.4498	-0.4095	-4.32
3.220	0.4493	-0.4086	-4.33
3.225	0.4486	-0.4082	-4.34
3.230	0.4481	-0.4078	-4.35
3.235	0.4477	-0.4080	-4.36
3.240	0.4469	-0.4080	-4.36
3.245	0.4470	-0.4080	-4.36
3.250	0.4464	-0.4080	-4.37
3.255	0.4459	-0.4081	-4.37
3.260	0.4459	-0.4084	-4.37
3.265	0.4455	-0.4087	-4.37
3.270	0.4455	-0.4085	-4.37
3.275	0.4454	-0.4087	-4.37
3.280	0.4453	-0.4087	-4.37

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Frequency (GHz)	**Re(S_{11})**	**Im(S_{11})**	**$\lvert S_{11}\rvert$ (dB)**
3.285	0.4451	-0.4089	-4.37
3.290	0.4450	-0.4091	-4.37
3.295	0.4448	-0.4092	-4.37
3.300	0.4448	-0.4092	-4.37
3.305	0.4446	-0.4091	-4.38
3.310	0.4444	-0.4093	-4.38
3.315	0.4445	-0.4095	-4.37
3.320	0.4444	-0.4092	-4.38
3.325	0.4443	-0.4094	-4.38
3.330	0.4443	-0.4093	-4.38
3.335	0.4442	-0.4092	-4.38
3.340	0.4442	-0.4092	-4.38
3.345	0.4443	-0.4089	-4.38
3.350	0.4441	-0.4087	-4.39
3.355	0.4439	-0.4085	-4.39
3.360	0.4440	-0.4081	-4.39
3.365	0.4440	-0.4078	-4.40
3.370	0.4440	-0.4076	-4.40
3.375	0.4434	-0.4071	-4.41
3.380	0.4434	-0.4067	-4.41
3.385	0.4431	-0.4063	-4.42
3.390	0.4425	-0.4059	-4.43
3.395	0.4424	-0.4055	-4.44
3.400	0.4417	-0.4050	-4.45
3.405	0.4415	-0.4045	-4.45
3.410	0.4409	-0.4039	-4.47
3.415	0.4403	-0.4034	-4.48
3.420	0.4398	-0.4032	-4.49
3.425	0.4391	-0.4027	-4.50
3.430	0.4386	-0.4024	-4.51
3.435	0.4377	-0.4021	-4.52
3.440	0.4370	-0.4017	-4.53
3.445	0.4361	-0.4016	-4.54
3.450	0.4353	-0.4010	-4.56
3.455	0.4346	-0.4007	-4.57
3.460	0.4335	-0.4007	-4.58
3.465	0.4330	-0.4007	-4.58
3.470	0.4320	-0.4007	-4.59
3.475	0.4312	-0.4006	-4.60
3.480	0.4306	-0.4009	-4.61
3.485	0.4299	-0.4014	-4.61
3.490	0.4294	-0.4014	-4.62
3.495	0.4287	-0.4015	-4.62
3.500	0.4288	-0.4021	-4.61
3.505	0.4285	-0.4024	-4.62
3.510	0.4283	-0.4025	-4.62

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Frequency (GHz)	**Re(S_{11})**	**Im(S_{11})**	**$\lvert S_{11}\rvert$ (dB)**
3.515	0.4285	-0.4024	-4.62
3.520	0.4285	-0.4023	-4.62
3.525	0.4284	-0.4022	-4.62
3.530	0.4290	-0.4020	-4.61
3.535	0.4293	-0.4014	-4.62
3.540	0.4297	-0.4003	-4.62
3.545	0.4296	-0.3990	-4.64
3.550	0.4296	-0.3976	-4.65
3.555	0.4293	-0.3958	-4.67
3.560	0.4285	-0.3943	-4.70
3.565	0.4272	-0.3928	-4.73
3.570	0.4256	-0.3913	-4.76
3.575	0.4241	-0.3898	-4.79
3.580	0.4215	-0.3885	-4.83
3.585	0.4190	-0.3884	-4.86
3.590	0.4164	-0.3885	-4.89
3.595	0.4141	-0.3895	-4.91
3.600	0.4124	-0.3910	-4.91
3.605	0.4114	-0.3927	-4.90
3.610	0.4111	-0.3941	-4.89
3.615	0.4115	-0.3956	-4.87
3.620	0.4119	-0.3965	-4.86
3.625	0.4127	-0.3978	-4.83
3.630	0.4137	-0.3972	-4.83
3.635	0.4146	-0.3968	-4.82
3.640	0.4152	-0.3963	-4.82
3.645	0.4156	-0.3952	-4.83
3.650	0.4157	-0.3946	-4.83
3.655	0.4157	-0.3937	-4.84
3.660	0.4156	-0.3928	-4.85
3.665	0.4151	-0.3920	-4.87
3.670	0.4150	-0.3911	-4.88
3.675	0.4148	-0.3902	-4.89
3.680	0.4145	-0.3893	-4.90
3.685	0.4138	-0.3885	-4.92
3.690	0.4133	-0.3875	-4.94
3.695	0.4125	-0.3867	-4.95
3.700	0.4117	-0.3860	-4.97
3.705	0.4105	-0.3851	-4.99
3.710	0.4092	-0.3845	-5.01
3.715	0.4079	-0.3839	-5.03
3.720	0.4062	-0.3837	-5.06
3.725	0.4047	-0.3838	-5.07
3.730	0.4023	-0.3839	-5.10
3.735	0.4008	-0.3849	-5.10
3.740	0.3994	-0.3865	-5.10

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Frequency (GHz)	**Re(S_{11})**	**Im(S_{11})**	**$\lvert S_{11}\rvert$ (dB)**
3.745	0.3985	-0.3879	-5.10
3.750	0.3982	-0.3898	-5.08
3.755	0.3983	-0.3916	-5.06
3.760	0.3994	-0.3935	-5.03
3.765	0.4011	-0.3948	-4.99
3.770	0.4032	-0.3949	-4.97
3.775	0.4062	-0.3935	-4.95
3.780	0.4075	-0.3931	-4.94
3.785	0.4086	-0.3900	-4.96
3.790	0.4079	-0.3878	-4.99
3.795	0.4071	-0.3860	-5.02
3.800	0.4057	-0.3849	-5.05
3.805	0.4044	-0.3839	-5.07
3.810	0.4028	-0.3835	-5.10
3.815	0.4014	-0.3831	-5.12
3.820	0.4000	-0.3832	-5.13
3.825	0.3989	-0.3836	-5.14
3.830	0.3979	-0.3837	-5.15
3.835	0.3975	-0.3864	-5.12
3.840	0.3970	-0.3867	-5.13
3.845	0.3969	-0.3867	-5.13
3.850	0.3964	-0.3867	-5.13
3.855	0.3962	-0.3867	-5.14
3.860	0.3959	-0.3868	-5.14
3.865	0.3955	-0.3867	-5.14
3.870	0.3953	-0.3869	-5.14
3.875	0.3949	-0.3867	-5.15
3.880	0.3946	-0.3868	-5.15
3.885	0.3943	-0.3866	-5.16
3.890	0.3938	-0.3867	-5.16
3.895	0.3932	-0.3868	-5.17
3.900	0.3928	-0.3868	-5.17
3.905	0.3923	-0.3866	-5.18
3.910	0.3917	-0.3867	-5.19
3.915	0.3911	-0.3869	-5.19
3.920	0.3906	-0.3869	-5.20
3.925	0.3901	-0.3870	-5.20
3.930	0.3897	-0.3871	-5.20
3.935	0.3889	-0.3870	-5.21
3.940	0.3883	-0.3871	-5.22
3.945	0.3876	-0.3871	-5.23
3.950	0.3871	-0.3872	-5.23
3.955	0.3864	-0.3872	-5.24
3.960	0.3857	-0.3872	-5.25
3.965	0.3851	-0.3872	-5.25
3.970	0.3844	-0.3872	-5.26

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Frequency (GHz)	**Re(S_{11})**	**Im(S_{11})**	**$\lvert S_{11}\rvert$ (dB)**
3.975	0.3836	-0.3872	-5.27
3.980	0.3829	-0.3874	-5.28
3.985	0.3822	-0.3873	-5.29
3.990	0.3815	-0.3872	-5.29
3.995	0.3806	-0.3874	-5.30
4.000	0.3798	-0.3875	-5.31
4.005	0.3791	-0.3874	-5.32
4.010	0.3782	-0.3874	-5.33
4.015	0.3774	-0.3872	-5.34
4.020	0.3765	-0.3873	-5.35
4.025	0.3757	-0.3873	-5.36
4.030	0.3749	-0.3873	-5.37
4.035	0.3740	-0.3872	-5.38
4.040	0.3731	-0.3872	-5.39
4.045	0.3723	-0.3872	-5.40
4.050	0.3714	-0.3870	-5.41
4.055	0.3706	-0.3871	-5.42
4.060	0.3696	-0.3870	-5.43
4.065	0.3687	-0.3871	-5.44
4.070	0.3679	-0.3869	-5.45
4.075	0.3670	-0.3869	-5.46
4.080	0.3661	-0.3869	-5.47
4.085	0.3653	-0.3869	-5.48
4.090	0.3643	-0.3868	-5.49
4.095	0.3636	-0.3868	-5.50
4.100	0.3626	-0.3867	-5.51
4.105	0.3615	-0.3866	-5.53
4.110	0.3607	-0.3868	-5.53
4.115	0.3599	-0.3870	-5.54
4.120	0.3589	-0.3869	-5.55
4.125	0.3581	-0.3870	-5.56
4.130	0.3570	-0.3874	-5.57
4.135	0.3558	-0.3873	-5.58
4.140	0.3547	-0.3875	-5.59
4.145	0.3534	-0.3872	-5.61
4.150	0.3524	-0.3875	-5.62
4.155	0.3511	-0.3873	-5.63
4.160	0.3501	-0.3871	-5.65
4.165	0.3492	-0.3867	-5.66
4.170	0.3482	-0.3865	-5.68
4.175	0.3476	-0.3863	-5.68
4.180	0.3469	-0.3862	-5.70
4.185	0.3463	-0.3859	-5.70
4.190	0.3457	-0.3858	-5.71
4.195	0.3450	-0.3857	-5.72
4.200	0.3446	-0.3857	-5.73

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Frequency (GHz)	**Re(S_{11})**	**Im(S_{11})**	**$\lvert S_{11}\rvert$ (dB)**
4.205	0.3441	-0.3857	-5.73
4.210	0.3435	-0.3856	-5.74
4.215	0.3432	-0.3855	-5.75
4.220	0.3427	-0.3854	-5.75
4.225	0.3422	-0.3853	-5.76
4.230	0.3419	-0.3854	-5.76
4.235	0.3415	-0.3853	-5.77
4.240	0.3410	-0.3852	-5.77
4.245	0.3407	-0.3852	-5.78
4.250	0.3405	-0.3853	-5.78
4.255	0.3401	-0.3852	-5.78
4.260	0.3399	-0.3852	-5.79
4.265	0.3396	-0.3854	-5.79
4.270	0.3396	-0.3854	-5.79
4.275	0.3394	-0.3854	-5.79
4.280	0.3392	-0.3854	-5.79
4.285	0.3391	-0.3855	-5.79
4.290	0.3389	-0.3853	-5.80
4.295	0.3388	-0.3853	-5.80
4.300	0.3386	-0.3853	-5.80
4.305	0.3384	-0.3852	-5.80
4.310	0.3382	-0.3852	-5.81
4.315	0.3380	-0.3852	-5.81
4.320	0.3379	-0.3851	-5.81
4.325	0.3377	-0.3851	-5.81
4.330	0.3376	-0.3851	-5.81
4.335	0.3373	-0.3853	-5.81
4.340	0.3372	-0.3855	-5.81
4.345	0.3371	-0.3856	-5.81
4.350	0.3370	-0.3859	-5.81
4.355	0.3369	-0.3860	-5.81
4.360	0.3368	-0.3862	-5.81
4.365	0.3369	-0.3864	-5.80
4.370	0.3370	-0.3867	-5.80
4.375	0.3370	-0.3869	-5.80
4.380	0.3373	-0.3870	-5.79
4.385	0.3377	-0.3875	-5.78
4.390	0.3379	-0.3874	-5.78
4.395	0.3382	-0.3877	-5.77
4.400	0.3384	-0.3877	-5.77
4.405	0.3388	-0.3877	-5.77
4.410	0.3391	-0.3877	-5.76
4.415	0.3395	-0.3876	-5.76
4.420	0.3396	-0.3878	-5.76
4.425	0.3400	-0.3874	-5.76
4.430	0.3401	-0.3874	-5.76

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Frequency (GHz)	**Re(S_{11})**	**Im(S_{11})**	**$\lvert S_{11}\rvert$ (dB)**
4.435	0.3402	-0.3871	-5.76
4.440	0.3403	-0.3869	-5.76
4.445	0.3404	-0.3866	-5.76
4.450	0.3405	-0.3865	-5.76
4.455	0.3406	-0.3864	-5.76
4.460	0.3406	-0.3863	-5.76
4.465	0.3409	-0.3862	-5.76
4.470	0.3413	-0.3862	-5.76
4.475	0.3418	-0.3863	-5.75
4.480	0.3422	-0.3865	-5.74
4.485	0.3427	-0.3869	-5.73
4.490	0.3433	-0.3873	-5.72
4.495	0.3439	-0.3877	-5.71
4.500	0.3444	-0.3881	-5.70
4.505	0.3449	-0.3883	-5.69
4.510	0.3455	-0.3886	-5.68
4.515	0.3462	-0.3886	-5.67
4.520	0.3467	-0.3887	-5.67
4.525	0.3471	-0.3885	-5.66
4.530	0.3474	-0.3883	-5.66
4.535	0.3479	-0.3882	-5.66
4.540	0.3480	-0.3879	-5.66
4.545	0.3482	-0.3876	-5.66
4.550	0.3483	-0.3875	-5.66
4.555	0.3486	-0.3874	-5.66
4.560	0.3486	-0.3873	-5.66
4.565	0.3487	-0.3871	-5.66
4.570	0.3489	-0.3869	-5.66
4.575	0.3488	-0.3865	-5.67
4.580	0.3489	-0.3864	-5.67
4.585	0.3488	-0.3862	-5.67
4.590	0.3490	-0.3858	-5.68
4.595	0.3488	-0.3857	-5.68
4.600	0.3487	-0.3856	-5.68
4.605	0.3486	-0.3853	-5.69
4.610	0.3484	-0.3849	-5.69
4.615	0.3481	-0.3848	-5.70
4.620	0.3479	-0.3845	-5.70
4.625	0.3480	-0.3844	-5.70
4.630	0.3476	-0.3843	-5.71
4.635	0.3475	-0.3842	-5.71
4.640	0.3474	-0.3840	-5.72
4.645	0.3472	-0.3837	-5.72
4.650	0.3471	-0.3835	-5.73
4.655	0.3469	-0.3833	-5.73
4.660	0.3468	-0.3829	-5.74

(continued on next page)

Frequency (GHz)	**Re(S_{11})**	**Im(S_{11})**	**$\lvert S_{11}\rvert$ (dB)**
4.665	0.3470	-0.3827	-5.74
4.670	0.3465	-0.3827	-5.74
4.675	0.3465	-0.3823	-5.75
4.680	0.3466	-0.3821	-5.75
4.685	0.3467	-0.3820	-5.75
4.690	0.3466	-0.3818	-5.75
4.695	0.3467	-0.3816	-5.75
4.700	0.3464	-0.3813	-5.76
4.705	0.3466	-0.3806	-5.77
4.710	0.3465	-0.3808	-5.77
4.715	0.3465	-0.3803	-5.77
4.720	0.3462	-0.3795	-5.79
4.725	0.3461	-0.3794	-5.79
4.730	0.3459	-0.3789	-5.80
4.735	0.3454	-0.3789	-5.80
4.740	0.3453	-0.3781	-5.81
4.745	0.3444	-0.3779	-5.83
4.750	0.3441	-0.3775	-5.83
4.755	0.3431	-0.3772	-5.85
4.760	0.3424	-0.3771	-5.86
4.765	0.3418	-0.3769	-5.87
4.770	0.3409	-0.3769	-5.88
4.775	0.3405	-0.3767	-5.89
4.780	0.3403	-0.3764	-5.89
4.785	0.3399	-0.3768	-5.89
4.790	0.3399	-0.3765	-5.90
4.795	0.3396	-0.3762	-5.90
4.800	0.3398	-0.3759	-5.90
4.805	0.3397	-0.3755	-5.91
4.810	0.3398	-0.3750	-5.92
4.815	0.3398	-0.3743	-5.92
4.820	0.3398	-0.3739	-5.93
4.825	0.3398	-0.3733	-5.94
4.830	0.3397	-0.3727	-5.95
4.835	0.3397	-0.3722	-5.95
4.840	0.3398	-0.3717	-5.96
4.845	0.3395	-0.3712	-5.97
4.850	0.3395	-0.3709	-5.97
4.855	0.3396	-0.3708	-5.97
4.860	0.3395	-0.3703	-5.98
4.865	0.3394	-0.3703	-5.98
4.870	0.3392	-0.3701	-5.99
4.875	0.3393	-0.3701	-5.98
4.880	0.3393	-0.3701	-5.98
4.885	0.3392	-0.3700	-5.99
4.890	0.3392	-0.3702	-5.98

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Frequency (GHz)	**Re(S_{11})**	**Im(S_{11})**	**$\lvert S_{11}\rvert$ (dB)**
4.895	0.3391	-0.3702	-5.99
4.900	0.3391	-0.3703	-5.98
4.905	0.3392	-0.3705	-5.98
4.910	0.3391	-0.3706	-5.98
4.915	0.3391	-0.3706	-5.98
4.920	0.3390	-0.3708	-5.98
4.925	0.3390	-0.3709	-5.98
4.930	0.3389	-0.3710	-5.98
4.935	0.3388	-0.3712	-5.98
4.940	0.3389	-0.3713	-5.97
4.945	0.3389	-0.3715	-5.97
4.950	0.3388	-0.3716	-5.97
4.955	0.3389	-0.3717	-5.97
4.960	0.3388	-0.3718	-5.97
4.965	0.3387	-0.3719	-5.97
4.970	0.3389	-0.3720	-5.96
4.975	0.3387	-0.3723	-5.96
4.980	0.3386	-0.3723	-5.96
4.985	0.3388	-0.3727	-5.96
4.990	0.3387	-0.3726	-5.96
4.995	0.3387	-0.3729	-5.96
5.000	0.3387	-0.3731	-5.95
5.005	0.3388	-0.3732	-5.95
5.010	0.3387	-0.3733	-5.95
5.015	0.3388	-0.3735	-5.95
5.020	0.3388	-0.3735	-5.95
5.025	0.3391	-0.3737	-5.94
5.030	0.3390	-0.3736	-5.94
5.035	0.3391	-0.3736	-5.94
5.040	0.3393	-0.3736	-5.94
5.045	0.3397	-0.3737	-5.93
5.050	0.3400	-0.3732	-5.94
5.055	0.3401	-0.3733	-5.93
5.060	0.3405	-0.3730	-5.93
5.065	0.3408	-0.3730	-5.93
5.070	0.3410	-0.3726	-5.93
5.075	0.3412	-0.3726	-5.93
5.080	0.3417	-0.3722	-5.93
5.085	0.3417	-0.3721	-5.93
5.090	0.3421	-0.3718	-5.93
5.095	0.3422	-0.3716	-5.93
5.100	0.3423	-0.3713	-5.93
5.105	0.3424	-0.3712	-5.93
5.110	0.3425	-0.3710	-5.94
5.115	0.3427	-0.3708	-5.94
5.120	0.3428	-0.3707	-5.93

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Frequency (GHz)	**Re(S_{11})**	**Im(S_{11})**	**$\lvert S_{11}\rvert$ (dB)**
5.125	0.3429	-0.3706	-5.94
5.130	0.3431	-0.3705	-5.94
5.135	0.3430	-0.3703	-5.94
5.140	0.3434	-0.3703	-5.93
5.145	0.3436	-0.3702	-5.93
5.150	0.3437	-0.3702	-5.93
5.155	0.3437	-0.3701	-5.93
5.160	0.3438	-0.3700	-5.93
5.165	0.3436	-0.3699	-5.94
5.170	0.3436	-0.3698	-5.94
5.175	0.3435	-0.3697	-5.94
5.180	0.3433	-0.3694	-5.95
5.185	0.3431	-0.3695	-5.95
5.190	0.3428	-0.3692	-5.95
5.195	0.3424	-0.3690	-5.96
5.200	0.3422	-0.3688	-5.97
5.205	0.3419	-0.3686	-5.97
5.210	0.3415	-0.3683	-5.98
5.215	0.3411	-0.3681	-5.99
5.220	0.3407	-0.3679	-6.00
5.225	0.3402	-0.3676	-6.00
5.230	0.3398	-0.3672	-6.02
5.235	0.3394	-0.3668	-6.02
5.240	0.3389	-0.3664	-6.04
5.245	0.3384	-0.3661	-6.05
5.250	0.3378	-0.3657	-6.06
5.255	0.3372	-0.3652	-6.07
5.260	0.3367	-0.3648	-6.08
5.265	0.3363	-0.3646	-6.09
5.270	0.3355	-0.3640	-6.11
5.275	0.3348	-0.3638	-6.12
5.280	0.3341	-0.3634	-6.13
5.285	0.3334	-0.3630	-6.14
5.290	0.3330	-0.3628	-6.15
5.295	0.3319	-0.3623	-6.17
5.300	0.3316	-0.3623	-6.18
5.305	0.3310	-0.3616	-6.19
5.310	0.3302	-0.3617	-6.20
5.315	0.3295	-0.3616	-6.21
5.320	0.3287	-0.3612	-6.23
5.325	0.3285	-0.3610	-6.23
5.330	0.3279	-0.3606	-6.24
5.335	0.3272	-0.3603	-6.25
5.340	0.3267	-0.3602	-6.26
5.345	0.3262	-0.3599	-6.27
5.350	0.3257	-0.3595	-6.28

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Frequency (GHz)	**Re(S_{11})**	**Im(S_{11})**	**$\lvert S_{11}\rvert$ (dB)**
5.355	0.3250	-0.3590	-6.30
5.360	0.3246	-0.3586	-6.31
5.365	0.3238	-0.3582	-6.32
5.370	0.3232	-0.3577	-6.34
5.375	0.3227	-0.3574	-6.35
5.380	0.3217	-0.3566	-6.37
5.385	0.3210	-0.3562	-6.38
5.390	0.3200	-0.3558	-6.40
5.395	0.3193	-0.3556	-6.41
5.400	0.3185	-0.3553	-6.43
5.405	0.3178	-0.3553	-6.44
5.410	0.3170	-0.3552	-6.45
5.415	0.3164	-0.3547	-6.46
5.420	0.3155	-0.3548	-6.47
5.425	0.3148	-0.3546	-6.48
5.430	0.3141	-0.3543	-6.49
5.435	0.3134	-0.3543	-6.50
5.440	0.3128	-0.3541	-6.51
5.445	0.3122	-0.3540	-6.52
5.450	0.3116	-0.3539	-6.53
5.455	0.3110	-0.3535	-6.54
5.460	0.3107	-0.3537	-6.54
5.465	0.3101	-0.3535	-6.55
5.470	0.3096	-0.3532	-6.56
5.475	0.3089	-0.3529	-6.58
5.480	0.3084	-0.3526	-6.59
5.485	0.3077	-0.3525	-6.60
5.490	0.3070	-0.3522	-6.61
5.495	0.3061	-0.3518	-6.63
5.500	0.3056	-0.3517	-6.63
5.505	0.3049	-0.3516	-6.64
5.510	0.3040	-0.3515	-6.66
5.515	0.3033	-0.3515	-6.66
5.520	0.3026	-0.3515	-6.67
5.525	0.3022	-0.3516	-6.68
5.530	0.3017	-0.3518	-6.68
5.535	0.3011	-0.3518	-6.69
5.540	0.3006	-0.3520	-6.69
5.545	0.3003	-0.3520	-6.69
5.550	0.2998	-0.3521	-6.70
5.555	0.2994	-0.3522	-6.70
5.560	0.2990	-0.3522	-6.71
5.565	0.2986	-0.3523	-6.71
5.570	0.2982	-0.3524	-6.71
5.575	0.2977	-0.3525	-6.72
5.580	0.2973	-0.3525	-6.72

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Frequency (GHz)	**Re(S_{11})**	**Im(S_{11})**	**$\lvert S_{11}\rvert$ (dB)**
5.585	0.2968	-0.3524	-6.73
5.590	0.2962	-0.3525	-6.74
5.595	0.2956	-0.3527	-6.74
5.600	0.2949	-0.3530	-6.74
5.605	0.2945	-0.3532	-6.75
5.610	0.2938	-0.3535	-6.75
5.615	0.2934	-0.3540	-6.75
5.620	0.2929	-0.3545	-6.75
5.625	0.2926	-0.3550	-6.74
5.630	0.2920	-0.3555	-6.74
5.635	0.2918	-0.3560	-6.74
5.640	0.2915	-0.3564	-6.74
5.645	0.2913	-0.3568	-6.73
5.650	0.2911	-0.3572	-6.73
5.655	0.2910	-0.3576	-6.73
5.660	0.2908	-0.3579	-6.72
5.665	0.2905	-0.3581	-6.72
5.670	0.2903	-0.3588	-6.72
5.675	0.2902	-0.3590	-6.71
5.680	0.2898	-0.3595	-6.71
5.685	0.2896	-0.3599	-6.71
5.690	0.2894	-0.3602	-6.71
5.695	0.2894	-0.3606	-6.70
5.700	0.2892	-0.3609	-6.70
5.705	0.2892	-0.3613	-6.69
5.710	0.2891	-0.3615	-6.69
5.715	0.2890	-0.3618	-6.69
5.720	0.2888	-0.3623	-6.68
5.725	0.2889	-0.3625	-6.68
5.730	0.2888	-0.3628	-6.68
5.735	0.2888	-0.3631	-6.67
5.740	0.2886	-0.3632	-6.67
5.745	0.2887	-0.3634	-6.67
5.750	0.2888	-0.3637	-6.66
5.755	0.2887	-0.3638	-6.66
5.760	0.2886	-0.3640	-6.66
5.765	0.2887	-0.3642	-6.66
5.770	0.2886	-0.3641	-6.66
5.775	0.2887	-0.3643	-6.65
5.780	0.2885	-0.3643	-6.66
5.785	0.2885	-0.3644	-6.65
5.790	0.2885	-0.3644	-6.66
5.795	0.2882	-0.3643	-6.66
5.800	0.2883	-0.3642	-6.66
5.805	0.2882	-0.3640	-6.66
5.810	0.2879	-0.3639	-6.67

(continued on next page)

Frequency (GHz)	**Re**(S_{11})	**Im**(S_{11})	$\lvert S_{11}\rvert$ **(dB)**
5.815	0.2878	-0.3637	-6.67
5.820	0.2875	-0.3635	-6.68
5.825	0.2872	-0.3634	-6.69
5.830	0.2871	-0.3631	-6.69
5.835	0.2868	-0.3629	-6.70
5.840	0.2866	-0.3627	-6.70
5.845	0.2862	-0.3624	-6.71
5.850	0.2859	-0.3620	-6.72
5.855	0.2857	-0.3619	-6.73
5.860	0.2853	-0.3616	-6.73
5.865	0.2850	-0.3611	-6.74
5.870	0.2843	-0.3610	-6.75
5.875	0.2839	-0.3607	-6.76
5.880	0.2835	-0.3604	-6.77
5.885	0.2828	-0.3601	-6.79
5.890	0.2823	-0.3599	-6.79
5.895	0.2820	-0.3596	-6.80
5.900	0.2814	-0.3593	-6.81
5.905	0.2811	-0.3590	-6.82
5.910	0.2807	-0.3592	-6.82
5.915	0.2801	-0.3590	-6.83
5.920	0.2798	-0.3587	-6.84
5.925	0.2794	-0.3582	-6.85
5.930	0.2791	-0.3580	-6.86
5.935	0.2787	-0.3574	-6.87
5.940	0.2782	-0.3573	-6.88
5.945	0.2777	-0.3565	-6.90
5.950	0.2772	-0.3563	-6.91
5.955	0.2769	-0.3559	-6.92
5.960	0.2762	-0.3553	-6.94
5.965	0.2750	-0.3549	-6.96
5.970	0.2745	-0.3545	-6.97
5.975	0.2737	-0.3545	-6.98
5.980	0.2725	-0.3541	-7.00
5.985	0.2718	-0.3538	-7.01
5.990	0.2713	-0.3540	-7.01
5.995	0.2706	-0.3540	-7.02
6.000	0.2700	-0.3538	-7.03